NE350

Radiological Engineering Design

NE350 Radiological Engineering Design

ABET eCourseBook AY2025-2026

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United States Military Academy

NE350
Radiological Engineering Design
Course Syllabus
AY 26-2

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Contents

Part I

Course Administrative Information

1 Course Introduction

NE350 Radiological Engineering Design focuses on nuclear engineering systems including radiation protection, shielding, and the uses of radioactive sources in industrial processes. Specific topics emphasize the operation of radiation detectors, shielding principles, health effects of radiation, radiological dispersion devices, and nuclear incidents. A semester-long design project applies the concepts presented in this course to the solution of practical problems. Groups will be able to apply engineering and mathematics while ensuring regulatory requirements, social and political acceptance, and economic parameters are met.

2 Course Objectives

2.1 Course Direction

Cadets assigned to the course will successfully complete one Written Partial Review (WPR), one Term End Examination (TEE), one Course Wide Writ, one Design Project with associated Intermediate Progress Reviews (IPRs), one Lab, and all assigned instructor assignments to prepare for follow-on missions at the United States Military Academy and as officers in the U.S. Army of the 21st Century.

2.2 Nuclear Engineering Program Educational Objectives

The nuclear engineering program educational objectives describe the career and professional accomplishments that the NE program is preparing you to achieve after graduation. The Department of Physics & Nuclear Engineering provides an integral and essential capability supporting the Academy's mission to develop leaders of the Army of the 21st Century. You are enabled through these objectives:

1. As Army leaders, graduates **solve complex, multi-disciplinary problems** for the Army and the Nation.

2. Graduates demonstrate the necessary **leadership and teamwork** skills to lead and work on multi-disciplinary teams.

3. Graduates are prepared to provide **nuclear and radiological engineering expertise** to the Army.

4. Graduates **communicate** effectively, orally and in writing; providing clear instructions to subordinates and feedback to supervisors.

5. Graduates continue to **grow intellectually and professionally**—as Army officers and as engineers.

2.3 Nuclear Engineering ABET Student Outcomes

Cadets must attain:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

3. An ability to communicate effectively with a range of audiences.

4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

6. An ability to develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.

7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

2.4 Assessment of Student Outcomes

1. The Department of Physics and Nuclear Engineering has identified that the NE student outcomes 1, 2, 3, 4, and 5 are best supported by NE350. They are highlighted in Table 1 below with the primary embedded indicators that will be used to assess the outcomes against established rubrics.

Table 1: Primary embedded indicators for nuclear engineering student outcomes.

Primary Embedded Indicators	1	2	3	4	5
TEE Problem	X				
NE Design Project IPRs 2 and 3					X
Problem Set	X				
NE Design Project IPR 1				X	X
NE Design Project Team Grade					X
Final NE Design Project Brief			X		X
Final NE Design Report Submission		X			X

2. Assessed NE student outcomes are achieved through the accomplishment of course block objectives. The specific course block objectives for NE350 are:

a. Understand the biological effects of ionizing radiation and design systems that incorporate radiation protection.

b. Understand and implement the nuclear engineering design process.

c. Understand how radiation detection devices function and how to interpret their outputs.

d. Understand medical and industrial applications for radiation technology other than commercial power generation.

e. Understand non-proliferation and how the concerns of security and safety play into the global discussion about the safe use of nuclear technology.

3. Additionally, NE350 will be used to assess cadet attainment of the USMA Academic Program Goals (APG) and What Graduates Can Do (WGCD) statements for the Core

Engineering Sequence (CES). Specifically, NE350 will assess attainment of the following WGCD statements:

5.4.1 In an environment of uncertainty and change, identify needs that can be fulfilled via engineered solutions.

5.4.3 Develop appropriate alternatives to solve an engineering problem.

5.4.5 Communicate an engineering solution to both technical and non-technical audiences and recommend an engineering solution to a decision maker.

Table 2: Primary evidence for attainment of WGCD statements.

Primary Direct Evidence	WGCD		
	5.4.1	5.4.3	5.4.5
NE Design Project IPR 1	X		
NE Design Project IPRs 2 and 3		X	
NEDP Final Submission / Presentation			X

3 Conduct and Expectations

3.1 What You Can Expect from the Course Director

This course is a “Design” course. As an engineer, it is my obligation to my students that I be strict on your computations and calculations throughout the term. I will provide instruction and guidance to facilitate your understanding of the concepts in this course with the intent of ensuring you are capable of producing a feasible, acceptable design. This course will enforce a learning model in which each class has a purpose. I will be fair in all aspects of grading, providing rubrics for your project submissions to match levels exhibiting “Failure” to “Excellence.” All other graded assignments or in-class testable submissions will have an associated cut sheet. As needed or as directed, I will provide additional instruction to supplement your learning.

3.2 Cadet Expectations

To perform well in this course, cadets must demonstrate proficiency in the Nuclear Engineering Student Outcomes. In supporting these outcomes and in preparation for each assigned lesson, cadets may be tasked to **Study, Review,** or **Read**. Cadets are required to know the **Learning Objectives**, study and know the **Lesson Notes**, know the **Vocabulary**, and complete the **Homework**. The WPR, problem sets, the Design Project, the Lab Report, and the TEE must be completed to receive course credit. Specific definitions of some of these tasks are listed below.

1. “**Study”** means that cadets are required to read these sections and are expected to recall the material and demonstrate the ability to comprehend, analyze, and synthesize the material by solving the assigned homework problems and discussing and evaluating common applications.

2. “**Review”** means that cadets are required to read these sections that have been assigned previously. Cadets are expected to recall this material and demonstrate the ability to comprehend, analyze, and synthesize the material by solving the assigned homework problems and discussing and evaluating common applications.

3. **“Read”** means that cadets are required to read these sections and are expected to gain insights from the material to help them better understand “study” and “review” requirements. Read assignments are typically given to provide further depth, clarity, and examples to assigned learning objectives.

4. Know the **Lesson Learning Objectives (LOs).** The learning objectives provide an outline with which to **study and prepare** for each lesson. Studying nuclear engineering means much more than simply reading the words. Contained within each text reading assignment are derivations and sample problems. After understanding the derivations presented, it is beneficial to work through the sample problems to grasp the usefulness of the equations presented. All lesson sheets in the course outline have learning objectives that are designed to guide cadet study of the assignments. The learning objectives use two types of highlighted action verbs that are defined as follows:

a. **List** or **Explain** or **Sketch** or **Draw** or **Describe** is used to indicate an objective that will be evaluated by short answer questions. In short answer questions, cadets must be able to correctly state pertinent facts about a given situation and then state a conclusion. Cadets may be required to sketch a graph to help illustrate the conclusion.

b. **Calculate** or **Construct** or **Solve** or **Derive** is used to indicate an objective that will be evaluated by the methods used above or by performing algebraic, diagrammatic, and/or numerical calculations. Cadets may be required to synthesize multiple concepts in order to solve a problem.

3.3 Sexual Harassment Policy

Prevention and elimination of sexual harassment and sexual assault is the number one priority at the United States Military Academy. The Army "I. A. M. STRONG" campaign combats sexual harassment and sexual assault by engaging all Soldiers to "Intervene," "Act," and "Motivate" others in preventing sexual assault before they occur.

1. Sexual harassment and sexual assault are reprehensible and hurt other team members and wound our Army. It is a betrayal of the trust inherent in serving in the Profession of Arms. Your instructor is dedicated to fostering a positive SHARP environment.

2. If you need to speak to anyone about SHARP, contact your instructor, the unit representative for Cadets Against Sexual Harassment/Assault (CASH/A), the West Point Sexual Assault 24/7 Helpline: (845) 659-7467, or the DoD Safe Helpline: (877) 995-5247.

3.4 Chain of Command

Below is the Chain of Command for NE350. The Department of Physics and Nuclear Engineering has an open-door policy. Feel free to address problems or concerns through the Chain of Command.

1. NE350 Course Director: MAJ Mashl, BH472, ext.

2. Nuclear Engineering Program Director: Prof. Allen, BH472B, ext. 3548

3. Head of the Department: COL Chapman, BH480B, ext. 3901

4 Course Grading and Requirements

Course block objectives and nuclear engineering student outcomes will be accomplished through lesson learning objectives over a 30-lesson course of instruction. Course block objectives will be evaluated at multiple times throughout the course with multiple assessment tools to include homework, Course Wide Writs, the WPR, a Lab, briefs, and the TEE. Through multiple evaluations, an assessment will be made as to whether or not cadets are accomplishing the course block objectives and if the course of instruction is sufficiently preparing cadets for success. Assessment of course block objectives will encompass evaluations of **reinforcement** and **extension** of concepts, principles, and problem solving.

Reinforcement. Cadets are required to know the concepts and principles and how to solve problems using these concepts and principles. Cadets demonstrate reinforcement knowledge by knowing the base-line knowledge outlined in the lesson learning objectives, lesson vocabulary, and lesson notes. Evaluation mechanisms will be primarily writs, WPRs, and the TEE.

Extension. Synthesizing multiple concepts and principles beyond reinforcement is extension. Successful extension demonstrates a higher understanding of the material and requires creative and critical thinking to solve more complex problems. Evaluation mechanisms will be primarily graded homework, the design project, and bonus problems.

4.1 Course Point Allocation

The following Table shows the Marks distribution for this course.

Table 3: NE350 Marks Distribution

Basis of Grade	Possible Marks	Percentage of Total
Course Wide Writ	100	5%
Problem Sets (3 @ 75)	225	11.25%
Course Written Partial Review	200	10%
Fundamental Knowledge Writs (2 @ 60 each)	120	6%
NE Design Project IPRs (3 @ 100 each)	300	15%
NE Design Project Final Submission	150	7.5%
NE Design Project Presentation	100	5%
NE Design Project Team Grade	50	2.5%
Laboratory Program (1 Exercise)	170	8.5%
TEE	500	25%
Instructor Marks	85	4.25%
Total Marks Available	**2000**	**100%**

1. Homework. Suggested daily homework problem(s) are assigned for each model class lesson. Cadets are not required to submit suggested homework problem(s); however, it is necessary that cadets do them. Cadets may be asked to present solutions to suggested homework problems in class. These presentations may be graded and included in part of the instructor marks for class participation. Graded assignments (e.g. course problem sets, design

submissions, laboratory report, etc.) are due NLT the date time group indicated on the Course Schedule on Canvas or as a hard copy turn-in as applicable. Cadets must arrange to turn in homework assignments if absent.

Discussion is encouraged and collaboration is authorized on all homework assignments. Collaboration must be documented IAW the USMA policies described in the pamphlet from the Office of the Dean entitled *Documentation and Acknowledgement of Academic Work (DAAW)*. For each individual problem or exercise that requires the documentation of any assistance or source, cadets must provide the documentation using endnote style. A single page of endnotes for the entire problem set is allowed. Note that copying, even though properly documented, will result in a 50% grade reduction from the achieved score.

2. Writs. If cadets have an authorized absence during a lesson in which either an instructor writ or FNEK Writ is administered, they will not be penalized. There are no “make-ups” for a missed writ. Rather, the total marks available will be reduced by the appropriate marks.

3. Fundamental Nuclear Engineering Knowledge (FNEK) writs. During the course cadets will be administered two 60-mark fundamental knowledge writs. The first will be given during the first five lessons of the course and the second during the final five lessons. The fundamental knowledge writs are designed to assess cadet retention of key nuclear engineering facts and equations they should know without the use of reference materials. As such, the fundamental knowledge writs will be reference-free with no calculator use permitted.

4. WPR. If cadets have an authorized absence during a lesson in which a WPR is administered, they will arrange for a make-up, make-ahead or a take-out exam. Cadets will notify their instructor as soon as possible for coordination of a make-up of a WPR.

5. Design Project. Design project submissions are due NLT the date time group indicated on the Course Schedule or as directed by your instructor. **All design project IPRs and the final submission must be turned in to receive course credit**.

6. Design Project Brief. Cadets will present their design project to the class during the last week of the semester. This is mandatory and cadets will ensure they schedule their brief when their entire design team is available. Cadets will also provide feedback to other groups’ presentations.

7. Laboratory. There will be one laboratory experiment. Cadets will turn in their laboratory worksheet NLT the date time group indicated on the course schedule. The laboratory and subsequent worksheet are mandatory for course credit. If cadets have an authorized absence during the lab period, they will arrange for a make-up with their instructor as soon as possible.

8. Bonus Marks. Bonus marks may be offered at the instructor’s discretion. If assistance is received when completing bonus problems, the documentation must conform to the standards in the *DAAW*. No marks will be awarded for copying bonus problems.

4.2 Grade Cutoffs

Numerical marks will be awarded for graded work during the semester. Grades will normally be made available to cadets through Canvas as soon as they are posted by the instructor and approved by the Head of the Department. Percentage to letter grade conversion is given in Table 4.

Table 4. Grade Cutoffs

Grade	Percentage
A+	94.0%
A	90.5%
A-	87.0%
B+	83.5%
B	80.0%
B-	76.5%
C+	73.0%
C	69.5%
C-	66.0%
D	60.0%
F	< 60.0%

4.3 Policy for Late Assignments

I am responsible for developing you as a leader of character. A key and essential aspect of developing cadets as leaders of character is to hold them accountable for completing academic work on time. The habits of mind that cadets develop at the Academy will follow them when they are commissioned. NLT when an assignment is due, cadets must notify their instructor that their assignment will be late. Otherwise, the cadet earns no credit for the assignment. With prior coordination, the standard academic cut for each academic day (24 hours) or part thereof that the assignment is late is 10%. After 4 academic days, the cadet earns no credit for the assignment.

5 Course Resources

5.1 Course Text

The textbooks for this course are listed below:

1. Nuclear Energy: An Introduction to the Concepts, Systems, and Applications of Nuclear Processes, (8^{th} Edition, 2015) by Raymond L. Murray and Keith E. Holbert

5.2 Course Outline

Learning model. Lessons are organized so that cadets progress through a “model/coach/fade” process of learning the fundamental concepts of the block.

1. Model Class [M]. In a model class, cadets build on knowledge gained from reading and studying the textbook and other materials. Additionally, cadets do homework problems to help gain an understanding of the basic concepts from the assignment. The instructor explains the fundamental concepts and demonstrates how the fundamental concepts can be used to solve problems. Cadets can expect to go to the boards and work problems during every model class.

2. Coach Class [C]. In a coach class, cadets build on conceptual and problem-solving abilities by applying concepts introduced in a model class. The instructor, acting as a coach, offers opportunities to apply the fundamental concepts to solve problems and coaches when problem-solving has stalled. Cadets will frequently work problems at the board during a coach class, alone or in groups. All problem-solving sessions are coach classes.

3. Fade Class [F]. In a fade class, cadets demonstrate mastery by applying the fundamental concepts to solve problems or to work on the design project. The instructor's role transitions to the role of assessor. Cadets are expected to know their strengths and weaknesses and work on those weaknesses. The fade class normally culminates with a writ or WPR, which evaluates understanding of course learning objectives.

5.3 Calculators

The only calculators authorized for use during graded events are calculators approved for use on the *Fundamentals of Engineering Examination*. If in doubt, use the cadet issued TI-30XIIS calculator or TI-36X Pro. It is each cadet's responsibility to ensure they have the calculator present during class. A cadet may not share a calculator with any other cadet during a graded event.

5.4 Canvas

All course material for NE350 can be accessed online using Canvas. Canvas is a combat multiplier. It provides valuable information throughout the course and is updated frequently. On Canvas, cadets will find course information to include any updates, course administration, lesson information, blank and solution homework sheets, and links to other interesting nuclear and radiation sites.

5.5 Examination Materials

The WPR and the TEE will be closed-book examinations. The NE350 Reference Data Card (RDC) and the Nuclear Engineering Reference and Data Pamphlet (NERDP) will be provided for each examination. The reference cards and the NERDP contain the equations, data, and charts that cadets will need as reference for all examinations. It is recommended cadets study, do homework, and do in-class boardwork with their copy of the reference cards and the NERDP. Cadets are encouraged to make notes (identify parameters, write conditions, write units, etc.) on their reference cards and NERDP as they study. This will provide cadets a greater degree of familiarity with these materials when they receive a clean copy for the examinations. Used correctly, the reference cards and NERDP are invaluable.

5.6 Additional Instruction

Additional instruction (AI) can be scheduled with the instructor and is highly encouraged. The primary purpose of AI is to get answers to specific questions cadets may have after doing homework, or to get assistance with homework problems they have already attempted on their own.

6 Documentation and Acknowledgement of Academic Work

6.1 Documentation and Acknowledgement of Work

1. The purpose of documentation is to give credit where credit is due. This is good scholarship. Proper documentation identifies the source(s) of borrowed ideas, quotations, and assistance in the preparation of assignments (*DAAW*, Section I).

2. Any homework prepared outside of class that is turned in for grade must be documented in accordance with the *DAAW*. The Department of Physics & Nuclear Engineering requires cadets to use end-notes for documented submissions.

3. Collaboration is highly encouraged; however, cadets must document all substantive assistance received. Endnotes should clearly distinguish cadet effort from collaboration, such as in the following example:

> George D. Martin, C-4, '14, Mary M. Brown, B-4, '14, Paul P. Smart, H-3 '14, collaboration with the author, verbal and written discussion, West Point, NY, 14 August 2012. We worked through the derivation of the flux for the diffusion equation as an informal group. CDT Martin set up the general equation. CDT Brown added the proper form of the Laplacian. Cadet Smart determined the correct source boundary condition.

4. Properly documented copying is subject to an academic cut of no less than 50% of the achieved score for that portion of the work that is copied. For bonus problems, copying will result in no credit for the problem(s) that were copied.

5. If a cadet reuses or resubmits work in accordance with course policies, he or she must still take efforts to avoid what is known as "self-plagiarism": the reuse of your own previous work in another venue while presenting it as new work. Cadets are required to cite and document that work like any other source cited or assistance received IAW the DAW.

6.3 Writing Format Standards

The following are the formatting standards for individual written reports prepared as homework. Cadet writing must conform to the standards in *The Standards for Technical Reports*. Written submissions will be evaluated on substance, organization, style, and correctness.

1. Font: Times New Roman 12 point
2. Margins: Left margin of 1.5" and all other margins 1.0"
3. Paragraph Indentation: 3/8 inches
4. Line Spacing: double space
5. Equations: neatly hand-written equations or typed electronically and numbered
6. Documentation: MLA format

Part II (Enclosure 1)
Course Schedule

LSN	Day 1	Day 2	Lesson Name	Remarks	Assignments Given	Assignments Due	MARKS	WEIGHT
01	7-Jan	6-Jan	Introduction to NE350 and the Design Process					
02	12-Jan	8-Jan	Charged Particle Interactions	FNEK 1	Problem Set 1		60	3%
03	14-Jan	13-Jan	Neutral Particle Interactions	NEDP teams chosen	IPR 1			
04	20-Jan	15-Jan	Biological Effects of Radiation	NEDP topics chosen				
05	22-Jan	21-Jan	Radiation Protection from Neutral Particles					
06	28-Jan	27-Jan	Radiation Protection from Charged Particles					
07	30-Jan	29-Jan	Problem-Solving Session			Problem Set 1	75	4%
08	2-Feb	3-Feb	Internal Radiation Exposure and Standards					
09	4-Feb	5-Feb	Design Project Briefs - In-Progress Review 1		IPR 2	IPR 1	125	6%
10	9-Feb	12-Feb	Course-Wide Writ	Conducted in class			100	5%
11	11-Feb	17-Feb	Information from Isotopes		Problem Set 2			
12	18-Feb	19-Feb	Useful Radiation Effects					
13	25-Feb	24-Feb	Medical Applications of Nuclear Technology					
14	2-Mar	26-Feb	Radiation Safety, Nuclear Accidents, and Lessons Learned					
15	4-Mar	3-Mar	Problem-Solving Session		Problem Set 3	Problem Set 2	75	4%
16	9-Mar	10-Mar	Written Partial Review	Conducted in class			200	10%
17	11-Mar	12-Mar	Gas Counters and Neutron Detectors					
18	16-Mar	17-Mar	Scintillation Counters and Solid-State Detectors					
19	18-Mar	19-Mar	Counting Statistics and Detector Comparisons		Uncertainty HW			
20	23-Mar	24-Mar	Design Project Briefs - In-Progress Review 2		IPR 3	IPR 2	125	6%
21	25-Mar	26-Mar	Gamma Spectroscopy Practical Exercise	In-Class Exercise	Lab Report, Pre-lab	Uncertainty Homework	50	10%
22	9-Apr	8-Apr	Spectroscopy Lab	Conducted in BHB47		Pre-Lab due at start of lab	20	1%
23	13-Apr	10-Apr	Problem-Solving Session			Problem Set 3	75	4%
24	15-Apr	14-Apr	Introduction to Non-Proliferation & Policy	FNEK 2			60	3%
25	20-Apr	16-Apr	Nuclear Waste and Waste Management		NEDP Final Submission	IPR 3 (slides only)	50	3%
26	22-Apr	21-Apr	Current Events in Nuclear Security	In-class exercise		Lab Report Due	110	6%
27	27-Apr	27-Apr	STERIS TRIP SECTION			STERIS Reflection Paper	25	1%
28	29-Apr	30-Apr	Design Project Briefs					
29	4-May	5-May	Design Project Briefs			Final Submission	250	13%
30	6-May	7-May	Problem-Solving Session (TEE Review)	Course Survey			10	1%

In-class Eval
Turn-in/Brief
Lab Exercise
Miscellaneous
Trip Section
Tentative Date

NEDP Team Grade	50	3%
Instructor Marks	40	2%
Term End Exam	500	25%
Total	2000	100%

NE350 Course Calendar AY 26-2 v4

JANUARY 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
28	29	30	31	01	02	03
04 WK 1 Corps Returns	05 Reorgy Day	06 2-1 LSN 01 1st Day of Classes	07 1-1 LSN 01 DEAN	08 2-2 LSN 02	09 1-2 DROP CMDT	10 B 500th Night
11 WK 2	12 1-3 LSN 02 CMDT	13 2-3 LSN 03	14 1-4 LSN 03 CMDT	15 2-4 LSN 04	16 2-5 DROP DEAN-PRIME	17 B
18 WK 3	19 MLK No Classes	20 1-5 LSN 04 CMDT	21 2-6 LSN 05	22 1-6 LSN 05 DEAN; Majors Open House	23 2-7 DROP	24 B YWW
25 WK 4	26 1-7 DROP CMDT	27 2-8 LSN 06	28 1-8 LSN 06 DEAN	29 2-9 LSN 07	30 1-9 LSN 07 CMDT	31 A SAMI

FEBRUARY 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
01 WK 5	02 1-10 LSN 08 CMDT	03 2-10 LSN 08	04 1-11 LSN 09 DEAN	05 2-11 LSN 09	06 2-12 DROP DEAN-PRIME	07 B 100th Night
08 WK 6	09 1-12 LSN 10 CMDT	10 2-13 DROP	11 1-13 LSN 11 CMDT	12 2-14 LSN 10	13 1-14 DROP DEAN	14 B
15 WK 7	16 Washington's B-Day No Classes	17 2-15 LSN 11	18 1-15 LSN 12 DEAN	19 2-16 LSN 12	20 1-16 DROP CMDT	21 B
22 WK 8	23 1-17 DROP CMDT	24 2-17 LSN 13	25 1-18 LSN 13 CMDT; Post Night	26 2-18 LSN 14	27 2-19 DROP DEAN-PRIME	28 A

MARCH 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
01 WK 9	02 1-19 LSN 14 DEAN	03 2-20 LSN 15	04 1-20 LSN 15 CMDT; End of Rnd 1	05 2-21 DROP MCC - 1CL	06 1-21 DROP CMDT	07 B
08 WK 10	09 1-22 LSN 16 CMDT	10 2-22 LSN 16	11 1-23 LSN 17 CMDT	12 2-23 LSN 17	13 2-24 DROP DEAN-PRIME	14 B
15 WK 11	16 1-24 LSN 18 CMDT	17 2-25 LSN 18	18 1-25 LSN 19 DEAN	19 2-26 LSN 19	20 1-26 DROP CMDT	21 A
22 WK 12	23 1-27 LSN 20 CMDT	24 2-27 LSN 20	25 1-28 LSN 21 DEAN	26 2-28 LSN 21 Upperclass Released	27 2-29 DROP Class for 4CL PPW	28 PPW
29 PPW WK 13	30 Spring Break	31 Spring Break	01 Spring Break	02 Spring Break	03 Spring Break	04 Spring Break

APRIL 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
29 WK 13	30 Spring Break	31 Spring Break	01 Spring Break	02 Spring Break	03 Spring Break	04 Spring Break
05 Easter WK 14	06 No Class Day	07 1-29 DROP	08 2-30 LSN 22 CMDT	09 1-30 LSN 22	10 2-31 LSN 23 PRIME	11 A Sandhurst Val
12 WK 16	13 1-31 LSN 23	14 2-32 LSN 24	15 1-32 LSN 24	16 2-33 LSN 25	17 1-33 DROP CMDT	18 B
19 WK 17	20 1-34 LSN 25 CMDT	21 2-34 LSN 26	22 1-35 LSN 26 DEAN	23 Projects Day No Classes	24 2-35 DROP	25 A
26 WK 18	27 1-36 LSN 27 STERIS TRIP SECTION	28 2-36 DROP	29 1-37 LSN 28 DEAN	30 2-37 LSN 28	01	02 A

MAY 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
26 WK 18	27	28	29	30	01 Sandhurst No Classes	02 A Sandhurst
03 WK 19	04 1-38 LSN 29 CMDT	05 2-38 LSN 29	06 1-39 LSN 30 DEAN	07 2-39 LSN 30	08 1-40 DROP CMDT	09 B
10 WK 20	11 2-40 DROP Reading Day	12 TEE	13 TEE	14 TEE	15 TEE	16 TEE End of Term
17 WK 21 Grad Wk	18 Grad Wk	19 Grad Wk	20 Grad Wk	21 Grad Wk	22 Grad Wk	23 Graduation CL26
24	25	26	27	28	29	30

Part III (Enclosure 2)

Nuclear Engineering Design Project

NE350 Radiological Engineering Design Project
AY 26-2

1 Design Project Introduction

Your group will use radiation technology to solve an engineering problem. This will be completed by the end of the semester. By completing this project, your group will internalize the design and its associated problem-solving processes while fostering creativity and critical thinking in problem-solving.

2 Design Project Objective

2.1 Direction of Project

The purpose of this project is to apply your knowledge and ingenuity towards designing a solution to a radiological engineering problem. Design projects may be chosen from the list below ONLY:

1. Design a deployable device that uses radiation to inhibit sprouting in carrots, onions, and potatoes that are to be fed to Soldiers at forward operating bases while ensuring adequate radiation protection for the Soldier. (Reference: Murray and Holbert, Section 14.2)

2. Design a gamma-ray sterilization system for surgical equipment. Ensure a sterility assurance level (SAL) of 10^{-6} for an initial bio-burden of 1000 microorganisms per unit. The decimal reduction dose, D_{10}, for bacillus pumilis, the most radiation-resistant bacterium of the contaminating population, is 0.3 Mrads (megarads). Ensure adequate radiation protection for a radiation worker. (Reference: Murray and Holbert, Sections 14.2-14.3)

3. Design a laboratory device that uses and successfully applies a sterile insect technique (SIT). Assume the sterilization will be applied to male flies that resemble that of tsetse flies. Ensure adequate radiation protection for the laboratory technicians. (Reference: Murray and Holbert, Section 14.6)

4. Design a thickness gauge system for rolls of 3' wide aluminum sheeting 0.75" thick with less than 1% error. Ensure adequate radiation protection for a radiation worker. (Reference: Murray and Holbert, Section 13.9)

5. Design a water level gauge using a radioisotope for a stainless-steel tank 2 meters tall x 2 meters in diameter. Ensure adequate radiation protection for the general population. (Reference: Murray and Holbert, Section 13.9)

6. You work for a radioisotope production facility. Design a shielded shipping container to send 650-Ci amounts of Co-60 to a company that manufactures gamma radiography equipment.

7. Design a device capable of delaying the spoilage of fresh fruits to support a local farmers market. This device must be fixed inside a facility. Use the approved dose by the Food and Drug Administration while ensuring adequate radiation protection for a radiation worker.

8. Design a set of augmentation armor for the Department of Defense's Joint Light Tactical Vehicle (JLTV) which must be able to protect its occupants for prolonged periods of time in a nuclear environment. Ensure adequate radiation protection for Soldiers so that they receive no more than 75 rem over the course of 24 hours.

9. Design a set of removable augmentation armor for the HH-60M Blackhawk Medevac Helicopter which must be able to conduct life-saving operations in a nuclear environment. Ensure adequate radiation protection for the helicopter crew so that they receive no more than 5 rem for 10 minutes of on-ground time.

2.2 Concept Map

This project will be assessed using the Engineering Design Process as outlined in the *Nuclear Engineering Design Supplement* (NEDS) dated August 2017. Intermediate progress reviews (IPRs) will focus on problem recognition, project proposal, implementation, and will ultimately be communicated in the final design. When completing these IPRs, the team leader will schedule a time when all team members will be present.

1. Problem Recognition (IPR 1). The IPR 1 presentation is a 10-minute brief to your instructor where you will present your progress and receive feedback. You will follow the procedure outlined in Appendix 1 of the NEDS. The goal of this IPR is to develop a revised problem statement, a thorough list of specified, implied, and essential tasks, a timeline for project completion (Gantt chart), a list of restrictions, limitations, and constraints, a research topic list, and a list of Requests for Information (RFIs).

2. Project Proposal (IPR 2). IPR 2 presentation is a 10-minute brief to your instructor where you will present your progress and receive feedback since IPR 1. You will follow the procedure outlined in Appendix 2 of the NEDS. The goal of this IPR is an analysis of and recommendation of a design concept with reasonable computation indicating best choice of design.

3. Implementation (IPR 3). The IPR 3 is a digital-only submission to your instructor where you will present your progress and receive feedback since IPR 2. You will follow the procedure outlined in Appendix 3 of the NEDS. The goal of this IPR is the refined and optimized design that was selected from IPR 2 supplemented with complete calculations and specifications. Treat IPR 3 as a rehearsal to your final presentation.

4. Communication of the Final Design (Final Report and Presentation). During Lessons 28 and 29, a classroom presentation of the design project will be communicated. A <u>written</u> final design will be submitted to your instructor by the indicated deadline and IAW standards in the *NEDS* grading rubric for the oral presentation and the final report posted to Canvas.

3 Design Project Formatting

Use the design and analysis process described in detail in the *Nuclear Engineering Design Supplement* (NEDS) when developing your project initial concept.

3.1. Requirements

1. A project journal and design notebook (as described in the NEDS) are <u>not required</u> for this project. No written reports will be collected for IPRs 1 through 3. Only the final report will require submission of a written document.

2. All IPR and final submissions will be submitted over Canvas by the specified deadline. The NEDP final report submission will be IAW the format in the NEDS. For margins, font, spacing, etc. refer to *Standards for Technical Reports*, under "Specifics" Paragraph three. Go to the course Canvas site for a copy of these documents.

3.2. Grading

A grading rubric will be posted for each submission to Canvas. Follow and address the requirements in each category to the best of your group's ability. A peer evaluation component will be assessed for each submission and will be added to the graded event total.

3.3. Formal Groups

Your design project group is a designated formal group. Formal group work only applies to IPR and final design submissions and require the appropriate documentation IAW *Documentation and Acknowledgment of Academic Work.*

Part IV (Enclosure 3)
Lesson Outline

United States Military Academy

NE 350: Radiological Engineering Design
Course Outline by Lesson
AY 26-2

Nuclear Engineering Program Director:
Professor Kenneth Allen

Course Director:
MAJ Joshua Mashl

January 2026

Lesson 1: Introduction to NE350 and the Design Process [Model]

Assignment
Read: Nuclear Engineering Design Supplement, pp. 1-7, Appendices 1, 2, and 5.

Learning Objectives:

1. Understand the scope of the course and set expectations for the completion of requirements throughout the semester.

2. Implement the engineering design process as outlined in the Nuclear Engineering Design Supplement.

3. Conduct problem recognition by identifying objectives, specified tasks, implied tasks, essential tasks, user requirements, constraints, limitations, and restrictions for the given problem statement and environmental considerations. Use these items to create an objectives tree.

4. Develop a request for information list, research topic list and a revised problem statement.

5. Conduct research of scholarly sources, experimentation, and/or modeling to generate appropriate courses of action.

6. Develop and select an appropriate initial design concept, project risk assessment, and understand the value of sample calculations to gain client approval for a project proposal.

7. Establish functions/roles amongst the design project team members. Identify the value your personal background will bring to the team.

8. Use a Gantt chart as a project management tool.

9. Recognize the graded requirements for the Nuclear Engineering Design Project (NEDP).

Notes:
This class introduces the primary intent and underlying theme of the course. In addition, course administrative information will be presented.

Lesson 2: Charged Particle Interactions (Review) / FNEK 1 [Model]

Assignment
Study: Murray and Holbert, 3.2 – 3.3, 5.1 – 5.3

Learning Objectives:

1. List and describe the ways that charged particles interact with matter.

2. Calculate the range of α and β-particles in different materials.

Vocabulary:

ionization	excitation	Cherenkov radiation
effective atomic mass	range	collisional energy loss
radiative energy loss	flux	bremsstrahlung

RDC Equations:

$$R_{max}\left[\frac{g}{cm^2}\right] = \begin{cases} 0.412E_\beta^{\,1.265-0.0954\ln(E_\beta)} & 0.01 \le E \le 2.5\ MeV \\ 0.530E_\beta - 0.106 & 2.5 \le E \le 20\ MeV \end{cases}$$

$$R_{air}\ [cm\ ] = \begin{cases} 0.56E_\alpha & E_\alpha < 4\ MeV \\ 1.24E_\alpha - 2.62 & 4 \le E_\alpha \le 8\ MeV \end{cases}$$

$$\frac{R_1}{R_2} = \frac{\rho_2}{\rho_1}\sqrt{\frac{M_1}{M_2}}$$

$$\sqrt{M_{eff}} = \sum_i \gamma_i\sqrt{M_i}$$

$$\frac{1}{\sqrt{M_{eff}}} = \sum_i \frac{\omega_i}{\sqrt{M_i}}$$

$$R = 3.2\times 10^{-4}\frac{\sqrt{M}}{\rho}R_{air}$$

Suggested Homework Problems:
Murray and Holbert 5.5, 5.7, 5.20

Notes: A 60-mark (15 minute) Fundamentals of Nuclear Engineering Knowledge writ will be administered. The writ will cover the following topics:

- On the Fundamentals of Nuclear Engineering Knowledge Card
  - Physical Constants & Unit Conversions
  - Units
  - Graphs and their significance
  - Fundamentals of Nuclear Engineering (NE300)
- On the Fundamentals of Nuclear Engineering Knowledge Concepts Card
  - Fundamentals of Nuclear Engineering (NE300)

Lesson 3: Neutral Particle Interactions (Review) [Model/Fade]

<u>Assignment</u>
Study: Murray and Holbert, 5.4 – 5.5

Learning Objectives:

1. List and describe the ways that neutrons and gamma-rays interact with matter.

2. Describe how photon energy affects the cross section for each type of gamma-ray/matter interaction.

3. Apply conservation of energy and conservation of momentum principles to gamma-ray/matter interactions to calculate energy, scattering angle, velocity, or mass of the particles involved.

4. Calculate the kinetic energy of the recoil electron in photoelectron scattering.

5. Calculate the unattenuated photon flux as a function of penetration depth into a material.

Vocabulary:

linear energy transfer (LET)	photoelectric effect	pair production
Compton effect	energy-absorption coefficient	mass-attenuation coefficient
scattering	absorption/capture	

RDC Equations:

$$\mu = N\sigma$$

$$\lambda = \frac{1}{\mu}$$

$$N\left[\frac{atoms}{cm^3}\right] = \frac{\rho N_A}{M}$$

$$E_K^e = E - E'$$

$$E_K^e = E_\gamma - I$$

$$\phi(x) = \phi_0 e^{-\mu x}$$

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2}[1 - \cos(\theta)]$$

$$y - y_1 = \left[\frac{y_2 - y_1}{x_2 - x_1}\right](x - x_1)$$

Suggested Homework Problems:
Murray and Holbert 5.6, 5.11, 5.12, 5.13

Lesson 4: Biological Effects of Radiation [Model]

Assignment
Study: Murray and Holbert 10.1 – 10.6

Learning Objectives:

1. Describe the biological effects of ionizing radiation on different types of tissue.

2. Calculate absorbed dose given an amount of energy imparted to an exposed mass.

3. Calculate dose equivalent given an absorbed dose and be able to convert units as necessary.

4. Describe the relationship between radiation dose and effect, to include the linear no threshold (LNT), linear quadratic, and hormesis models.

5. List the dose equivalent limits adopted by the NRC for radiation workers and the general public.

6. Calculate the effective half-life given the biological and physical half-lives and the associated fraction of a substance that decays within the body.

7. Describe the effects of a radiological dispersion device (RDD).

Vocabulary:

absorbed dose	dose equivalent	somatic
Gray (Gy)	Sievert (Sv)	genetic
rad	rem	teratogenic
dose rate	quality factor (QF)	free radical
biological half-life	effective half-life	ALARA
acute exposure	chronic exposure	LD 50/30

RDC Equations:

$$D = \frac{\Delta E}{m}$$

$$D = \dot{D} \cdot t$$

$$H = D \cdot QF$$

$$\lambda_E = \lambda + \lambda_B$$

$$\lambda = \frac{\ln 2}{t_H}$$

$$A = \lambda n$$

$$n\ [atoms] = \frac{mN_A}{M}$$

Suggested Homework Problems:
Murray and Holbert 10.3, 10.5, 10.6, 10.7, 10.10, 11.8

Lesson 5: Radiation Protection from Neutral Particles [Model]

<u>Assignment</u>
Read: Murray and Holbert 11.1
Study: Murray and Holbert 11.2 – 11.3

Learning Objectives:

1. Explain the effect of time, distance, and shielding on radiation intensity.

2. Calculate the dose rate at a point some distance from various source geometries of monoenergetic gamma-rays.

3. Explain the process for shielding from neutrons and gamma rays and describe what material properties are desirable for either.

4. Calculate the total flux (including buildup) of point-source gamma rays after passing through a shield.

5. Calculate the shielding thickness required to reduce the total gamma-ray flux (including buildup) to a particular value.

Vocabulary:

inverse square law	external dose	isotropic
attenuation	uncollided flux	buildup factor
removal cross-section		

RDC Equations:

$$\dot{D} = \phi E_\gamma \left(\frac{\mu_{en}}{\rho}\right)$$

$$\phi = \frac{S}{4\pi r^2}$$

$$S = f \cdot A$$

$$\phi_u = \frac{S\, e^{-\mu R}}{4\pi r^2}$$

$$\phi = B\phi_u$$

$$\dot{D} = \dot{D}_0 e^{-\Sigma_R t}$$

Suggested Homework Problems:
Murray and Holbert 11.1, 11.2, 11.4, 11.5, 11.17

Lesson 6: Radiation Protection from Charged Particles [Model]

Assignment
Review: Murray and Holbert, 3.2 – 3.3, 5.1 – 5.3

Learning Objectives:

1. Discuss the properties of a good charged particle shield

2. Calculate the range of various charged particles

3. Calculate the average beta particle energy

4. Identify the various depths the NRC uses to identify external dose

5. Calculate the beta absorption coefficients and discuss how it's related to photons

6. Calculate the dose from an external beta source

Vocabulary:

maximum beta energy	average beta energy	shallow dose
eye dose	whole body (deep tissue) dose	
beta absorption coefficient		

RDC Equations:

$$E_{\beta,avg} \cong \frac{E_{\beta,max}}{3}$$

$$\dot{D}_\beta = \phi_\beta \left[\frac{\beta}{cm^2\, s}\right] \bar{E}_\beta [MeV]\, \mu_\beta \left[\frac{cm^2}{g}\right]$$

$$\dot{D}_\beta \left[\frac{mGy}{hr}\right] = 2.9 \times 10^{-4}\, C_a\, \bar{E}\, \mu_{\beta,t}\, e^{-(\mu_{\beta,t}(\rho_t x_t))}$$

$$\dot{D}_\beta \left[\frac{mGy}{hr}\right] = 3.6 \times 10^{-4}\, C_a\, \bar{E}\, \mu_{\beta,t}\, e^{-\left(\mu_{\beta,air}(\rho_a x_a)\right)} e^{-(\mu_{\beta,t}(\rho_t x_t))}$$

$$\mu_{\beta,air} = 16\left(E_{\beta,max} - 0.036\right)^{-1.4}$$

$$\mu_{\beta,tissue} = 18.6\left(E_{\beta,max} - 0.036\right)^{-1.37}$$

$$\mu_{\beta,other} = 17\,(E_{\beta,max})^{-1.14}$$

Suggested Homework Problems:
Complete the board problems

Lesson 7: Problem-Solving Session [Fade/Coach]

Assignment
Review: Lessons 1-6.

Learning Objective:

Know learning objectives, vocabulary, notes, and homework problems from lessons 1-6.

Notes:
This class provides time for students to work on the first problem set.

Lesson 8: Internal Radiation Exposure and Standards [Model]

<u>Assignment</u>
Study: Murray and Holbert 11.4, 11.8
Read: Murray and Holbert 11.5 – 11.7

Learning Objectives:

1. Explain the factors that result in an internal radiation dosage.

2. Define the Maximum Permissible Concentration (MPC) of radioisotopes.

3. Explain the role of the EPA in determining radionuclide contaminant limits.

4. Explain where radon comes from and why it can be a problem.

5. Calculate effective dose equivalent and dose limits based off contemporary radiation standards (10CFR20).

Vocabulary:

Maximum Permissible Concentration (MPC)
Annual Limit of Intake (ALI)
deterministic effects
ICRP
10CFR20
committed effective dose equivalent
Derived Air Concentration (DAC)
stochastic effects
NCRP
ALARA

RDC Equations:

$$\sum_i \frac{C_i}{(MPC)_i} \leq 1$$

$$n = n_0 e^{-\lambda_E t}$$

$$n = n_0 \left(\frac{1}{2}\right)^{t/t_E}$$

$$n_E = n_0 - n$$

$$n_D = n_E \left(\frac{t_E}{t_H}\right)$$

$$H_{50} = \frac{n_D E_d (QF)}{m}$$

$$(H_{50})_E = \sum_T \omega_T (H_{50})_T$$

Suggested Homework Problems:
Murray and Holbert 11.11, 11.12, 11.13

Lesson 9: Design Project Briefs – In-Progress Review 1 [Model/Coach]

<u>**Assignment**</u>
Read: Nuclear Engineering Design Supplement, Appendices 2 – 4.

Learning Objectives:

1. Recognize the graded requirements for the Nuclear Engineering Design Project (NEDP).

2. Perform the initial calculations needed for your specific project in preparation for IPR 2.

Notes:
For this class, design project groups will brief their IPRs in-class. Brief times are short, usually around 10 minutes per group.

Lesson 10: Course-Wide Writ [Fade]

Assignment
CWW

1. Questions and problems on the Course Wide Writ will come from the lesson learning objectives, vocabulary terms, lesson notes, fundamental knowledge, and homework problems from **lessons 1 through 8.**

2. The examination will be given in your normal classroom **during your assigned class period for lesson 10.** The NE350 reference card and the Nuclear Engineering Reference and Data Pamphlet will be provided. You will be authorized to use your cadet-issued (TI 30 or TI 36X Pro) calculator.

Lesson 11: Information from Isotopes [Model/Coach]

Assignment
Study: Murray and Holbert 13.6 – 13.9
Read: Murray and Holbert 13.1 – 13.2
Shultis and Faw 13.3, 14.3

Learning Objectives:

1. Perform dating calculations using radiometric dating techniques for organic matter (carbon dating) or minerals containing uranium.

2. Calculate the Compton edge for a given incident photon energy or calculate the incident photon energy given the Compton edge.

3. Describe the physics of radiation gauges and the employment of this device in measurements.

Vocabulary:

radioisotope	Compton edge
neutron activation analysis	radiography

RDC Equations:

$$N_{\mathrm{Pb}}(t) = N_{\mathrm{U}}(0) - N_{\mathrm{U}}(t)$$

$$N_U(t) = N_U(0)e^{-\lambda t}$$

$$t = \frac{\ln\left[1 + \frac{N_{\mathrm{Pb}}(t)}{\mathrm{N_U}(t)}\right]}{\lambda}$$

$$t = \frac{\ln\left[\frac{N_{\mathrm{C-14}}(t)}{N_{\mathrm{C-14}}\,(t_0)}\right]}{-\lambda}$$

$$S = f \cdot A$$

$$n\ [atoms] = \frac{mN_A}{M}$$

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2}\left[1 - \cos(\theta)\right]$$

$$E_K^e = E - E'$$

Suggested Homework Problems:
Murray and Holbert 13.5, 13.6, 13.19, 13.20

Lesson 12: Useful Radiation Effects [Model/Coach]

Assignment
Study: Murray and Holbert 14.3 and 14.8
Read: Murray and Holbert 14.1, 14.2, 14.4 – 14.7

Learning Objectives:

1. Calculate the required dose to achieve a specified sterility assurance level (SAL).

2. Explain how radiation can be used to prevent food spoilage and estimate the doses required.

3. Explain the concerns about irradiated food products and discuss their validity.

4. Describe various other applications for radiation including insect control, pathogen reduction, and crop mutations.

5. Explain the irradiation reactions involved in the transmutation doping of semiconductors and calculate the dopant concentration of its phosphorous impurities.

Vocabulary:
sterility assurance level (SAL) decimal reduction dose (D_{10})
sterile insect technique (SIT) neutron transmutation doping (NTD)

RDC Equations:

$$S = f \cdot A$$

$$n\,[atoms] = \frac{mN_A}{M}$$

$$D = nD_{10}$$

$$n = \log_{10}\frac{initial\ bioburden}{final\ bioburden}$$

$$\Sigma = N\sigma$$

$$R = \phi\Sigma$$

$$n = RVt$$

$$N\left[\frac{atoms}{cm^3}\right] = \frac{\rho N_A}{M}$$

$$N_{\mathrm{P}} = aN_{\mathrm{Si}}\sigma\phi t$$

Suggested Homework Problems:
Murray and Holbert 14.3, 14.5, 14.9

Lesson 13: Medical Applications of Nuclear Technology [Model]

Assignment
Study: Murray and Holbert 13.4, 14.1
Shultis and Faw 14.0 – 14.1.1, 14.1.5, 14.1.7 – 14.1.8, 14.5 – 14.5.11
Read: Murray and Holbert 13.3

Learning Objectives:

1. Describe how radioisotopes can be used for medical imaging to include SPECT and PET.

2. Describe the difference between brachytherapy, teletherapy, and radionuclide therapy.

3. Explain how an X-ray is produced and how an X-ray can be used in medical applications.

4. Describe the purpose of a collimator and the characteristics of a good collimating material.

Vocabulary:

Radiopharmaceuticals | tomography | brachytherapy | teletherapy
radionuclide therapy | collimators

Single Photon Emission Computed Tomography (SPECT)
Positron Emission Tomography (PET)

Lesson 14: Radiation Safety, Nuclear Accidents, and Lessons Learned [Model]

<u>Assignment</u>
Study: Murray and Holbert 21.1 and 21.3
Read: Murray and Holbert 21.2, 21.4, 21.6 – 21.9, 21.11

Learning Objectives:

1. Describe the systems by which safety is assured and categorized.

2. Explain the role of the Nuclear Regulatory Commission (NRC) and their use of the principal reference, the *Code of Federal Regulations Title 10, Energy*, in maintaining nuclear safety and security.

3. Describe the failure mechanisms involved in the Three Mile Island accident, Chernobyl accident, the Fukushima Daiichi accident, and Goiania.

4. Discuss the importance of nuclear security in terms of the protection of nuclear facilities from adverse actions.

Vocabulary:

assurance of safety	Nuclear Regulatory Commission (NRC)	
Fukushima Daichi accident	Chernobyl accident	Three-Mile Island accident
design basis accident	emergency core cooling and containment	
QA/QC	design basis threat	

Suggested Homework Problems:
Murray and Holbert: 21.4

Lesson 15: Problem-Solving Session [Fade/Coach]

Assignment
Review: Previous lessons.

Learning Objective:

Know learning objectives, vocabulary, notes, and homework problems from previous lessons.

Notes:
This class provides time for students to work on the second problem set.

Lesson 16: Written Partial Review [Fade]

<u>Assignment</u>
WPR

1. Questions and problems on the Written Partial Review will come from the lesson learning objectives, vocabulary terms, lesson notes, fundamental knowledge, and homework problems from previous lessons.

2. The examination will be in class. The NE350 reference card and the Nuclear Engineering Reference and Data Pamphlet will be provided. You will be authorized to use your cadet-issued (TI 30 or TI 36X Pro) calculator.

Lesson 17: Gas Counters and Neutron Detectors [Model]

<u>Assignment</u>
Study: Murray and Holbert 12.1 – 12.3

Learning Objectives:

1. Describe how the configuration of the gas counter and source affects the efficiency.

2. Describe the principles of operation of a gas-filled detector.

3. Describe the regions of the curve representing the charge collected as a function of applied voltage to the gas-filled detector.

4. Compare and contrast the characteristics of ionization chambers, proportional counters and Geiger-Muller counters.

5. Define exposure and its units and convert to absorbed dose or dose equivalent.

6. Describe how a boron counter functions.

7. Describe how a fission chamber functions.

8. Calculate the number density of BF_3 to achieve a desired efficiency.

Vocabulary:

anode	cathode	ionization chamber
proportional counter	Geiger-Mueller counter	exposure
boron counter	fission chamber	

RDC Equations:

$$S = f \cdot A$$

$$\varepsilon_g = \frac{\phi A_d}{S}$$

$$\varepsilon = \frac{R}{S}$$

$$\varepsilon = \varepsilon_g \varepsilon_i$$

$$X = \frac{\Delta Q}{m}$$

$$N\left[\frac{atoms}{cm^3}\right] = \frac{\rho N_A}{M}$$

$$\Sigma = N\sigma$$

$$\phi(x) = \phi_0 e^{-\Sigma d}$$

$$Q = -\Delta mc^2$$

$$D_{med} = X\,\frac{\left(\frac{\mu_{en}}{\rho}\right)_{med}}{\left(\frac{\mu_{en}}{\rho}\right)_{air}}$$

Suggested Homework Problems:
Murray and Holbert 12.1, 12.12, 12.13

Lesson 18: Scintillation Counters and Solid-State Detectors [Model]

<u>Assignment</u>
Study: Murray and Holbert 12.4 – 12.6

Learning Objectives:

1. Describe the principal method of scintillation detection including necessary detector components.

2. Describe how a photomultiplier tube increases the amplitude of the detected pulse.

3. Approximate a scintillation detector's intrinsic efficiency given its thickness parallel to a beam of gamma rays.

4. Explain how a thermo-luminescent dosimeter (TLD) functions.

5. Describe semiconductors and electron-hole pairs and explain their utility in a radiation detector.

6. Describe n-type, p- type and junction semiconductor devices.

Vocabulary:

photomultiplier tube (PMT)	dynode	photocathode
phosphor	scintillation	hole
n-type	p-type	depletion region

RDC Equations:

$\varepsilon_i \approx 1 - e^{-\mu d}$

Suggested Homework Problems:
Murray and Holbert 12.11, 12.16

Lesson 19: Counting Statistics and Detector Comparisons

<u>Assignment</u>
Study: Murray and Holbert 12.7 – 12.8
Read: Murray and Holbert 12.9 – 12.11

Learning Objectives:

1. Explain when it is appropriate to use the binomial, Poisson and Gaussian distributions to interpret the statistical nature of the decay of radioactive materials.

2. Explain the significance of the standard deviation and how it is used to estimate the uncertainty in a radiation count.

3. Given the sample count, background count and counting time, calculate the net sample count rate and uncertainty.

4. Explain how a single-channel analyzer and multi-channel analyzer (MCA) can be used to determine the energy spectrum from detected radiation.

5. Describe how radiation detectors can be used in protecting against terrorism.

6. Given the FWHM and gamma-ray energy, calculate the energy resolution.

Vocabulary:

binomial distribution	Poisson distribution	Gaussian (or normal) distribution
standard deviation	discriminator	
single-channel analyzer	multi-channel analyzer	Gamma-ray spectroscopy
Dead time	full width at half maximum (FWHM)	
energy resolution		

RDC Equations:

$$\sigma = \sqrt{\bar{x}}$$

$$p = 1 - e^{-\lambda t}$$

$$R_N = \frac{\bar{x}_S - \bar{x}_B}{T} \pm \frac{\sqrt{\bar{x}_S + \bar{x}_B}}{T}$$

$$E_R = \frac{\Delta E}{E}$$

Suggested Homework Problems:
Murray and Holbert 12.8, (see exercise 12.6 and 12.7 for the binomial and Poisson distribution formulas)

Lesson 20: Design Project Work Period [Model/Coach]

Assignment
Read: Nuclear Engineering Design Supplement, Appendices 3 – 4.

Learning Objectives:

1. Perform the initial calculations needed for your specific project in preparation for IPR 3.

2. Develop initial project designs.

Notes:
For this class, design project groups will brief their IPRs in-class. Brief times are short, usually around 10 minutes per group.

Lesson 21: Course Drop for STERIS Trip Section [TENTATIVE]

Notes: This is a designed drop for our trip section to STERIS. A short writing assignment will be assigned for the trip section.

This date for this trip section is currently tentative for all NE350 sections. Details for the trip section will be provided in-class.

Lesson 22: Gamma Spectroscopy Practical Exercise [Coach]

Assignment

Today we'll be using a NaI detector to discuss gamma-spectroscopy. The data manipulation we perform today will be directly applicable to your lab next lesson.

Download the excel spreadsheet from Canvas. You will use it as a template to help us pull data from the detector.

The goals of today's PE are to:

* graph the raw data relating counts to channel
* relate channel to energy
* graph the data relating counts to energy

Lesson 23: Spectroscopy Laboratory [Fade]

<u>Assignment</u>
Study: Pre-Lab/ Lab handout
Review: Standards for Technical Reports (Canvas)
Science Laboratory Analysis Manual (SLAM): Chapter 3, Paragraph 4C4, Appendix G – Error Propagation, and Appendix H – Graphing, Linear Regression, and Curve Fitting.

Learning Objectives:

1. Experimentally determine the energy resolution and intrinsic efficiency for a NaI detector.

2. Experimentally determine the unknown source using gamma ray spectroscopy.

Notes: Complete the Pre-lab prior to class. I will check your derived equations at the start of the lab period.

Lesson 24: Problem-Solving Session [Fade/Coach]

<u>Assignment</u>
Review: Lessons 17-23.

Learning Objective:

Know learning objectives, vocabulary, notes, and homework problems from lessons 18-23.

Notes:
This class provides time for students to work on the second problem set.

Lesson 25: Introduction to Non-Proliferation and Policy / FNEK 2 [Coach/Fade]

Assignment
Read supplements on Canvas

Think about the following questions. We'll discuss them in class.

1. What is nuclear proliferation?

2. Who are the nuclear weapon states (NWS)?

3. What are the methods/routes a state can/have taken to become an NWS?

4. What is the governing policy against the proliferation of nuclear weapons technology?

Notes:
A 60-mark (15 minute) Fundamentals of Nuclear Engineering Knowledge writ will be administered. The writ will cover the following topics:

- On the Fundamentals of Nuclear Engineering Knowledge Card
  - Physical Constants & Unit Conversions
  - Units
  - Graphs and their significance
  - Fundamentals of Nuclear Engineering (NE300)
  - Radiological Engineering Design (NE350)
- On the Fundamentals of Nuclear Engineering Knowledge Concepts Card
  - Fundamentals of Nuclear Engineering (NE300)
  - Radiological Engineering Design (NE350)

Lesson 26: Nuclear Waste [Coach]

<u>Assignment</u>
Review: Nuclear Engineering Design Supplement Standards for Technical Reports
Review: Provided articles and literature on nuclear waste (see Canvas)

Learning Objectives (Nuclear Waste):

1. Understand how nuclear waste develops

2. Understand the public's concern with nuclear waste

3. Understand the pathways for nuclear waste in the US

Learning Objectives (IPR 3):

1. Recognize the graded requirements for the Nuclear Engineering Design Project (NEDP). See the rubric.

2. Refine your calculations needed for your specific project in preparation for your final submission.

Vocabulary:

vitrification	repository	storage pool
dry storage cask	low-level waste	transuranic waste
high-level waste	spent fuel	radwaste

Notes:
IPR 3 for the NEDP will be digitally submitted on the day of this class.

Lesson 27: Current Events in Nuclear Security [Coach/Fade]

<u>**Assignment**</u>
Read supplements and web links on Canvas

Think about the following questions. We'll discuss them in class.

1. What is required for a nuclear weapon?
2. Uranium vs plutonium: is one better for a clandestine weapons program?
3. How does radioactive waste play into non-proliferation?
4. Why is material control and accountability important?

Lessons 28/29: Design Project Briefs [Fade]

Assignment

1. Brief your team's design project to the class. You will brief for 20 minutes followed by a 5-minute question and answer period. The assignment rubric will be posted to the course website.

2. Provide feedback and ask questions of other groups during their design project briefs.

Lesson 30: Problem-Solving Session [Coach/Fade]

<u>Assignment</u>
Review: Lessons 1-27.

Learning Objective:

Know learning objectives, vocabulary, notes, and homework problems primarily from lessons 1 through 27. Other problems from throughout the semester may be worked in preparation for the Term End Examination.

Notes:
This lesson will serve as a term end exam review. Students will take an end of course survey, as well.

SECTION I

COURSE ASSESSMENTS

TAB A AY2026 COURSE ASSESSMENT

NE350 Radiological Engineering Design
AY 26-2

ARMY WEST POINT™
NUCLEAR ENGINEERING

MAJ Joshua Mashl
NE350 Course Director

Agenda

AY 26-2 NE350 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE350 Assessment Process
- AY26 Course Assessment
- Recommended Changes for AY27

AY25 Program Director's Recommendations (for AY26)

Adopt CAT recommendations and

- Properly complete ABET record year data collection
- Integrate new junior and senior faculty into program
- Continue to work on assessment tools that correctly identify any lack of fundamental skills to include time management (i.e. late turn-ins of homework)
- Continue assessment SO (1)-(7)

Standardization of briefings

- Continue standardized briefs and reports for NE-specific topics
- Continue use of Overleaf

Effective use of NE labs and spaces

- Finish cleaning and organizing subcrit, BH50 and accelerator spaces
- Integrate PROLAB into curriculum (especially NE355)
- Continue registration for cluster (VM) with RSIC and meet all compliance for use
- Work with IT for replacement for VMware

AY25 Program Director's Recommendations (for AY26)

(2 of 3)

Complete updates and integration of text for NE450

Integrate new version of *Nuclear Engineering Design Supplement*

- Use document for NE350 and NE495/496 and note errors
- Complete instruction for Project Management using document

Sustain internal and external relations:

- Include ANS Club, 1 activity per month
- Trip sections for courses (STERIS)

Continue to develop and engage lab tech in the program

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Continue to improve FE exam preparation between NE400 and NE496

AY25 Program Director's Recommendations (for AY26)

(3 of 3)

NE Working Group Topics

- Shift leadership responsibilities between senior faculty to include new AP
- ABET record year
- Work on future manning for near and sustained term with department leadership
- Continue work with OCE and other outside organizations for research and manning
- Continue to improve use of advanced computational tools such as Attila4MC throughout the program
- Annual update of internal and external websites – ensure compliance with ABET APPM

Course Director’s Areas of Focus
Academic Year 2026

NE350

- NE Outcomes are: 2, 5
- Implement CAT recommendations
- Add a third problem set to capture course concepts covered in class after problem set 2 is submitted. **(Complete)**
- Increase maximum design project group size from 3 to 4 to allow more time for briefs for IPRs. **(Complete)**
- Sustain points make-up policy for problem sets and writs (incentivizes non-major students to engage with course material) **(Complete)**
- Implement new design supplement **(Complete)**
- Add nuclear waste lesson in place of the last design project work period **(Complete)**

NE Student Outcomes

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

2. **an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors**

3. an ability to communicate effectively with a range of audiences

4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

5. **an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives**

6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Concept Map: Key

Understand and implement the nuclear engineering design process

Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection

Understand medical and industrial applications for radiation technology other than commercial power generation

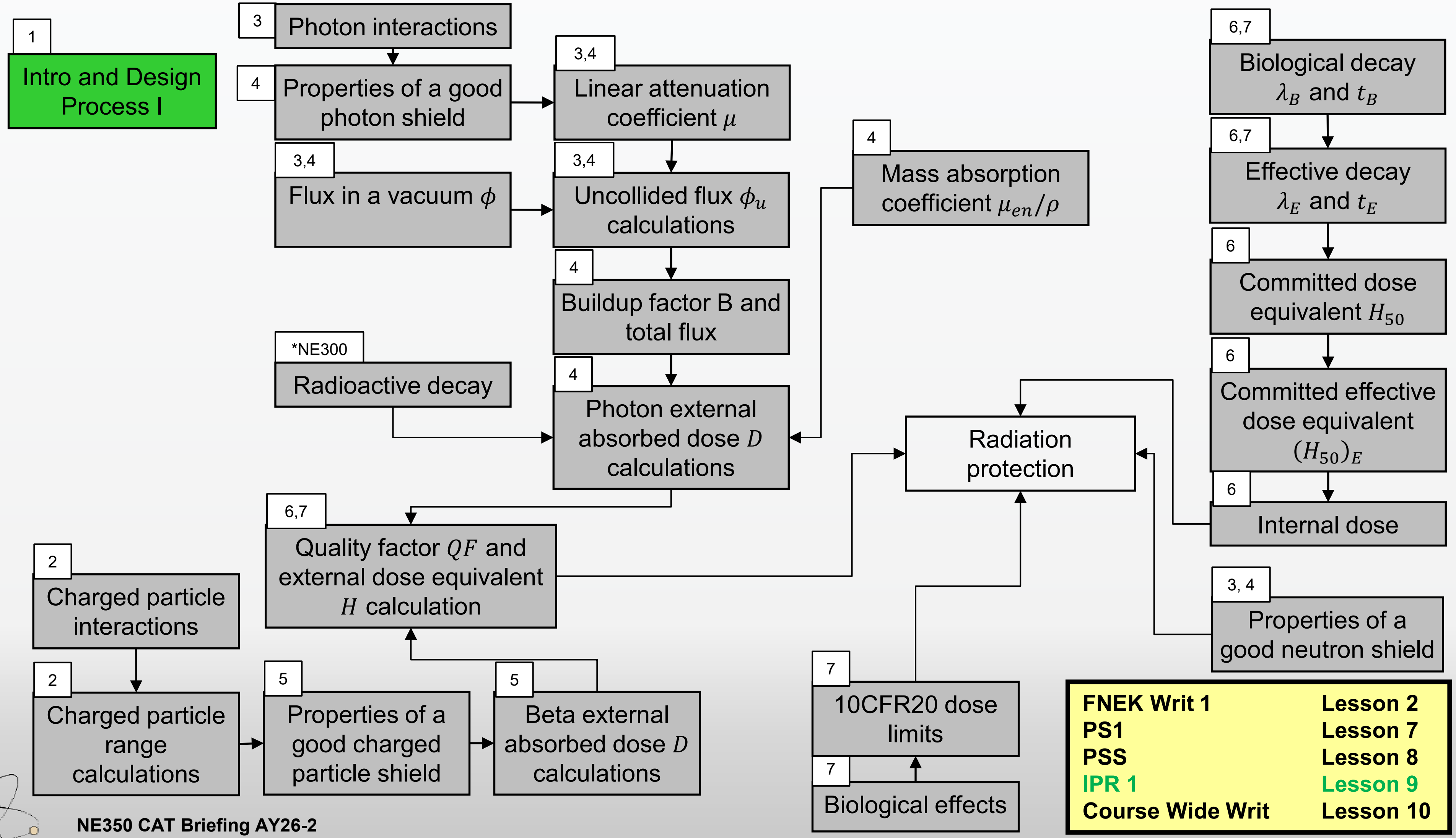
Understand how radiation detection devices function and how to interpret their outputs

Understand non-proliferation and how the concerns of security and safety play into the global discussion about the safe use of nuclear technology

Concept Map for NE350, CWW, Block 1

Understand and implement the nuclear engineering design process

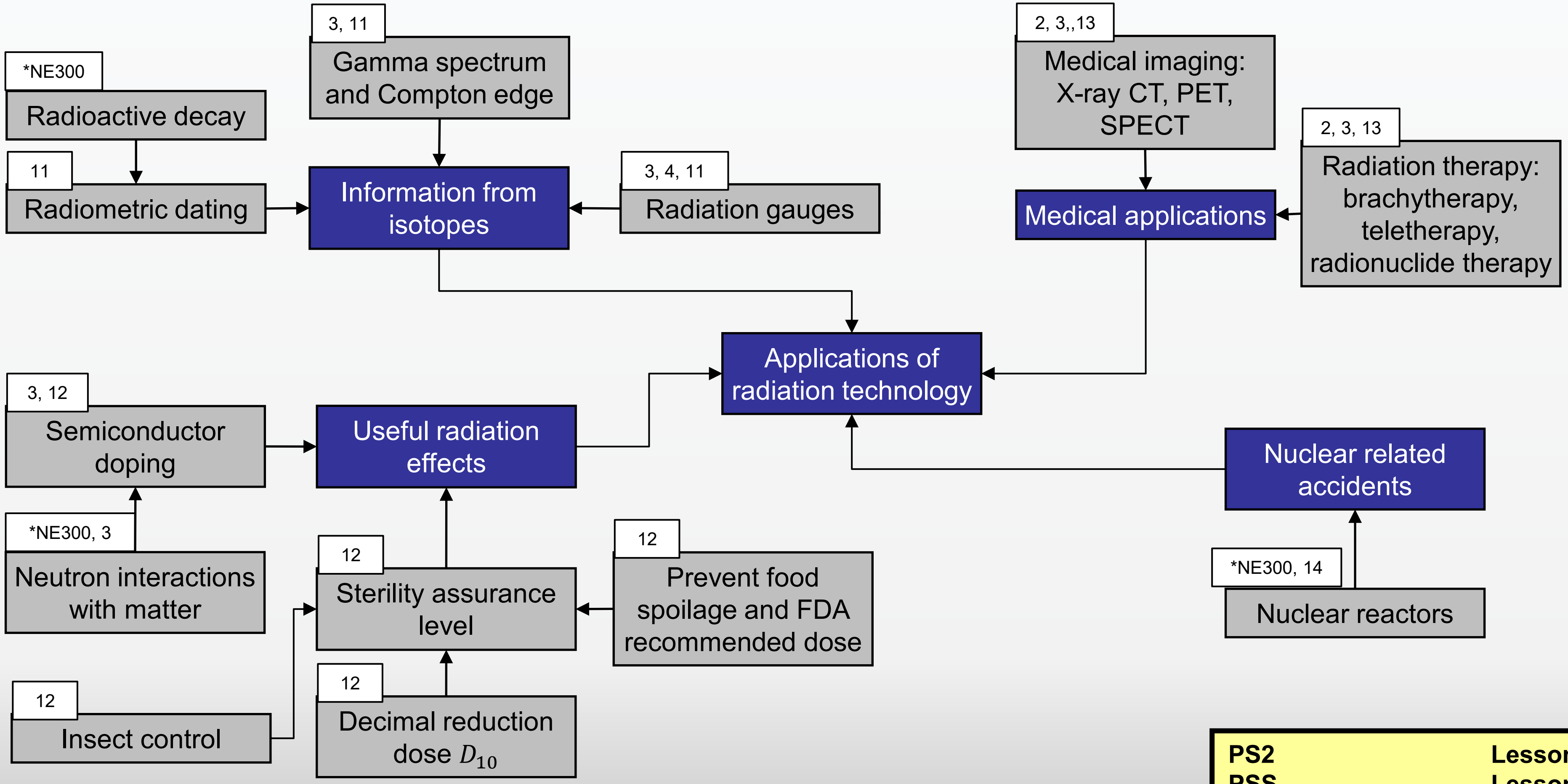
Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection



Concept Map for NE350, WPR, Block 2

Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection

Understand medical and industrial applications for radiation technology other than commercial power generation

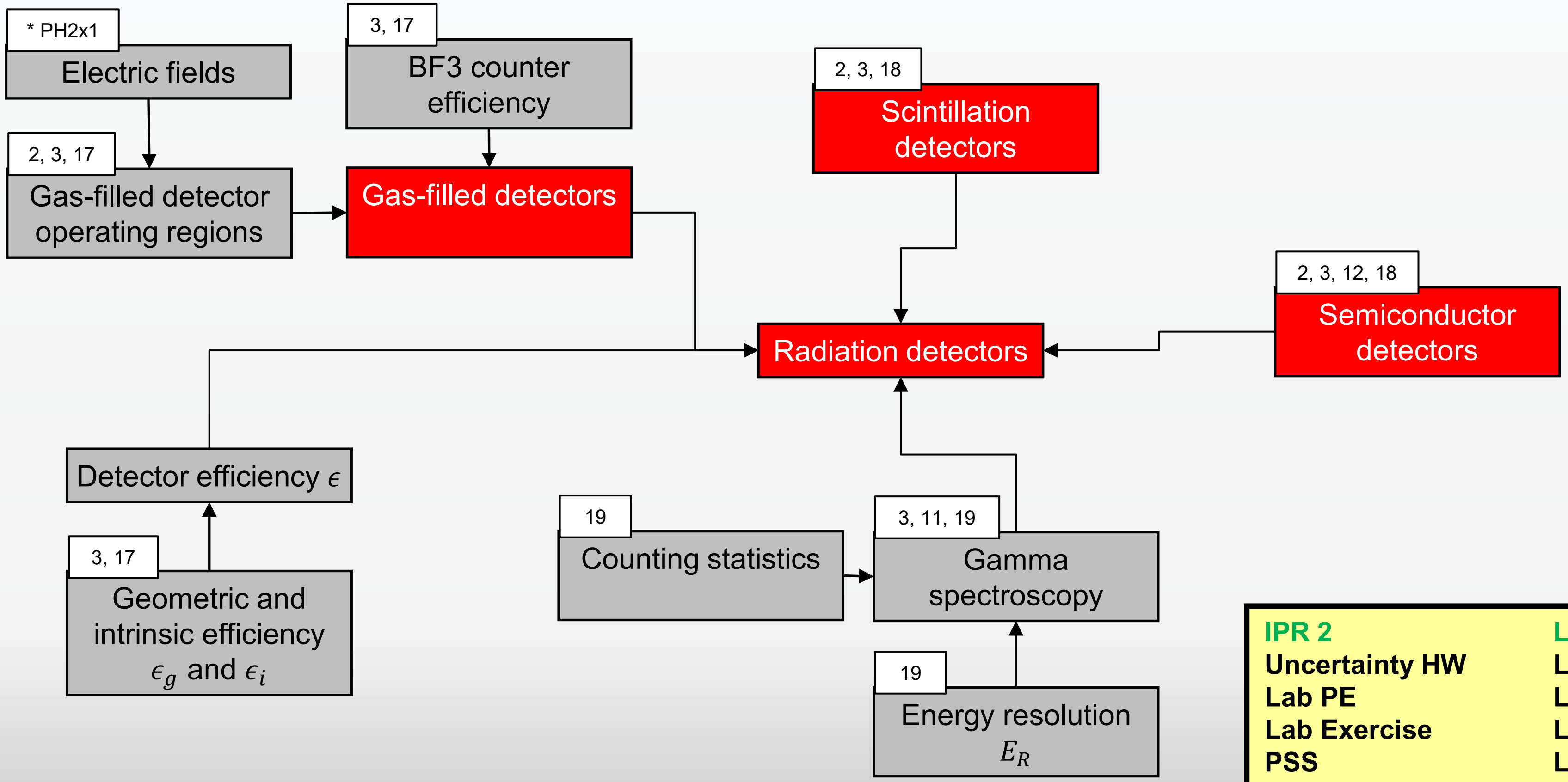


PS2	**Lesson 15**
PSS	**Lesson 15**
WPR	**Lesson 17**
Steris Trip	**Lesson 16**

Concept Map for NE350, Lab, Block 3

Understand and implement the nuclear engineering design process

Understand how radiation detection devices function and how to interpret their outputs

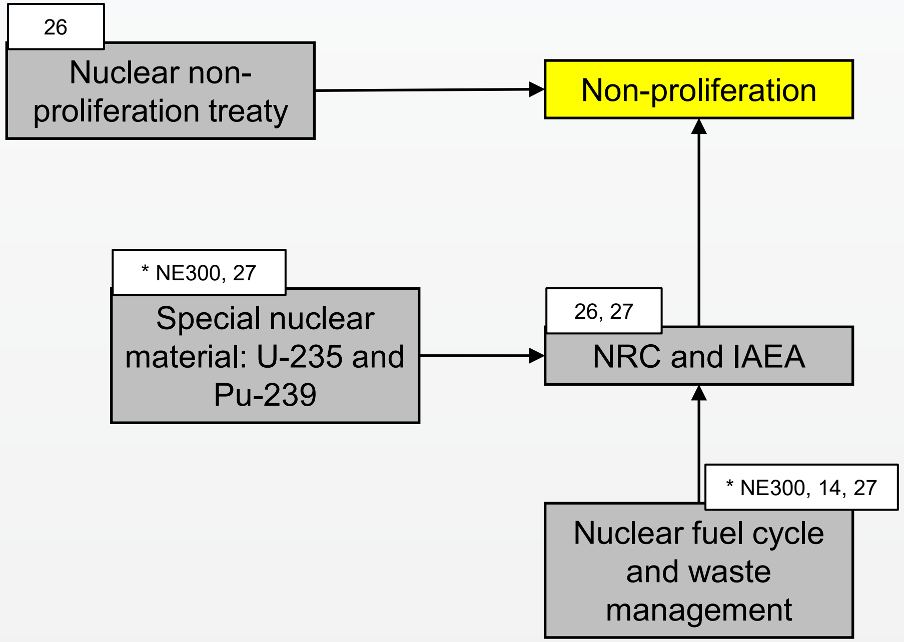


IPR 2	Lesson 21
Uncertainty HW	Lesson 22
Lab PE	Lesson 22
Lab Exercise	Lesson 23
PSS	Lesson 24
IPR 3	Lesson 25
Lab Due	Lesson 27

Concept Map for NE350, Project/TEE, Block 4

Understand and implement the nuclear engineering design process

Understand non-proliferation and how the concerns of security and safety play into the global discussion about the safe use of nuclear technology



FNEK Writ 2	Lesson 26
Project presentations	L 28/29
PSS	Lesson 30
Project write up due	L29
TEE	

4-Step Evaluation & Assessment Process

1. Identify the connections between:
   - Course block objectives and student outcomes
   - Lesson learning objectives and course block objectives by developing a concept map

2. Select multiple embedded indicators to measure against established rubrics that will:
   - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
   - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)

3. Assess the level of attainment of student outcomes based on the population distribution of cadet performance on the embedded indicators as:

Descriptor	Color Code	Likert Rating	Criteria
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Embedded Indicators (Tools)	NE Student Outcomes						
	1	2	3	4	5	6	7
Problem sets	X						X
Laboratory Exercise					X	X	
Exams	X						
NEDP IPRs		X			X		X
NEDP Presentation		X	X	X	X		
NEDP Report		X		X	X		

Assessment Instruments
(for each student outcome)
Red Indicates Best Instruments

- SO 2 – NEDP Report
  - NEDP IPRs
  - NEDP Presentation

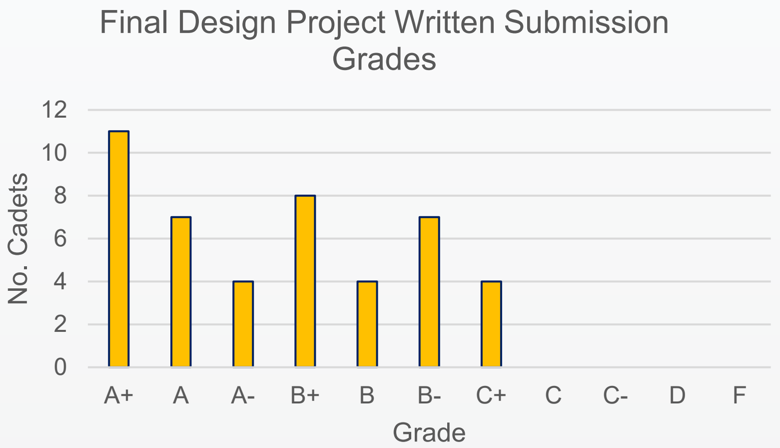
- SO 5 – NEDP IPRs
  - NEDP Presentation
  - NEDP Report
  - Radiation Measurement Lab

NE Student Outcome Assessment – Embedded Indicators

SO 2: NEDP Final Report

26-2

5



26-1

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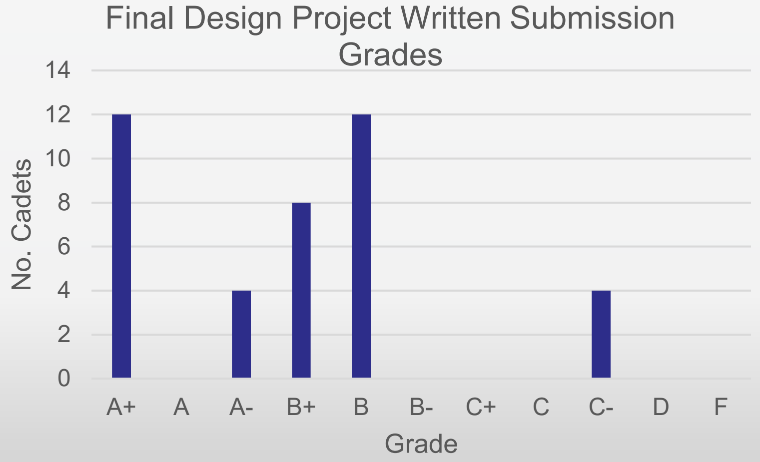

SO 5: NEDP IPRs

26-2

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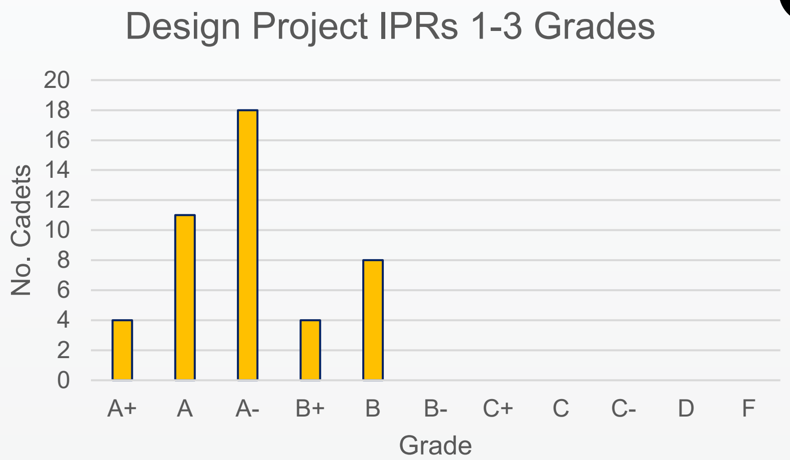

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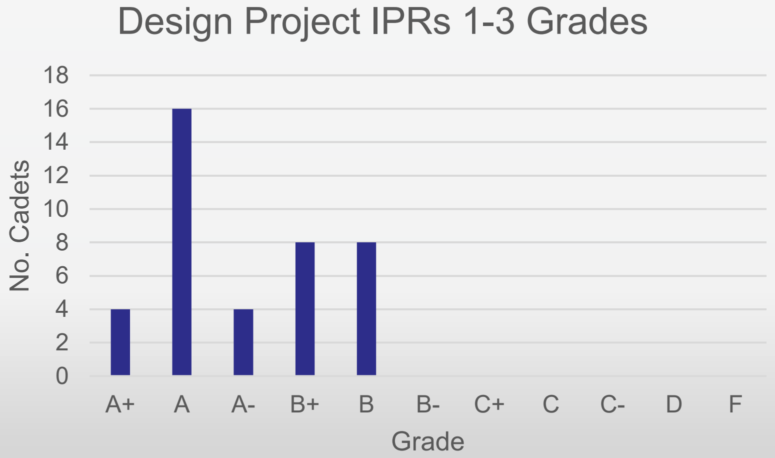
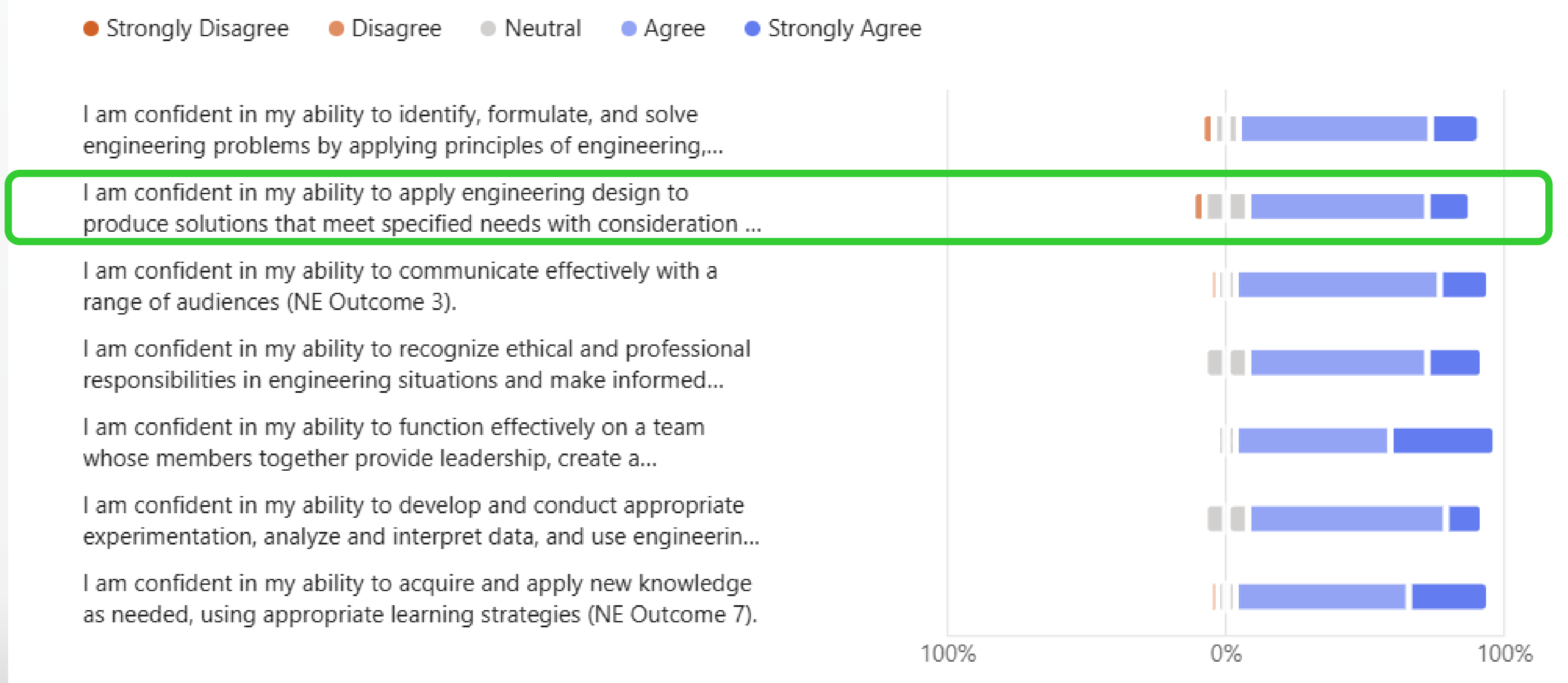


Course Feedback
(NE Student Outcome 2)

Course-Level Feedback

I am confident in my ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.



26-2 Responses; Score: 3.91

I am confident in my ability to apply engineering design to produce...

Disagree	4.4%
Neutral	15.6%
Agree	64.4%
Strongly Agree	15.6%

26-1 Responses; Score: 3.93

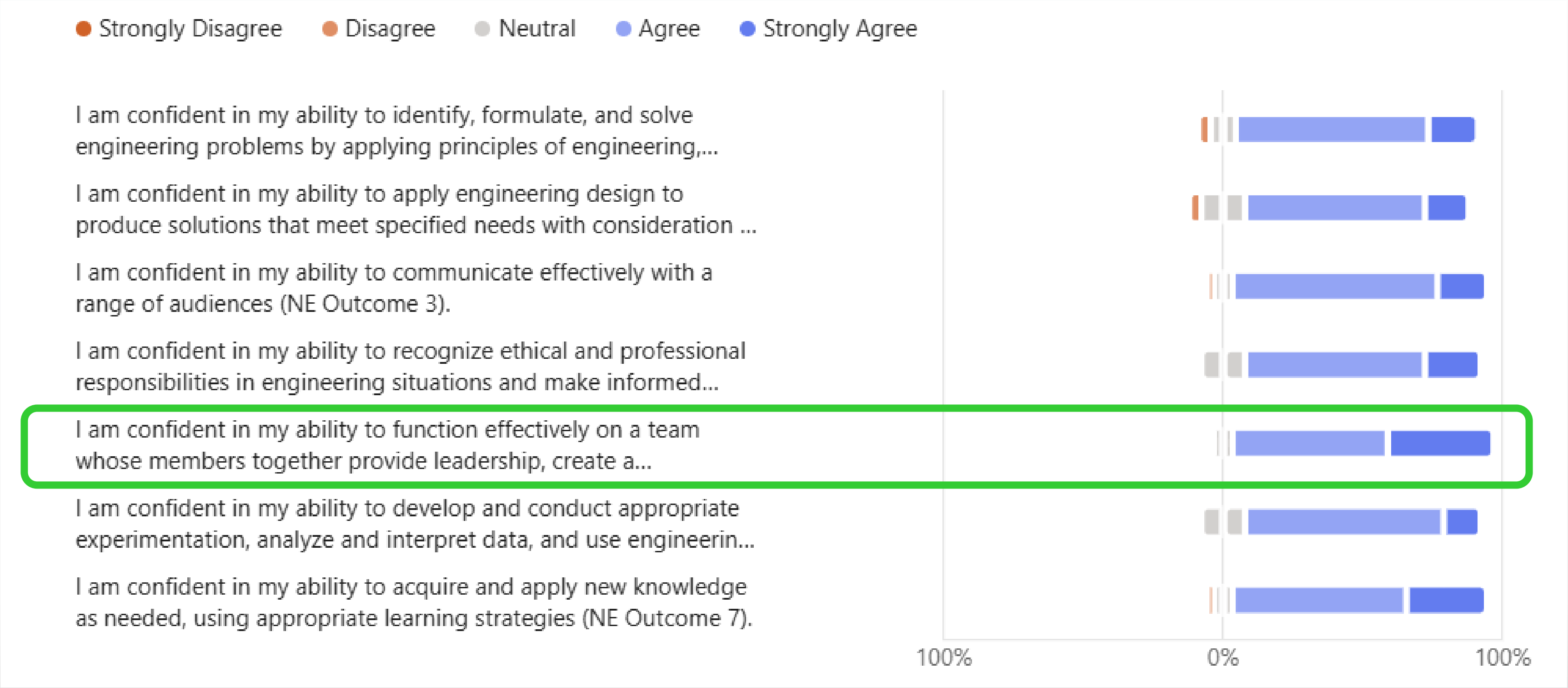
I am confident in my ability to apply engineering design to produce...

Disagree	2.4%
Neutral	24.4%
Agree	51.2%
Strongly Agree	22%

Course Feedback (NE Student Outcome 5)

Course-Level Feedback

I am confident in my ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.



26-2 Responses; Score 4.32

I am confident in my ability to function effectively on a team whos...

Neutral	6.7%
Agree	55.6%
Strongly Agree	37.8%

26-1 Responses; Score: 4.44

I am confident in my ability to function effectively on a team whos...

Neutral	4.9%
Agree	46.3%
Strongly Agree	48.8%

NE350 Course Director's Assessment – AY26-1

Student Outcomes	NE350	Indicators	Surveys	CD
1. Solve problems				
2. Engineering design	X	5.0	3.9	3.5
3. Communicate effectively				
4. Ethical responsibilities				
5. Function on team	X	5.0	4.3	3.7
6. Conduct experiments				
7. Apply knowledge				

This semester, I continued to receive improved design projects as most groups used CAD software for their designs. One major issue with this is that most groups only had one person focus on the design portion. The design project serves as a forcing function for teams to work together, though there are always some outliers. Most groups commented that they enjoyed the project and found it fulfilling. The indicator scores improved for both outcomes this semester, signifying improvement in the course.

Redbook Entry

NE350 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE350	RADIOLOGICAL ENGR DESIGN	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on nuclear engineering systems including radiation protection, shielding, and the uses of radioactive sources in industrial processes. Specific topics emphasize the operation of radiation detectors, shielding principles, health effects of radiation, radiological dispersion devices, and nuclear incidents. A design project applies the concepts presented in this course to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

One design project.

NE350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE					
	NE300	2004	1	1	Y

NE350 (Version 2019-2) COURSE OFFERINGS

AYT	#SECT/SIZE		CPBLTY	ENRLD	WAIT	SEATS	CLOSED	DETAILS
2026 - 1	3	18	54	45	0	9	N	Hours
2026 - 2	5	19	95	46	0	49	N	Hours
2027 - 1	3	18	54	34	0	20	N	Hours
2027 - 2	3	18	54	2	0	52	N	Hours
2028 - 1	3	18	54	1	0	53	N	Hours
2028 - 2	3	18	54	0	0	54	N	Hours

No changes to the Redbook suggested.

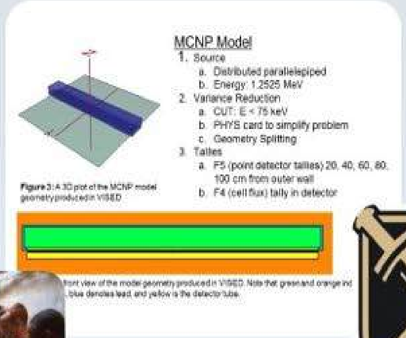
AY26-2 Recommended Changes (For AY-27)

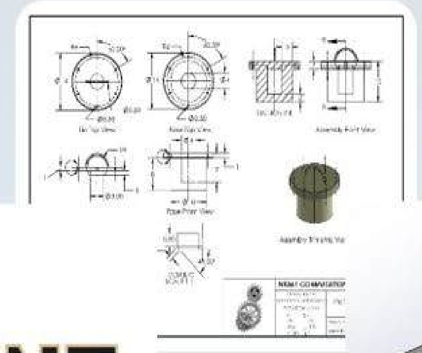
- Sustain encouragement of CAD software for design projects
- Sustain points-back policy for CWW and problem sets
- Sustain groups of 4 for the design project
- Sustain the inclusion of the nuclear waste lesson
- Incorporate more nuclear waste and waste protection into the course; it complements the design aspect and rad protection focus of the course
- Implement GradeScope grading
- Increase rigor of TEE cutsheet; the problems are fine but the cutsheet is too lenient on mistakes

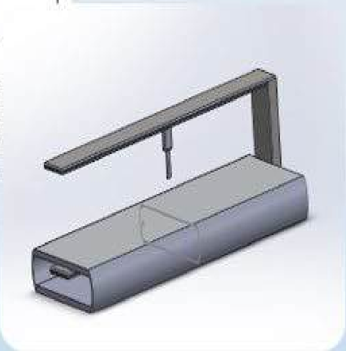
Resource Requirements

- None

Questions?







ARMY WEST POINT™
NUCLEAR ENGINEERING

TAB B AY2025 COURSE ASSESSMENT

NE350 Radiological Engineering Design
AY 25-2

ARMY WEST POINT™
NUCLEAR ENGINEERING

MAJ Joshua Mashl
NE350 Course Director

Agenda

AY 25-2 NE350 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE350 Assessment Process
- AY25 Course Assessment
- Recommended Changes for AY26

AY24 Program Director's Recommendations (for AY25)

(1 of 3)

Adopt CAT recommendations and

- Fully use Canvas to its highest potential
- Continue to work on assessment tools that correctly identify any lack of fundamental skills to include time management (i.e. late turn-ins of homework)
- Continue assessment SO (1)-(7)

Standardization of briefings

- Continue standardized briefs and reports for NE-specific topics
- Continue use of Overleaf

Effective use of NE labs and spaces

- Finish cleaning and organizing subcrit, BH50 and accelerator spaces
- Continue Orientation for 2^{nd} class for BH347 as collaborative space
- Continue Orientation for 1^{st} class for BH66B as collaborative space
- Continue registration for cluster (VM) with RSIC and meet all compliance for use
- Work with IT for replacement for Vmware

AY24 Program Director's Recommendations (for AY25)

(2 of 3)

Complete updates and integration of text for NE450

Update *Nuclear Engineering Design Supplement*

- Include format for a standard template for peer-reviewed journal and conference poster and presentation
- Change "restrictions" to "standards"
- Change IPR2 and remove "new start"

Sustain internal and external relations:

- Include ANS Club, 1 activity per month
- Trip sections for courses (STERIS)

Continue to develop and engage lab tech in the program

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Continue to improve FE exam preparation between NE400 and NE496

AY24 Program Director's Recommendations (for AY25)

NE Working Group Topics

- Divide leadership responsibilities between senior faculty
- Begin preparations for ABET record year
- Work on future manning for near and sustained term with department leadership
- Continue work with OCE and other outside organizations for research and manning
- Continue to improve use of advanced computational tools such as Attila4MC throughout the program
- Annual update of internal and external websites – ensure compliance with ABET APPM

Program Director's Areas of Focus
Academic Year 2025

NE350

- NE Outcomes are: 2, 5
- Implement CAT recommendations
- Add a third problem set to capture course concepts covered in class after problem set 2 is submitted. **(Complete)**
- Increase maximum design project group size from 3 to 4 to allow more time for briefs for IPRs. **(Complete)**
- Sustain points make-up policy for problem sets and writs (incentivizes non-major students to engage with course material) **(Complete)**
- Implement new design supplement **(Not Complete)**
- Add nuclear waste lesson in place of the last design project work period **(Not Complete)**

NE Student Outcomes

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

2. **an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors**

3. an ability to communicate effectively with a range of audiences

4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

5. **an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives**

6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Concept Map: Key

Understand and implement the nuclear engineering design process

Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection

Understand medical and industrial applications for radiation technology other than commercial power generation

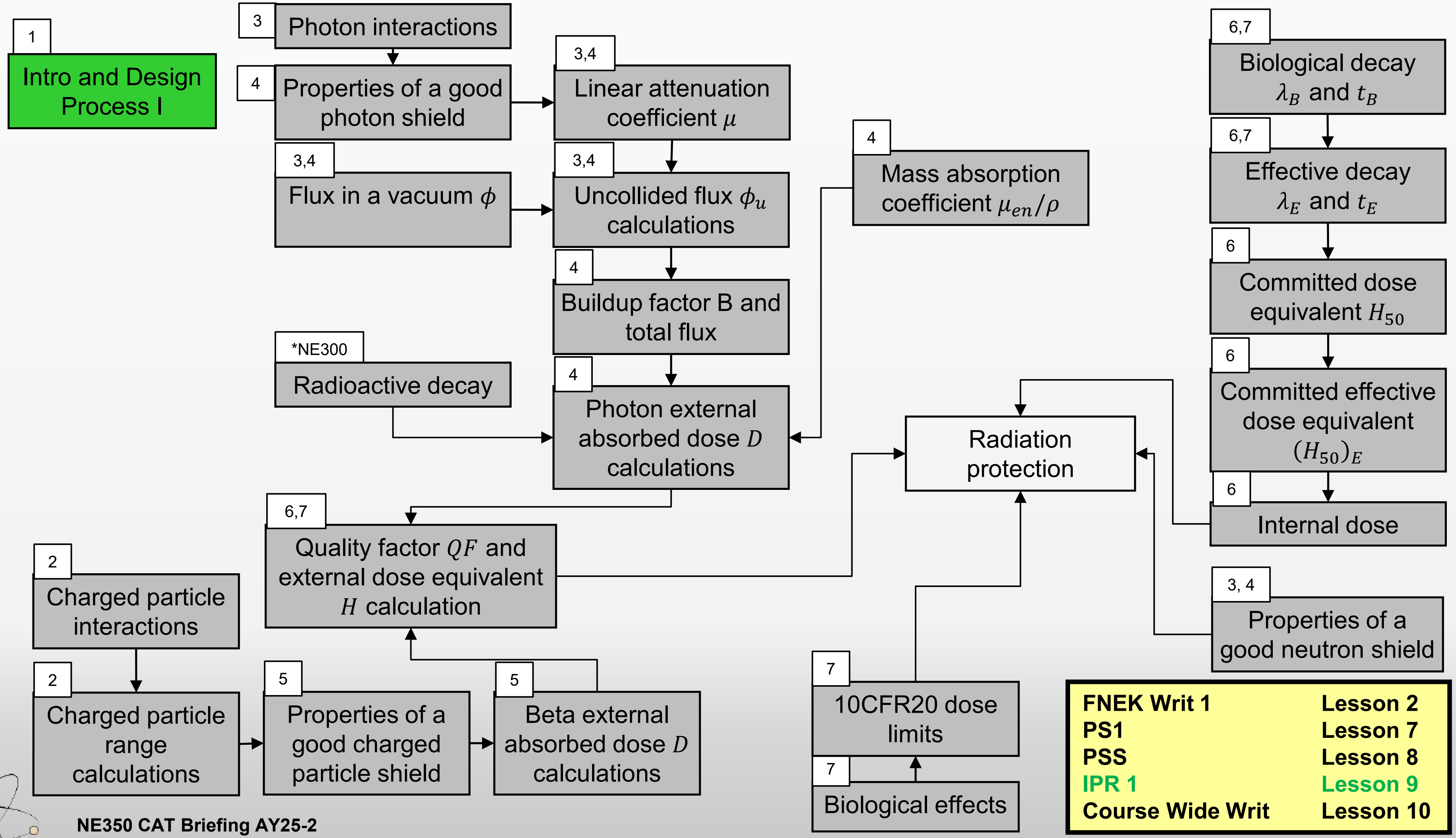
Understand how radiation detection devices function and how to interpret their outputs

Understand non-proliferation and how the concerns of security and safety play into the global discussion about the safe use of nuclear technology

Concept Map for NE350, PLW/CWW, Block 1

Understand and implement the nuclear engineering design process

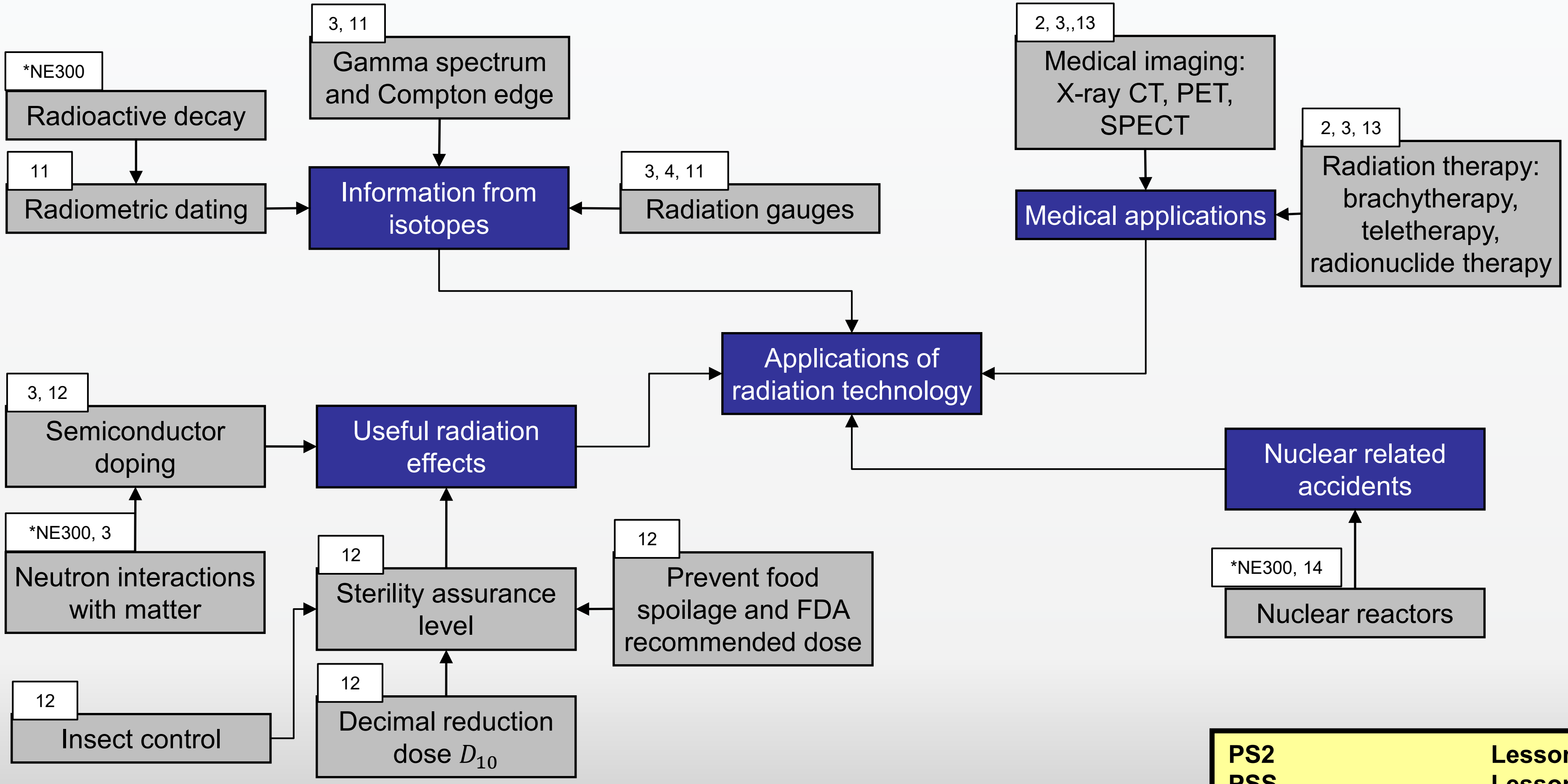
Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection



Concept Map for NE350, WPR, Block 2

Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection

Understand medical and industrial applications for radiation technology other than commercial power generation

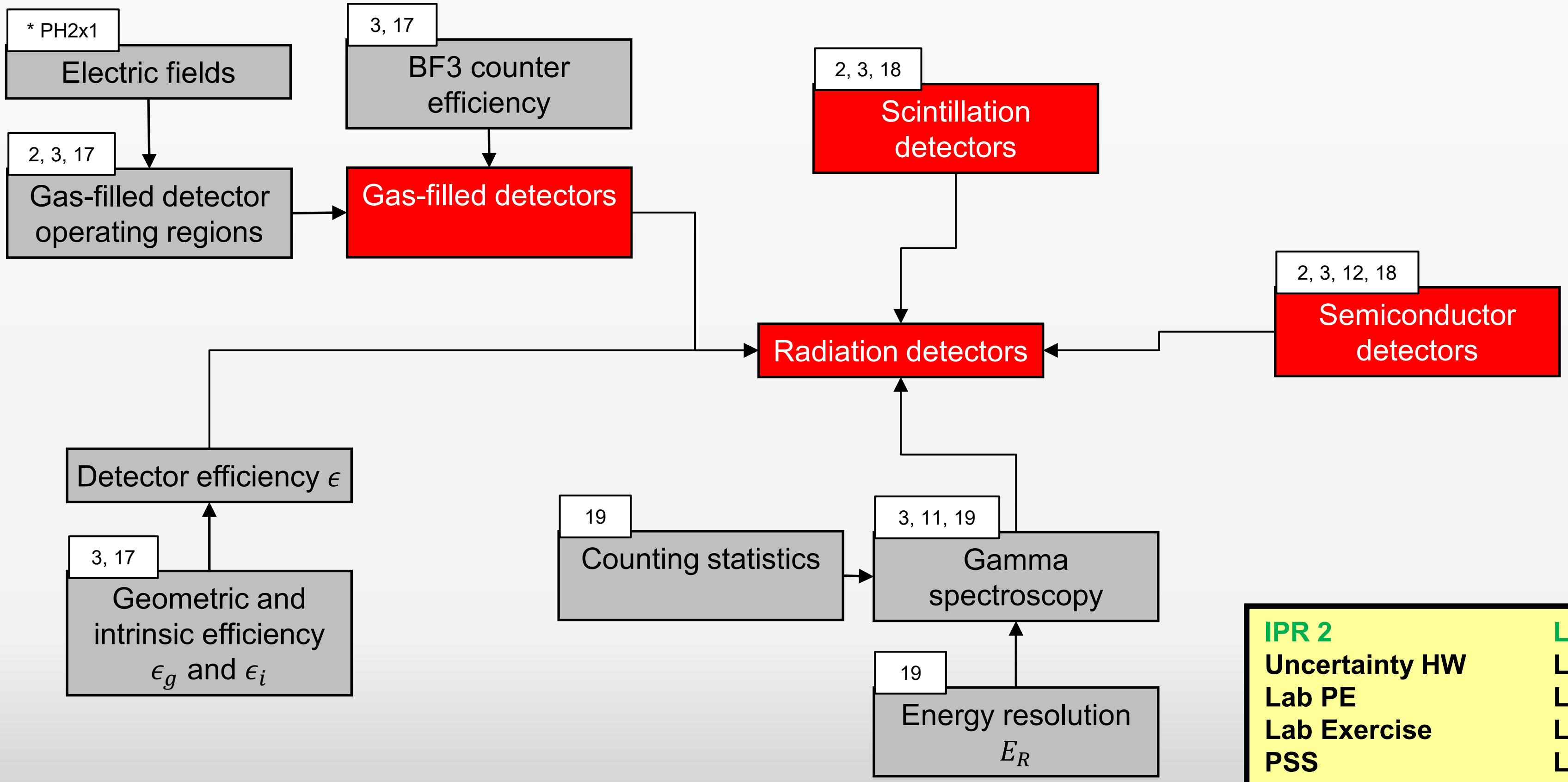


PS2	**Lesson 15**
PSS	**Lesson 15**
WPR	**Lesson 17**
Steris Trip	**Lesson 16**

Concept Map for NE350, Lab, Block 3

Understand and implement the nuclear engineering design process

Understand how radiation detection devices function and how to interpret their outputs

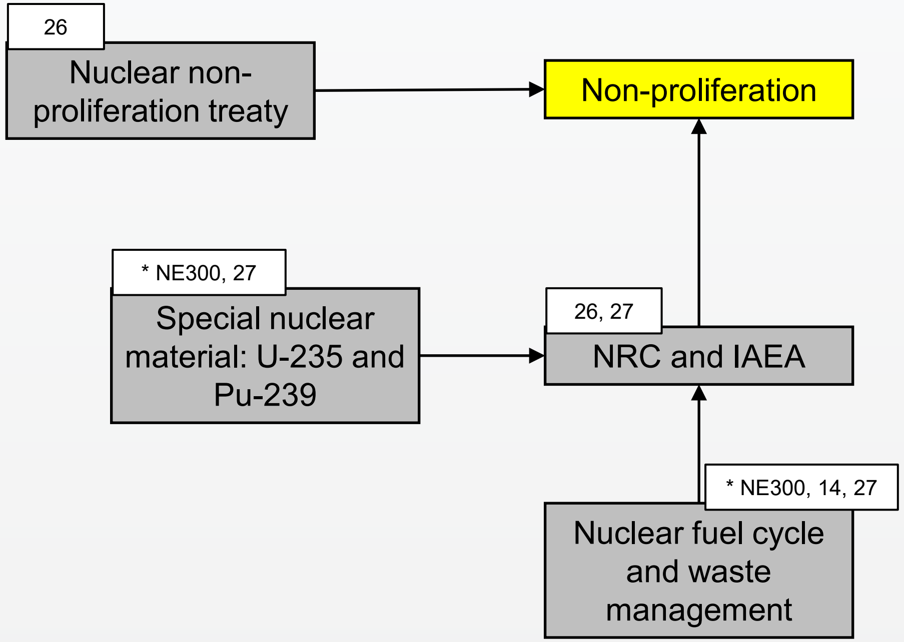


IPR 2	Lesson 21
Uncertainty HW	Lesson 22
Lab PE	Lesson 22
Lab Exercise	Lesson 23
PSS	Lesson 24
IPR 3	Lesson 25
Lab Due	Lesson 27

Concept Map for NE350, Project/TEE, Block 4

Understand and implement the nuclear engineering design process

Understand non-proliferation and how the concerns of security and safety play into the global discussion about the safe use of nuclear technology



FNEK Writ 2	**Lesson 26**
Project presentations	**L 28/29**
PSS	**Lesson 30**
Project write up due	**L29**
TEE	

4-Step Evaluation & Assessment Process

1. Identify the connections between:
   - Course block objectives and student outcomes
   - Lesson learning objectives and course block objectives by developing a concept map

2. Select multiple embedded indicators to measure against established rubrics that will:
   - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
   - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)

3. Assess the level of attainment of student outcomes based on the population distribution of cadet performance on the embedded indicators as:

Descriptor	Color Code	Likert Rating	Criteria
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Embedded Indicators (Tools)	NE Student Outcomes						
	1	2	3	4	5	6	7
Problem sets	X						X
Laboratory Exercise					X	X	
Exams	X						
NEDP IPRs		X			X		X
NEDP Presentation		X	X	X	X		
NEDP Report		X		X	X		

Assessment Instruments
(for each student outcome)
Red Indicates Best Instruments

- SO 2 – NEDP Report
  - NEDP IPRs
  - NEDP Presentation

- SO 5 – NEDP IPRs
  - NEDP Presentation
  - NEDP Report
  - Radiation Measurement Lab

NE Student Outcome Assessment – Embedded Indicators

SO 2: NEDP Final Report

25-1

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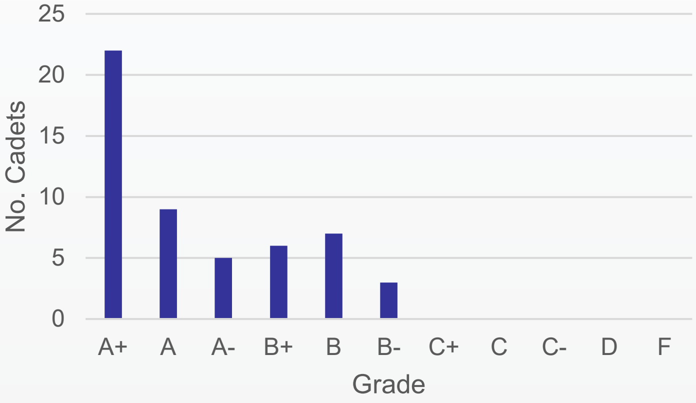

25-2

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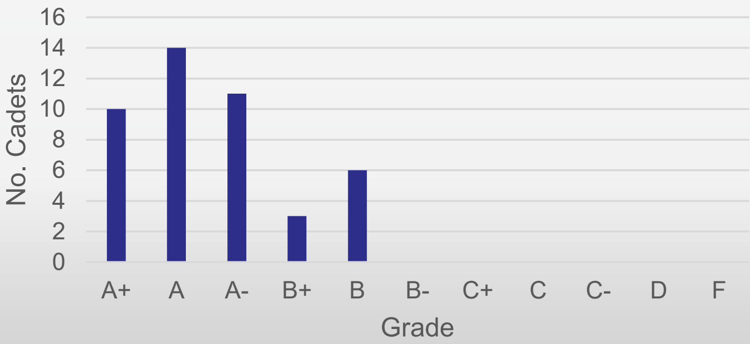
Final Design Project Written Submission



SO 5: NEDP IPRs

25-1

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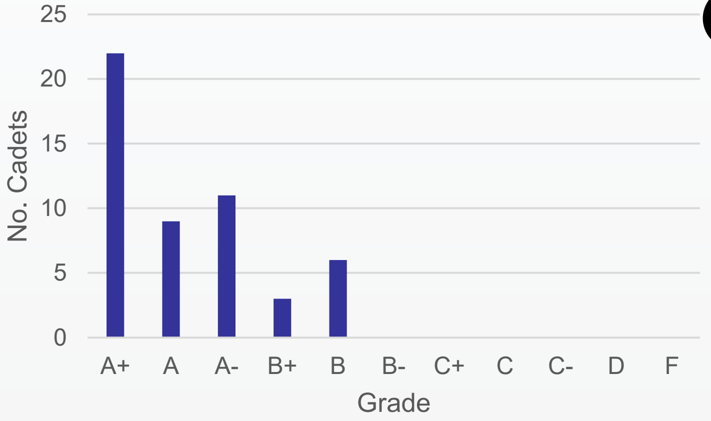

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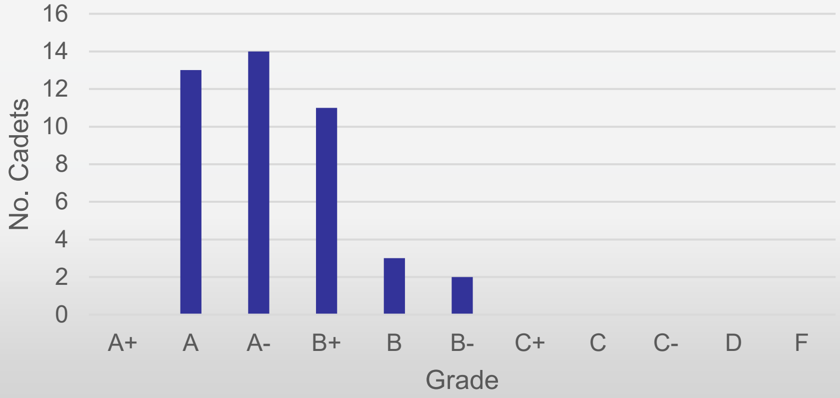
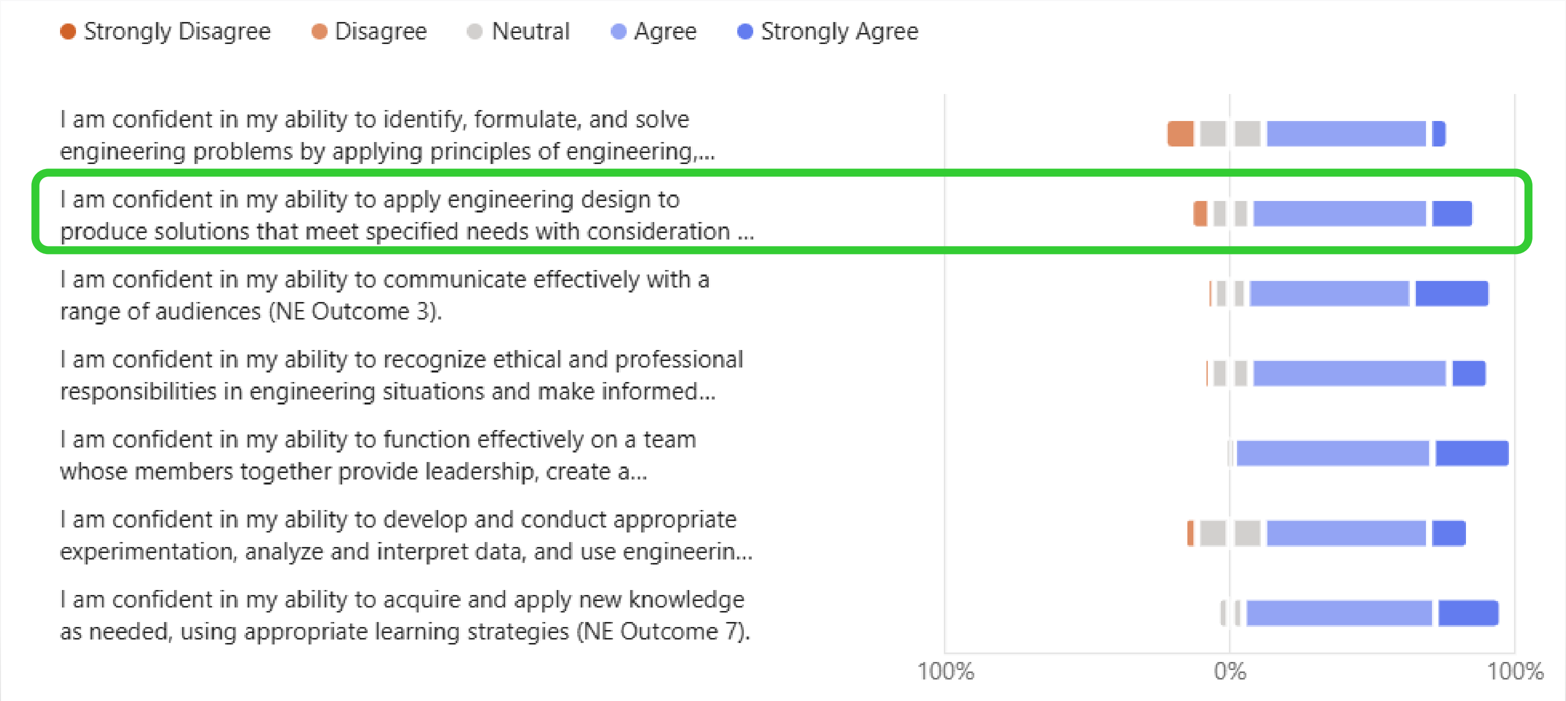
Design Project IPRs 1-3



Course Feedback (NE Student Outcome 2)

Course-Level Feedback

I am confident in my ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.



25-1 Responses; Score: 3.92

I am confident in my ability to apply engineering design to produce...

Disagree	1.9%
Neutral	9.6%
Agree	82.7%
Strongly Agree	5.8%

25-2 Responses; Score: 3.89

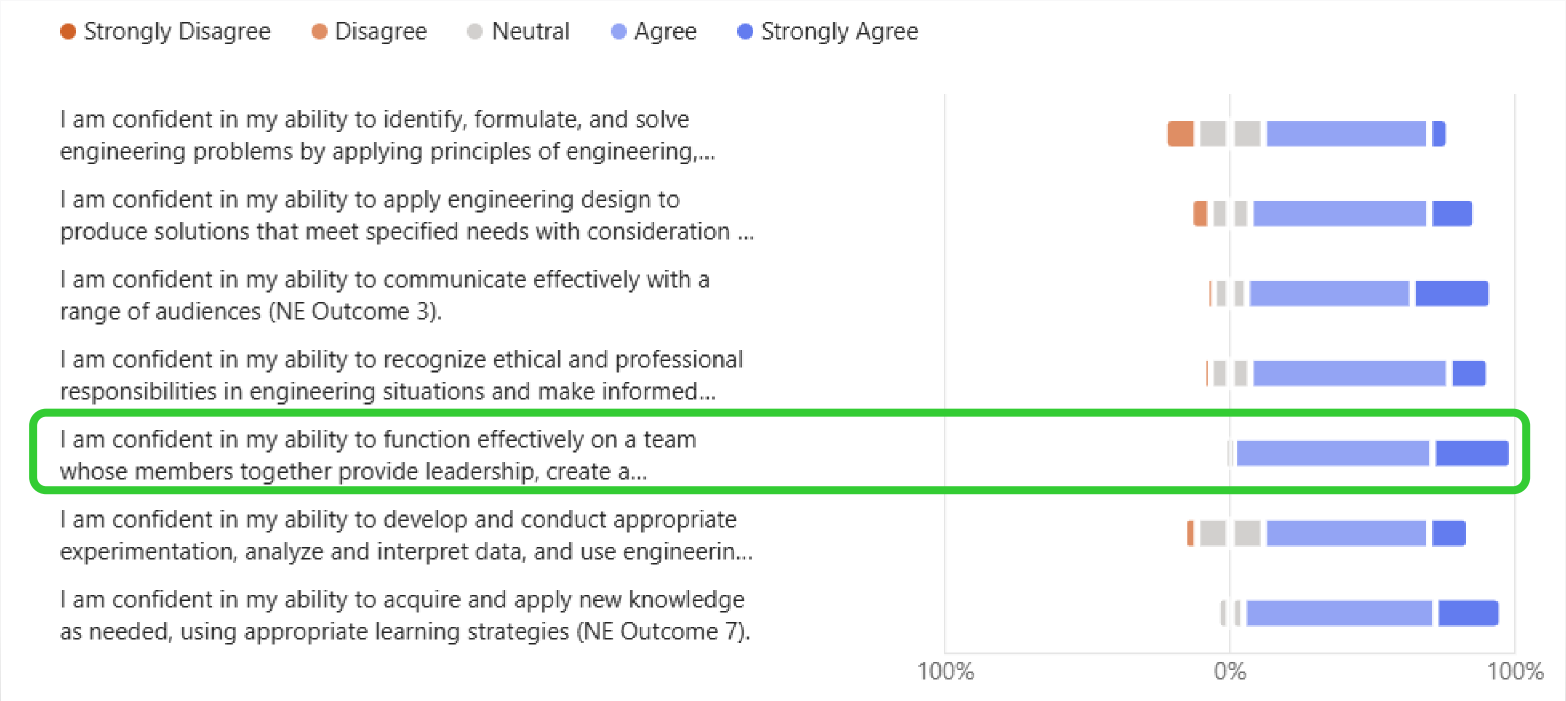
I am confident in my ability to apply engineering design to produce...

Disagree	7%
Neutral	14%
Agree	62.8%
Strongly Agree	16.3%

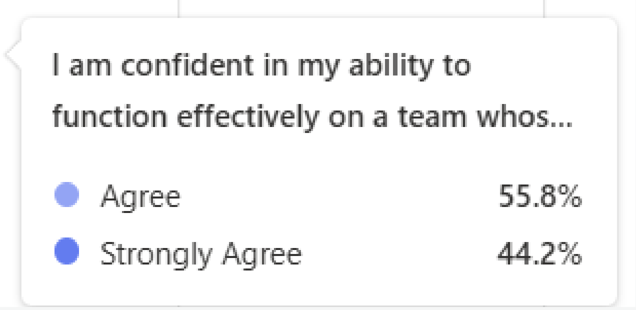
Course Feedback (NE Student Outcome 5)

Course-Level Feedback

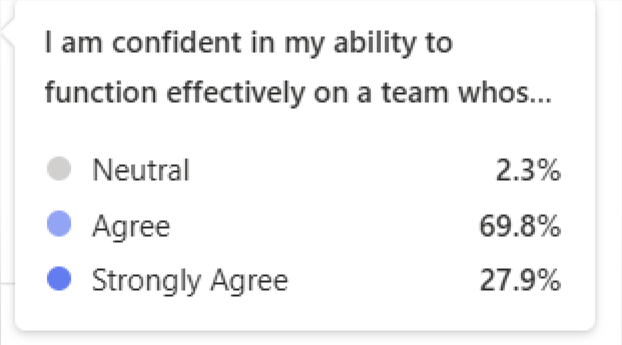
I am confident in my ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.



25-1 Responses; Score 4.42



25-2 Responses; Score: 4.26



NE350 Course Director's Assessment – AY25

Student Outcomes	NE350	Indicators	Surveys	CD
1. Solve problems				
2. Engineering design	X	5.0	3.9	3.9
3. Communicate effectively				
4. Ethical responsibilities				
5. Function on team	X	5.0	4.3	3.8
6. Conduct experiments				
7. Apply knowledge				

This semester, we saw an improvement design project deliverables. I pushed the cadets harder and graded more harshly, but the products themselves are better. Although the indicators did not change, the survey results decreased slightly. I think this was due to the difficulty of the project which many students do not believe is appropriate for a non-major engineering sequence course. Another gripe cadets have is the level of guidance provided for the project, which I agree needs improvement. I hope to give this improved guidance with a new design supplement and new project topics.

Redbook Entry

NE350 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE350	RADIOLOGICAL ENGR DESIGN	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on nuclear engineering systems including radiation protection, shielding, and the uses of radioactive sources in industrial processes. Specific topics emphasize the operation of radiation detectors, shielding principles, health effects of radiation, radiological dispersion devices, and nuclear incidents. A design project applies the concepts presented in this course to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

One design project.

NE350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE					
	NE300	2004	1	1	Y

NE350 (Version 2019-2) COURSE OFFERINGS

AYT	#SECT/SIZE		CPBLTY	ENRLD	WAIT	SEATS	CLOSED	DETAILS
2026 - 1	3	18	54	45	0	9	N	Hours
2026 - 2	5	19	95	46	0	49	N	Hours
2027 - 1	3	18	54	34	0	20	N	Hours
2027 - 2	3	18	54	2	0	52	N	Hours
2028 - 1	3	18	54	1	0	53	N	Hours
2028 - 2	3	18	54	0	0	54	N	Hours

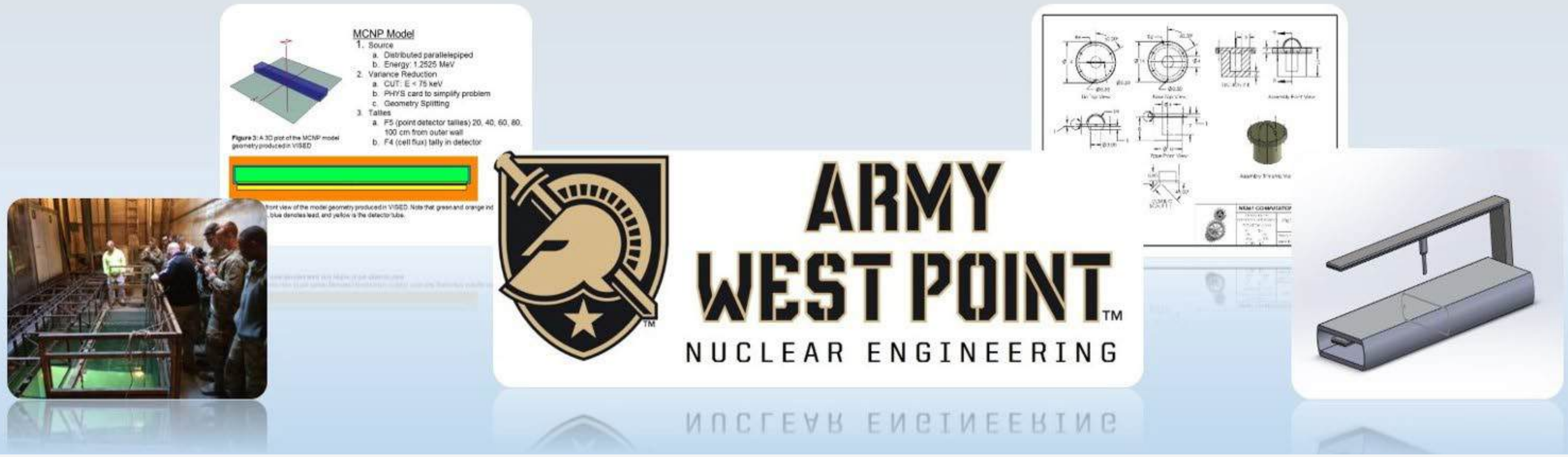
AY25-2 Recommended Changes (For AY26-1)

- **Adjust points make-up policy for only problem sets (incentivizes non-major students to engage with course material)**
- **Implement new design supplement**
- **Add nuclear waste lesson in place of the last design project work period**

Resource Requirements

- **None**

Questions?


MCNP Model
1. Source
a. Distributed parallelepiped
b. Energy: 1.2525 MeV
2. Variance Reduction
a. CUT: E < 75 keV
b. PHYS card to simplify problem
c. Geometry Splitting
3. Tallies
a. F5 (point detector tallies) 20, 40, 60, 80, 100 cm from outer wall
b. F4 (cell flux) tally in detector
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TAB C AY2024 COURSE ASSESSMENT

NE350: Radiological Engineering Design AY24

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NUCLEAR ENGINEERING

Agenda: AY24 NE350 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE350 Assessment Process
- AY24 Course Assessment
- Recommended Changes for AY25

AY23 Program Director's Recommendations for AY24 (1 of 3)

Adopt CAT recommendations and:

- Fully use Canvas to its highest potential
- Implement Canvas assessment into courses and program
- Work on assessment tools that correctly identify any lack of fundamental skills to include time management (i.e. late turn-ins of homework)
- Continue assessment SO (1)-(7)

Standardization of briefings:

- Continue to work with Core Program and CD Academy for standard WPR and grades briefs
- Continue standardized briefs and reports for NE-specific topics
- Continue use of Overleaf

Effective use of NE labs and spaces:

- Finish cleaning and organizing subcrit, BH50 and accelerator spaces
- Continue Orientation for 2nd class for BH347 as collaborative space
- Continue Orientation for 1st class for BH66B as collaborative space
- Complete registration for cluster (VM) with RSIC and meet all compliance for use

AY23 Program Director's Recommendations for AY24 (2 of 3)

Work on a comprehensive text for NE450

Update *Nuclear Engineering Design Supplement:*

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Include ANS Club, 1 activity per month
- Trip sections for courses (STERIS)

Integrate new lab tech into program

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Coordinate a better FE exam preparation between NE400 and NE496

AY23 Program Director's Recommendations for AY24 (3 of 3)

NE Working Group Topics:

- Work on future manning for near and sustained term with department leadership
- Continue and expand sponsored research on accelerator with outside organizations
- Continue to improve use of advanced computational tools such as Attila4MC throughout the program
- Annual update of internal and external websites – ensure compliance with ABET APPM

Program Director's Areas of Focus AY24

NE350:

- NE Outcomes are: 2, 5
- Implement CAT recommendations
- Reinstate in-person trip section to STERIS – complete
- Work with future instructor for positive hand-over in fall of 25
- Look at possibly having outside persons receive cadet briefings – complete in 24-1

AY23 Recommended Changes for AY24

- Additional demos beyond lectures to explain the physics – ongoing
- Review the Standards for Technical Reports and Nuclear Engineering Design Supplement – ongoing
- Create reading supplement for readings that are not in Murray/Holbert – ongoing
- Potentially replace beta dosimetry lesson with a design project work period – change recomendation

NE Student Outcomes

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Redbook Entry

NE350 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE350	RADIOLOGICAL ENGR DESIGN	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on nuclear engineering systems including radiation protection, shielding, and the uses of radioactive sources in industrial processes. Specific topics emphasize the operation of radiation detectors, shielding principles, health effects of radiation, radiological dispersion devices, and nuclear incidents. A design project applies the concepts presented in this course to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

One design project.

NE350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE					
	NE300	2004	1	1	Y

Concept Map: Key

Understand and implement the nuclear engineering design process

Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection

Understand medical and industrial applications for radiation technology other than commercial power generation

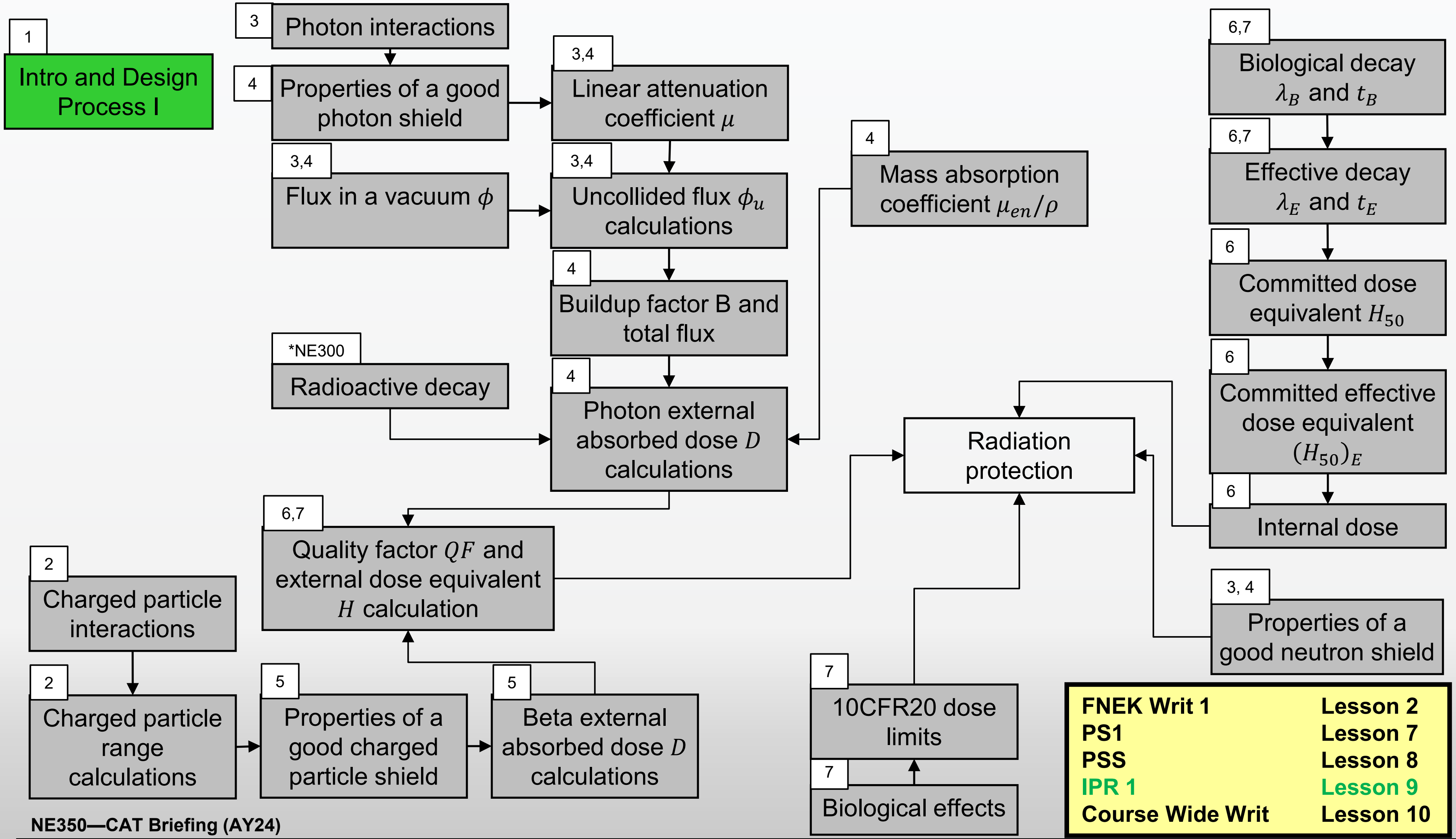
Understand how radiation detection devices function and how to interpret their outputs

Understand non-proliferation and how the concerns of security and safety play into the global discussion about the safe use of nuclear technology

Concept Map for NE350, PLW/CWW, Block 1

Understand and implement the nuclear engineering design process

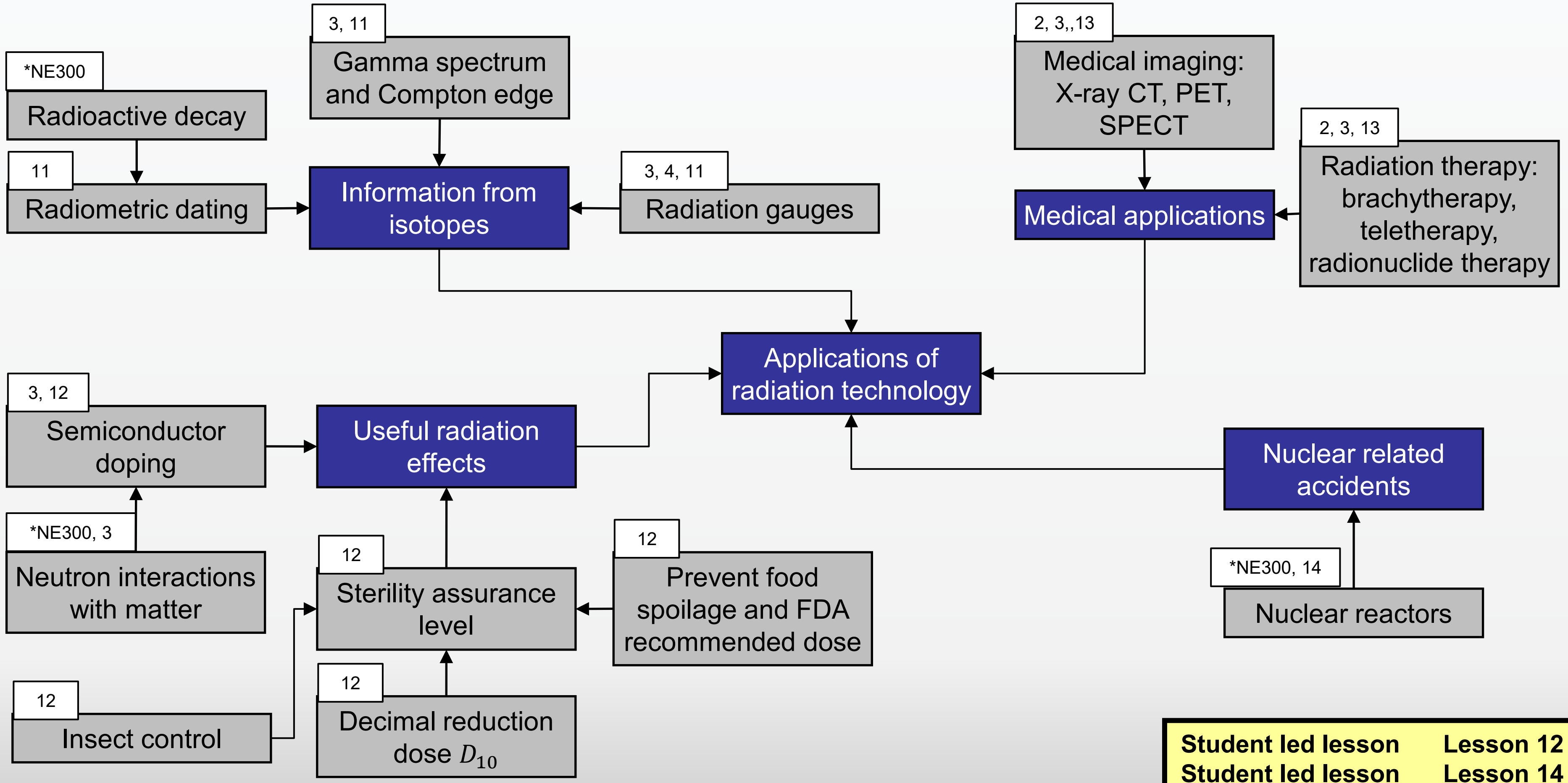
Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection



Concept Map for NE350, WPR, Block 2

Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection

Understand medical and industrial applications for radiation technology other than commercial power generation

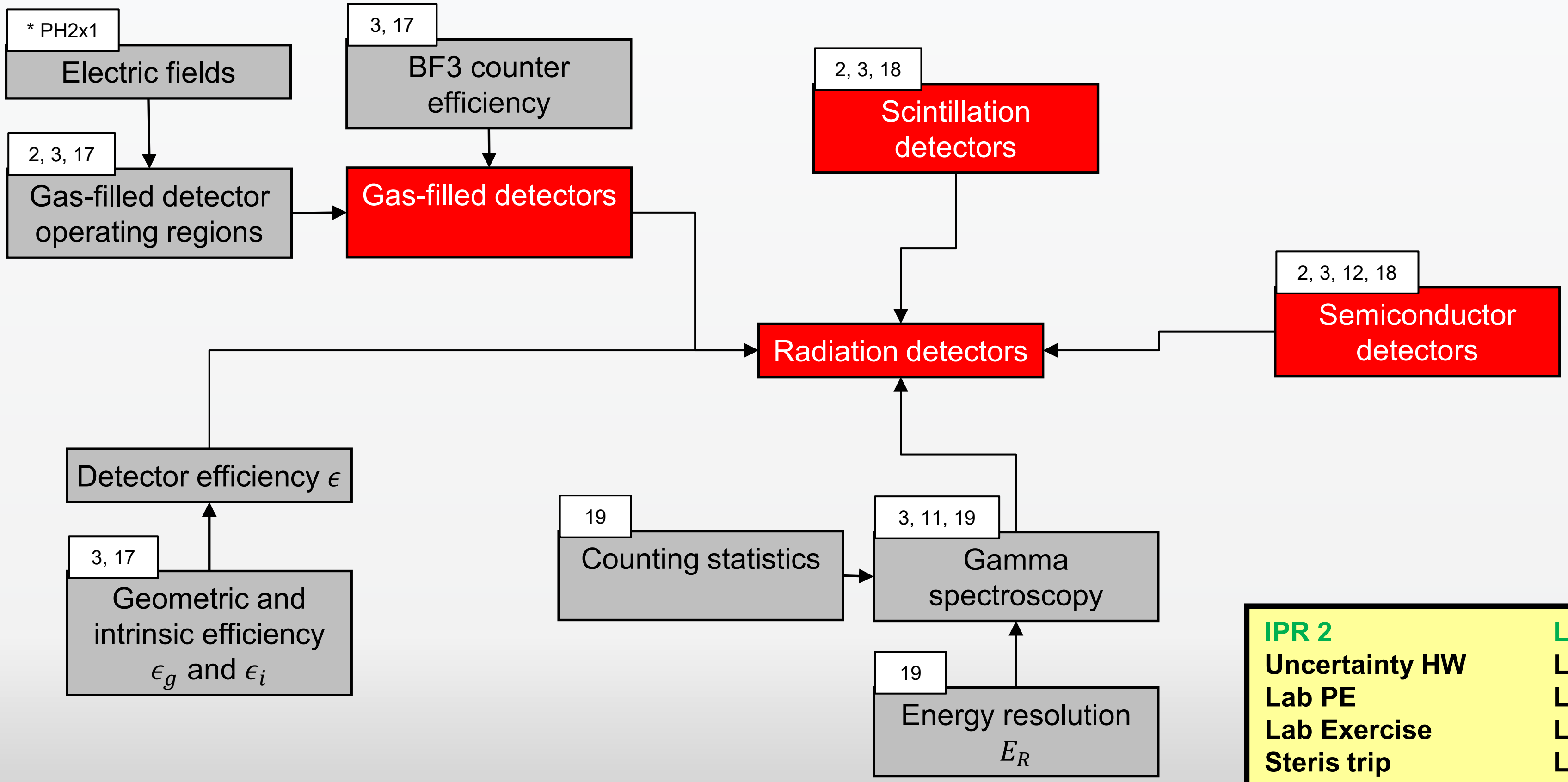


Student led lesson	**Lesson 12**
Student led lesson	**Lesson 14**
PS2	**Lesson 14**
PSS	**Lesson 15**
WPR	**Lesson 16**

Concept Map for NE350, Lab, Block 3

Understand and implement the nuclear engineering design process

Understand how radiation detection devices function and how to interpret their outputs

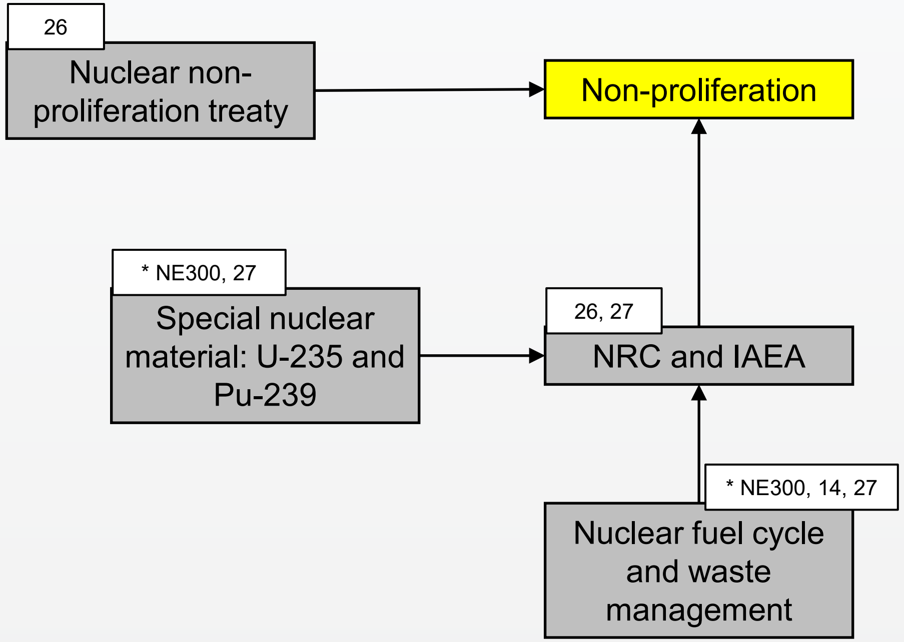


IPR 2	Lesson 20
Uncertainty HW	Lesson 21
Lab PE	Lesson 21
Lab Exercise	Lesson 22
Steris trip	Lesson 23
Lab due	Lesson 24
PSS	Lesson 24
IPR 3	Lesson 25

Concept Map for NE350, Project/TEE, Block 4

Understand and implement the nuclear engineering design process

Understand non-proliferation and how the concerns of security and safety play into the global discussion about the safe use of nuclear technology



FNEK Writ 2	**Lesson 27**
Project presentations	**L 28/29**
PSS	**Lesson 30**
Project write up due	**after L30**
TEE	

4-Step Evaluation & Assessment Process

1. Identify the connections between:
   - Course block objectives and program outcomes
   - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
   - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
   - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of program outcomes based on the population distribution of cadet performance on the embedded indicators as:

Descriptor	Color Code	Likert Rating	Criteria
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Embedded Indicators (Tools)	NE Student Outcomes						
	1	2	3	4	5	6	7
Problem sets	X						X
Laboratory Exercise					X	X	
Exams	X						
NEDP IPRs		X			X		X
NEDP Brief		X	X	X	X		
NEDP Report		X		X	X		

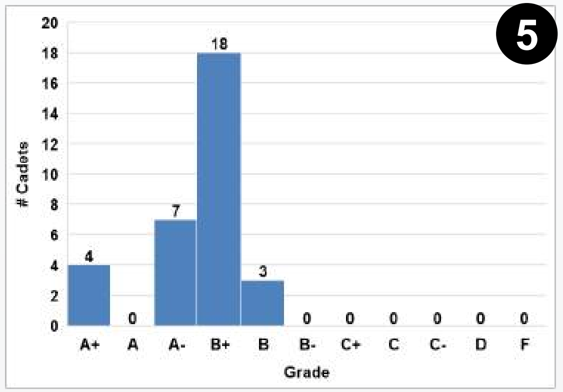
Best Assessment Instrument

- SO 2 – NEDP Final Report
- SO 5 – NEDP IPRs

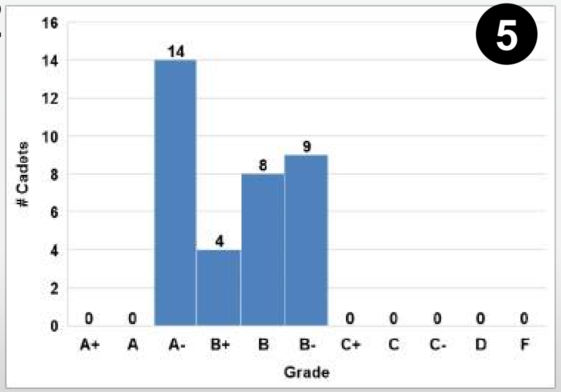
NE Student Outcome Assessment – Embedded Indicators

SO 2: NEDP Report

24-1

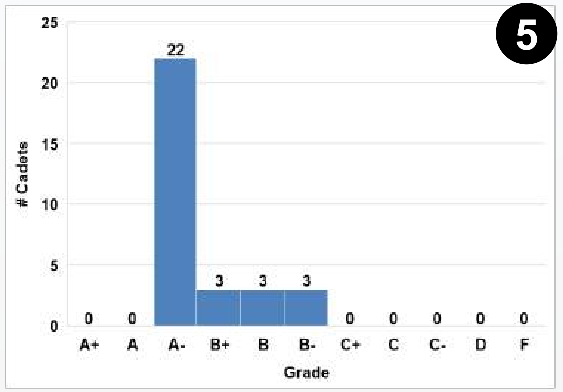


24-2

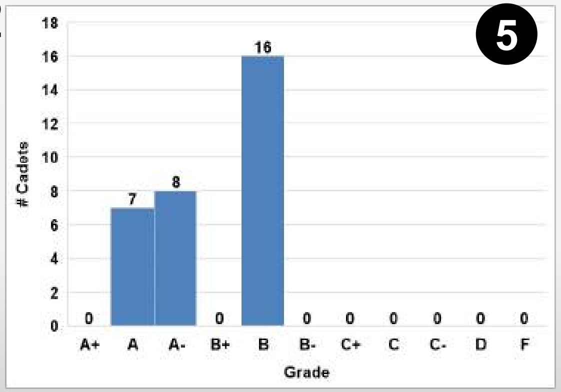


SO 5: NEDP IPRs

24-1



24-2



NE Student Outcome Assessment – Cadet Surveys

#	Question	SA	A	N	D	SD	N	Mean	Std Dev
Q2	I am confident in my ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors (NE Outcome 2).	12	47	4	0	1	64	4.1	0.6
Q5	I am confident in my ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives (NE Outcome 5).	21	38	4	0	1	64	4.2	0.7

NE350 Course Director's Assessment – Assessments for AY24

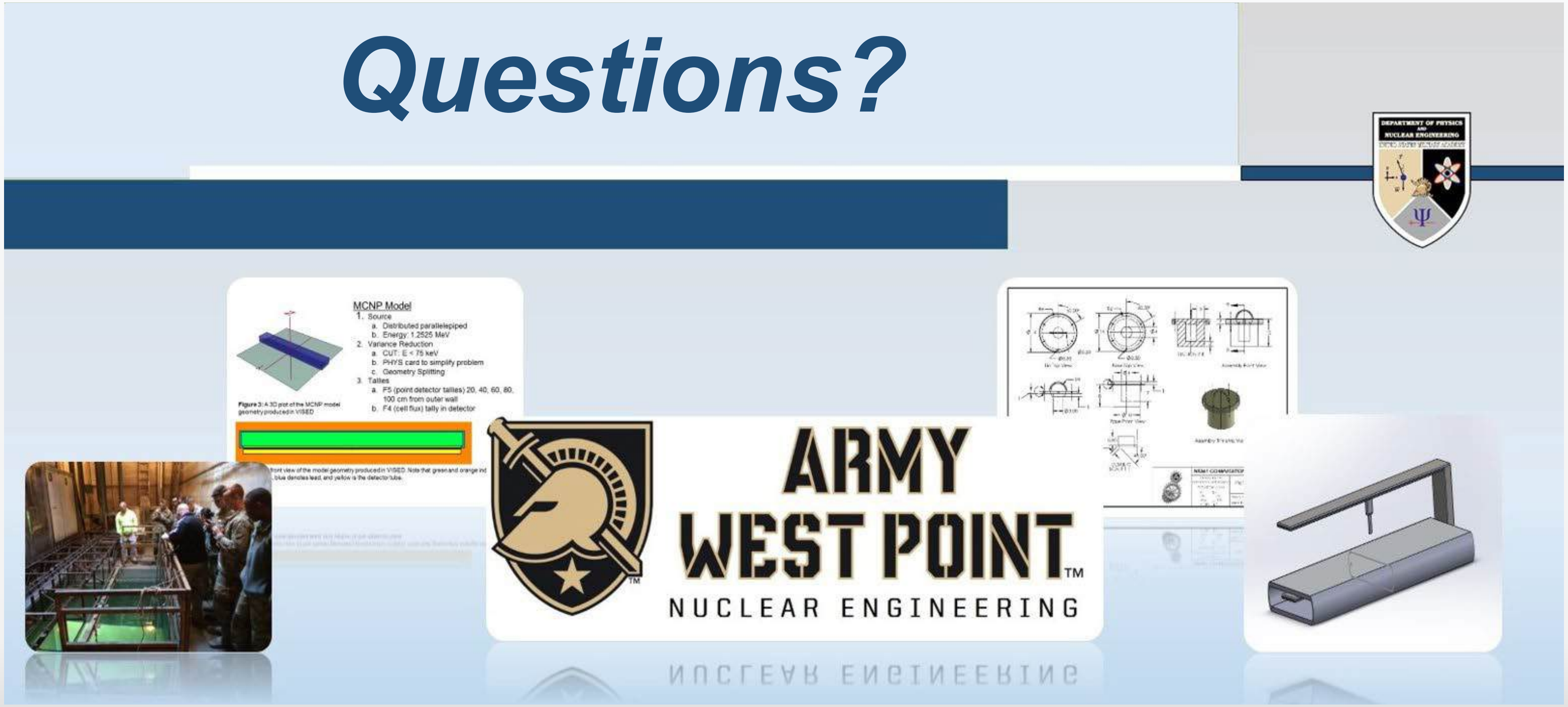
Student Outcomes	NE350	Indicators	Surveys	CD
1. Solve problems				
2. Engineering design	X	5.0	4.1	4.5
3. Communicate effectively				
4. Ethical responsibilities				
5. Function on team	X	5.0	4.2	4.5
6. Conduct experiments				
7. Apply knowledge				

Recommended Changes for AY25

- Create additional design projects
- Devote more time to explaining the design process
- Update IPR templates
- Consider template for design report
- Make board problems re: STERIS operations

Resource Requirements

- Requirements for next AY
  - N/A


Questions?
DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
MCNP Model
1. Source
a. Distributed parallelepiped
b. Energy: 1.2525 MeV
2. Variance Reduction
a. CUT: E < 75 keV
b. PHYS card to simplify problem
c. Geometry Splitting
3. Tallies
a. F5 (point detector tallies) 20, 40, 60, 80, 100 cm from outer wall
b. F4 (cell flux) tally in detector
Figure 3: A 3D plot of the MCNP model geometry produced in VISED
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TAB D AY2023 COURSE ASSESSMENT

NE 350 Radiological Engineering Design
AY 23-2

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MAJ Annie M. Arauz
NE350 Course Director

Agenda
AY 23-2 NE350 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE350 Assessment Process
- AY23 Course Assessment
- Recommended Changes
- Redbook Entry

AY22 Program Director's Recommendations (for AY23)

(1 of 3)

Adopt CAT recommendations and

- Assess and work on improving basic skills lost to COVID
- Work on assessment tools that correctly identify any lack of fundamental skills to include time management (i.e. late turn-ins of homework)
- Continue assessment SO (1)-(7)

Standardization of briefings

- Continue to work with Core Program and CD Academy for standard WPR and grades briefs
- Continue standardized briefs and reports for NE-specific topics
- Continue use of Overleaf

Effective use of NE labs and spaces

- Work on cleaning and organizing subcrit, BH50 and accelerator spaces
- Orientation for 2nd class for BH347 as collaborative space
- Orientation for 1st class for BH66B as collaborative space
- Cadets and faculty leverage remote use VMs

AY22 Program Director's Recommendations (for AY23)

Emphasize assumptions in design process and assess the impact to a design throughout the program.

Update *Nuclear Engineering Design Supplement*

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Include ANS Club, 1 activity per month
- Trip sections for courses (STERIS)

Maintain flexibility and continue level of instruction during period of reduced faculty and missing lab tech.

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- More external review within courses when possible
- Coordinate a better FE exam preparation between NE400 and NE496

AY22 Program Director's Recommendations (for AY23)

(3 of 3)

NE Working Group Topics

- Work on future manning for near and sustained term with department leadership
- Refine NE program concept map.
- Perform sponsored research on accelerator with outside organizations
- Continue to improve use of advanced computational tools such as Attila4MC throughout the program
- Annual update of internal and external websites – ensure compliance with ABET APPM

Program Director's Areas of Focus
Academic Year 2023

NE350

- ✓ NE Outcomes assessment is: **2, 5**
- ✓ Implement CAT recommendations **Complete**
- ✓ Reinstate in-person trip section to STERIS **Complete**
- ✓ Work with future instructor for positive hand-over in spring of 23 **Complete**
- ✓ Look at possibly having outside persons receive cadet briefings **Ongoing**

AY22-2 Recommended Changes (For AY-23)

- ~~**Update material cost estimates tables for NEDP**~~
- ~~**Continue to develop additional project topics to increase variability**~~
- **Additional demos beyond lectures to explain the physics (Ongoing)**
- **Review the Standards for Technical Reports and Nuclear Engineering Design Supplement (Ongoing)**
- **Create reading supplement for readings that are not in Murray/Holbert (Ongoing)**
- **Move IPR 1 due date to after block 1, maximize time for IPR 2 (Complete)**

NE Student Outcomes

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. **an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors**
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. **an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives**
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Concept Map: Key

Understand and implement the nuclear engineering design process

Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection

Understand medical and industrial applications for radiation technology other than commercial power generation

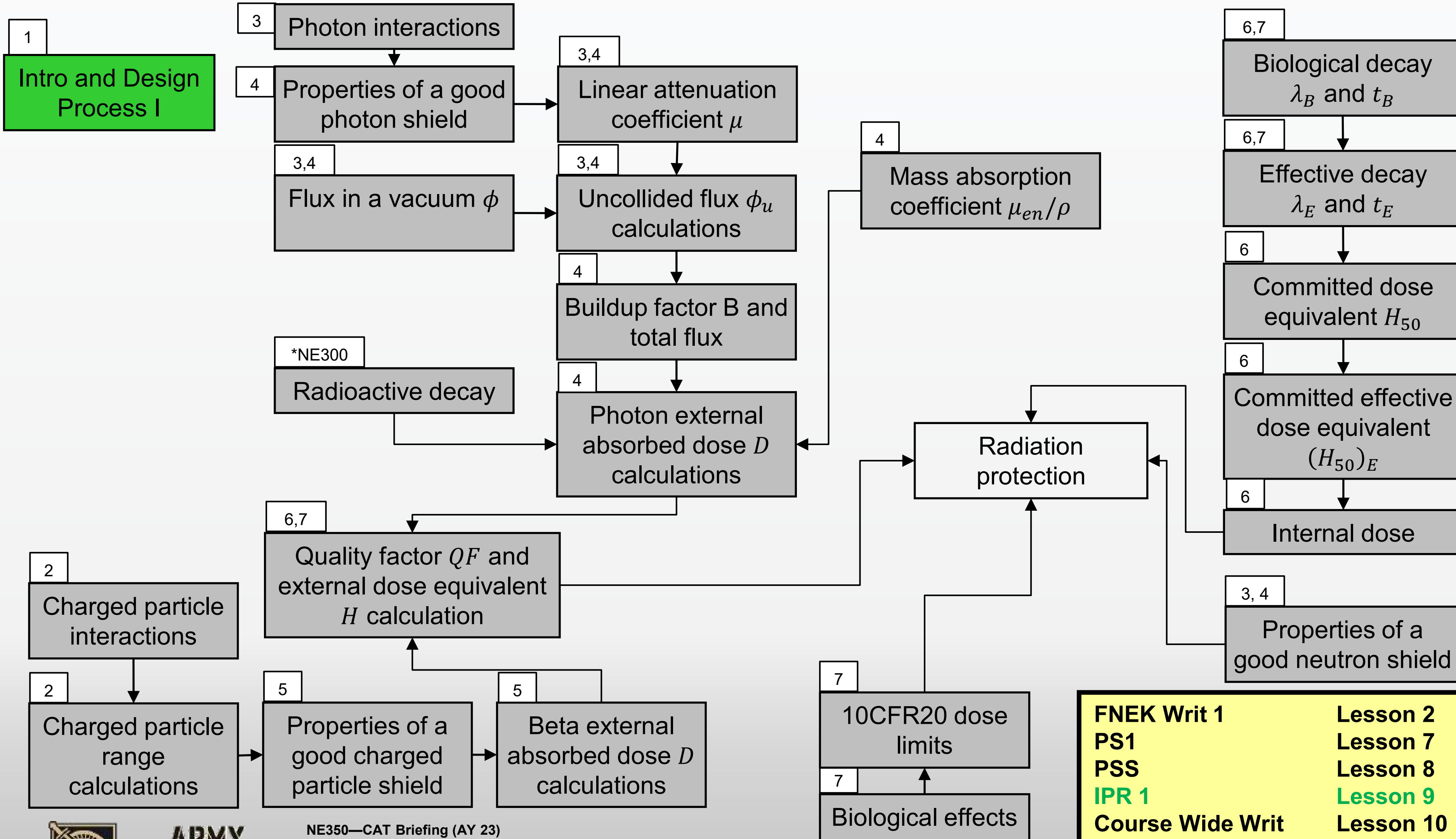
Understand how radiation detection devices function and how to interpret their outputs

Understand non-proliferation and how the concerns of security and safety play into the global discussion about the safe use of nuclear technology

Concept Map for NE350, PLW/CWW, Block 1

Understand and implement the nuclear engineering design process

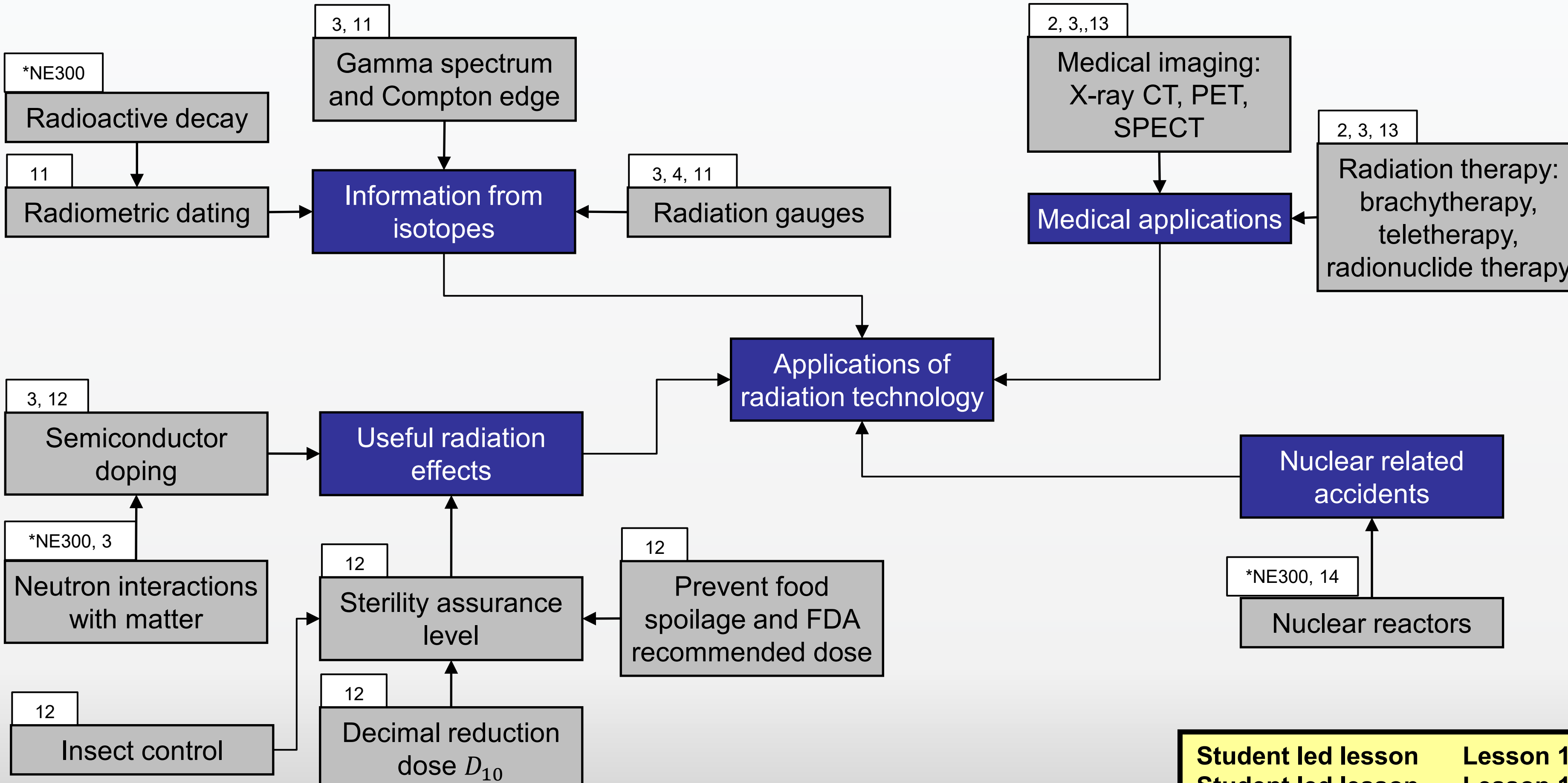
Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection



Concept Map for NE350, WPR, Block 2

Understand the biological effects of ionizing radiation, and design systems that incorporate radiation protection

Understand medical and industrial applications for radiation technology other than commercial power generation

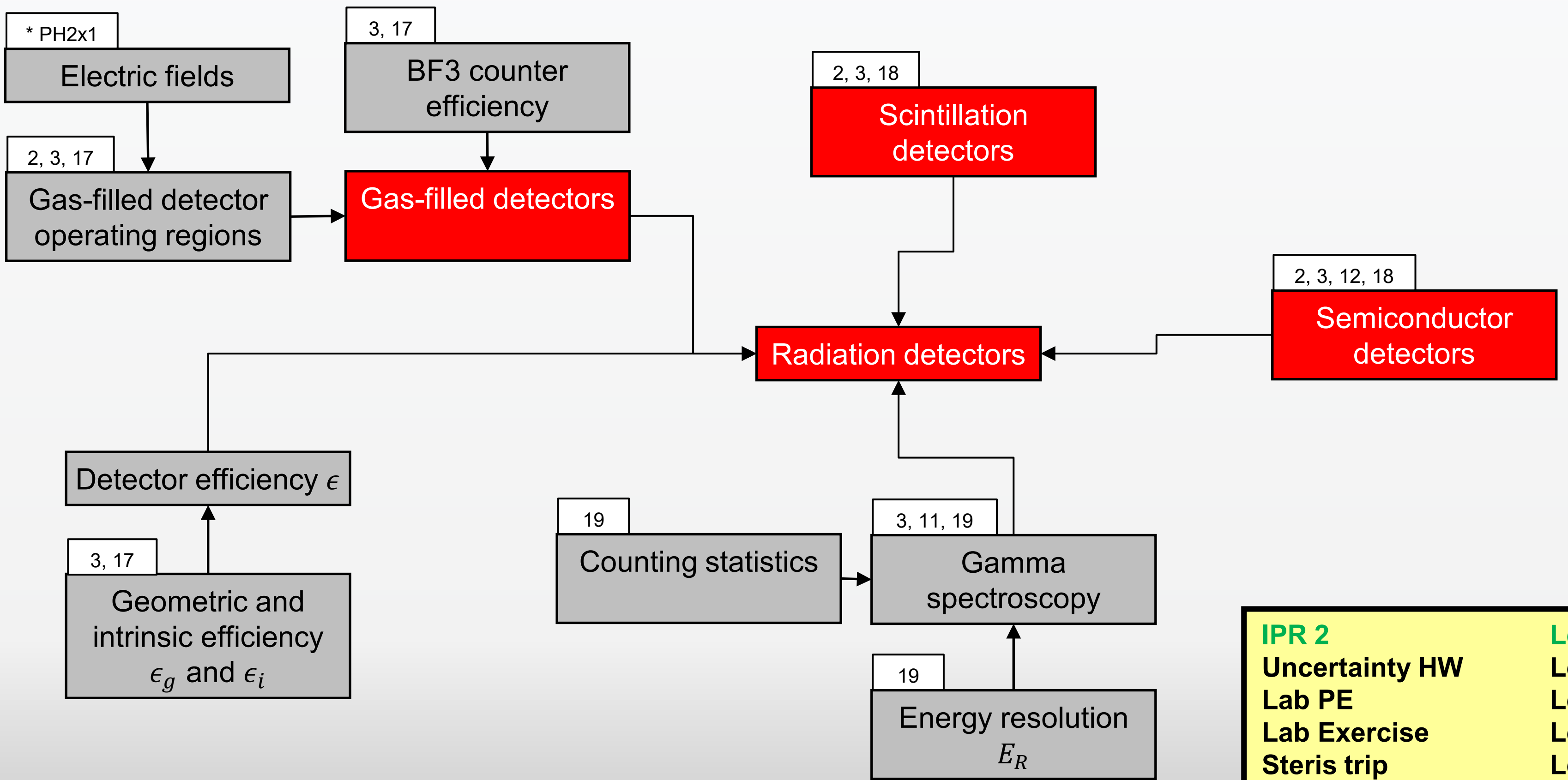


Student led lesson	**Lesson 12**
Student led lesson	**Lesson 14**
PS2	**Lesson 14**
PSS	**Lesson 15**
WPR	**Lesson 16**

Concept Map for NE350, Lab, Block 3

Understand and implement the nuclear engineering design process

Understand how radiation detection devices function and how to interpret their outputs

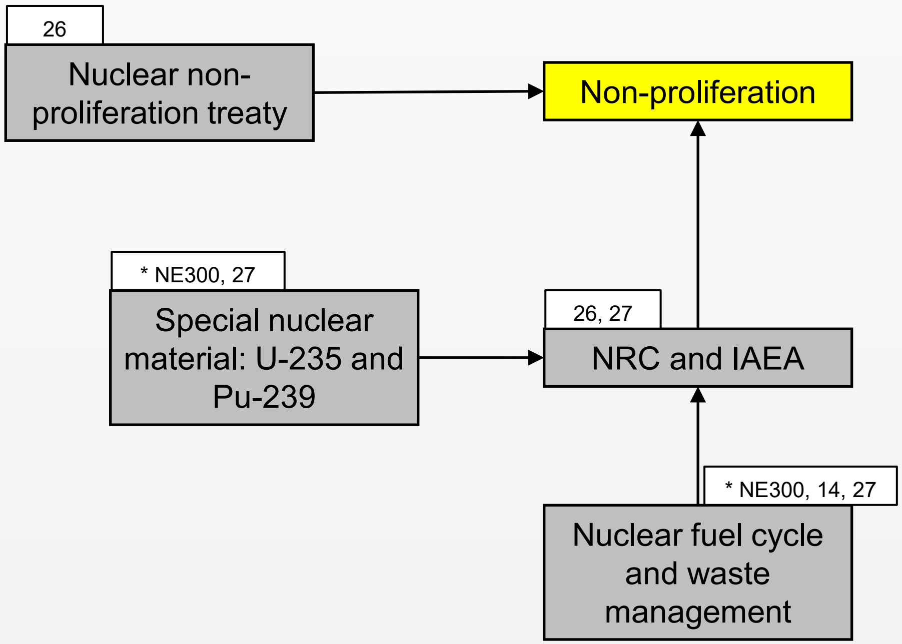


IPR 2	**Lesson 20**
Uncertainty HW	**Lesson 21**
Lab PE	**Lesson 21**
Lab Exercise	**Lesson 22**
Steris trip	**Lesson 23**
Lab due	**Lesson 24**
PSS	**Lesson 24**
IPR 3	**Lesson 25**

Concept Map for NE350, Project/TEE, Block 4

Understand and implement the nuclear engineering design process

Understand non-proliferation and how the concerns of security and safety play into the global discussion about the safe use of nuclear technology



FNEK Writ 2	**Lesson 27**
Project presentations	**L 28/29**
PSS	**Lesson 30**
Project write up due	**after L30**
TEE	

4-Step Evaluation & Assessment Process

1. Identify the connections between:
   - Course block objectives and student outcomes
   - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
   - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
   - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of student outcomes based on the population distribution of cadet performance on the embedded indicators as:

Descriptor	Color Code	Likert Rating	Criteria
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

NE350							
Embedded Indicators (Tools)	**NE Outcome**						
	1	**2**	**3**	**4**	**5**	**6**	**7**
Problem Sets	X						X
Laboratory Exercise					X	X	
Exams (CWWs, WPR, TEE)	X						
Design Project IPR 1		X		X	X		X
Design Project IPR 2		X			X		X
Design Project IPR 3		X			X		X
Design Project Final Report		X		X	X		
Design Project Brief		X	X	X	X		
FNEK Writs							
Instructor Writs	X						

Best Assessment Instrument

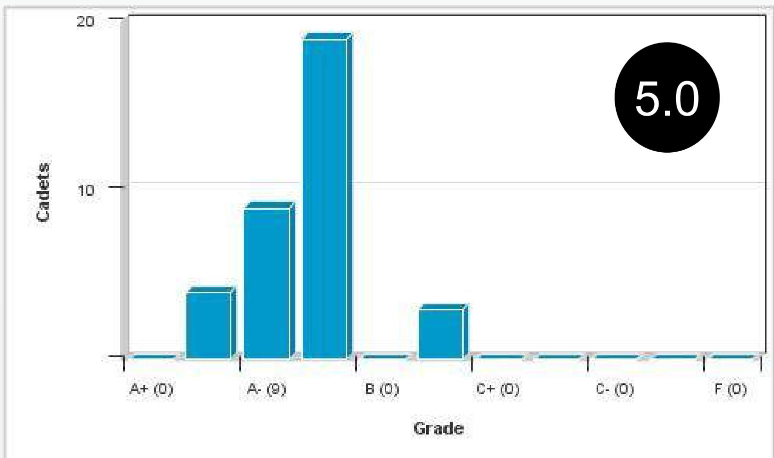
- SO 2 – NEDP Final Submission (Report)
- SO 5 – NEDP (IPRs 1-3)

An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors (NE SO 2)

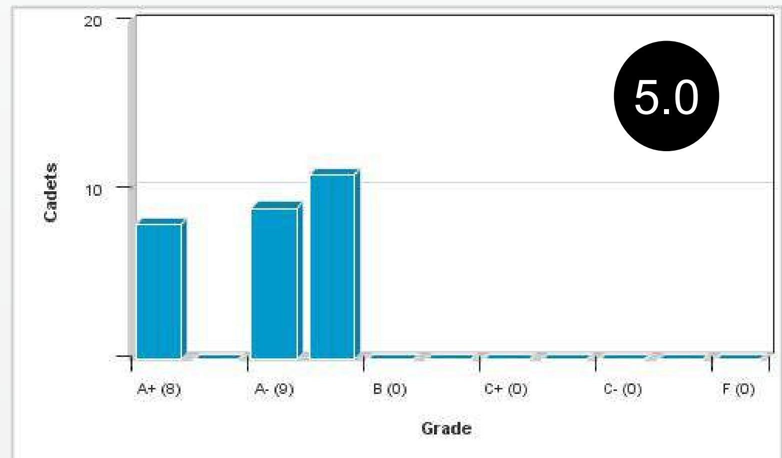
NEDP Final Report

NE SO 2 5.0

AY 23-1



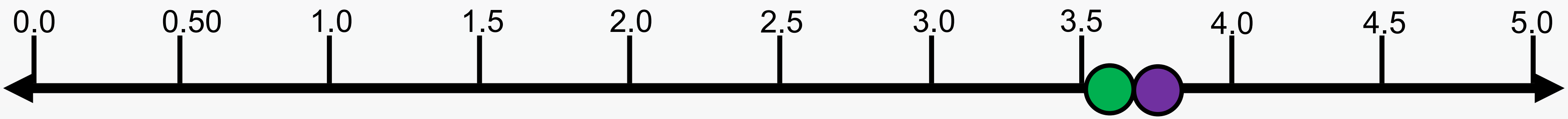
AY 23-2



<u>Good</u>. Most design groups fully understand the design process as outlined in the Nuclear Engineering Design Supplement. Overall, groups struggle to put their ideas and concepts into a well written technical document. Only a few reports were formatted correctly, IAW the course syllabus and Standards for Technical Reporting. In the methodology section, equations were expertly presented but most groups lacked prose, indicating minimal understanding of the physics concepts used. Some groups included extraneous detail while others left out necessary background information to frame the problem/design.

Course Feedback
(NE Student Outcome 2)

I am confident in my ability to apply both analysis and synthesis in the engineering design process, resulting in designs that meet desired needs (NE Outcome 2)



AY 23-1

AY 23-2

AY 23-1 Mean = 3.75

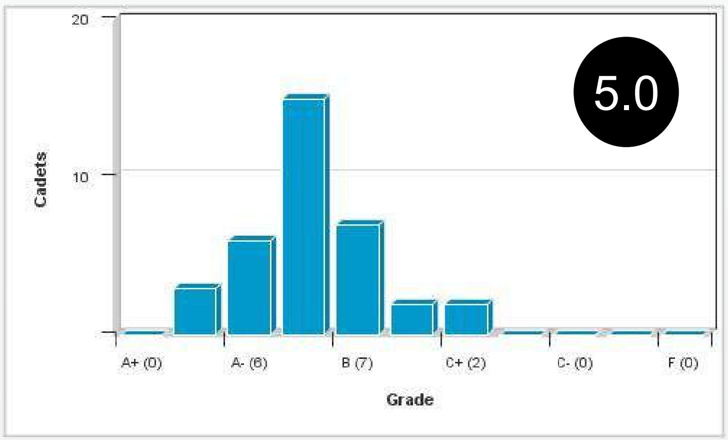
AY 23-2 Mean = 3.57

An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives (NE SO 5)

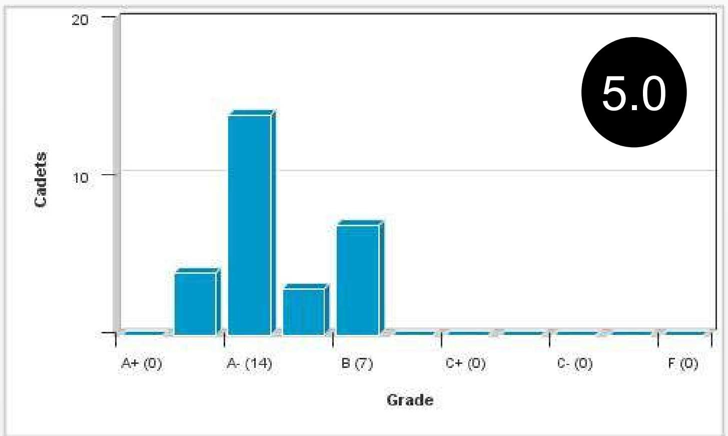
NEDP (IPRs 1-3)

NE SO 5 — 5.0

AY 23-1



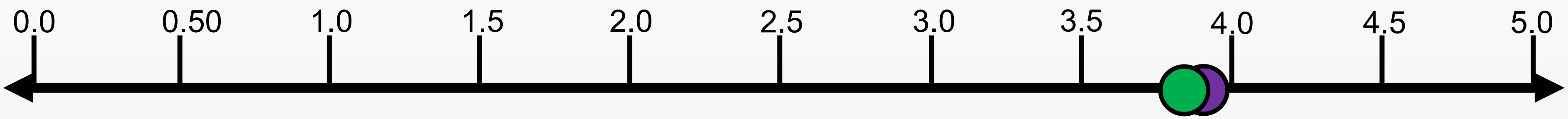
AY 23-2



Good. This accounts for the total average of all NEDP IPR submissions (IPR1, IPR2, IPR3). For IPR 1, most groups did well in defining the design problem. For IPR 2, some groups struggled with identifying quantifiable and qualitative criteria for COA comparison, but they did generally well deriving equations. For IPR 3, most groups did well with optimizing their final design, with minor errors.

Course Feedback
(NE Student Outcome 5)

I am confident in my ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts (NE Outcome 5).



AY 23-1

AY 23-2

AY 23-1 Mean = 3.9

AY 23-2 Mean = 3.88

Outcomes Assessment (AY 22-1)

NE SO	Embedded Indicator	CDT Survey (Course Level) AYT 23-1 / 23-2	CD	Course Director Comments
2	5.0	3.75 / 3.57	4.5	Overall the cadets are grasping the design process but they continue to struggle with putting their ideas into a logical, easy to understand brief or report.
5	5.0	3.9 / 3.88	4.5	The cadets tend to work together effectively, but the ones who understood the calculations usually carried the team.

KEY – Cadet Survey

5 – Strongly Agree
4 – Agree
3 – Neutral
2 – Disagree
1 – Strongly Disagree

KEY – Embedded Indicators and CD

5 -- Excellent
4 -- Good
3 -- Acceptable
2 -- Marginally Acceptable
1 -- Unacceptable

Redbook Entry

NE350 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE350	RADIOLOGICAL ENGR DESIGN	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on nuclear engineering systems including radiation protection, shielding, and the uses of radioactive sources in industrial processes. Specific topics emphasize the operation of radiation detectors, shielding principles, health effects of radiation, radiological dispersion devices, and nuclear incidents. A design project applies the concepts presented in this course to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

One design project.

NE350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE					
	NE300	2004	1	1	Y

No changes to the Redbook

AY23 Recommended Changes (For AY-24)

- **Additional demos beyond lectures to explain the physics**
- **Review the Standards for Technical Reports and Nuclear Engineering Design Supplement**
- **Create reading supplement for readings that are not in Murray/Holbert**
- **Potentially replace beta dosimetry lesson with a design project work period**

Resource Requirements

- **None**

Questions?

MCNP Model
1. Source
a. Distributed parallelepiped
b. Energy: 1.2525 MeV
2. Variance Reduction
a. CUT: E < 75 keV
b. PHYS card to simplify problem
c. Geometry Splitting
3. Tallies
a. F5 (point detector tallies) 20, 40, 60, 80, 100 cm from outer wall
b. F4 (cell flux) tally in detector
ARMY
WEST POINT
NUCLEAR ENGINEERING

UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB E AY2022 COURSE ASSESSMENT

NE 350 Radiological Engineering Design
AY 22-2

ARMY
WEST POINT™
NUCLEAR ENGINEERING

MAJ Annie M. Arauz
NE350 Course Director

Agenda

AY 22-2 NE350 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE350 Assessment Process
- AY22 Course Assessment
- Recommended Changes

AY21 Program Director's Recommendations (for AY22)

(1 of 3)

Adopt CAT recommendations and

- Continue to assess effect of COVID response for students in follow-on courses
- Assess impact of MC311 taken in STAP for Class 22
- Continue assessment SO (1)-(7)

Standardization of briefings

- Work with Core Program and CD Academy for standard WPR and grades briefs
- Continue standardized briefs and reports for NE-specific topics
- Work on use of Overleaf

Standard use of Academic Terms

- Where appropriate use terms standard in academia (i.e. course-wide writ vs. PLW)

Effective use of NE labs and spaces

- Orientation for 2nd class for BH347 as collaborative space
- Orientation for 1st class for BH66B as collaborative space
- Cadets and faculty leverage remote use VMs

AY21 Program Director's Recommendations (for AY22)

Emphasize assumptions in design process and assess the impact to a design throughout the program.

Update *Nuclear Engineering Design Supplement*

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Sustain ANS Club, 1 activity per month
- Sustain relations with D/Math (MA255 and MA364) and D/CME
- Reestablish trip sections for courses (STERIS)

Identify a backup instructor for every course and ensure they are current in the material

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Continue to develop laboratory tech assigned to the program
- More external review within courses when possible

AY21 Program Director's Recommendations (for AY22)

(3 of 3)

NE Working Group Topics

- Develop funded research to help bring research dollars to department in support of Department Head guidance
- Refine NE program concept map.
- Get accelerator operational and back in program.
- Continue to expand use of accelerator and sub-crit in curriculum beyond research.
- Annual update of internal and external websites – ensure compliance with ABET APPM

Program Director's Areas of Focus
Academic Year 2022

NE350

- ✓ NE Outcomes assessment is: **2, 5** **Completed**
- ✓ Implement CAT recommendations **Completed**
- ✓ Prepare for in-person classes for AY22 at 75 min **Completed**
- ✓ Maintain ability to use remote format for cadets on quarters **Completed**
- ✓ Continue to assess impact of NE350 no longer having NE majors in the course **Ongoing**
- ✓ Look at possibly having outside persons receive cadet briefings **Ongoing**

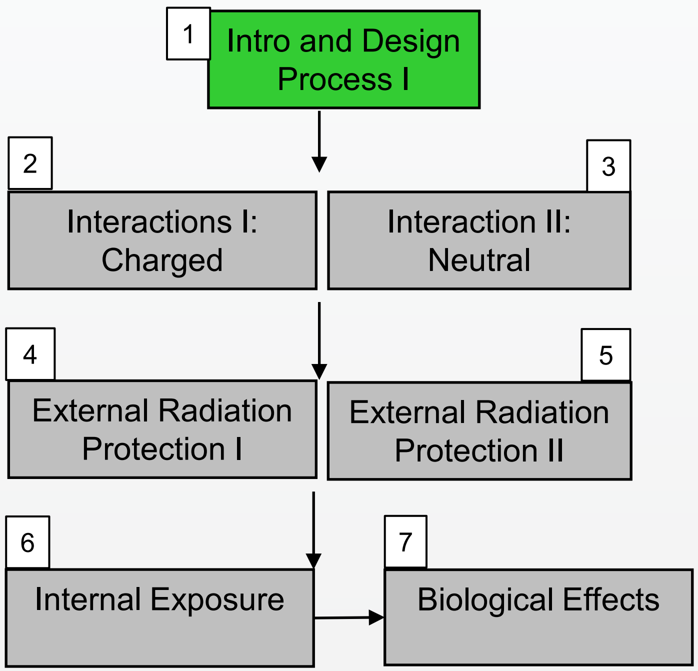
NE Student Outcomes

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. **an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors**
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. **an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives**
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Concept Map: Lessons 1-15

Radiation Protection, Radiation Applications, Reactor Safety

The Design Process: Problem Recognition



FNEK Writ 1	**Lesson 3**
PS 1	**Lesson 5**
NEDP IPR 1	**Lesson 7**
Problem Solving Session	**Lesson 8**
PS 2	**Lesson 8**
Design Project Period	**Lesson 9**
Course Wide Writ	**Lesson 10**

Concept Map: Lessons 1-15

Radiation Protection, Radiation Applications, Reactor Safety

The Design Process: Problem Recognition

11

Radiation Applications I: Useful Radiation Effects in Industry

12

Radiation Applications II: Medical Applications

13

Radiation Safety, Security, and Lesson's learned

NEDP IPR 2	**Lesson 13**
Design Project Period	**Lesson 14**
Problem Solving Session	**Lesson 15**
PS 3	**Lesson 15**
WPR	**Lesson 17**

Concept Map: Lessons 16-30

Radiation Detectors, Nuclear Reactors

The Design Process: Proposal, Implementation, and Communication of Final Design

16
Detectors I: Gas Filled and Neutron Detectors

↓

18
Detectors II: Scintillation Counters,
Personnel Dosimeters,
Solid State
Detectors

↓

19
Detectors III: Statistics, Pulse Height Analysis

↓

20
Drop for Steris

↓

21
Detectors IV: NaI and PE

NEDP IPR 3	**Lesson 22**
Lab Exercise	**Lesson 22**
Problem Solving Session	**Lesson 23**
Design Project Period	**Lesson 24**

Concept Map: Lessons 16-30

Radiation Detectors, Nuclear Reactors

The Design Process: Proposal, Implementation, and Communication of Final Design

Design Project Period	**Lesson 24**
Lab Report	**Lesson 24**

25

Non-Proliferation: Intro and Policy

↓

26

Non Proliferation II: Weapon Programs and U and Pu

FNEK Writ 2	**Lesson 26**
Design Project Period	**Lesson 27**
Design Project Briefings	**Lesson 28/29**
Problem Solving Session	**Lesson 30**

4-Step Evaluation & Assessment Process

1. Identify the connections between:
   - Course block objectives and student outcomes
   - Lesson learning objectives and course block objectives by developing a concept map

2. Select multiple embedded indicators to measure against established rubrics that will:
   - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
   - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)

3. Assess the level of attainment of student outcomes based on the population distribution of cadet performance on the embedded indicators as:

Descriptor	Color Code	Likert Rating	Criteria
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

NE350							
Embedded Indicators (Tools)	**NE Outcome**						
	1	**2**	**3**	**4**	**5**	**6**	7
Problem Sets					X		
Laboratory Exercise					X		
Exams (CWWs, WPR, TEE)							
Design Project IPR 1		X			X		
Design Project IPR 2		X			X		
Design Project IPR 3		X			X		
Design Project Final Report		X					
Design Project Brief		X					
FNEK Writs							
Instructor Writs							

Best Assessment Instrument

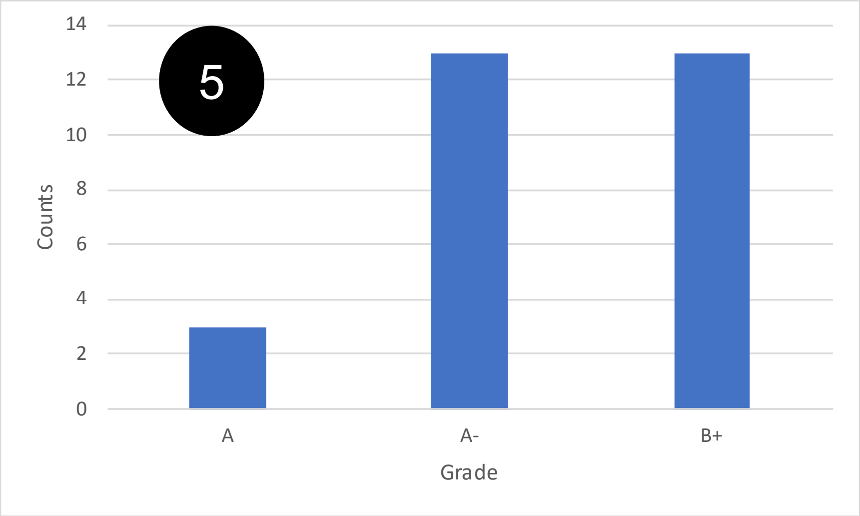
- SO 2 – NEDP Final Submission (Report)
- SO 5 – NEDP (IPRs 1-3)

An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors (NE SO 2)

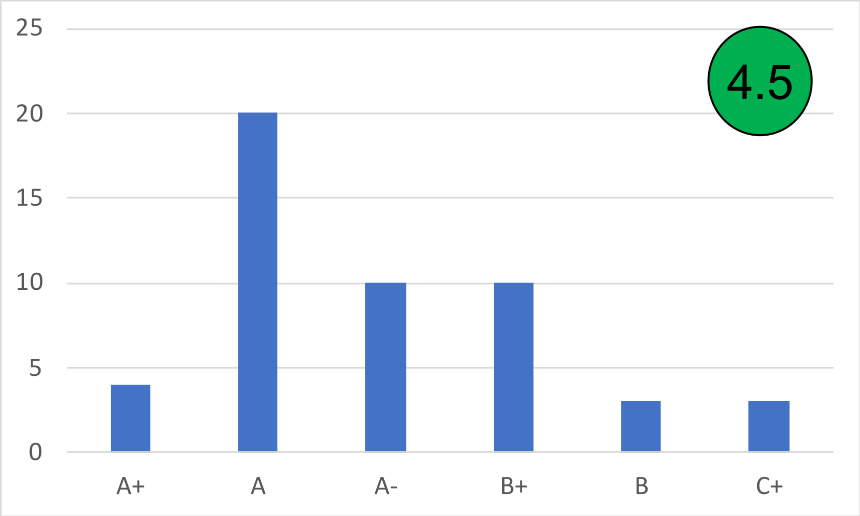
NEDP Final Report

NE SO 2 4.5

AY 22-1

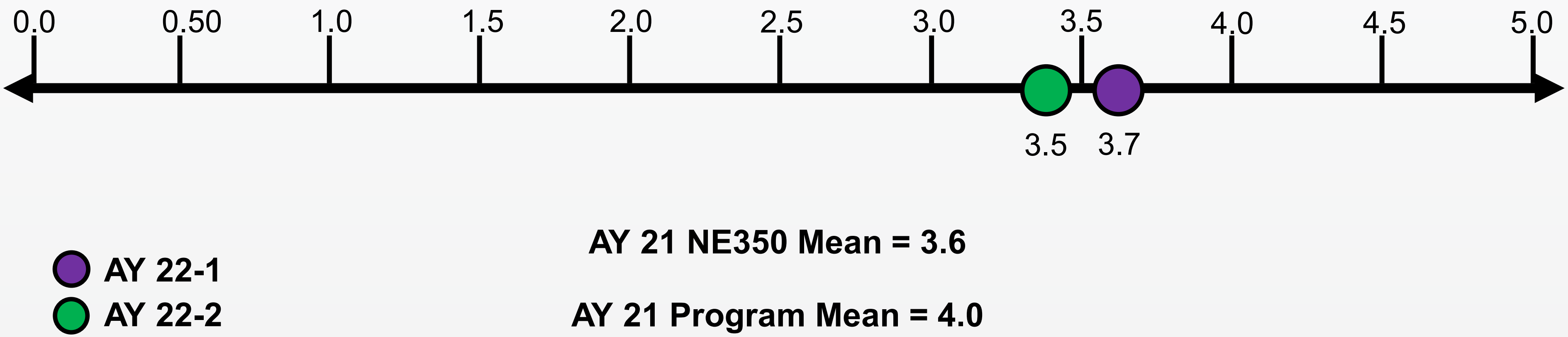


AY 22-2



<u>Good</u>. The cadets fully understand their projects but struggle to put their ideas and concepts into a well written technical document. They struggle with formatting and determining what to add in what section. Cadets also had difficulty managing their time with the brief and write up being due on the same day.

I am confident in my ability to apply both analysis and synthesis in the engineering design process, resulting in designs that meet desired needs (NE Outcome 2)



AY 21 NE350 Mean = 3.6

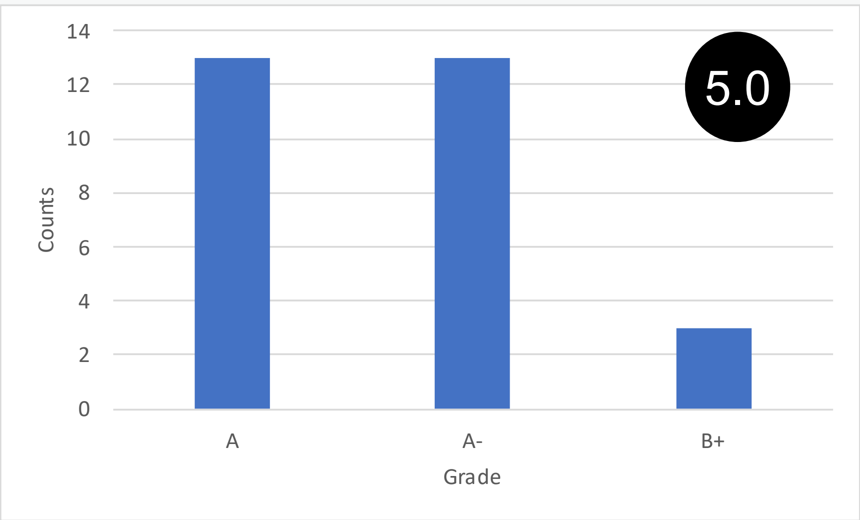
AY 21 Program Mean = 4.0

An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives (NE SO 5)

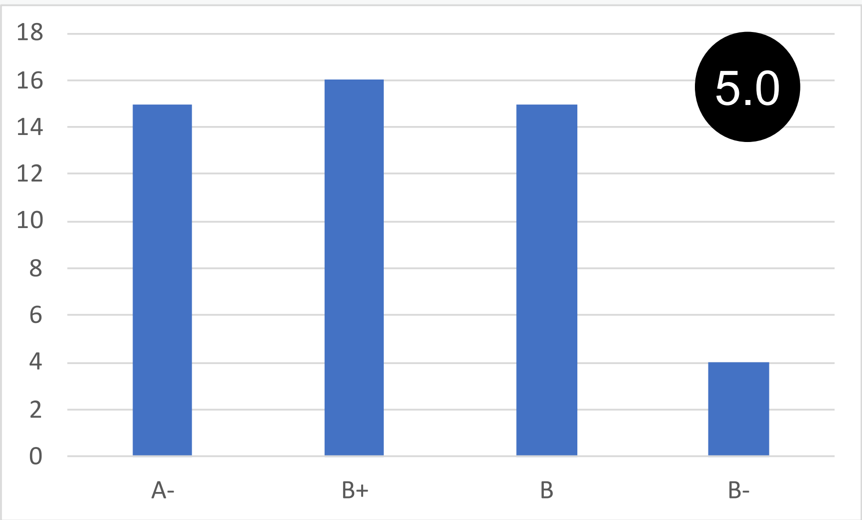
NEDP (IPRs 1-3)

NE SO 5 | 5.0

AY 22-1



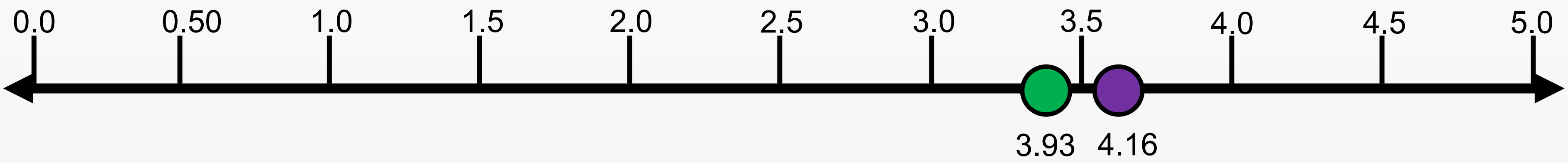
AY 22-2



Excellent. This accounts for the total average of all NEDP IPR submissions (IPR1, IPR2, IPR3). The cadet's initially struggle with grasping the concept of their design (IPR1) and optimizing their final design (IPR3) but overall they do well with calculations for IPR2. They just need to learn how to present their designed COAs in a manner that the CO can make a decision.

Course Feedback
(NE Student Outcome 5)

I am confident in my ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts (NE Outcome 5).



AY 22-1
AY 22-2

AY 22 NE350 Mean = 4.05

AY 21 Program Mean = 4.20

Outcomes Assessment (AY 22)

NE SO	BI (AY 22-1/ AY 22-2)	CDT Survey (AY 22-1/ AY 22-2)	CD	Course Director Comments
2	5.0/ 4.5	3.7/ 3.5	4.5	Overall the cadets are grasping the design process but they continue to struggle with putting their ideas into a logical, easy to understand brief or report.
5	5.0/ 5.0	4.16/ 3.93	4.5	The cadets tend to work together effectively, but the ones who understood the calculations usually carried the team.

The changes to the lesson plan (aggressive timeline for IPRs) did not significantly alter the cadet's ability to complete assignments or digest the material. The BI and Survey results remained steady across the semesters.

KEY – Cadet Survey

5 – Strongly Agree
4 – Agree
3 – Neutral
2 – Disagree
1 – Strongly Disagree

KEY – Embedded Indicators and CD

5 -- Excellent
4 -- Good
3 -- Acceptable
2 -- Marginally Acceptable
1 -- Unacceptable

Redbook Entry

NE350 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE350	RADIOLOGICAL ENGR DESIGN	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on nuclear engineering systems including radiation protection, shielding, and the uses of radioactive sources in industrial processes. Specific topics emphasize the operation of radiation detectors, shielding principles, health effects of radiation, radiological dispersion devices, and nuclear incidents. A design project applies the concepts presented in this course to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

One design project.

NE350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE					
	NE300	2004	1	1	Y

No changes to the Redbook

AY22-2 Recommended Changes (For AY-23)

- Update material cost estimates tables for NEDP (Ongoing)
- Continue to develop additional project topics to increase variability (Ongoing)
- Additional demos beyond lectures to explain the physics (Ongoing)
- Review the Standards for Technical Reports and Nuclear Engineering Design Supplement (Ongoing)
- Incorporate more examples of various flux geometries (Complete)
- Create reading supplement for readings that are not in Murray/Holbert (Ongoing)
- Break PS down in smaller PS/HW assignments (Complete)
- Move IPR 1 due date to after block 1, maximize time for IPR 2 (Complete)

Resource Requirements

- **NSTR**

Questions?


MCNP Model
1. Source
a. Distributed parallelepiped
b. Energy: 1.2525 MeV
2. Variance Reduction
a. CUT: E < 75 keV
b. PHYS card to simplify problem
c. Geometry Splitting
3. Tallies
a. F5 (point detector tallies) 20, 40, 60, 80, 100 cm from outer wall
b. F4 (cell flux) tally in detector
ARMY
WEST POINT
NUCLEAR ENGINEERING

UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB F AY2021 COURSE ASSESSMENT

NE 350 Radiological Engineering Design
AY 21-2

ARMY WEST POINT™
NUCLEAR ENGINEERING

MAJ Joshua P. Herrera
NE350 Course Director

Agenda
AY 21-2 NE350 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE350 Assessment Process
- AY21 Course Assessment
- Assessment of Remote Teaching for COVID
- Recommended Changes for AY22
- Redbook Entry

AY20 Program Director’s Recommendations (for AY21)

(1 of 3)

Adopt CAT recommendations and

- Continue to assess effect of remote classes for students in follow-on courses
- Continue assessment SO (1)-(7)

Standard use of Blackboard

- Look to see if there are some of the resources used in the remote environment that will benefit the course content even when taught in person

Upgrades for new BH177B computer center and teaching of NE361 and use in other courses

- Finish upgrade this space NLT beginning of 21-1
- Every NE cadet has a unique IP address for a remote machine
- Cadets and faculty leverage in-class and remote use of BH177B and BH347 to continue to build on computational thread within program

Integrate NE461 as a required course in the NE program

Create a Program Drop Schedule to post to NE SharePoint

AY20 Program Director's Recommendations (for AY21)
(2 of 3)

Update *Nuclear Engineering Design Supplement*

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Sustain ANS Club, 1 activity per month
- Sustain relations with D/Math (MA255 and MA364) and D/CME
- DTRA,MSKCC, NDU, STERIS (continue modified trip sections because of 75-min classes)

Identify a backup instructor for every course and ensure they are current in the material

Other Items:

- Execute ABET visit in Nov remotely
- Develop a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Train and establish a relationship with new laboratory tech assigned to the program

AY20 Program Director's Recommendations (for AY21)

NE Working Group Topics

- Refine NE program concept map.
- Look at overlap and redundancy and where NE350 fits in program
- Continue to expand use of accelerator and sub-crit in curriculum beyond research.
- Complete all administrative edits to Redbook
- Complete update of internal and external websites – ensure compliance with ABET APPM

Program Director's Areas of Focus
Academic Year 2021

NE350

- ✓ NE Outcomes assessment is: **2, 5**
- ✓ Implement CAT recommendations
- ✓ Be prepared for ABET accreditation visit in Nov
- ✓ Prepare for in-person, remote or hybrid classes for AY21
- ✓ Assess use of remote technology to enhance course even when taught in-person such as asynchronous video lessons for "flipped classrooms" or video AI for cadets who are on quarters or on trips.
- ✓ Assess impact of NE350 no longer having NE majors in the course

AY20-2 Recommended Changes (For AY21)

- Update material cost estimates tables for NEDP (Ongoing)
- Continue to develop additional project topics to increase variability (Ongoing)
- Drop PLW 2 and replace it with a research paper (Complete)
- Incorporate second lab, GM counting, into the lesson plan (Complete)
- Update FNEK Knowledge card to reflect the changes to the NE350 lesson plan (Ongoing)
- Re-design RDC to reflect changes to lesson plan (Complete)
- Additional demos beyond lectures to explain the physics (Ongoing)
- Review the Standards for Technical Reports and Nuclear Engineering Design Supplement (Ongoing)

NE Student Outcomes

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. **an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors**
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. **an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives**
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Concept Map: Lessons 1-14

Radiation Protection, Radiation Applications, Reactor Safety

The Design Process: Problem Recognition

1. Drop for Steris
2. Intro and Design Process I
3. Interactions I: Charged
4. Interaction II: Neutral
5. External Radiation Protection I
6. External Radiation Protection II
7. Internal Exposure
8. Biological Effects

FNEK Writ 1	**Lesson 4**
Problem Solving Session	**Lesson 9**
NEDP IPR 1	**Lesson 8**
Design Project Period	**Lesson 10**
PLW	**Lesson 11**

Concept Map: Lessons 1-14

Radiation Protection, Radiation Applications, Reactor Safety

The Design Process: Problem Recognition

12
Radiation Applications I: Useful Radiation Effects in Industry

13
Radiation Applications II: Medical Applications

14
Radiation Safety, Security, and Lesson's learned

15
Design Project Work Period

Problem Solving Session Lesson 16

PS 1 Lesson 16

WPR Lesson 17

Concept Map: Lessons 15-30

Radiation Detectors, Nuclear Reactors

The Design Process: Proposal, Implementation, and Communication of Final Design

18
Detectors I: Gas Filled and Neutron Detectors

19
Detectors II: Scintillation Counters,
Personnel Dosimeters,
Solid State
Detectors

20
Detectors III: Statistics, Pulse Height Analysis

21
Detectors IV: NaI and PE

NEDP IPR 2 — **Lesson 18**

Lab Exercise — **Lesson 22**

Problem Solving Session — **Lesson 23**

NEDP IPR 3 — **Lesson 23**

Design Project Period — **Lesson 24**

Lab Report — **Lesson 25**

Concept Map: Lessons 15-30

Radiation Detectors, Nuclear Reactors

The Design Process: Proposal, Implementation, and Communication of Final Design

25
Non-Proliferation: Intro and Policy

26
Non Proliferation II: Weapon Programs and U and Pu

27
Case Study: DPRK

Problem Solving Session	**Lesson 28**
FNEK Writ 2	**Lesson 28**

Design Project Briefings	**Lesson 29/30**

4-Step Evaluation & Assessment Process

1. Identify the connections between:
   - Course block objectives and student outcomes
   - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
   - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
   - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of student outcomes based on the population distribution of cadet performance on the embedded indicators as:

Descriptor	Color Code	Likert Rating	Criteria
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

NE350							
Embedded Indicators (Tools)	**NE Outcome**						
	1	**2**	**3**	**4**	**5**	**6**	7
Problem Sets					X		
Laboratory Exercise					X		
Exams (PLWs, WPR, TEE)							
Design Project IPR 1		X			X		
Design Project IPR 2		X			X		
Design Project IPR 3		X			X		
Design Project Final Report		X					
Design Project Brief							
FNEK Writs							
Instructor Writs							

Best Assessment Instrument

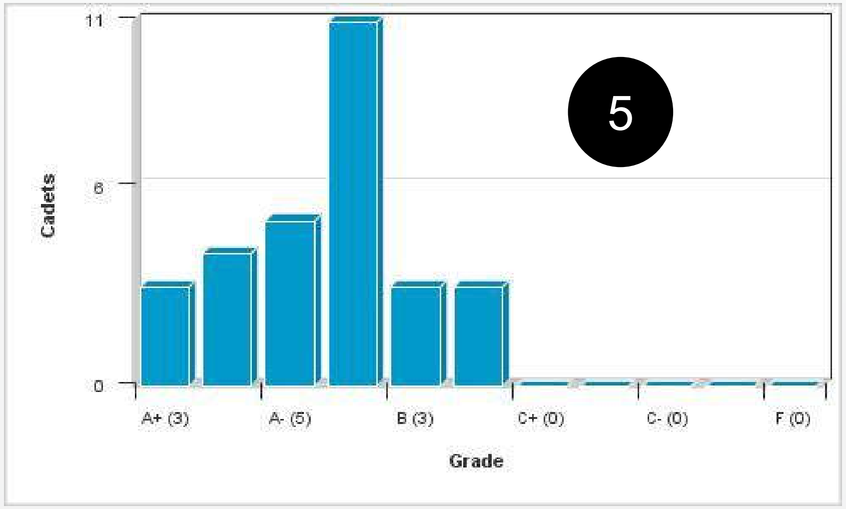
- SO 2 – NEDP Final Submission (Report)
- SO 5 – NEDP (IPRs 1-3)

An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors (NE SO 2)

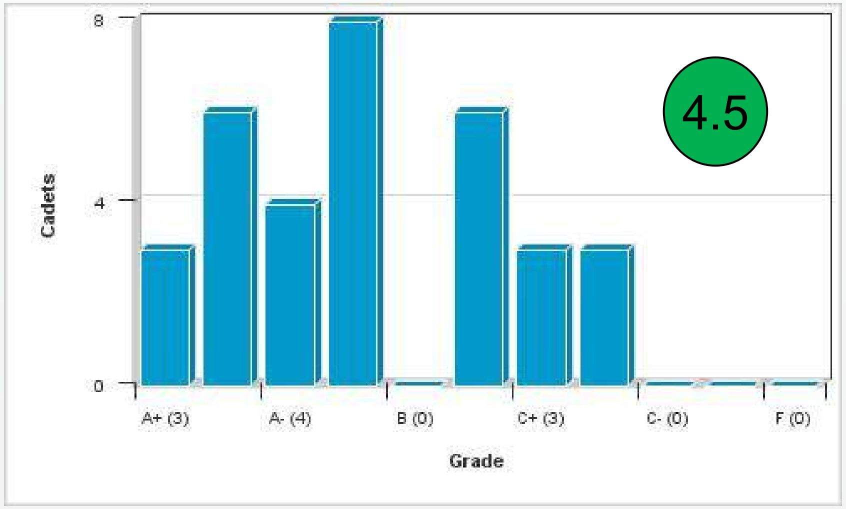
NEDP Final Report

NE SO 2 4.5

AY 21-1



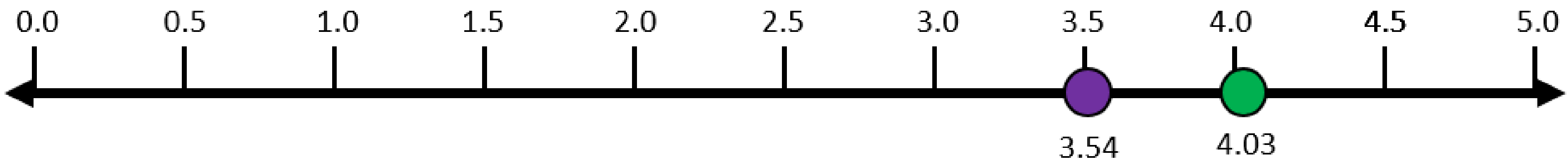
AY 21-2



Good. The cadets fully understand their projects but struggle to put their ideas and concepts into a well written technical document. They struggle with formatting and determining what to add in what section.

Course Feedback
(NE Student Outcome 2)

I am confident in my ability to apply both analysis and synthesis in the engineering design process, resulting in designs that meet desired needs (NE Outcome 2)



AY 21 NE350 Mean = 3.78

AY 21 Program Mean = 4.13

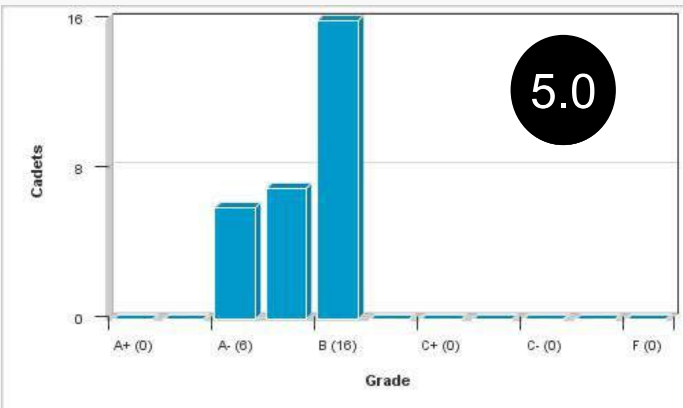
- **AY 21-1**
- **AY 21-2**

An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives (NE SO 5)

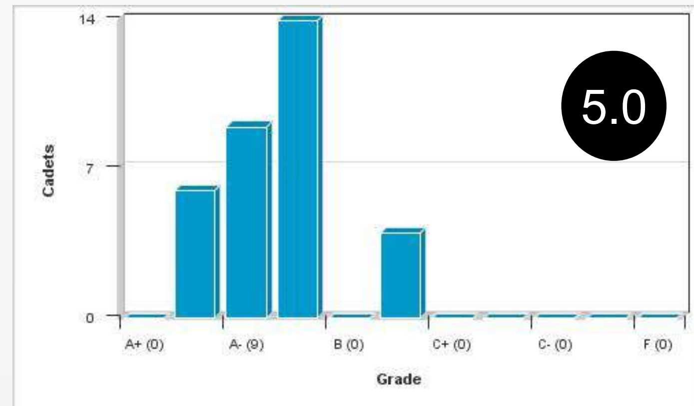
NEDP (IPRs 1-3)

NE SO 5 **5.0**

AY 21-1



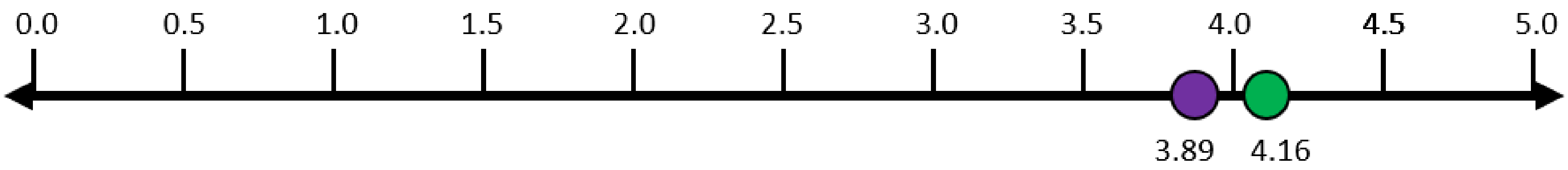
AY 21-2



Excellent. This accounts for the total average of all NEDP IPR submissions (IPR1, IPR2, IPR3). The cadet's initially struggle with grasping the concept of their design (IPR1) and optimizing their final design (IPR3) but overall they do well with calculations for IPR2. They just need to learn how to present their designed COAs in a manner that the CO can make a decision.

Course Feedback
(NE Student Outcome 5)

I am confident in my ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts (NE Outcome 5).



AY 21-1
AY 21-2

AY 21 NE350 Mean = 4.03

AY 21 Program Mean = 4.20

Outcomes Assessment (AY21)

NE SO	BI (fall/spring)	CDT Survey (fall/spring)	CD	Course Director Comments
2	5.0/4.5	3.54/4.03	4.5	Overall the cadets are grasping the design process but they continue to struggle with putting their ideas into a logical, easy to understand brief or report.
5	5.0/5.0	3.89/4.16	4.5	The cadets tend to work together effectively. The cadets in AY21-1 had some remote time which made it difficult for the groups to work effectively as a team.

The changes to the lesson plan due to COVID-19 did not significantly alter the cadet's ability to complete assignments or digest the material. The BI and Survey results remained steady across the semesters.

KEY – Cadet Survey

5 – Strongly Agree
4 – Agree
3 – Neutral
2 – Disagree
1 – Strongly Disagree

KEY – Embedded Indicators and CD

5 -- Excellent
4 -- Good
3 -- Acceptable
2 -- Marginally Acceptable
1 -- Unacceptable

Redbook Entry

NE350 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE350	RADIOLOGICAL ENGR DESIGN	2019	2	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course focuses on nuclear engineering systems including radiation protection, shielding, and the uses of radioactive sources in industrial processes. Specific topics emphasize the operation of radiation detectors, shielding principles, health effects of radiation, radiological dispersion devices, and nuclear incidents. A design project applies the concepts presented in this course to the solution of practical problems.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

One design project.

NE350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE REQUISITE					
	NE300	2004	1	1	Y

No changes to the Redbook

AY21-2 Recommended Changes (For AY-22)

- Update material cost estimates tables for NEDP
- Continue to develop additional project topics to increase variability
- Additional demos beyond lectures to explain the physics
- Review the Standards for Technical Reports and Nuclear Engineering Design Supplement
- Incorporate more examples of various flux geometries
- Create reading supplement for readings that are not in Murray/Holbert
- Break PS down in smaller PS/HW assignments
- Create an example lab report to demonstrate proper formatting and technical writing.

Resource Requirements

- Purchase new Co60 sources

Questions?

MCNP Model
1. Source
a. Distributed parallelepiped
b. Energy: 1.2525 MeV
2. Variance Reduction
a. CUT: E < 75 keV
b. PHYS card to simplify problem
c. Geometry Splitting
3. Tallies
a. F5 (point detector tallies) 20, 40, 60, 80, 100 cm from outer wall
b. F4 (cell flux) tally in detector
ARMY
WEST POINT
NUCLEAR ENGINEERING

SECTION II

EXAMINATIONS

TAB G WRITTEN PARTIAL REVIEW #1

Instructor Solution

118	59%	64	32%	18	9 %	10	10 Bonus

CUTSHEET

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE350, RADIOLOGICAL ENGINEERING DESIGN
WRITTEN PARTIAL REVIEW, VERSION 1

Name: ______________________________ **Section:** ____________________

Date: ____________________________ **Time of Completion:** __________

Instructions:

1. This examination is worth **200 marks** and consists of **eight problems** and **two bonus problems** on **six numbered pages**. Check your exam for completeness and notify your instructor if you are missing any problems or pages.
2. An instructor-provided **NE350 Reference Data Card** and **Nuclear Engineering Reference and Data Pamphlet (NERD-P)**, and your cadet-issued (TI-30 or TI-36x Pro) calculator are the **only authorized references** for this examination.
3. **Show all work**; partial credit may be given for correct work shown. For full credit, you must cite all figure numbers and table numbers used.
4. Take up to **75 minutes** to complete the examination. Record your time of completion above if you finish early.
5. Your instructor will begin the examination with the command *"begin work"* and end it with the command *"cease work."* You are to make **no marks** on your examination after the command to cease work.

Grading Summary *(for instructor use only)*:

PROBLEM	MARKS	SCORE
1	15	
2	10	
3	25	
4	10	
5	35	
6	10	
7	55	
8	40	
Bonus	5	
Bonus	5	
TOTAL	[redacted]	
GRADE		

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CUTSHEET

Name: ____________________

After an unusual amount of winter weather this semester, Army investigators have discovered the origin: the US Naval Academy. It appears that in an effort to slow down the academic progress of the superior military academy, the midship-people have turned to nefarious, supernatural means to disrupt West Point's operations. As an aggressive winter storm covers your Rockbound Highland Home in thick snow, you seek shelter in the comfort of your cozy NE350 classroom. As you're working on board problems, a mid-snowman with glowing eyes breaks into the classroom and attacks. You and your classmates manage to overpower it, and as it melts into a puddle, it leaves behind a strange mineral with a computer chip attached to it.
(Note: Express all answers to 3 significant figures.)

Wt.	**No.**	
15	**1.**	You suspect the mineral may contain uranium, so MAJ Mashl directs you to perform radiometric dating on it to determine its age. If the sample contains 0.600 moles of U-238 (natural uranium) and 1.18 moles of Pb-206 (stable lead), **calculate** the age of the sample in *years*.

6 $t = \dfrac{\ln\left[1+\frac{N_{Pb}(t)}{N_U(t)}\right]}{\lambda_U}$

3 $\lambda = \dfrac{\ln(2)}{t_{1/2}}$

Lookup: $t_{1/2,U} = 4.468\cdot10^{9}\,yr$ — 1 Lookup for uranium half-life

$\lambda_U = \dfrac{\ln(2)}{t_{1/2,U}} = 1.5514\cdot10^{-10}\,\frac{1}{yrs}$ — 2 Solves for decay constant

$M_{Pb} = 207.19\,\frac{g}{mol}$
$M_U = 238.03\,\frac{g}{mol}$ — 1 Looks up each molar mass

$t = \dfrac{\ln\left[1+\frac{1.18(207.19)}{.6(238.03)}\right]}{1.5514\times10^{-10}\,\frac{1}{yr}} = 6.43\times10^{9}\,yrs$ ANS — 2 Value and units

Your calculations verify that the sample is quite ancient. What is that dastardy Naval Academy up to? Perhaps the uranium is used to power the computer chip.

10	**2.**	A classmate covers the uranium sample with a 0.00750 cm thick aluminum can to block the alpha particles. **Calculate** the maximum range of U-238 alpha particles in this shield and **state** whether the can is an appropriate shield. *Assume there is no attenuation due to air.*

2 Eqn. $R_{\alpha,air}[cm] = \begin{cases} 0.56E_\alpha & E_\alpha < 4\,MeV \\ 1.24E_\alpha - 2.62 & 4 \le E_\alpha \le 8\,MeV \end{cases}$

Lookups: $\rho_{Al} = 2.699\,\left[\frac{g}{cm^3}\right]$, $M_{Al} = 26.98\,\left[\frac{g}{mol}\right]$ — 2 Correct lookup

1 Correct peace wise

$R_{\alpha,air} = 1.24(4.198) - 2.62 = 2.5855\,[cm]$

1 Uses max alpha energy

3 Eqn. $R_\alpha = 3.2\times10^{-4}\dfrac{\sqrt{M}}{\rho}R_{\alpha,air} = 1.59\cdot10^{-3}\,[cm]$ ANS

$= 3.2\cdot10^{-4}\dfrac{\sqrt{26.98}}{2.699}(2.5855)$

1 Correct value and units

CUTSHEET
Name: ______________________

MAJ Mashl says his stomach is grumbling. He orders you to go to Grant Hall to pick up some of their delicious mozzarella sticks. As you brave the outdoors, you're ambushed by an abominable-shipman! He hits you square in the face with a radioactive snowball, causing you to inhale a plume of radioisotopes.

25 **3.** The radioactive material you inhaled from the snowball contains iodine-131 and strontium-90. You're a little concerned you're going to exceed your annual dose equivalent limit. The concentration of ^{90}Sr you inhaled was $1.59 \times 10^{-5} \frac{\mu Ci}{mL}$ whereas the concentration of ^{131}I was $2.40 \times 10^{-4} \frac{\mu Ci}{mL}$. Although you only breathed in the radioisotopes for 5.00 minutes, your breathing rate was elevated to $2.00 \times 10^{6} \frac{mL}{hr}$. If you've already received 200 mrem of dose equivalent this year, **calculate** the fraction of your annual dose equivalent limit thus far. *Assume you are a radiation worker.*

Lookup: ALI for Sr-90 : 20 μCi
I-131 : 50 μCi

3 Looks up ALI
1 for attempt
2 for correct

$\boxed{13}$ $\sum_i \frac{C_i}{MPC_i} \le 1$

$1.59 \cdot 10^{-5} \frac{\mu Ci}{\cancel{mL}} \cdot 2 \cdot 10^{6} \frac{\cancel{mL}}{\cancel{hr}} \cdot \left(\frac{5}{60}\right) = 2.65\ \mu Ci$

$2.40 \cdot 10^{-4} \frac{\mu Ci}{\cancel{mL}} \cdot 2 \cdot 10^{6} \frac{\cancel{mL}}{\cancel{hr}} \cdot \left(\frac{5\ \cancel{min}}{60\ \cancel{min}}\right) = 400\ \mu Ci$

3 Calculates amount of inhalation hazard
1 for attempt
2 for correct

Internal (Sr-90) Internal (I-131) External

$\frac{2.65\ \mu Ci}{20\ \mu Ci} + \frac{400\ \mu Ci}{50\ \mu Ci} + \frac{200\ mrem}{5000\ mrem} = \underline{0.973}$ ANS

3 2 1 Value and units

3 Incorporates all three components

On your way back from Grant, you encounter a fellow cadet who was injured by an evil snow creature.

10 **4.** The injured cadet appears to have a broken arm. Someone suggests that he needs to go the hospital for a PET scan. Someone else suggests he needs an X-ray or CT scan instead. **State** which cadet is correct and **explain** your reasoning.

$\boxed{5}$ Correctly chooses x-ray/CT

$\boxed{2}$ Provides a poor explanation

3 Complete explanation

The cadet who suggested an x-ray or CT scan is correct. A PET scan is not appropriate because it shows function over anatomy, meaning areas of high metabolic activity. For an injury such as a broken bone, medical professionals will want to see the physical details of the injury, which an x-ray or CT scan will provide.

CUTSHEET

Name: ______________________

Once you return to your classroom, you realize that the radioisotopes you inhaled are going to stay in your body for decades! You leap to your desk to start working calculations.

35 **5.** The radioactive material you inhaled from the snowball contains iodine-131 and strontium-90. The strontium collects on the surfaces of your bones whereas the iodine is rapidly taken up by your thyroid. Over 50 years, you expect 8.00×10^{14} strontium atoms and 6.00×10^{13} iodine atoms to decay in your body. Your thyroid weighs 0.0300 kg and your total bone mass is 8.50 kg. **Calculate** your committed effective dose equivalent in Sv from this incident. For the ^{131}I particles, assume the following weighted average energies: $\bar{E}_\gamma = 0.380$ MeV and $\bar{E}_\beta = 0.193$ MeV.

8 $H_{50} = \dfrac{n_D E_\lambda (QF)}{m}$

7 $(H_{50})_E = \sum_T \omega_T (H_{50})_T$

3 $H = D(QF)$

3 $\bar{E}_\beta = \dfrac{E_{\beta,max}}{3}$

Lookup: 1 $\omega_{T_{Thyroid}} = 0.03$

$\omega_{T_{Bones}} = 0.03$

2 Lookups; 1 ea

$\bar{E}_\beta = \dfrac{0.546\,\text{MeV}}{3} = 0.182$ MeV

1 $QF_\gamma = 1$

$QF_\beta = 1$

1 $E_{\beta,max,Sr} = 0.546$ [MeV]

3 Accounts for Sr-90, I-131 beta and gamma

Sr-90 β I-131 $(\beta+\gamma)$

$$H_{50} = \frac{(8.00\cdot10^{14})(0.182\,[\text{MeV}])(1)}{8.50\,[\text{kg}]} + \frac{(6.00\cdot10^{13})(0.380+0.193\,[\text{MeV}])(1)}{0.03\,[\text{kg}]}$$

$= 1.72\cdot10^{13}\ \frac{\text{MeV}}{\text{kg}}$ $1.146\cdot10^{15}\ \frac{\text{MeV}}{\text{kg}}$

1 Applies QF

$$\frac{1.72\cdot10^{13}\,\text{MeV}}{\text{kg}}\Big/\frac{1.6022\cdot10^{-13}\,\text{J}}{1\,\text{MeV}}\Big/\frac{1\,\text{Gy}}{1\,\text{J/kg}}\Big/\frac{1\,\text{Sv}}{1\,\text{Gy}}$$

$= 2.744\ \text{Sv}(0.03)$

$= 0.0823$ Sv

$$\frac{1.146\cdot10^{15}\,\text{MeV}}{\text{kg}}\Big/\frac{1.6022\cdot10^{-13}\,\text{J}}{1\,\text{MeV}}\Big/\frac{1\,\text{Gy}}{1\,\text{J/kg}}\Big/\frac{1\,\text{Sv}}{1\,\text{Gy}}$$

$= 183.6\ \text{Sv}(0.03)$

$= 5.51$ Sv

2 Attempts conversion

2 Correct conversion

$(H_{50})_E = 5.59$ Sv ANS

3 2 1 Value and units

Name: CUTSHEET

You spend more time investigating the computer chip. Somehow, the Naval Academy is powering these snow creatures with the uranium chunks and controlling them with the computer chips! You ask MAJ Mashl to pull out the biggest gamma source he can so you can see if you can destroy the computer chips through radiation damage.

10 **6.** Someone recommends using x-rays to damage the computer chip. **Explain**, in detail, how x-rays are produced in a traditional radiograph machine (one that does not use a radioisotope as its source).

$\boxed{4}$ Basic explanation, may contain misconceptions

$\boxed{1}$ Mentions voltage potential difference

$\boxed{1}$ Mentions Bremsstrahlung x-rays

2 Mentions tungsten filament

2 Mentions high-z target

To produce x-rays, we run an electrical current through an electron-rich filament, something like tungsten. This heats up the filament and the addition of kinetic energy into the atoms causes some of the electrons to become so energetic they overcome the binding energy of their orbital shells. Colloquially, we refer to this as the atoms "boiling off" electrons. We apply a voltage across a vacuum gap, which accelerates the electrons toward an anode (positively-charged) target, another dense, high-Z material. When the electrons interact with with the anode material, they will deposit their energy through a series of interactions including ionization events, excitation events, and Bremsstrahlung radiation. In many cases, the electrons will dislodge electrons in the target material, causing characteristic x-rays to be produced. The collection of Bremsstrahlung and characteristic x-rays make up a majority of the photons that are directed at a patient to create the x-ray image.

Name: CUTSHEET

55 **7.** You ask MAJ Mashl to pull out the biggest gamma source possible so you can try to destroy the computer chip with radiation damage. He finds a 1.80-Ci source that emits 2.00 MeV gammas with a frequency of emission of 43.0%. You realize that if you're going to irradiate the chip, you need to protect yourself, so you erect a wall using lead bricks. Given a wall thickness of 9 cm, **calculate** the distance in cm that a person would need to stand away from the source to receive a dose equivalent rate of $5.00\frac{rad}{yr}$. *Assume the source to be point-like and isotropic.*

$\boxed{14}$ Any dose eq.

$\boxed{15}$ $\dot{D}_\gamma = \phi E_\gamma \frac{\mu_{en}}{\rho}$

$\boxed{6}$ $\phi = B\phi_u$

$\boxed{6}$ $\phi_u = \frac{Se^{-\mu x}}{4\pi r^2}$

$\boxed{3}$ $S = fA$

$\boxed{3}$ $y = \left[\frac{y_2 - y_1}{x_2 - x_1}\right](x - x_1) + y_1$

Lookups: $\frac{\mu}{\rho_{Pb}} = 0.0457 \left[\frac{cm^2}{g}\right]$

$\frac{\mu_{en}}{\rho} = 0.0256 \left[\frac{cm^2}{g}\right]$ $\rho_{Pb} = 11.34 \left[\frac{g}{cm^3}\right]$

$\mu = 0.5182 \left[\frac{1}{cm}\right]$

$\mu x = 4.664$

3 Correct lookups 1 ea

2 Solves for ux

Find B: Interpolate

$B = \frac{3.66 - 2.51}{7 - 4}(4.66 - 4) + 2.51$

$B = 2.76$

3 Interpolates for B

3 Correct B

2 Converts to Bq

$1\ Ci = 3.7 \cdot 10^{10}\ Bq$

2 Converts dose rate

$5.00 \frac{rad}{yr} \Big/ \frac{1}{.5052} \frac{MeV/(gs)}{rad/yr} = 9.897 \frac{MeV}{gs}$

$\dot{D} = B \frac{fAe^{-\mu x}}{4\pi r^2} E_\gamma \frac{\mu_{en}}{\rho} \longrightarrow r = \sqrt{\frac{BfAe^{-\mu x} E_\gamma \frac{\mu_{en}}{\rho}}{4\pi \dot{D}}}$

$r = \sqrt{\frac{2.76\,(0.43)(6.66 \cdot 10^{10} \frac{1}{s}) e^{-4.664} (2.00\ MeV)(0.0256 \frac{cm^2}{g})}{4\pi\,(9.897 \frac{MeV}{gs})}}$

2 Attempts algebraic solution

$r = 5.54 \cdot 10^{2}\ cm$ ANS

5 (3, 2) Value and units

Name: CUTSHEET

The gamma source is promising, but you think it'll take too long to destroy the chip. You imagine pouring a solution of beta-emitters will do the job more rapidly.

40 **8.** You find some liquid phosphorus-32 to use as your beta-emitter. As you're walking over to pour it on the computer chip, you slip on the puddle from the mid-snowman and accidentally spill some of the solution on your exposed legs (you're wearing jorts, naturally). If the source you spilled on your legs has an activity of 9.00 μCi and is spread over a 37.0 cm^2 area, **calculate** the time in seconds that you have to clean your legs before you receive a dose greater than 0.05 Gy. *Note: the density-thickness of your skin is 0.007* $\frac{g}{cm^2}$.

10 Any beta eq.

12 $\dot{D}_\beta = 2.9 \cdot 10^{-4} C_a \bar{E}_\beta \mu_{\beta,t} e^{-(\mu_{\beta,t} \rho_t x_t)}$

8 $\mu_{\beta,t} = 18.6(E_{\beta,max} - 0.036)^{-1.37}$

1 $\bar{E}_\beta = \frac{E_{\beta,max}}{3}$

3 $D = \dot{D} t$

2 Lookups

Lookup: $E_{\beta Max} = 1.7106 [MeV]$

Given: $\rho_t x_t = 0.007 \left[\frac{g}{cm^2}\right]$

Convert: $\frac{9 \mu Ci}{} \Big/ \frac{1 Ci}{1E6 \mu Ci} \Big/ \frac{3.7 \cdot 10^{10} Bq}{1 Ci} = 3.33 \cdot 10^5 \; Bq$

3 Converts to Bq

$\mu_{\beta,t} = 18.6(1.71066 - 0.036)^{-1.37}$
$= 9.178 \left[\frac{cm^2}{g}\right]$

$\bar{E}_\beta = \frac{1.71066}{3} = 0.5702 [MeV]$

2 Calculates average beta energy

$$\dot{D}_\beta = 2.9 \cdot 10^{-4} \left(\frac{3.33 \cdot 10^5 Bq}{37 cm^2}\right)(0.5702 [MeV])\left(9.178 \left[\frac{cm^2}{g}\right]\right) e^{\left(-9.178 \left[\frac{cm^2}{g}\right] 0.007 \frac{g}{cm^2}\right)}$$

$.05 Gy = 12.8089 \frac{Gy}{hr} t$

2 Solves for time
1 for attempt
1 for correct

$t = \frac{0.05 Gy}{12.8089 \frac{Gy}{hr}}$

$= 3.9035 \cdot 10^{-3} hrs \Big/ \frac{3600 s}{1 hr}$

3 Conversions
1 for attempt
2 for correct

$= 14.1 [s]$

ANS

4 2 2 Value and units

Name: CUTSHEET

Congratulations! With the knowledge you've gained, you are able to fight back and eventually eradicate the Navy's army of snow-creatures! As a reward, your CDT leadership awards you with big bites!

5 Bonus **9.** **Describe** the primary concern for nuclear reactors during an accident scenario. **Explain** why?

3 Good explanation

2 Great explanation

The primary concern is the release of radionuclides into the environment which is most likely to occur in a meltdown. Meltdowns are most likely due to loss of coolant accidents (LOCAs). In a LOCA, if the fuel cladding heats up too much, it can melt, causing the fuel to melt together, which introduces many complications.

5 Bonus **10.** **Describe** the decimal reduction dose.

3 Good explanation

2 Great explanation

The decimal reduction dose is the dose required for a given microbe to reduce its population by 90%, or 1/10th of the original population.

Figure 1: A vituperative mid-snowman

Figure 2: A pugnacious abominable-shipman

TAB H TERM END EXAM

Instructor Solution

$\boxed{288}$	57.6%	120	24.0%	92	18.4%

Time of Completion ________ Cadet CUTSHEET

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE350, Radiological Engineering Design
Term End Examination Version I

1. Instructions:

   a. The weight of this examination is 500 marks.

   b. You have 3.5 hours to complete the examination. The examination consists of 18 questions on 8 pages (not including this cover sheet). Check your examination for completeness.

   c. Authorized references are the NE350 Reference Data Card (provided by the instructor), the Nuclear Engineering Reference Data Pamphlet (provided by the instructor), your cadet-issued TI30 or TI36 calculator, and a straight edge.

   d. Show all work. Partial credit may be given for correct work shown.

2. Grading Summary:

PROBLEM	WEIGHT	SCORE	PROBLEM	WEIGHT	SCORE
1	10		11	20	
2	10		12	20	
3	10		13	20	
4	10		14	25	
5	25		15	25	
6	25		16	65	
7	20		17	80	
8	25		18	65	
9	20			Total	
10	25			GRADE	

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Wt. No. Cadet ____________________________

10 1. Circle the letter next to the correct tissue weighting factor for dose to the lungs.

10

a. 0.01
b. 0.04
c. 0.08
d. 0.12

10 2. Circle the letter next to the name of the relationship between radiation dose and biological effect (harm) that takes the most conservative approach to protecting the radiation worker.

10

a. Hormesis
b. Linear No Threshold
c. Quadratic
d. Threshold
e. Cubic

10 3. True/False: High-Z materials are good neutron shields.

10

a. True
b. False

10 4. Circle the letter next to the following detector(s) that utilize a photomultiplier tube.

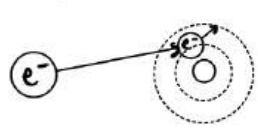
10 +5 1 right
1 wrong
+5 1 right

a. Solid state detector
b. Gas-filled detector
c. Scintillation counters

25 5. Using words and figures, describe two of the following particle interactions: *excitation*, *ionization*, and *Bremsstrahlung*. Clearly label your figures.

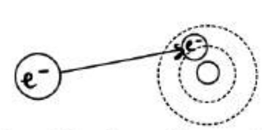
5 attempt
10 5 ea. words/figures present
4 +2 reasonable
+2 reasonable
6 +3 complete
+3 complete

Excitation



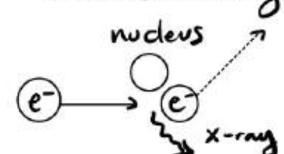
incident e⁻ undergoes a collisional energy loss, raising an inner shell e⁻ to a higher energy shell

Ionization



incident e⁻ undergoes a collisional energy loss, completely removing an e⁻ from the atom

Bremsstrahlung



incident e⁻ undergoes a radiative energy loss; coulombic force of the positively charged nucleus causes the e⁻ to undergo radial acceleration, losing energy in the form of an X-ray

Wt. No. Cadet ____________________________

25 6. A gauge uses Y-90 to emit β-particles with a maximum energy of 2.280 MeV to measure the thickness of aluminum sheets. Calculate the maximum thickness in cm of the aluminum sheet that would be practical to measure with a Y-90 gauge to three significant figures.

10 R eqn

6 correct Rmax eqn

5 $\rho_{Al} = 2.699\ g/cm^3$

2 ANS

2 +1 ea units/sf

$$R_{max} = 0.412 E^{1.265 - 0.0954 \ln(E)}$$

$$= 1.095 \quad g/cm^2$$

$$\rho_{Al} = 2.699\ g/cm^3 \quad [\text{tab } 2\text{-}7]$$

$$R_{Al} = \frac{R_{max}}{\rho_{Al}}$$

$$= \underline{\underline{0.406\ cm}}$$

ANS

20 7. Scientists transmute silicon to phosphorous in research reactors to create useful semiconductors. Si-30 with an abundance of 3.10% and neutron capture cross section of $\sigma =$ 0.108 barns is bombarded with a thermal neutron flux of $\phi = 2.89 \text{ x } 10^{13}\ \mathrm{n/(cm^2 s)}$ to produce an impurity content of 8.00 parts per billion. Calculate the total required irradiation time of Si-30 in the reactor. Report your answer in hours.

7 any reactor equation

5 $N_p = a N_{Si} \sigma \phi t$

3 $\frac{N_P}{N_{Si}} = 8.00 \times 10^{-9}$

2 solve/alg for t

1 convert from s to hr

1 ANS

1 units + sf

$$N_P = a N_{Si} \sigma \phi t$$

$$\Rightarrow t = \frac{N_P / N_{Si}}{a \sigma \phi}$$

$$= \frac{8 \times 10^{-9}}{0.031 (1.08 \times 10^{-25} cm^2)(2.89 \times 10^{13} cm^{-2} s^{-1})}$$

$$= 82681\ s \left(\frac{h}{3600\ s}\right)$$

$$= \underline{\underline{23.0\ h}}$$

ANS

$a = 0.031$

$\sigma = 0.108\ b$

$= 1.08 \times 10^{-25}\ cm^2$

$\phi = 2.89 \times 10^{13}\ cm^{-2} s^{-1}$

$N_p / N_{Si} = 8 \times 10^{-9}$

Wt. No. Cadet ____________________________

25 8. Over the course of 5 x 8-hr working days, an employee receives an external dose of 230 mrem and inhales air consisting of 1.9×10^{-8} μCi/mL of I-131 and 8.3×10^{-8} μCi/mL of Cs-137. Calculate the ratio of the dose received to the annual dose equivalent limit. (A standard breathing rate is 1.2 m^3/hr and 10^6 mL = 1 m^3)

$\boxed{10}$ attempt ratio

$\boxed{6}$ $\sum_i \frac{C_i}{(MPC)_i}$ or shows comparison to dose limit, ALI, or DAC

(6) +2 ea correct values (5 rem, 50 μCi, 200 μCi)

$\triangle 1$ conversion attempt

$\triangle 1$ ANS

$\triangle 1$ units

	i	C_i	$(MPC)_i$
	ext	230 mrem	5000 mrem
I-131	int 1	0.912 μCi	50 μCi
Cs-137	int 2	3.984 μCi	200 μCi

$$\sum_i = \frac{C_i}{(MPC)_i}$$

$$= \underline{\underline{0.08416}}_{ANS}$$

$t = 40$ h

I-131: $1.9 \times 10^{-8} \frac{\mu Ci}{mL} \times 1.2 \times 10^{6} \frac{mL}{hr} \times 40\ hr$

$= 0.912\ \mu Ci$

Cs-137: $8.3 \times 10^{-8} \frac{\mu Ci}{mL} \times 1.2 \times 10^{6} \frac{mL}{hr} \times 40\ hr$

$= 3.984\ \mu Ci$

20 9. A 2.754 MeV photon is Compton-scattered and reduced in energy to 1.451 MeV. Calculate the angle that the photon was scattered.

$\boxed{8}$ any E_r

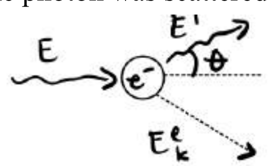
$\boxed{4}$ $\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2}[1-\cos\theta]$

(3) $m_e c^2 = 0.511$ MeV

(2) alg or solve for θ

$\triangle 1$ ANS

$\triangle 2$ +1 ea, units + sf



$E = 2.754$ MeV

$E' = 1.451$ MeV

$m_e c^2 = 0.511$ MeV

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2}(1-\cos\theta)$$

$$\Rightarrow \theta = \cos^{-1}\left[1 - m_e c^2\left(\frac{1}{E'} - \frac{1}{E}\right)\right]$$

$$= \underline{\underline{33.55^\circ}}_{ANS} \quad \text{or} \quad 0.586 \text{ radians}$$

25 10. Define build-up and discuss why it is important to account for it in radiation shielding.

$\boxed{15}$ +10 attempt to define
+5 attempt to explain signifiance

(19) +5 reasonable
+5 contributes to dose

$$\phi = B\phi_u$$

$$\Rightarrow B = \frac{\phi}{\phi_u}$$

important, bc there is greater dose than ϕ_u accounts for

- when shielding is considered, buildup is the ratio between the actual and uncollided flux
- buildup accounts for the Compton scattered photon that contributes to the flux
- it accounts for the 2 x 511 keV photons produced from the annihilation of a β^+ + a free e^- (β^+ produced from pair production)
- it also accounts for the secondary photons photons produced when an atom is ionized via photoelectric/Compton effects (secondary photons are the X-rays/characteristic X-rays)

Wt. No. Cadet ______________________________

20 11. Calculate the required number density of molecules of BF_3 in a detector with a 5.00-cm diameter to ensure that 98.0% of the thermal neutrons incident along a diameter are absorbed (σ_a for natural boron is 760 barns).

$\boxed{8}$ any ϕ, ℓ, N, detec. eqn.
$\boxed{4}$ $\phi = \phi_0 e^{-\Sigma_a d}$
(2) $\Sigma_a = N\sigma_a$
(2) $1 - \frac{\phi}{\phi_0} = 98\%$ or $\frac{\phi}{\phi_0} = 2\%$
(2) alg or solve for N
$\triangle 1$ ANS
$\triangle 1$ units + sf

$$\phi = \phi_0 e^{-\Sigma_a d}$$
$$\Rightarrow \Sigma_a = \frac{\ln(\phi/\phi_0)}{-d}$$
$$\Sigma_a = N\sigma_a$$
$$\Sigma_a = N\sigma_a$$
$$\Rightarrow N = \frac{\ln(\phi/\phi_0)}{-d\,\sigma_a}$$
$$= 1.03 \times 10^{21}\ cm^{-3}$$
ANS

$$1 - \frac{\phi}{\phi_0} = 98\%$$
$$\Rightarrow \frac{\phi}{\phi_0} = 0.02$$
$d = 5$ cm
$\sigma_a = 760$ b
$= 760 \times 10^{-24}\ cm^2$

20 12. A laboratory study with Staphylococcus has found that a radiation dose of 4.84 kGy reduces the initial population of 10^7 organisms down to 30 organisms. Calculate the decimal reduction dose, D_{10}, to three significant figures.

+6 $D = nD_{10}$
$\boxed{12}$ +6 n eq.
(6) +3 solve for n
+3 combine 2 Eq.
$\triangle 2$ +1 value, 3 SF
+1 units

$$D = nD_{10} \Rightarrow D_{10} = \frac{D}{n}$$
$$n = \log_{10}\left[\frac{\text{initial bioburden}}{\text{final bioburden}}\right]$$

$$n = \log_{10}\left[\frac{10^7}{30}\right]$$
$$n = 5.523$$
$$D_{10} = \frac{4.84\ kGy}{5.523} = 0.8763545$$
$$D_{10} = 0.876\ kGy$$
ANS

20 13. Explain the difference between geometric and intrinsic efficiency in detectors.

$\boxed{12}$ attempt
(4) +2 ea reasonable
$\triangle 4$ +2 ea complete/correct

geometric efficiency accounts for the detectors orientation with respect to the source

$\varepsilon_g = \frac{\phi A_d}{S}$ where A_d is the active detector surface area normal to the source

intrinsic accounts for the fact the radiation may or may not interact as it passes through the detector's active volume; for a scintillation counter, it may be estimated as

$\varepsilon_i \sim 1 - e^{-\mu d}$ where d is the depth of interaction

Wt. No. Cadet ______________________________

25 14. What is the Nuclear Non-Proliferation Treaty (NPT)? What are the pillars of the treaty and why does it matter?

12 attempt

4 reasonable

6 +3 ea 1st/2nd

3 3rd pillar

Landmark international treaty whose objective is to prevent the spread of nuclear weapons and weapons technology, to promote cooperation in the peaceful uses of nuclear energy, and to further the goal of achieving nuclear disarmament and general and complete disarmament. The Treaty represents the only binding commitment in a multilateral treaty to the goal of disarmament by the nuclear-weapon States. Opened for signature in 1968, the Treaty entered force in 1970. On 11 May 1995, the Treaty was extended indefinitely. A total of 191 States have joined the Treaty, including the five nuclear-weapon States. More countries have ratified the NPT than any other arms limitation and disarmament agreement.

Pillars:
1) non-proliferation
2) disarmament
3) peaceful use of nuclear energy

25 15. The military decision-making process (MDMP) is used by battalion and above echelons to allow commanders, staffs, and others to think critically and creatively while planning. The steps are listed below. How do they relate to the Nuclear Engineering Design process you went through in this course?

I. Receipt the mission
II. Mission Analysis
III. COA Development
IV. COA Analysis
V. COA Comparison
VI. COA Approval
VII. Orders Production, Dissemination, and Transition

12 attempt

4 reasonable

6 +3 ea relate 2

3 relate 3

I. Receipt the mission	desired need / ID problem
II. Mission Analysis	define problem
III. COA Development	list of possible solutions
IV. COA Analysis	feasible solutions
V. COA Comparison	decide how to evaluate & compare / perform eval/comp.
VI. COA Approval	acceptible solution
VII. Orders Production, Dissemination, and Transition	communicate results

Wt. No. Cadet ____________________________

65 16. You have a gamma-ray detector with an intrinsic efficiency of 31.3% for Cs-137. Your 5.00 μCi Cs-137 source is placed 11.5 cm away from your detector that is measuring 1325 counts per second. You measured your background to be 129 counts per second. Calculate the detector surface area that is normal to the gamma flux.

24 any detector eqn

10 $\varepsilon = \varepsilon_g \varepsilon_i$

8 $\varepsilon_g = \frac{\phi A_d}{S}$

6 $\varepsilon = \frac{R}{S}$

3 R = gross - background

4 $\phi = \frac{S}{4\pi r^2}$

3 $S = fA$

2 alg/solve for A_d

2 conversion

1 ANS

2 +1 ea sf & units

$\varepsilon_i = 0.313$

$A = 5\,\mu Ci$

$= 1.85 \times 10^5$ Bq

$r = 11.5$ cm

$R_N = \frac{\bar{X}_S}{T_S} - \frac{\bar{X}_B}{T_B}$

$= (1325 - 129)$ cps

$= 1196$ cps

$f = 0.851$ γ/dec [tab 1-3]

task: calculate A_d

physics:

1 $\varepsilon = \varepsilon_g \varepsilon_i$

2 $\varepsilon = \frac{R}{S}$

3 $\varepsilon_g = \frac{\phi A_d}{S}$

4 $\phi = \frac{S}{4\pi r^2}$

5 $S = fA$

solve: sub 4 & 3 into 1

$$\varepsilon = \frac{A_d}{4\pi r^2} \varepsilon_i$$

set 1 = 2

$$\frac{A_d}{4\pi r^2} \varepsilon_i = \frac{R}{S}$$

sub 5 & solve for A_d

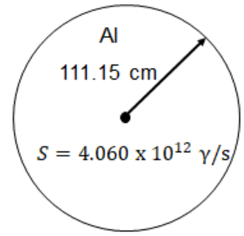
$$\Rightarrow A_d = \frac{R(4\pi r^2)}{f A \varepsilon_i} \left[\frac{\gamma/s \cdot cm^2}{\gamma/dec \cdot \frac{dec}{s}} = cm^2 \right]$$

$= 40.3\ cm^2$

ANS

Wt. No. Cadet ____________________________

80 17. As shown in the figure below, a 1.500-MeV γ-ray isotropic point source, with source strength S = 4.060 x 10^{12} γ/s, is at the center of a sphere of aluminum with a radius of 111.15 cm. Calculate the maximum dose equivalent rate a person would receive (in mrem/hr) at a point on the outside surface to four significant figures. You must account for buildup.



* Not drawn to scale

$\boxed{15}$ any ϕ or dose eqn

$\boxed{10}$ $\dot{H} = \dot{D} \cdot QF$

$\boxed{15}$ $\dot{D} = \phi E_\gamma (\mu_{en}/\rho)$

$\boxed{6}$ $\phi = B\phi_u$

$\boxed{5}$ $\phi_u = \frac{Se^{-\mu R}}{4\pi r^2}$

5 $\mu = 0.13495\ cm^{-1}$

5 [NERD-P table 4-6] B

3 interpolation attempt

3 correct B

4 $\mu_{en}/\rho = 0.0276\ cm^2/g$

2 correct subs

2 conversion attempt

3 ANS

2 +1 ea sf + units

task: calculate $\dot{H}$ [mrem/hr]

physics: $\dot{H} = \dot{D}(QF)$

$\dot{D} = \phi E_\gamma (\mu_{en}/\rho)$

$\phi = B\phi_u$

$\phi_u = \frac{S}{4\pi r^2} e^{-\mu R}$

givens: $E_\gamma = 1.5$ MeV

$S = 4.06 \times 10^{12}\ \gamma/s$

$r = R$

$= 111.15$ cm

QF: $1 \frac{mrem}{mrad}$ [tab. 4-2]

$\mu_{Al} = (\mu/\rho)\rho$ [tab 2-1 + 2-7]

$= 0.05\ cm^2/g\ (2.699\ g/cm^3)$

$= 0.13495\ cm^{-1}$

$(\mu_{en}/\rho)_t = 0.0276\ cm^2/g$ [tab 2-2]

$B(E_\gamma, \mu R)$ [tab 4-6]

$\mu R = 0.13495\ cm^{-1} (111.15\ cm)$

$= 14.9997$ MFPs

~ 15 MFPs

$B(1.5\ MeV, 15\ MFP)_{Al}$ = average between 37.9 + 18.7

= 28.5

$$\dot{H} = B\frac{S}{4\pi r^2} e^{-\mu R} E_\gamma (\mu_{en}/\rho)(QF)$$

$$\left[\frac{\cancel{\gamma}}{\cancel{cm^2} s} \cdot \frac{MeV}{\cancel{\gamma}} \cdot \frac{\cancel{cm^2}}{g} \cdot \frac{mrem}{mrad} = \frac{MeV}{s \cdot g} \cdot \frac{mrem}{mrad}\right] \left(\frac{1.602 \times 10^{-13} J}{MeV} \Big| \frac{10^3 g}{kg} \Big| \frac{10^5 mrad}{J/kg} \Big| \frac{3600 s}{hr}\right)$$

$$= 9.375627\ \frac{MeV}{g \cdot s} \cdot \frac{mrem}{mrad}$$

7

$$= 0.5407\ \frac{mrem}{hr}$$

ANS

Wt. No. Cadet ______________________________

65 18. While working with Sr-90, you accidently spill a vial onto your lab bench resulting in a 1364 Bq/cm^2 contamination zone. Calculate the beta absorbed dose rate to your skin at a height of 1.30 meters above the contaminated area in rads/hr. Remember, your skin has a range density of 0.007 g/cm^2. Report your answer to three significant figures.

$\boxed{18}$ any $\dot{D}$ or D

$\boxed{23}$ $\dot{D}_\beta$

(4) $\mu_{\beta,a} = 16\,(E_{\beta,max} - 0.036)^{-1.4}$

(3) $\mu_{\beta,a}$ correct val.

(4) $\mu_{\beta,t} = 18.6(E_{\beta,max} - 0.036)^{-1.37}$

(3) $\mu_{\beta,t}$ correct val.

(3) $\bar{E} = E_{\beta,max}/3$

(2) ρ_a [tab 2-7]

△2 conversion correct

△1 ans

△2 +1 ea sf & units

task: calculate $\dot{D}$ [rad/hr]

physics: $\dot{D}_\beta = 3.6 \times 10^{-4} \left[\frac{mGy/hr}{MeV/(g \cdot s)}\right] C_a \bar{E}\, \mu_{\beta,t}\, e^{-\mu_{\beta a} \rho_a x_a - \mu_{\beta t} \rho_t x_t}$

givens: $C_a = 1364\ \beta/cm^2 \cdot s$

$\rho_t x_t = 7 \times 10^{-3}\ g/cm^2$

$x_a = 130\ cm$

$E_{\beta,max} = 0.546$ MeV [tab. 1-3]

$\bar{E} = \frac{E_{\beta,max}}{3}$

$= 0.182$ MeV

$\mu_{\beta t} = 18.6(E_{\beta,max} - 0.036)^{-1.37}$

$= 46.7888\ cm^2/g$

$\mu_{\beta a} = 16\,(E_{\beta,max} - 0.036)^{-1.4}$

$= 41.0697\ cm^2/g$

$\rho_a = 1.20479 \times 10^{-3}\ g/cm^3$

solve: $\dot{D}_\beta = 4.84743 \times 10^{-3}\ mGy/hr \left(\frac{10^2\ rads}{10^3\ mGy}\right)$

$= 4.85 \times 10^{-4}$ rads/hr or 7.76×10^{-4} rads/hr

ANS

Outstanding Student Work

Time of Completion 2035 Cadet Kyra Dunnivine

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE350, Radiological Engineering Design
AY 26-2 Term End Examination, Version 1

1. Instructions:

    a. The weight of this examination is 500 marks.

    b. You have 3.5 hours to complete the examination. The examination consists of 18 questions on 8 pages (not including this cover sheet). Check your examination for completeness.

    c. Authorized references are the NE350 Reference Data Card (provided by the instructor), the Nuclear Engineering Reference Data Pamphlet (provided by the instructor), your cadet-issued TI30 or TI36 calculator, and a straight edge.

    d. Show all work. Partial credit may be given for correct work shown.

2. Grading Summary:

PROBLEM	WEIGHT	SCORE
1	10	10
2	10	10
3	10	10
4	10	0
5	25	25
6	25	25
7	20	20
8	25	25
9	20	20
10	25	25

PROBLEM	WEIGHT	SCORE
11	20	20
12	20	20
13	20	20
14	25	25
15	25	25
16	65	65
17	80	80
18	65	65
	Total	490
	GRADE	98.0%

A+

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Wt. No. Cadet Kyra Dunnivine

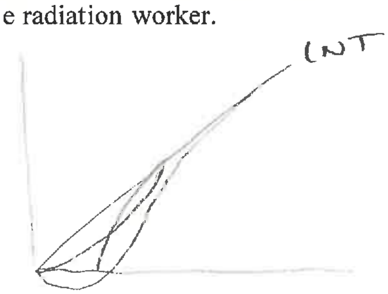
10 1. **Select** the tissue weighting factor for dose absorbed in the lungs.

a. 0.01
b. 0.04
c. 0.08
d. 0.12

Table 4-1

10 2. **Select** the relationship between radiation dose and biological effect (harm) that takes the most conservative approach to protecting the radiation worker.

a. Hormesis
b. Linear No Threshold
c. Quadratic x
d. Threshold
e. Cubic y



10 3. True/False: High-Z materials are good neutron shields.

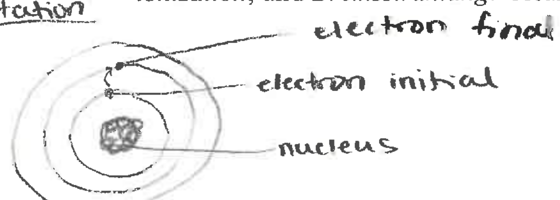
a. True
b. False

~~10~~ 0 4. **Select** the detector type or types that utilize a photomultiplier tube.

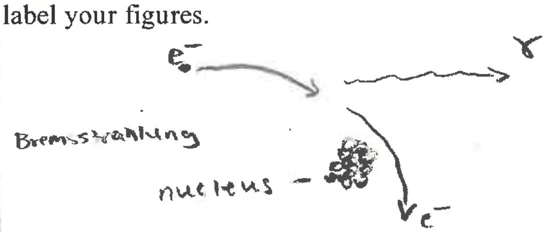
a. Solid state detectors X
b. Gas-filled detectors
c. Scintillation detectors

25 5. Using words **and** figures, describe two of the following particle interactions: *excitation*, *ionization*, and *Bremsstrahlung*. Clearly label your figures.

excitation



Excitation is a charged particle interaction that occurs when an electron in an atom has enough energy to jump from one energy level to the next, as seen above. This process does not immeaditely release energy, but when the electron goes to a lower energy level again, energy is then realeased. However, excitation is only the jump from one energy level to a higher one after the electron gains enough energy.



Bremsstrahlung occurs when a charged particle (electron) is decelerated/slows down, and moves towards a nucleus. It moves past the nucleus (no collision), but the decrease in speed releases energy in the form of a photon. This is a charged particle interaction.

Wt. No. Cadet Kyra Dunnimre

25 6. An industrial thickness gauge uses Y-90 to emit β-particles with a maximum energy of 2.280 MeV to measure the thickness of aluminum sheets. **Calculate** the maximum thickness in cm of the aluminum sheet that would be practical to measure with a Y-90 gauge to three significant figures.

$$R_{\beta,max}\left[\frac{g}{cm^2}\right] = \begin{cases} 0.412\, E_\beta^{1.265-0.0954\ln(E_\beta)} & 0.01 \le E_\beta < 2.5\text{ MeV} \\ 0.530\, E_\beta - 0.106 & 2.5 \le E_\beta \le 20\text{ MeV} \end{cases}$$ *

$E_\beta = 2.280$ MeV

$$R_{\beta,max}\left[\frac{g}{cm^2}\right] = 0.412 E_\beta^{1.265-0.0954\ln(E_\beta)}$$

$$= 0.412(2.280)^{1.265-0.0954\ln(2.280)}$$

$$= 1.095321699\ \frac{g}{cm^2}$$ → convert to cm by dividing by ρ

ρ of aluminum: $2.699\ \frac{g}{cm^3}$

$$\frac{1.095321699\ \frac{g}{cm^2}}{2.699\ \frac{g}{cm^3}} = 0.4058\text{ cm}$$

$$= 0.406\text{ cm}$$

$$\frac{g}{cm^2} \cdot \frac{cm^3}{g} = cm$$

20 7. A sample of silicon is placed in a reactor and bombarded with a thermal neutron flux of ϕ = 2.89 x 10^{13} n/(cm^{2}s) to transmute silicon to phosphorus. The silicon target has a Si-30 abundance of 3.10% and a neutron capture cross section, σ, of 0.108 barns. **Calculate** the irradiation time required to produce a semiconductor with an impurity content of 8.00 parts per billion (8 phosphorus atoms per 1 billion silicon atoms). Report your answer in hours.

3 sig figs

$\phi = 2.89\times10^{13}\ \frac{n}{cm^2\cdot s}$

$a = 0.0310$

$\sigma = 0.108\text{ b} \cdot \frac{1.0\times10^{-24}\text{ cm}^2}{\text{b}} = 1.08\times10^{-25}\text{ cm}^2$

$\frac{N_P}{N_{Si}} = \frac{8}{1{,}000{,}000{,}000}$

$1{,}000{,}000{,}000 = 1\times10^9$

$N_P = a N_{Si}\sigma\phi t$

$\frac{N_P}{a N_{Si}\sigma\phi} = t \Rightarrow t = \frac{N_P}{N_{Si}}\frac{1}{a\sigma\phi}$

$$t = \left(\frac{8}{1\times10^9}\right)\left(\frac{1}{(0.0310)(2.89\times10^{13})(1.08\times10^{-25})}\right) = 82681.19\text{ seconds}$$

$$82681.19\ \cancel{s} \cdot \frac{1\ \cancel{min}}{60\ \cancel{s}} \cdot \frac{1\text{ hr}}{60\ \cancel{min}} = 22.97\text{ hrs} = 23.0\text{ hrs}$$

inhilation

Wt.	No.		40 hrs	Cadet Kyra Dunniwine

25 8. Over the course of 5 × 8-hr working days, an employee receives an external dose of 230 mrem and inhales air consisting of 1.9 x 10^{-8} µCi/mL of I-131 and 8.3 x 10^{-8} µCi/mL of Cs-137. **Calculate** the ratio of the dose received to the annual dose equivalent limit. (A standard breathing rate is 1.2 m^3/hr and 10^6 mL = 1 m^3)

2 sig figs

I-131

$$1.9\times10^{-8}\frac{\mu Ci}{\cancel{mL}}\cdot\frac{1.0\times10^{6}\cancel{mL}}{1\cancel{m^3}}\cdot\frac{1.2\cancel{m^3}}{hr}=0.0228\frac{\mu Ci}{\cancel{hr}}\cdot 40\cancel{hrs}=0.912\ \mu Ci$$

Denominator: ALI [µCi] for inhilation of I-131: 50 µCi

Cs-137

$$8.3\times10^{-8}\frac{\mu Ci}{\cancel{mL}}\cdot\frac{1.0\times10^{6}\cancel{mL}}{1\cancel{m^3}}\cdot\frac{1.2\cancel{m^3}}{hr}=0.0996\frac{\mu Ci}{\cancel{hr}}\cdot 40\cancel{hrs}=3.984\ \mu Ci$$

Denominator: ALI [µCi] for inhilation of Cs-137: 200 µCi

$$\Sigma_i \frac{C_i}{(MPC)_i}\le 1$$

ratio, so unitless

$$=\underbrace{\frac{230\ mrem}{5000\ mrem}}_{external}+\underbrace{\frac{0.912\ \mu Ci}{50\ \mu Ci}+\frac{3.984\ \mu Ci}{200\ \mu Ci}}_{internal\ (inhilation)}=0.08416$$

$$=0.084$$

20 9. A 2.754-MeV photon undergoes Compton scattering and is reduced in energy to 1.451 MeV. **Calculate** the angle that the photon was scattered.

$$\frac{1}{E'}-\frac{1}{E}=\frac{1}{m_ec^2}[1-\cos(\theta)]$$

↑ photon final ↑ photon initial

E (photon initial) = 2.754 MeV

E' (photon final) = 1.451 MeV

θ = ?

m_ec^2 = 0.510998902 MeV $\cancel{c^2}$

$$\left(\frac{1}{E'}-\frac{1}{E}\right)m_ec^2=1-\cos\theta$$

$$1-\left(\frac{1}{E'}-\frac{1}{E}\right)m_ec^2=\cos\theta$$

$$\theta=\cos^{-1}\left[1-\left(\frac{1}{E'}-\frac{1}{E}\right)m_ec^2\right]$$

$$\rightarrow \theta=\cos^{-1}\left[1-\left(\frac{1}{1.451}-\frac{1}{2.754}\right)0.510998902\right]$$

θ = 33.5527°

θ = 33.55° ← 4 sig figs

25 10. **Define** build-up and **discuss** why it is important to account for it in radiation shielding.

Buildup is the secondary radiation that occurs when there is scattering due to shielding. Therefore, when there is radiation shielding it is important to account for build-up with the build-up factor (B) to determine the correct true flux: $\Phi = B\Phi_u$. Without the buildup factor, the estimated flux would be lower than the actual, making the estimated/calculated dose rate lower than the actual recieved, which can be a major safety concern for the radiation workers and impact the object being irradiated. Again, build-up is the radiation that is scattered from the shielding and has a secondary impact

/ 3 sig figs

Wt. No. Cadet Kyra Dunnivine

20 11. **Calculate** the required number density of molecules of BF_3 in a detector with a 5.00-cm diameter to ensure that 98.0% of the thermal neutrons incident along a diameter are absorbed (σ_a for natural boron is 760 barns). #/cm³

↳2% go through

$\phi(d) = \phi_o e^{-\Sigma_a d} \Rightarrow \frac{\phi(d)}{\phi_o} = e^{-\Sigma_a d}$

$\frac{\phi(d)}{\phi_o} = 0.02$

$\ln\left(\frac{\phi(d)}{\phi_o}\right) = -N\sigma d$

$d = 5.00\text{ cm}$

$N = \frac{\ln\left(\frac{\phi(d)}{\phi_o}\right)}{-\sigma d}$

$N = \frac{\ln(0.02)}{-(7.6\times10^{-22})(5)}$

$= 1.02948 \times 10^{21}$

$= 1.03 \times 10^{21}$ molecules/cm³

$\sigma_a = 760\text{ b} \cdot \frac{1.0\times10^{-24}\text{ cm}^2}{1\text{ b}} = 7.6\times10^{-22}\text{ cm}^2$

$\Sigma = N\sigma \Rightarrow \Sigma_a = N(7.6\times10^{-22}\text{ cm}^2)$

20 12. A laboratory study with Staphylococcus has found that a radiation dose of 4.84 kGy reduces the initial population of 10^7 organisms down to 30 organisms. **Calculate** the decimal reduction dose, D_{10}, to three significant figures.

$D = nD_{10} \Rightarrow D_{10} = \frac{D}{n}$

$n = \log_{10}\left[\frac{\text{initial bioburden}}{\text{final bioburden}}\right]$

initial bioburden $= 10^7$

final bioburden $= 30$

D (dose) = 4.84 kGy

$n = \log_{10}\left[\frac{10^7}{30}\right] = 5.52288$

$D_{10} = \frac{D}{n}$

$= \frac{4.84\text{ kGy}}{5.52288}$

$= 0.87635$ kGy

$D_{10} = 0.876$ kGy

20 13. **Explain** the difference between geometric efficiency and intrinsic efficiency in detectors.

Geometric efficiency is defined with the following equation: $\varepsilon_g = \frac{\phi A_d}{S}$, where $\phi = \frac{S}{4\pi r^2}$, meaning $\varepsilon_g = \frac{A_d}{4\pi r^2}$, where A_d is the area of the detector and r is the distance to the source. ε_d only accounts for the surface area and shape of the detector, and the distance from the source when considering its efficiency. Intrinsic efficiency is defined with the following equation: $\varepsilon_i \approx 1-e^{-\mu d}$, where d is the thickness of the detector and μ is the attenuation coefficient for the material the detector is made of. Therefore, ε_i only accounts for the detector material and thickness when considering efficiency. A combination of geometric efficiency and intrinsic efficiency gives the total efficiency of the detector, as ε_g and ε_i each cover the efficiency of different components. Therefore, $\varepsilon = \frac{R}{S} = \varepsilon_g \varepsilon_i$, where R is the counts per second detected by the detector (excluding background), and S is the source strength (S = fA).

Non-proliferation
Disarmament
Peaceful use of nuclear technology

Wt. No. Cadet Kyra Dunnivine

25/25 14. **Describe** the Nuclear Non-Proliferation Treaty (NPT). In your response, **provide** the pillars of the treaty and **explain** why they are important.

The Nuclear Non-Proliferation Treaty (NPT) is a treaty signed by multiple countries in an attempt to regulate nuclear power and protect the world from nuclear weapons. There are three pillars of the treaty: non-proliferation, disarmament, and peaceful use of nuclear technology. These pillars are important as their goal is to prevent nuclear disasters and the use of nuclear weapons. Nuclear power has devestating impacts when misused and can very easily kill many people. The effects of the use of nuclear weapons have been seen in WWII, and it is in the world's best interest to avoid the use of these deadly weapons in the future. As a result, the purpose/goal of the NPT with its three pillars is extremely important for the safety of the world.

25/25 15. The Military Decision-Making Process (MDMP) is used by battalion and above echelons to allow commanders, staffs, and others to think critically and creatively while planning. The steps are listed below. **Describe** how the MDMP steps relate to the individual Engineering Design process steps you applied throughout the development of your design project.

Engineering Design Process steps:

I. Receipt the mission → Identify problem
II. Mission Analysis → Define problem
III. COA Development → Develop potential ideas; Develop feasible ideas
IV. COA Analysis → Determine how ideas/designs will be compared
V. COA Comparison → Evaluate + compare
VI. COA Approval → Decide on best design
VII. Orders Production, Dissemination, and Transition → Communicate results

The individual engineering design process steps (briefly summarized above) directly parallel the MDMP steps on how to create a result/orders from a given mission/problem. The first step of both is recieving/identifying the mission/problem. Then, this problem is defined/analyzed. Then, the engineering design process develops potential and feasible ideas (2 steps), which parallels COA development in the MDMP. The determination of how ideas will be compared parallels COA analysis. The evaluation and comparison of ideas parallels COA comparison. Deciding on the best design parallels COA approval. Finally, the communication of the results parallels orders production, dissemination, and transition. The MDMP steps are very similar to the individual engineering design process steps, as each step corresponds with one (or two for the development steps) steps in the other process. In both process, a problem or mission is given, options/solutions are explored, narrowed, then selected, then the final selection is disseminated/communicated.

, 3 sig figs

Wt.	No.	Cadet Kyra Dunnivin
65	16.	You have a detector with an intrinsic efficiency of 31.3% for Cs-137. A 5.00-µCi Cs-137 source is placed 11.5 cm away from your detector. Your detector measures 1325 counts per second. The measured background count rate is 129 counts per second. **Calculate** the detector surface area that is normal to the gamma flux.

A_d

only gamma, no β^-

$\therefore f$ (of Cs-137 γ) $= 0.8510$

$\varepsilon_i = 0.313$

$A = 5.00 \times 10^{-6}\, Ci \cdot \frac{3.7 \times 10^{10}\, Bq}{Ci} = 1.85 \times 10^{5}\, Bq$

$r = 11.5$ cm

$R = 1325 - 129 = 1196$

$$\varepsilon_g = \frac{\phi A_d}{S}$$

$$\phi = \frac{S}{4\pi r^2}$$

$$> \varepsilon_g = \frac{\cancel{S}}{4\pi r^2} \frac{A_d}{\cancel{S}} = \frac{A_d}{4\pi r^2}$$

$$\varepsilon = \frac{R}{S} = \varepsilon_g \varepsilon_i$$

$$\frac{R}{S} = \frac{A_d}{4\pi r^2} \varepsilon_i$$

$$\frac{R(4\pi r^2)}{S \varepsilon_i} = A_d$$

$$A_d = \frac{R(4\pi r^2)}{f A \varepsilon_i} = \frac{(1196)(4\pi)(11.5)^2}{(0.8510)(1.85 \times 10^5)(0.313)} = 40.336\ cm^2$$

$$= \underline{40.3\ cm^2}$$

maximum dose occurs right outside sphere, so right at edge of thickness

Wt. No. Cadet Kyra Dunnmire

80 17. As shown in the figure below, a 1.500-MeV isotropic γ-ray point source, with source strength S = 4.060 x 10^{12} γ/s, is at the center of a solid sphere of aluminum with a radius of 111.15 cm. **Calculate** the maximum dose equivalent rate a person would receive (in mrem/hr) at a point on the outside surface of the sphere to four significant figures.

$\dot{D}_\gamma = \phi E_\gamma \left(\frac{\mu_{en}}{\rho}\right)$

$\phi = B\phi_u$

$\phi_u = \frac{Se^{-\mu x}}{4\pi r^2}$

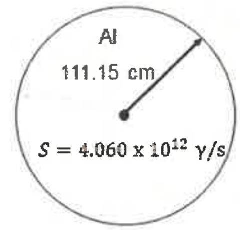
Given variables

$S = 4.060 \times 10^{12} \frac{\gamma}{s}$

$r = 111.15$ cm

$E = 1.500$ MeV

$x = 111.15$ cm



* Not drawn to scale

Finding μ (of Al)

$\frac{\mu}{\rho} = 0.0500 \frac{cm^2}{g}$ (when E=1.50 MeV)

ρ (of Al) $= 2.699 \frac{g}{cm^3}$

$\mu = \frac{\mu}{\rho}\rho = (0.0500)(2.699) = \underbrace{0.13495 \frac{1}{cm}}_{\mu}$

$\dot{D}_\gamma = B\left(\frac{Se^{-\mu x}}{4\pi r^2}\right)E_\gamma\left(\frac{\mu_{en}}{\rho}\right)$

Finding $\frac{\mu_{en}}{\rho}$ (for tissue)

$\frac{\mu_{en}}{\rho} = 0.0276 \frac{cm^2}{g}$

$\dot{D}_\gamma = (28.3)\left(\frac{4.060 \times 10^{12} e^{-(0.13495)(111.15)}}{4\pi(111.15)^2}\right)(1.5)(0.0276)$

$= 9.375627 \frac{MeV}{\cancel{g}\cdot s} \cdot \frac{0.5052 \frac{rad}{yr}}{\frac{MeV}{\cancel{g}\cdot s}}$

Finding B (isotropic)

$\mu x = (0.13495)(111.15)$

$= 14.99969$

$\mu x = 15.000$

$(1.0, 37.9)$ (x_1, y_1)

$(2.0, 18.7)$ (x_2, y_2)

$x = 1.500$

$y - y_1 = \left[\frac{y_2 - y_1}{x_2 - x_1}\right](x - x_1)$

$y = y_1 + \left[\frac{y_2 - y_1}{x_2 - x_1}\right](x - x_1)$

$= 37.9 + \left[\frac{18.7 - 37.9}{2.0 - 1.0}\right](1.5 - 1)$

$= 28.3 \Rightarrow B = 28.3$

$= 4.73657 \frac{\cancel{rad}}{yr} \cdot \frac{1000\ m\cancel{rad}}{1\ \cancel{rad}} \cdot \frac{1\ rem}{rad}$

b/c QF=1 for γ

H = D·QF

$= 4736.5668 \frac{mrem}{\cancel{yr}} \cdot \frac{1\ \cancel{yr}}{365\ \cancel{days}} \cdot \frac{1\ \cancel{day}}{24\ hrs}$

$= 0.5407 \frac{mrem}{hr}$ 4 sig figs

3 sig figs

on flat surface

Wt. No. Cadet Kyra Dunnirvine

65 18. While working with Sr-90, you accidentally spill a vial onto your lab bench resulting in a contamination zone with a surface activity of 1364 Bq/cm^2. **Calculate** the beta absorbed dose rate to your skin in rads/hr if you were standing at a height of 1.30 meters above the contaminated area. Remember, your skin has a thickness/density of 0.007 g/cm^2. Assume your skin is not shielded by clothing. Report your answer to three significant figures.

$$\dot{D}_\gamma\left[\frac{mGy}{hr}\right] = 3.6\times10^{-4}\, C_a \bar{E}_\beta \mu_{\beta,t}\, e^{-(\mu_{\beta,air}(\rho_a x_a))}\, e^{-(\mu_{\beta,t}(\rho_t x_t))}$$

$C_a = 1364 \frac{Bq}{cm^2}$

$\rho_t x_t = 0.007 \frac{g}{cm^2}$

$E_{\beta,max} = 0.5460$ MeV

$\bar{E}_{\beta,avg} = \frac{E_{\beta,max}}{3} = 0.182$ MeV

$\mu_{\beta,t} = 18.6(E_{\beta,max} - 0.036)^{-1.37}$
$= 18.6(0.5460 - 0.036)^{-1.37}$
$= 46.7887888$

$\mu_{\beta,a} = 16(E_{\beta,max} - 0.036)^{-1.4}$
$= 16(0.5460 - 0.036)^{-1.4}$
$= 41.06971945$

$\rho_a = 0.00120479 \frac{g}{cm^3}$

$x_a = 1.30\text{ m} = 130\text{ cm}$

$\rho_a x_a = (0.00120479)(130)$
$= 0.1566227 \frac{g}{cm^2}$

$$\dot{D}_\gamma = \underbrace{(3.6\times10^{-4})}_{}\underbrace{(1364)}_{C_a}\underbrace{(0.182)}_{E_\beta}\underbrace{(46.7887888)}_{\mu_{\beta,t}}\, e^{-(\underbrace{41.06971945}_{\mu_{\beta,a}})(\underbrace{0.1566227}_{\rho_a x_a})}\, e^{-(\underbrace{46.7887888}_{\mu_{\beta,t}})(\underbrace{0.007}_{\rho_t x_t})}$$

$$= 0.00484743 \frac{\cancel{mGy}}{hr}\cdot\frac{1\,\cancel{Gy}}{1000\,\cancel{mGy}}\cdot\frac{100\text{ rad}}{1\,\cancel{Gy}}$$

$= 4.847\times10^{-4}\ \frac{rad}{hr}$

$= 4.85\times10^{-4}$ rad/hr

Poor Student Work

Time of Completion 2030

Cadet Thomas Blevins

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE350, Radiological Engineering Design
AY 26-2 Term End Examination, Version 1

1. Instructions:

   a. The weight of this examination is 500 marks.

   b. You have 3.5 hours to complete the examination. The examination consists of 18 questions on 8 pages (not including this cover sheet). Check your examination for completeness.

   c. Authorized references are the NE350 Reference Data Card (provided by the instructor), the Nuclear Engineering Reference Data Pamphlet (provided by the instructor), your cadet-issued TI30 or TI36 calculator, and a straight edge.

   d. Show all work. Partial credit may be given for correct work shown.

2. Grading Summary:

PROBLEM	WEIGHT	SCORE
1	10	10
2	10	10
3	10	10
4	10	0
5	25	12
6	25	7
7	20	14
8	25	24
9	20	15
10	25	15

PROBLEM	WEIGHT	SCORE
11	20	12
12	20	18
13	20	14
14	25	12
15	25	16
16	65	61
17	80	60
18	65	61
	Total	371
	GRADE	74.2%

C+

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Wt. No. Cadet Blevins

10 1. **Select** the tissue weighting factor for dose absorbed in the lungs.

a. 0.01
b. 0.04
c. 0.08
d. 0.12

10 2. **Select** the relationship between radiation dose and biological effect (harm) that takes the most conservative approach to protecting the radiation worker.

a. Hormesis
b. Linear No Threshold
~~c. Quadratic~~
d. Threshold
~~e. Cubic~~

10 3. True/False: High-Z materials are good neutron shields.

a. True
b. False

~~10~~ 0 4. **Select** the detector type or types that utilize a photomultiplier tube.

a. Solid state detectors
b. Gas-filled detectors
c. Scintillation detectors

~~25~~ 12 5. Using words **and** figures, describe two of the following particle interactions: *excitation*, *ionization*, and *Bremsstrahlung*. Clearly label your figures.

Brehmstrahlung occurs when an excited photon is sent into a high Z material and the interaction between the two produces Brehmstrahlung radiation as a result (most commonly in X Rays)

excitation is the particle interaction that occurs when a shell electron of a given material's energy is raised and the electron moves out to an outer electron shell

Wt.	No.	Cadet ______________________

~~25~~ 7

6. An industrial thickness gauge uses Y-90 to emit β-particles with a maximum energy of 2.280 MeV to measure the thickness of aluminum sheets. **Calculate** the maximum thickness in cm of the aluminum sheet that would be practical to measure with a Y-90 gauge to three significant figures.

$\rho = 2.699$

$Y\ 0.0432 \quad 0.0358$
$\times 2 \quad 3$

$$Y - 0.0432 = \frac{0.0358 - 0.0432}{3-2}(2.28-2)$$
$$Y - 0.0432 = -0.0074 \cdot (0.28)$$
$$Y - 0.0432 = -0.002072$$
$$Y = 0.041128$$

$$\mu/\rho = 0.041128\ \frac{cm^2}{g}$$

$$\mu = \mu/\rho \cdot \rho \quad \frac{cm^2}{g}\ \frac{g}{cm^3}$$
$$\mu = 0.041128 \cdot 2.699$$
$$\mu = 0.111004472\ cm^{-1}$$
$$\mu = 1.11004472\ cm$$
$$\boxed{\mu = 1.11\ cm}$$

~~20~~ 14

7. A sample of silicon is placed in a reactor and bombarded with a thermal neutron flux of $\phi = 2.89 \times 10^{13}$ n/(cm^{2}s) to transmute silicon to phosphorus. The silicon target has a Si-30 abundance of 3.10% and a neutron capture cross section, σ, of 0.108 barns. **Calculate** the irradiation time required to produce a semiconductor with an impurity content of 8.00 parts per billion (8 phosphorus atoms per 1 billion silicon atoms). Report your answer in hours.

$$N_P = a N_{Si} \sigma \phi t$$

$$\frac{N_P}{N_{Si}\, a \sigma \phi} = t$$

$$t = \frac{8}{(10^9)(0.031)(0.108)(2.89E13)} = 8.26811855e^{-20}\ s \,\Big|\, \frac{60\ s}{1\ min} \,\Big|\, \frac{60\ min}{1\ hr} = \boxed{0.00006135\ \text{hours}}$$

Wt. No. 40hr Cadet ____________________

~~25~~ 24 8. Over the course of 5 × 8-hr working days, an employee receives an external dose of 230 mrem and inhales air consisting of 1.9 x 10^{-8} μCi/mL of I-131 and 8.3 x 10^{-8} μCi/mL of Cs-137. **Calculate** the ratio of the dose received to the annual dose equivalent limit. (A standard breathing rate is 1.2 m^3/hr and 10^6 mL = 1 m^3) 1.2×10^6 mL/hr

$$\frac{230\,\text{mrem}}{5000\,\text{mrem}} + \frac{0.912\,\mu Ci}{50\,\mu Ci} + \frac{3.984\,\mu Ci}{200\,\mu Ci} = 0.046 + 0.01824 + 0.001992$$

$$= 0.066232$$

$\boxed{6.62\% \text{ of annual dose equivalent limit}}$

I131

$$\frac{1.2\times10^6\,\cancel{mL}}{\cancel{hr}}\left|\frac{1.9\times10^{-8}\,\mu Ci}{\cancel{mL}}\right|\frac{40\,\cancel{hr}}{} = 0.912\,\mu Ci$$

Cs137

$$\frac{1.2\times10^6\,\cancel{mL}}{\cancel{hr}}\left|\frac{8.3\times10^{-8}\,\mu Ci}{\cancel{mL}}\right|\frac{40\,\cancel{hr}}{} = 3.984\,\mu Ci$$

~~20~~ 15 9. A 2.754-MeV photon undergoes Compton scattering and is reduced in energy to 1.451 MeV. **Calculate** the angle that the photon was scattered.

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2}\left[1 - \cos(\theta)\right]$$

$$\frac{1}{1.451} - \frac{1}{2.754} = \frac{1}{931.494}\left[1 - \cos(\theta)\right]$$

$$0.68917987 6 - 0.3631082 06 = 0.001073544\left[1 - \cos\theta\right]$$

$$0.32607167 \div 0.001073544 = 1 - \cos\theta$$

$$0.00350052 = 1 - \cos\theta$$

$$\cos\theta = 0.99964998 48$$

$$\theta = \cos^{-1}(0.9996499948)$$

$$\theta = 1.51606067 1$$

$$\boxed{\theta = 1.52^\circ}$$

~~25~~ 15 10. **Define** build-up and **discuss** why it is important to account for it in radiation shielding.

Build up is the term that refers to the photons that are not attenuated by a shield material and still are present in the general area of concern. It is important to account for buildup in radiation shielding, as the unattenuated photons may have shifted direction when passing through shielding material and ones that were initially not going to reach the subject now may reach them.

Wt. No. Cadet ______________________________

~~20~~ 12 11. **Calculate** the required number density of molecules of BF_3 in a detector with a 5.00-cm diameter to ensure that 98.0% of the thermal neutrons incident along a diameter are absorbed (σ_a for natural boron is 760 barns). $2\% = 0{,}02$ $1-.98 = 0{,}02$

$R_c = \phi \Sigma_a$

$\Sigma = N\sigma$

$N\sigma = -\frac{\ln(0{,}02)}{d}$

$N = -\frac{\ln(0{,}02)\,\sigma}{d}$

$N = -\frac{\ln(0{,}02) \cdot 760}{5}$

$\boxed{N = 594{,}63 \frac{BF_3}{cm}}$

~~20~~ 18 12. A laboratory study with Staphylococcus has found that a radiation dose of 4.84 kGy reduces the initial population of 10^7 organisms down to 30 organisms. **Calculate** the decimal reduction dose, D_{10}, to three significant figures.

find

$D = n\, D_{10}$

$D_{10} = D/n$

$D_{10} = \frac{4.84\,KGy}{n}$

$\boxed{D_{10} = 1{,}452 \times 10^{-6}\,KGy}$

$n = \log_{10}\left[\frac{\text{initial}}{\text{final}}\right]$

$n = \log_{10}\left[\frac{10^7}{3}\right]$

$n = \log_{10}[3333333.333]$

$n = 3333333{,}333$

~~20~~ 14 13. **Explain** the difference between geometric efficiency and intrinsic efficiency in detectors.

Geometric and intrinsic efficiency are the two kinds of efficiency that detectors use. Geometric efficiency relies on the geometry of the material used in a detector while intrinsic efficiency relies on the given properties of the material used in the detector

Wt. No. Cadet ____________________________

~~25~~ 12 14. **Describe** the Nuclear Non-Proliferation Treaty (NPT). In your response, **provide** the pillars of the treaty and **explain** why they are important.

The Nuclear Non Proliferation Treaty created to prevent the proliferation of Nuclear weapons. The five pillars of this treaty are inspection, safe use of nuclear materials, disarming, halt of testing and peaceful sharing of nuclear tech for civil uses. The sharing allows for countries who seek nuclear energy to receive help from nuclear states. Halt of testing weapons halts the arms race for more powerful nuclear weapons. Disarming helps to reduce the number of warheads in the world. Safe use of nuclear power limits new states from acquiring nuclear weapons of their own. Inspections are important as they are used to ensure that the states party to the treaty are not violating any of the other pillars. All combined seek to make the world safer through limitation of nuclear weapons.

~~25~~ 16 15. The Military Decision-Making Process (MDMP) is used by battalion and above echelons to allow commanders, staffs, and others to think critically and creatively while planning. The steps are listed below. **Describe** how the MDMP steps relate to the individual Engineering Design process steps you applied throughout the development of your design project.

I. Receipt the mission
II. Mission Analysis
III. COA Development
IV. COA Analysis
V. COA Comparison
VI. COA Approval
VII. Orders Production, Dissemination, and Transition

The MDMP steps follow a similar path in which the NEDP did throughout the course of the design project. Step I follows receiving the initial problem/task to begin with. Next analysis of the problem and what we would need such as additional information from the client or what they want. Then the COA Development stage follows the prototyping stage where the group began coming up with initial designs and calculations for the project. Phase IV of COA analysis occurred at IPR 2 once prototypes were presented and critiqued giving areas for improvement. Then phase V would occur with IPR 3 and the further refinement of the groups project. Phases VI + VII tie into the final presentation of the product where the final designs were presented and put forward for approval, and pending that, the project would be put into production as outlined in phase VII.

Wt. No. Cadet ______________________

~~65~~ 61 16. You have a detector with an intrinsic efficiency of 31.3% for Cs-137. A 5.00-µCi Cs-137 source is placed 11.5 cm away from your detector. Your detector measures 1325 counts per second. The measured background count rate is 129 counts per second. **Calculate** the detector surface area that is normal to the gamma flux.

$\varepsilon_i = 31.3\%$ $\quad \varepsilon_g = \frac{\phi A_d}{S}$ ← find $\quad \phi = \frac{S}{4\pi r^2} = \frac{fA}{4\pi r^2}$ $\quad R = 1325 - 129 = 1196$

$\varepsilon = \varepsilon_g \varepsilon_i$ $\quad = \frac{15743500}{4\pi \cdot (11.5)^2}$ $\quad A = 5.00 \cdot 10^6 \cdot 3.7$

$\varepsilon = \frac{R}{S}$ $\quad A = 18500000$

$\frac{\phi A_d}{S} = \frac{R}{S} \cdot 0.313$ $\quad \phi = 9473.179003$ $\quad S = fA$

$S = 0.851 \cdot 18500000$

$\varepsilon_g = \frac{\varepsilon}{\varepsilon_i}$ $\quad S = 15743500$

$\phi A_d = \frac{R}{\cancel{S}} \cdot 31.3 \cdot \cancel{S}$

$A_d = \frac{R \cdot 31.3}{\phi}$

$A_d = \frac{1196 \cdot 31.3}{9473.179003}$

$A_d = \frac{37434.8}{9473.179003}$

$A_d = 3.951661843\ cm^2$

$\boxed{A_d = 3.952\ cm^2}$

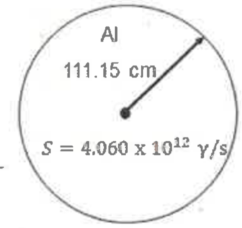
Wt. No. Cadet ______________________

~~80~~ 60 17. As shown in the figure below, a 1.500-MeV isotropic γ-ray point source, with source strength $S = 4.060 \times 10^{12}$ γ/s, is at the center of a solid sphere of aluminum with a radius of 111.15 cm. **Calculate** the maximum dose equivalent rate a person would receive (in mrem/hr) at a point on the outside surface of the sphere to four significant figures.

X tr

$\dot{D}_\gamma = \phi E_\gamma \left(\frac{\mu_{en}}{\rho}\right)$ 1.5 −4

$\phi = B_0 \frac{S e^{-\mu x}}{4\pi r^2}$



* Not drawn to scale

$\dot{D}_\gamma = 226.4670889 \cdot 1.5 \cdot 0.0248$

$\dot{D}_\gamma = 8.424575707 \frac{mGy}{hr}$

$\phi = B_0 \frac{S e^{-\mu x}}{4\pi r^2}$

$\phi = 28.3 \frac{4.06E^{12} \cdot e^{-0.13495 \cdot 111.15}}{4\pi (111.15)^2}$

$\phi = 28.3 \frac{4.06E^{12} \cdot 0.000000306}{155248.9952}$

$\phi = 28.3 \frac{1242360}{155248.9952}$

$\phi = 28.3 \cdot 8.002370633$

$\phi = 226.4670889$

$\frac{\mu}{\rho}(1.5) = 0.050$ ·2.699

$\mu = 0.13495$

$\frac{\mu_{en}}{\rho}(1.5) = 0.0248$

$\mu r = 14.9996925$

$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

$y - 37.9 = \frac{18.7 - 37.9}{2 - 1}(1.5 - 1)$

$y - 37.9 = -9.6$

$y = 28.3$

$B = 28.3$

−2

$\frac{8.424575707\ mGy}{hr} \cdot \frac{1\ Gy}{1000\ mGy} \cdot \frac{100\ rem}{1\ Gy} \cdot \frac{1000\ mrem}{1\ rem} = 842.4575707 \frac{mrem}{hr}$

$\boxed{\dot{D}_\gamma = 842.4576 \frac{mrem}{hour}}$

Wt.	No.	Cadet ______________________
~~65~~ 61	18.	While working with Sr-90, you accidentally spill a vial onto your lab bench resulting in a contamination zone with a surface activity of 1364 Bq/cm². **Calculate** the beta absorbed dose rate to your skin in rads/hr if you were standing at a height of 1.30 meters above the contaminated area. Remember, your skin has a thickness/density of 0.007 g/cm². Assume your skin is not shielded by clothing. Report your answer to three significant figures.

$E_\beta = \frac{E_{\beta max}}{3}$

$E_\beta = \frac{0.546}{3}$

$E_\beta = 0.1182$

$$\dot{D}_\beta \frac{mGy}{hr} = 3.6\times10^{-4}\, C_a\, \bar{E}_\beta\, \mu_{\beta t}\, \underbrace{e^{-(\mu_{\beta air}\rho_a x_a)}}_{0.0016085057}\, \underbrace{e^{-(\mu_{\beta t}(\rho_t t_t))}}_{0.72070778}$$

$\rho_t t_t = 0.007$

$\mu_{\beta air} = 16\,(E_{\beta max} - 0.036)^{-1.4}$

$= 16\,(0.546 - 0.036)^{-1.4}$

$= 41.06971945$

$\dot{D}_\beta = 3.6E^{-4} \cdot 1364 \cdot 0.1182 \cdot 46.7887888 \cdot 0.001608505 \cdot 0.72070778$

$\dot{D}_\beta = 0.003148167 \frac{mGy}{hr}$

$\mu_{\beta\, tissue} = 18.6\,(0.546 - 0.036)^{-1.37}$

$= 46.7887888$

$\rho_a x_a = 130cm \cdot 0.00120479$

$= 0.1566227$

$C_a = 1364$

$$0.003148167 \frac{\cancel{mGy}}{hr} \left| \frac{1000 Gy}{1 \cancel{mGy}} \right| \frac{100 rad}{1 Gy} = \dot{D}_\beta = 314.8167 \frac{rad}{hr}$$

$$\boxed{\dot{D}_\beta = 314.817 \frac{rad}{hr}}$$

TAB I FINAL COURSE GRADES

NE350, AY26-2
End of Term Grades

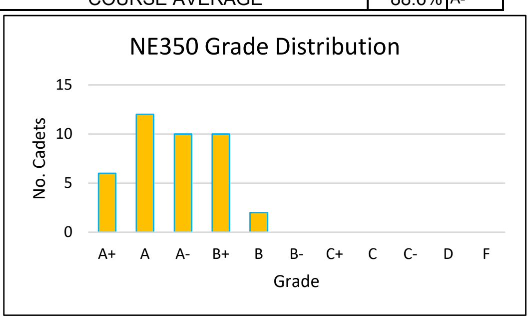
OML	Name	Grad Year	Max Points	Points Earned	Percent	Grade	FOS		CES	PS (%)	NEDP (%)	WPR (%)	TEE (%)	Points to Gain
1	Dunnirvine, Kyra	2027	2000	1925	96.3%	A+	SSC0		NE	100	92	97.5	98	
2	Wimer, Claire	2027	2000	1907	96.25%	A+	KIN1		NE	99.56	90.33	100	95	
3	Wilson, John	2027	2000	1851	96.25%	A+	CHM2A		NE	94.67	87.5	95	94.8	
4	Zang, Kaitlyn	2027	2000	1848	96.25%	A+	BIO0		NE	99.56	86.83	91	94.8	
5	Heckle, Carolyn	2027	2000	1884	94.2%	A+	PHY1	MSM0N	NE	100	92.17	98	94	
6	Singh, Samrath	2026	2000	1881	94.1%	A+	CEN2		NE	96.44	94.67	92	90.2	
7	Hockensmith, Silas	2027	1940	1818	93.7%	A	GIS2H		NE	94.67	89.83	101	98.4	5.6
8	Corbett, Richard	2027	2000	1874	93.7%	A	SSC0		NE	93.78	89.83	96.5	96	6
9	Smith, Riley	2027	2000	1869	93.5%	A	LLS3		NE	95.56	93.67	93.5	95.6	11
10	Levendoski, Lauren	2027	2000	1865	93.3%	A	SSC0		NE	97.78	92.17	87	93.6	15
11	Poland, Connor	2027	2000	1858	92.9%	A	CHM2A		NE	89.78	88.33	99	95.6	22
12	Cobb, Megan	2027	2000	1853	92.7%	A	SSC0		NE	99.11	86.83	89	94.4	27
13	Capriola, Jack	2027	2000	1851	92.6%	A	HNT3T		NE	93.33	94.33	90.5	93.2	29
14	Doherty, Rowan	2027	2000	1839	92.0%	A	PHY1	RSU1N	NE	97.78	94.33	87.5	91	41
15	Gerard, Allison	2027	2000	1839	92.0%	A	CHM2A		NE	90.67	89.83	95	96.6	41
16	Kim, Elaine	2027	2000	1825	91.3%	A	BIO0		NE	95.11	91.33	97.5	85	55
17	Whitfield, Audrey	2027	2000	1822	91.1%	A	BIO0		NE	95.56	85.67	89	90.6	58
18	Records, Benjamin	2027	2000	1819	91.0%	A	CHM2A		NE	90.67	88.5	92.5	96.2	61
19	Taysom, Rex	2027	2000	1808	90.4%	A-	DSS2		NE	96	90	91	87	2
20	Macune, Grace	2026	2000	1784	89.2%	A-	CEN2		NE	94.67	82.33	85.5	95.6	26
21	Zvik, Tal	2027	2000	1777	88.9%	A-	PIF0		NE	90.67	85.83	91.5	91.6	33
22	Littrell, Cooper	2027	2000	1776	88.8%	A-	PAP3	TST1N	NE	98.22	84.67	84	90.6	34
23	Morgan, Daniel	2026	2000	1767	88.4%	A-	SSC0	RSU1N	NE	99.11	84	86.5	85.2	43
24	Kump, Austyn	2027	2000	1757	87.9%	A-	LLS3		NE	94.67	85.67	84	94	53
25	Custodio, Christian	2026	2000	1756	87.8%	A-	PHY1	RSU1N	NE	84.89	86.67	76.5	94	54
26	Lataif, Philip	2027	2000	1754	87.7%	A-	PAP3		NE	99.56	84	92	90	56
27	Ross, James	2027	2000	1748	87.4%	A-	CHM2A		NE	84	90	86	89.8	62
28	Jacobs, Kyra	2027	2000	1742	87.1%	A-	PIF0		NE	96.89	85	98.5	84.4	68
29	Rodriguez, Eric	2026	2000	1734	86.7%	B+	CEN2		NE	92	75.33	98.5	91.4	6
30	Baldwin, Hugh	2026	2000	1733	86.7%	B+	PII0		NE	93.33	93.5	93	79	7
31	Nyhuis, Parker	2027	2000	1718	85.9%	B+	HMH3T		NE	89.33	83	96	86.8	22
32	Dolan, Patrick	2026	2000	1717	85.9%	B+	LLS3		NE	96.89	82.17	83	86.2	23
33	Jaworski, Preston	2027	2000	1715	85.8%	B+	MGN0		NE	92	90.17	82.5	78.4	25
34	Grauberger, Ashlyn	2027	2000	1713	85.7%	B+	EPS1	AST1N	NE	93.78	81.17	81	83.4	27
35	Holland, Noah	2027	2000	1702	85.1%	B+	GIS2		NE	95.11	92	79.5	81.6	38
36	Dickerson, Thomas	2026	2000	1698	84.9%	B+	SSC0		NE	86.22	84.33	85	89.2	42
37	Hall, Michael	2027	2000	1687	84.4%	B+	FLA2		NE	86.67	87.83	70	83.4	53
38	Hicks, Jackson	2027	2000	1681	84.1%	B+	ECN2	RSE1N	NE	92.44	89	68.5	77	59
39	Green, Henry	2027	2000	1665	83.3%	B	ECN2		NE	85.78	93.17	80	78.2	5
40	Morales, Esteban	2027	2000	1640	82.0%	B	CHM2A		NE	79.11	87	82	87.2	30
41	Schneider, Allen	2026	2000	1579.5	79.0%	B-	GIS2		NE	78.89	83	73.5	83.2	20.5
42	Blevins, Thomas	2027	2000	1570	78.5%	B-	LLS3	TST1N	NE	89.78	78.67	84.5	74.2	30
43	Olson, Zachariah	2027	2000	1558	77.9%	B-	PAP3		NE	85.78	82.17	75	78.8	42
44	Baez, Isaac	2026	2000	1500	75.0%	C+	MGN0		NE	81.78	82	52.5	72.4	30
	COURSE AVERAGE				88.6%	A-								

Grading Scale		# CDTs
A+	94.0%	6
A	90.5%	12
A-	87.0%	10
B+	83.5%	10
B	80.0%	2
B-	76.5%	3
C+	73.0%	1
C	69.5%	0
C-	66.0%	0
D	60.0%	0
F	<60%	0

Grade Distribution		
Graded Requirement	**Marks**	**% of Total**
TEE	500	25%
WPR	200	10%
NEDP (3 IPRs, 1 Final)	600	30%
Problem Sets (3)	225	11%
Lab	170	9%
Instructor Points	85	4%
Course-Wide Writ	100	5%
FNEK Writ (2)	120	6%
Total	2000	100%

Historical Data		
AY	**AVG**	**Grade**
26-1	87.19%	A-
25-2	87.67%	A-
25-1	89.35%	A-
24-2	89.96%	A-
24-1	86.46%	B+
23-2	84.18%	B+
23-1	86.25%	B+
22-2	82.11%	B
22-1	83.96%	B+
21-2	84.10%	B+
21-1	83.10%	B
20-2	81.50%	B

CDT Demographics	
0	NEN1
3	PHY1
6	SSC0
0	NSM0N
40	NE



Course Notes:

1. No TEE Failures; no course failures
2. Point Redemption continues to drive success in the course
3. I don't recommend any changes to grades

AY26-2 Course Changes:

1. Added 2 design project options; recommend next CD look over them
2 .Recommend nuclear waste course remain
3. Implemented improvements to the lab which I believe simplified and streamlined the lab report
4. I recommend an adjustment to the TEE grading rubric; it is simply too lenient.

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Course Director

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SECTION III

ASSIGNMENTS

TAB J PROBLEM SETS

UNITED STATES MILITARY ACADEMY
WEST POINT.

PROBLEM SET #1

38 50.7% 26 34.7 % 11 14.7%

CUTSHEET

NE350 Radiological Engineering Design, Problem Set 1, AY26-1
75 Marks / Submit digitally in Canvas by 2359 on Lesson 07 (08 or 09 SEP)

Marks	**Question**	
10	**1.**	Calculate the minimum thickness required to shield the particles below. Express your answers in units of cm to three significant figures.

a A 0.250 MeV electron in zinc

2 Eqn.

$R_{\beta,max}\left[\frac{g}{cm^2}\right] = \begin{cases} 0.412\,E_\beta^{\,1.265-0.0954\ln(E_\beta)} & 0.01 \le E_\beta < 2.5\text{ MeV} \\ 0.530\,E_\beta - 0.106 & 2.5 \le E_\beta \le 20\text{ MeV} \end{cases}$

Lookup: $\rho_{Zn} = 7.133\left[\frac{g}{cm^3}\right]$ 1 Correct lookup

Correct value and units

$R_{\beta,max} = 0.412(0.25)^{1.265-0.0954\ln(.25)}$

1 Correct setup $= 0.05938\ \frac{g}{cm^2} \rightarrow \frac{0.05938\left[\frac{g}{cm^2}\right]}{7.133\left[\frac{g}{cm^3}\right]} = 8.33\cdot10^{-3}\ [cm]$ ANS

b A 5.45 MeV α (alpha) particle in cadmium

1 Eqn. $R_{\alpha,air}[cm] = \begin{cases} 0.56\,E_\alpha & E_\alpha < 4\text{ MeV} \\ 1.24\,E_\alpha - 2.62 & 4 \le E_\alpha \le 8\text{ MeV} \end{cases}$

Lookups: $\rho_{Cd} = 8.65\left[\frac{g}{cm^3}\right]$

$M_{Cd} = 112.4\left[\frac{g}{mol}\right]$ 1 Correct lookup

$R_{\alpha,air} = 1.24(5.45) - 2.62 = 4.13\,[cm]$

1 Eqn. $R_\alpha = 3.2\times10^{-4}\,\frac{\sqrt{M}}{\rho}\,R_{\alpha,air} = 1.62\cdot10^{-3}\,[cm]$ ANS

1 Correct setup $= 3.2\cdot10^{-4}\,\frac{\sqrt{112.4}}{8.65}(4.13)$

Correct value and units

15	**2.**	Consider a beam of 1.90 MeV gamma rays incident on a slab of silicon. Calculate the fraction of the incident beam that passes through the slab without undergoing a collision at the thicknesses provided. Report your answer to three significant figures. Hint: this problem requires you to interpolate on Table 2-1 of the NERD-P.

6 Eqn.

a 0.100 cm

$I = I_o e^{-\mu x}$ $\rho_{Si} = 2.33\,\frac{g}{cm^3}$ $\frac{\mu}{\rho}(\text{interpolated}) = 0.0461\,\frac{cm^2}{g}$

$\mu = 0.0461 * 2.33 = 0.1074\ cm^{-1}$

2 Correct lookups 1 each

$I = 1\,e^{-.1074(.1)} = 0.989$ Correct value and units

b 1.00 cm

$I = 1\,e^{-.1074(1)} = 0.898$

2 Interpolated correctly

Correct value and units

2 Interpolation attempt

c 10.0 cm

$I = .342$

Correct value and units

10 **3.** A medical patient is given an injection of technetium-99m for medical imaging. The biological half-life of Tc-99m is 24.0 hours. Calculate the percentage of the sample that decays within the body. Report your answer to three significant figures.

$T_B = 24.0[hr]$ $\lambda = \frac{\ln(2)}{T_{1/2}}$ $\lambda_B = 0.02888\left[\frac{1}{hr}\right]$

$T_R = 6.01[hr]$ $\boxed{2}$ Eqn. $\lambda_R = 0.115\left[\frac{1}{hr}\right]$

3 Solve/convert for half lives or decay constants

$\boxed{2}$ Eqn. $\lambda_E = \lambda_B + \lambda_R = 0.1438\left[\frac{1}{hr}\right]$

Fraction Decay W/in body: $\frac{\lambda_R}{\lambda_E}$ $\boxed{2}$ Eqn.

$$\frac{0.115}{0.1438} = 0.7997$$

$$= 80.0\% \quad \text{ANS}$$

1 Correct value and units

10 **4.** A 654 keV photon undergoes Compton Scattering with an electron. Following the interaction, the recoil electron has 123 keV of kinetic energy. Calculate the angle at which the photon scattered away from its original path. Report your answer to three significant figures.

$\boxed{4}$ Eqn. $\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_ec^2}[1-\cos(\phi)]$ $E^e_k = E - E'$

$E = 654[keV]$ $123 = 654 - E'$

$E^e_k = 123[keV]$ $E' = 532[keV]$

2 Solve for E prime

$$\frac{1}{532} - \frac{1}{654} = \frac{1}{511}[1-\cos(\phi)]$$

$$\phi = 34.8^\circ \quad \text{ANS}$$

$$511\left(\frac{1}{532} - \frac{1}{654}\right) = 1-\cos(\phi)$$

2 Correct value and units

$$0.17918 = 1-\cos(\phi)$$

2 Correct setup and algebra

$$\cos(\phi) = 0.8208$$

$$\phi = \cos^{-1}(.8208) = 34.8^\circ$$

CUTSHEET

NE350 Radiological Engineering Design, Problem Set 1, AY26-1

30 **5.** You are a radiation worker. Your boss asks you to perform a job that requires you to spend a full working year (2000 hours) ~~300~~ 500 cm away from an isotropic ~~1.50~~ 0.75-Ci (Curie) Co-60 source. The source is shielded as it is stored in the center of a tin sphere with a thickness (radius) of ~~3.00~~ 12.cm cm. Report all answers to three significant figures.

4 Finds B using NERD-P

a Determine the buildup factor for this situation.

B lookup a function of $E_\gamma, \mu x$

Lookups: $\frac{\mu}{\rho}$ @ 1.25 MeV = 0.0501 $\left[\frac{cm^2}{g}\right]$ $\quad \mu_{Sn} = \frac{\mu}{\rho}\rho = 0.0501 \cdot 7.298 = 0.3656\left[\frac{1}{cm}\right]$

$\rho_{Sn} = 7.298\left[\frac{g}{cm^3}\right]$

$\mu x = 0.3656[cm^{-1}]12[cm] = 4.3876$

2 Correct lookups

1 Correct ux

Using bi-linear interpolation tool

2 Uses interpolation

$B = 3.9985$ ANS

1 Correct B value and units

b Calculate the dose equivalent you will ~~receive~~ received by the end of the working year. Express your answer in rem.

4 Eqn. $\dot{D} = \phi E_\gamma \frac{\mu_{en}}{\rho}$

Lookup: $\frac{\mu_{en}}{\rho}$(tis, 1.25 MeV) = 0.0288 $\left[\frac{cm^2}{g}\right]$

2 Correct lookups

$S = \frac{(.9985 + .9998)\, 0.75\,Ci \;/\; 3.7 \cdot 10^{10} Bq}{1\,Ci} = 5.545 \cdot 10^{10} Bq$

2 Eqn. $\phi_u = \frac{S e^{-\mu x}}{4\pi r^2}$

$\dot{D} = \frac{S e^{-\mu x}}{4\pi r^2}(B)(E_\gamma)\left(\frac{\mu_{en}}{\rho}\right) = \frac{5.545 \cdot 10^{10}\left[\frac{1}{s}\right] e^{-4.3872}}{4\pi(500^2)}(3.99)(1.25[MeV])\left(0.0288\left[\frac{cm^2}{g}\right]\right)$

2 Eqn. $\phi = B\phi_u$

$= 3.1527 \cdot 10^{1}\left[\frac{MeV}{g\,s}\right] \rightarrow \frac{3.1527 \cdot 10^{1}\,\cancel{MeV}}{\cancel{g\,s}} \Big/ \frac{0.5052\,\cancel{rad}/yr}{\frac{\cancel{MeV}}{\cancel{g\,s}}} \Big/ \frac{.2283\,\cancel{yr}}{1\,working\,\cancel{yr}} \Big/ \frac{1\,rem}{\cancel{rad}} = 3.64\,rem$ ANS

1 Eqn. $S = fA$

2 Correct conversions

2 Correct value and units

c Based on your calculations and NRC dose equivalent limits for radiation workers, state whether you should proceed with this job. Explain your answer.

5 Eqn.

3.64 rem < 5 rem, so yes, you can proceed.

PROBLEM SET #2

41 | 54.6% | 23 | 30.7% | 11 | 14.7%

CUTSHEET

NE350 Radiological Engineering Design, Problem Set 2, AY26-1
75 Marks / Digital submission due by midnight on the day of Lesson 18 (20 or 21 MAR) / Show all work and report all answers to 3 significant figures

Marks **Question**

5 **1.** Over fifteen 10.00-hour working days, a radiation worker received an external dose equivalent of 5.8×10^{-3} Sv and, throughout the work period, inhaled air containing $9.55 \times 10^{-8} \frac{\mu Ci}{mL}$ of ^{131}I. Assuming a standard breathing rate of $1.20 \times 10^{6} \frac{mL}{hr}$, **calculate** the decimal fraction of the dose received to the annual effective dose equivalent limit for a radiation worker.

1 Lookup ALI

1 Solves for uCi inhaled correctly

1 $\sum \frac{C_i}{MPC_i} \le 1$

$ALI = 50\ \mu Ci$

$\frac{10\ hr}{1\ d} \cdot \frac{15\ d}{} = 150\ hr$

$\frac{.0058\ Sv}{} \cdot \frac{100\ rem}{1\ Sv} \cdot \frac{1000\ mrem}{1\ rem} = 580\ mrem$

$\frac{9.55E{-}8\ \mu Ci}{mL} \cdot \frac{1.2E6\ mL}{hr} \cdot \frac{150\ hr}{} = 17.19\ \mu Ci$

1 Converts to mrem correctly

Ext + Int

$\frac{580\ mrem}{5000\ mrem} + \frac{17.19\ \mu Ci}{50\ \mu Ci} = 0.4598 \rightarrow \underline{\underline{0.460}}$ ANS

1 Correct value and units

15 **2.** A Soldier maneuvering through a radiation field accidentally ingests some earth containing ^{90}Sr and ^{41}Ar. The strontium quickly collects on the Soldiers bone surface whereas the argon deposits throughout his entire body (whole body). Over 50 years, we expect 3.3×10^{14} ^{90}Sr atoms and 1.9×10^{13} ^{41}Ar atoms to decay in his body (these are n_D values). The man weighs 102 kg and has a total bone mass of 14 kg. **Calculate** his committed effective dose equivalent in rem from this incident. Assume every argon decay results in one beta and one gamma emission.

From Table 4-1

4 $H_{50} = \frac{n_D E_d (QF)}{m}$

4 $(H_{50})_E = \sum_T \omega_T (H_{50})_T$

$E_{\beta,Sr} = 0.546\ [MeV]$

$\bar{E}_{\beta,Sr} = \frac{0.546}{3} = 0.182\ [MeV]$

$E_{\beta,Ar} = 1.1984\ [MeV]$

$\bar{E}_{\beta,Ar} = \frac{1.1984}{3} = 0.3995\ [MeV]$

$E_{\gamma,Ar} = 1.2936\ [MeV]$

$\omega_{whole} = 1.0$

1 Used correct tissue factors

$\omega_{bones} = 0.03$

1 Used avg beta E

1 Used correct energies

1 Summed 3 dose equivalents

Sr-β-bones, Ar-γ-whole, Ar-β-whole

$(H_{50})_E = \frac{(3.3E14)(0.182\ [MeV])}{14\ [kg]}(0.03) + \frac{(1.9E13)(1.2936\ MeV)}{102\ [kg]}(1) + \frac{(1.9E13)(0.3995\ MeV)}{102\ [kg]}(1)$

Conversion: $\frac{\cancel{MeV}}{\cancel{kg}} \cdot \frac{1.602E{-}13\ \cancel{J}}{1\ \cancel{MeV}} \cdot \frac{100\ rad}{1\ \cancel{J/kg}} = 1.602E{-}11$

$2.062\ \cancel{rad}\left(\frac{1\ rem}{1\ \cancel{rad}}\right) + 3.86\ \cancel{rad}\left(\frac{1\ rem}{1\ \cancel{rad}}\right) + 1.192\ \cancel{rad}\left(\frac{1\ rem}{1\ \cancel{rad}}\right)$

1 Converted to rad, then rem

$(H_{50})_E = \underline{\underline{7.11\ rem}}$ ANS

2 (1, 1) Correct value and units

5 **3.** A laboratory study with Streptococcus agalactiae determined that an absorbed dose of 12.0 kGy delivered to the bacteria reduces the original population from 4.00×10^7 to 1000.0. **Calculate** the decimal reduction dose, D_{10}, in kGy.

$\boxed{1}$ $D = nD_{10}$

$\boxed{1}$ $n = \log_{10}\left[\frac{\text{init. bio}}{\text{final bio}}\right]$

Given: $D = 12.0$ kGy

init bio = 4.0E7

final bio = 1E3

1 Solves for D_10

Solve: $n = \log_{10}\left[\frac{4.0E7}{1E3}\right]$

1 Solves for n

$= 4.602$

$12.0[\text{kGy}] = 4.602\, D_{10}$

$D_{10} = \frac{12\text{ kGy}}{4.602}$

$D_{10} = 2.6075$ kGy

2.61 [kGy] ANS

1 Correct value and units

5 **4.** **Calculate** the energy of the Compton Edge of a gamma emitted from ^{99m}Tc.

$\boxed{1}$ $\frac{1}{E'} - \frac{1}{E} = \frac{1}{M_e c^2}(1-\cos(\theta))$

$\boxed{1}$ $E_k^e = E - E'$

$\phi = 180$ for Compton Edge

E_k^e gives the E.

Lookup:

$E_\gamma = 0.1405$ MeV

1 Looks up correct E

$E' = \left[\frac{1}{M_e c^2}(1-\cos(\theta)) + \frac{1}{E}\right]^{-1}$

$E' = \left[\frac{1}{.511}(1-\cos(180)) + \frac{1}{.1405}\right]^{-1}$

$E' = 0.0907$ MeV

1 Solves for E prime correctly

$E_k^e = 0.1405 - 0.0907$

0.0498 MeV ANS

1 Correct value and units

5 **5.** Vindija Cave in Croatia is famous for containing the bones of Neanderthals. Using carbon dating, the bones found are estimated to be from around 26,000 BCE. If the ^{14}C to ^{12}C ratio was 1.20×10^{-12} when the neanderthals in the cave perished, **calculate** the current ^{14}C to ^{12}C ratio.

$\boxed{1}$ $t = \frac{\ln\left[\frac{N_{C-14}(t)}{N_{C-14}(t_0)}\right]}{-\lambda}$

Given: $N_{C-14}(t) = ?$

$N_{C-14}(t_0) = 1.2E{-}12 \frac{C14}{C12}$

$t = 2025 + 26000 = 28025$

Lookup: $t_{\frac{1}{2}, C-14} = 5700 [yr]$

$\boxed{1}$ Solve for lambda

Solve: $\lambda_{C14} = \frac{\ln(2)}{5700} = 1.216E^{-4}$

Solve:

$-t\lambda = \ln\left[\frac{x}{N_{C14}(t_0)}\right]$

$e^{-t\lambda} = \frac{x}{N_{C14}(t_0)}$

$e^{-28025(1.216E-4)} = \frac{x}{1.2E-12}$

$x = 3.9735 \cdot 10^{-14} \frac{C_{14}}{C_{12}}$

$= 3.97 \cdot 10^{-14} \frac{C_{14}}{C_{12}}$

ANS

2 Correct log math

1 Correct value and units

5 **6.** Scientists and engineers plan to use a research reactor to irradiate a semiconductor with neutrons to transmute silicon into phosphorus. The neutron capture cross-section of ^{30}Si, σ, is 0.097 barns. The abundance of ^{30}Si is 3.1%. **Calculate** the thermal flux required to produce an impurity content of 55 parts per billion for an irradiation period of 8 hours.

$\boxed{2}$ $N_P = a N_{Si} \sigma \phi t$

$t = \frac{8 hrs}{} \Big/ \frac{3600 s}{1 hr} = 28800 [s]$

$\boxed{1}$ Convert to s

$\phi = \frac{N_P}{N_{Si}\, a\, \sigma\, t}$

$\frac{N_P}{N_{Si}} = \frac{55}{1E9}$

1 Converted to cm^2

$\sigma = 0.097 b \Big/ \frac{1 \cdot 10^{-24} cm^2}{1 b} = 9.7 \cdot 10^{-26} cm^2$

$\phi = \frac{55}{(1E9)(0.031)(9.7E-26)[cm^2]\, 28800 [s]}$

$\phi = 6.35 \cdot 10^{14} \frac{n}{cm^2 s}$

ANS

1 Correct value and units

15 **7.** A worker accidentally spills a solution of ^{24}Na on a workbench (a planar surface). The solution has an areal concentration of 300.0 $\frac{Bq}{cm^2}$. Assuming the worker stands 40.0 cm from the spill, **Calculate** the dose rate from betas to the worker's exposed skin. The air is at standard temperature and pressure (STP). Express your answer in $\frac{mrad}{hr}$.

$\boxed{4}$ any $\dot{D}_\beta$ eq.

1 Correct lookup

Lookups: $\rho_a = 0.001205 \left[\frac{g}{cm^3}\right]$

$\boxed{5}$ $\dot{D}_\beta = 3.6\cdot10^{-4} C_a \bar{E} \mu_{\beta,t} e^{-\mu_{\beta,t}(x_t\rho_t)} e^{-\mu_{\beta,a}(x_a\rho_a)}$

1 Correct E_bar value

$\boxed{2}$ $\bar{E} = \frac{E_{\beta,max}}{3} \Rightarrow \frac{1.393\ MeV}{3} = .464\ [MeV]$

$C_a = 300.0\ \frac{1}{cm^2 s}$

$\boxed{1}$ $\mu_{\beta,t} = 18.6(E_{\beta,max}-0.036)^{-1.37}$ $\mu_{\beta,t} = 12.24 \left[\frac{cm^2}{g}\right]$

1 Correct values

$\boxed{1}$ $\mu_{\beta,a} = 16(E_{\beta,max}-0.036)^{-1.4}$ $\mu_{\beta,a} = 10.44 \left[\frac{cm^2}{g}\right]$

$$\dot{D}_\beta = 3.6\cdot10^{-4}\left(300\left[\frac{1}{cm^2 s}\right]\right)(0.464[MeV])\left(12.24\left[\frac{cm^2}{g}\right]\right)e^{-12.24(0.007)}e^{-10.44(40)(0.001205)}$$

$$= 0.3404\ \frac{mGy}{hr} \bigg/ \frac{100\ mrad}{1\ mGy} = 34.0\ \frac{mrad}{hr}$$ ANS

1 Correct conversion

2 Correct value and units 1 1

20 **8.** You are designing a spherical shield that will surround a 10-Ci ^{137}Cs source and want to limit the dose equivalent rate to humans to $2.00\frac{mrem}{hr}$ at a distance of 50.0 cm from the source. With lead as your shielding material, **calculate** the thickness required to meet this design criterion with a precision of less than or equal to 2%. Consider only the dose due to γ radiation and assume the attenuation due to air is negligible.

Note: Solving this problem requires computational software. You must include buildup in your calculations and use software such as Excel, Mathematica, Python, etc. Attach a copy of your code to your submission.

Physics:

$\boxed{1}$ $\dot{H} = \dot{D}(QF)$

$\boxed{5}$ $\dot{D} = \phi\left(\frac{\mu_{en}}{\rho}\right)E_\gamma$

$\boxed{2}$ $\phi = B\phi_u$

$\boxed{3}$ $\phi_u = \frac{Se^{-\mu R}}{4\pi r^2}$

$\boxed{1}$ $S = fA$

$\boxed{1}$ $\frac{\mu}{\rho}(\rho) = \mu$

Given: $A = 10\,[Ci]$

$r = 50\,[cm]$

Lookups: $E_{\gamma, Cs\text{-}137} = 0.662\,[MeV]$

$f = 0.851$

$\frac{\mu_{Pb}}{\rho_{Pb}} = 0.1046\,\left[\frac{cm^2}{g}\right]$ via linear interpolation Table 2-1

$\mu = 1.186\left[\frac{1}{cm}\right]$ $\rho_{Pb} = 11.34\left[\frac{g}{cm^3}\right]$

$\frac{\mu_{en,t}}{\rho_{,t}} = 0.0317\left[\frac{cm^2}{g}\right]$ via linear interpolation Table 2-2

2 +1 partially correct lookups
+1 complete lookups

3 +2 Used something that involved iterations
+1 some conversions

$R = 8.2\,[cm]$ yields $1.96\left[\frac{mrem}{hr}\right]$

2 +1 Correct value
+1 within 2%

PROBLEM SET #3

49 | $65.\bar{3}\%$ | 16 | $21.\bar{3}\%$ | 10 | $13.\bar{3}\%$

CUTSHEET

NE350 Radiological Engineering Design, Problem Set 3, AY25-2
75 Marks / Digital submission due by midnight on the day of Lesson 25 (16 or 17 APR) / Show all work and report all answers to 3 significant figures

Marks **Question**

15 **1.** Lesson 17: Gas Counters and Neutron Detectors covers the idea of radiation exposure (labeled as X on your reference card) measured in Roentgen as it pertains to radiation dosimetry.

a **Explain** the meaning of the classic term, 'exposure.' **Explain** why the use of the term has decreased in modern times.

2 Partially correct

2 Good explanation

Amount of charge liberated in air. We mostly care about does and energy deposition in tissue, so does or dose equivalent is more relevant for use in human radiation exposure.

b **Calculate** the amount of charge liberated in coulombs, C, by gammas in 1 cubic meter of air if the measured exposure,R, is 3 Roentgen. Assume the air is at STP.

4 Eq. $X = \frac{\Delta Q}{m}$ Lookups: $R = 2.58 \times 10^{-4} \frac{C}{kg}$ $\therefore$ $3R = 7.74 \times 10^{-4}\ C$ $\rho_{air} = 0.00120479 \frac{g}{cm^3}$ 1 Lookup density

Convert: $\frac{0.00120479\ g}{cm^3} \Big/ \frac{1\ kg}{1000\ g} \Big/ \frac{100^3\ cm}{1\ m^3} = 1.20479$ kg of air

Solve: $3R = \frac{\Delta Q}{m} \Rightarrow \frac{7.74 \cdot 10^{-4}\ C}{kg} = \frac{\Delta Q}{1.20479\ kg}$

1 Solves for mass of air

1 Correct value and units

$\Delta Q = 9.33 \times 10^{-4}\ C$ ANS

c Assume your answer in part b. was 7.62×10^{-3} (it shouldn't be). **Calculate** how many ion pairs are produced in the volume of air gas.

$\Delta Q = 7.62 \times 10^{-3}\ C$

2 Lookup IP/C

1 Converts from charge to IP

$\frac{7.62 \times 10^{-3}\ C}{} \Big/ \frac{1\ ip}{1.602 \times 10^{-19}\ C} = 4.76 \times 10^{16}$ ion pairs ANS

1 Correct value and units

5 **2.** **Calculate** the estimated instrinsic efficiency for a sodium iodide, NaI, detector for gammas emitted from Technetium-99m given a detector thickness of 0.25 cm. The density of NaI is 3.67 $\frac{g}{cm^3}$. Hint: sodium iodide is listed on Table 2-1.

2 Eq.

$\varepsilon_i = 1 - e^{-\mu d}$

Given: $d = 0.25\ cm$

Lookup: $E_\gamma = 0.140511\ MeV$

$\frac{\mu}{\rho} = 0.75816 \frac{cm^2}{g}$ (interpolated)

$\rho = 3.67 \frac{g}{cm^3}$

Solve: $\mu = \frac{\mu}{\rho}(\rho) = \left(0.75816 \frac{cm^2}{g}\right)\left(3.67 \frac{g}{cm^3}\right) = 2.78\ cm^{-1}$

1 Solve for Lin. Atten. Coefficient

$\varepsilon_i = 1 - e^{-(2.87\ cm^{-1})(0.25\ cm)}$

1 Lookup all

$\varepsilon_i = 0.501$

$= 0.501$ ANS

1 Correct value and units

CUTSHEET

15 **3.** **Calculate** the Q-values for the following nuclear reactions:

* Use $931.49 \left[\frac{MeV}{amu}\right]$ for c^2

a β-decay of Cobalt-60 to Nickel-60: ${}^{60}_{27}$Co $\rightarrow$ ${}^{60}_{28}$Ni + ${}^{0}_{-1}$e^{-1} + $\bar{v}$
Note: assume 0 mass and energy for the anti-neutrino, $\bar{v}$.

3 Eq. $Q = -\Delta mc^2$ Lookups: $M_{^{60}Co} = 59.9338222$ amu

$M_{^{60}Ni} = 59.9307906$ amu 1 Lookup all

$-(59.9307906 - 59.9338222)\left(931.49\left[\frac{MeV}{amu}\right]\right)$

$Q = 2.82$ MeV ANS 1 Correct value and units

b α-decay of Radon-222 to Po-218: ${}^{222}_{86}$Rn $\rightarrow$ ${}^{218}_{84}$Po + ${}^{4}_{2}$He

3 Eq. $Q = -\Delta mc^2$ Lookups: $M_{^{222}Rn} = 222.017565$ amu

$M_{^{218}Po} = 218.0089658$ amu 1 Lookup all

$M_{^{4}He} = 4.0026032497$ amu

$= -((218.0089658 + 4.0026032497) - 222.017565)\,931.49$

$Q = 5.59$ MeV ANS 1 Correct value and units

c neutron capture reaction of Boron-10: ${}^{10}_{5}$B + ${}^{1}_{0}$n $\rightarrow$ ${}^{7}_{3}$Li + ${}^{4}_{2}$He

3 Eq. $Q = -\Delta mc^2$ Lookups: $M_{^{10}B} = 10.012937$ amu

$M_{^{1}n} = 1.0086649233$ amu

$M_{^{7}Li} = 7.016004$ amu 1 Lookup all

$M_{^{4}He} = 4.0026032497$ amu

$= -((7.016004 + 4.0026032497) - (10.012937 + 1.0086649233))\,931.49$

$Q = 2.7895 \therefore Q = 2.79$ MeV ANS 1 Correct value and units

10 **4.** **Calculate** the required number density of molecules of BF_3 in a detector with a diameter of 4.5 cm to ensure that 75% of thermal neutrons incident along the full length of the diameter are absorbed. The microscopic absorption cross-section, σ_a, for natural boron is 760 barns.

3 Eq. $\Sigma_a = N\sigma_a$

Given: $x = 4.5$ cm

$\phi(x) = 0.25$ 1 Use .25 vs 0.75

$\sigma_a = 760\,b \therefore 760 \cdot 10^{-24}\,cm^2$ 2 Convert to cm^2

2 Eq. $\phi(x) = \phi_0 e^{-\Sigma_a x}$

$0.25 = 1e^{-N\sigma_a x}$

$\ln(0.25) = -N\sigma_a x$

$N = \frac{-\ln(0.25)}{\sigma_a x}$

1 Log algebra

$N = \frac{1.3863}{(760\times10^{-24}\,cm^2)(4.5\,cm)}$

$= 4.05\times10^{20}\,\frac{BF_3}{cm^3}$ ANS 1 Correct value and units

CUTSHEET

5 **5.** **Describe** something fun that you did over Spring Break.

Any relevant answer.

$\boxed{5}$

25 **6.** A high-purity germanium (HPGe) detector is situated a distance, r, away from an unshielded, point-like, isotropic source of Iodine-131 with activity 0.8 mCi. The face of the detector has an area of 15 cm^2 and the detector is 3.5 cm in length. The detector measures 2850 $\frac{\gamma}{min}$. Prior to the introduction of the source, the detector measured a background count rate of 150 $\frac{\gamma}{min}$. **Calculate** the distance, r, between the detector and the source. Assume there are no counts from β-particles. Assume the detector is pure germanium. Note: germanium is not listed in Table 2-1, so use the table provided below to assist in your work.

Germanium
$Z = 32$

ASCII format

	Energy (MeV)	μ/ρ (cm^2/g)	μ_{en}/ρ (cm^2/g)
	1.00000E-03	1.893E+03	1.887E+03
	1.10304E-03	1.502E+03	1.496E+03
	1.21670E-03	1.190E+03	1.185E+03
L3	1.21670E-03	4.389E+03	4.343E+03
	1.23215E-03	4.734E+03	4.684E+03
	1.24780E-03	4.974E+03	4.922E+03
L2	1.24780E-03	6.698E+03	6.622E+03
	1.32844E-03	6.348E+03	6.279E+03
	1.41430E-03	5.554E+03	5.495E+03
L1	1.41430E-03	6.287E+03	6.221E+03
	1.50000E-03	5.475E+03	5.420E+03
	2.00000E-03	2.711E+03	2.688E+03
	3.00000E-03	9.613E+02	9.527E+02
	4.00000E-03	4.497E+02	4.445E+02
	5.00000E-03	2.472E+02	2.433E+02
	6.00000E-03	1.509E+02	1.477E+02
	8.00000E-03	6.890E+01	6.660E+01
	1.00000E-02	3.742E+01	3.564E+01
	1.11031E-02	2.811E+01	2.653E+01
K	1.11031E-02	1.981E+02	1.157E+02
	1.50000E-02	9.152E+01	6.256E+01
	2.00000E-02	4.222E+01	3.178E+01
	3.00000E-02	1.385E+01	1.126E+01
	4.00000E-02	6.207E+00	5.152E+00
	5.00000E-02	3.335E+00	2.759E+00
	6.00000E-02	2.023E+00	1.642E+00
	8.00000E-02	9.501E-01	7.184E-01
	1.00000E-01	5.550E-01	3.803E-01
	1.50000E-01	2.491E-01	1.288E-01
	2.00000E-01	1.661E-01	6.865E-02
	3.00000E-01	1.131E-01	3.891E-02
	4.00000E-01	9.327E-02	3.193E-02
	5.00000E-01	8.212E-02	2.930E-02
	6.00000E-01	7.452E-02	2.790E-02
	8.00000E-01	6.426E-02	2.618E-02
	1.00000E+00	5.727E-02	2.489E-02
	1.25000E+00	5.101E-02	2.353E-02
	1.50000E+00	4.657E-02	2.242E-02
	2.00000E+00	4.086E-02	2.094E-02
	3.00000E+00	3.524E-02	1.977E-02
	4.00000E+00	3.275E-02	1.962E-02
	5.00000E+00	3.158E-02	1.987E-02
	6.00000E+00	3.107E-02	2.027E-02
	8.00000E+00	3.103E-02	2.120E-02
	1.00000E+01	3.156E-02	2.208E-02
	1.50000E+01	3.340E-02	2.364E-02
	2.00000E+01	3.528E-02	2.452E-02

$\boxed{3}$ Eq. $\varepsilon = \frac{R}{S} = \varepsilon_i \varepsilon_g$

$\boxed{3}$ Eq. $\varepsilon_g = \frac{\phi A_d}{S}$

$S = fA$ $\boxed{2}$ Eq.

$R_N = R_S - R_B$ $\boxed{2}$ Eq.

$\phi = \frac{S}{4\pi r^2}$ $\boxed{2}$ Eq.

$\varepsilon_i = 1 - e^{-\mu x}$ $\boxed{2}$ Eq.

Given:
$A = 0.8\,mCi$
$A_d = 15\,cm^2$
$R_S = 2850\,min^{-1}$
$R_B = 150\,min^{-1}$
$x = 3.5\,cm$

Lookups:

2 Solves for E_gamma

$$\bar{E}_\gamma = \frac{0.284305(.0612) + 0.364489(.815) + .636989(0.0716)}{(0.0612 + .815 + .0716)}$$

$\bar{E}_\gamma = 0.3778\,MeV$

$\boxed{1}$ All lookups

$\rho_{Ge} = 5.36\,\frac{g}{cm^3}$ 2 Uses plot/interpolates

$\frac{\mu}{\rho} = 0.097276\,\frac{cm^2}{g}$ (interpolate)

$f = (.0612 + .815 + .0716) = 0.9478$

Solve for r:

$R_N = 2850 - 150 = 2700\,\frac{\gamma}{min} \Big/ \frac{1\,min}{60\,s} = 45\,\frac{\gamma}{s}$

$S = (.9478)(0.8\,mCi) = 0.75824\,mCi \left(\frac{1\,Ci}{1000\,mCi}\right)\left(\frac{3.7\times10^{10}\,Bq}{1\,Ci}\right) = 2.805\times10^7\,Bq$

$\mu = \frac{\mu}{\rho}\rho = (.097276)(5.36) = 0.5215\,cm^{-1}$

$\varepsilon_i = 1 - e^{-(.5215)(3.5)} = 0.8388$

$\frac{R}{S} = \varepsilon_i \varepsilon_g \rightarrow \frac{R_N}{fA} = (1 - e^{-\mu x})\frac{\cancel{fA} A_d}{4\pi r^2 \cancel{fA}}$ 3 Sub in correctly

$\frac{R_N}{fA} = (1 - e^{-\mu x})\frac{A_d}{4\pi r^2}$

$r = \sqrt{\frac{(1 - e^{-\mu x}) A_d fA}{4\pi R_N}}$

$r = \sqrt{\frac{(0.8388)\,15\,cm^2\,(2.805\times10^7)\,s^{-1}}{4\pi(45\,s^{-1})}}$

$r = 790.\,cm$ ANS

+2 +1

3 Correct value and units

TAB K DESIGN PROJECT

DESIGN PROJECT: IPR #1

Revised Problem Statement

Take the project topic and turn it into a new mission statement and present this as your bottom line up front (BLUF). This will include the 5 W's and have quantifiable metrics. Consider this the last thing you write up after having completed your initial research.

Tasks

Specified Tasks

- These are tasks specified by the customer in the original design document.
- The customer will state requirements for the final product that must be accomplished
- These are NOT processes (i.e. complete design report or conduct IPR)

Tasks

Implied Tasks

- Tasks that are not specified by a higher echelon, but are recognized as tasks to be accomplished in support of the objective or mission.

Example: If given a task to conduct a mounted, tactical movement from the assembly area to an objective, conducting preventive maintenance checks and services on your vehicles prior to your movement would be an implied task.

Tasks

Essential Tasks

- Specified or implied tasks that must be accomplished to make a successful design.
- Included in your mission requirements of the revised problem statement

Constraints

1. Include your list of constraints here. Follow the definitions as outlined in the NEDS.

Limitations

1. Include your list of limitations here. Follow the definitions as outlined in the NEDS.

Restrictions

1. Include your list of restrictions here. Follow the definitions as outlined in the NEDS.

Assumptions

List and explain any assumptions necessary to formulate the problem statement and potential solutions. These must be valid and required to complete the design.

Previous Work in the Field

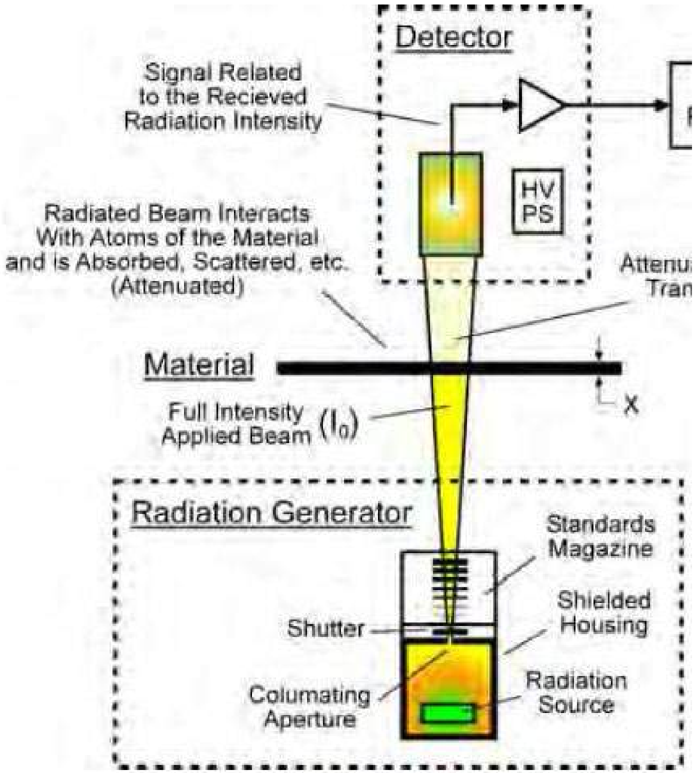


Figure 1: The basic generator-detector concept[2]

1. Indicate any previous work done and cite it; peer-reviewed articles, etc.
2. See example figure on left for what can be shown as previous work in the field
3. Show any existing apparati or machines where the design has already been done in the past (if available)

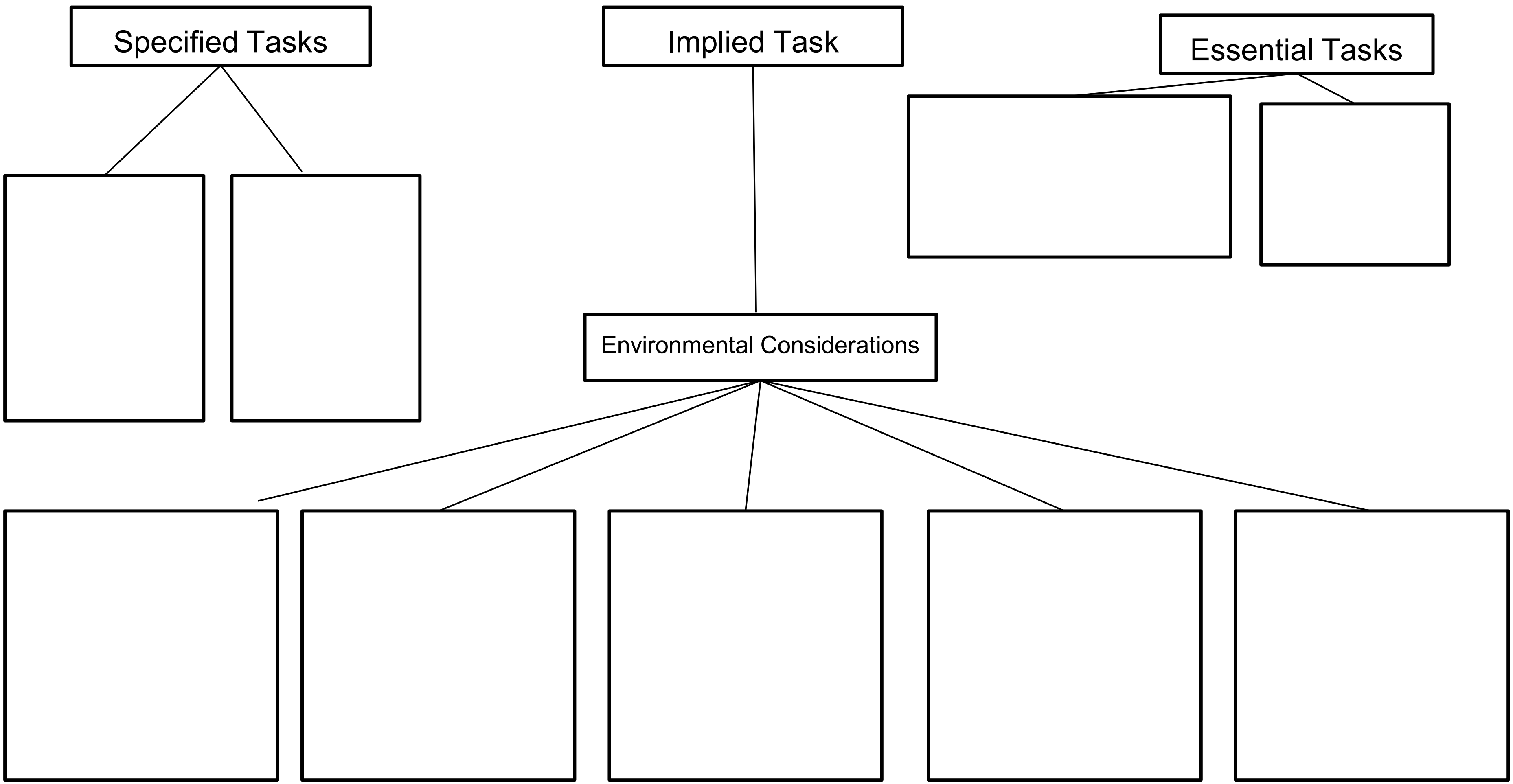
Requests for Information

1. List questions that you need answered to determine what the client really wants
2. The answers to your RFIs will be given to you by your client at the end of the briefing

Research Topic List

1. This is a list of tasks that you don't know enough information about and will require further research.
2. Example topics could include things like: available radioactive sources, regulatory requirements, existing designs, etc.
3. If you are unfamiliar with some of the background information needed to complete the design, begin to list them in this section.

Objective Tree Diagram Example



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Gantt Chart

Example is below:

				September					October					November					D	
Task Name	Duration (days)	Who	additional notes	1	4	11	18	25	2	9	16	23	30	1	6	13	20	27	1	4
IPR 1 problem recognition [MILESTONE]		NT, TO, AS						22-Sep												
Tab C								∇												
Revised Problem Statement	2	AS						20-22 Sep												
create Objectives Tree	2	TO	est possible research					23-25 Sep												
Specified Tasks	3	AS	make assumptions					23-Sep	26-Sep											
Implied Task research	5	TO, AS	lit review for						2-Oct	7-Oct										
constraints, restrictions, limitations	2	AS	state of the art						2-4 Oct											
research topic list	4	AS	expert info						2-Oct	6-Oct										
RFI from research	5	AS	codes/regs							6-Oct	11-Oct									
Essential Task finalized	2	NT, TO, AS	updated assumptions								11-12 Oct									
"New Start" Design Proposal	2	NT, TO, AS	proposal slides								13-15 Oct									
'roject Risk Management Strateg	1	NT									14-Oct									
Develop initial design concept	3	TO, AS	sample calculations								15-18 Oct									
Quad Chart	1	AS	describes concept								16-Oct									
Initial Design Concept Drawing	2	NT, TO, AS									17-19 Oct									
IPR 2 Proposal [MILESTONE]												19-Oct								
												∇								
Completing final design	7	NT, TO	format IAW supplement									23-Oct	30-Oct							
beta testing	2	AS											25-27 Oct							
Calculations to support design	4	NT, TO	tables, charts, drawings										28-Oct	2-Oct						
public hearings	1	NT, TO, AS	rehearsing												4-Oct					
3 IMPLEMENTATION [MILESTONE]															6-Nov					
															∇					
ollect appendix 2a. Requirement	2	NT, TO														16-Nov	18-Nov			
ıpload write up onto google docs	1	NT	sub task														19-Nov			
general approach	2	NT															19-20 Nov			
results and discussion	4	TO, AS															24-Nov	28-Nov		
Conclusions	3	TO, AS																28-Nov	1-Dec	
Revision to final paper	6	NT, TO, AS																	1-Dec	7-Dec
INAL REPORT DUE [MILESTONE]																				7-Dec
																				∇

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References

1. Properly cite all your references here.

NE350 NEDP IPR 1 Names: ________________________

Grading Rubric

Task	Failure (0-4)	Marginal (5-7)	Satisfactory (8-10)	Good (11-13)	Excellent (14-15)
Revised Problem Statement (15 points)	Fails to develop a revised problem statement.	Marginally specifies client's needs.	Succinctly specifies client's needs. Includes 5 W's. Not linked to essential tasks.	Succinctly specifies client's needs. Includes 5 W's. Linked to essential tasks.	Explains why project is important. Clearly quantifiable.
Task	Failure (0-11)	Marginal (12-15)	Satisfactory (16-20)	Good (21-23)	Excellent (24-25)
Specified, Implied, Essential Tasks (25 points)	Fails to describe tasks.	Describes some tasks.	Uses creativity, ingenuity, completeness in listing and describing some tasks. Marginally considers technical, social, political, economic, health & safety, ethical, security, and global issues. Identifies some essential tasks	Uses creativity, ingenuity, completeness in listing and describing most tasks. Considers technical, social, political, economic, health & safety, ethical, security, and global issues. Correctly identifies essential tasks.	Uses creativity, ingenuity, completeness in listing and describing all tasks. Considers technical, social, political, economic, health & safety, ethical, security, and global issues. Correctly identifies essential tasks.
Task	Failure (0-4)	Marginal (5-7)	Satisfactory (8-10)	Good (11-13)	Excellent (14-15)
Standards, Limitations, Constraints (15 points)	Fails to list and explain standards, limitations, constraints.	Lists some standards, limitations, constraints. Confuses the difference between standards, limitations, constraints. Doesn't consider environmental factors.	Uses creativity, ingenuity in listing and explaining few standards, limitations, constraints. Doesn't consider some important environmental factors (technical, social, political, economic, health & safety, ethical, security, global).	Uses creativity, ingenuity in listing and explaining standards, limitations, constraints. Considers environmental factors (technical, social, political, economic, health & safety, ethical, security, global).	Uses creativity, ingenuity, completeness in listing and explaining standards, limitations, constraints. Considers environmental factors (technical, social, political, economic, health & safety, ethical, security, global).
Task	Failure (0-4)	Marginal (5-7)	Satisfactory (8-10)	Good (11-13)	Excellent (14-15)
Assumptions (15 points)	Fails to list and explain any assumptions made.	Lists some assumptions that may assume away critical design requirements.	Lists some assumptions but excludes other necessary ones.	Provides well thought-out and valid assumptions that are necessary to complete the design project. May leave out a small number of assumptions	Provides a comprehensive list of valid, critical assumptions that are necessary to complete the project.
Task	Failure (0-4)	Marginal (5-7)	Satisfactory (8-10)	Good (11-13)	Excellent (14-15)
Previous Work in the Field (15 points)	Fails to describe previous research related to the problem.	Describes previous research unrelated to the problem. Scholarly article not used.	Describes previous research marginally related to the problem. At least one scholarly article used.	Describes previous research related to the problem. At least one scholarly article used.	Describes previous research related to the problem. More than one scholarly article used.

Task	Failure (0-3)	Marginal (4-5)	Satisfactory (6-8)	Good (9-10)	
Request for Information (10 points)	Fails to develop request for information.	Develops list with many questions that the team can answer by appropriate research.	Develops list with some questions that the team can answer by appropriate research.	Develops appropriate list void of questions that the team can answer by appropriate research.	
Task	Failure (0-2)	Marginal (3-4)	Good (5)		
Research Topic List (5 points)	Fails to identify any research requirements.	Identifies research requirements that do not lead to success of the project.	Anticipates and identifies research requirements that will lead to success of the project.		
Task	Failure (0-3)	Marginal (4-5)	Satisfactory (6-8)	Good (9-10)	
Gantt Chart (10 points)	Fails to submit a Gantt Chart.	Includes some tasks. Not properly formatted.	Includes most appropriate tasks and milestones with mostly appropriate time duration. Properly formatted IAW NEDS w/minor mistakes.	Includes all appropriate tasks and milestones with appropriate time duration. Properly formatted IAW NEDS.	
Task	Failure (0-4)	Marginal (5-7)	Satisfactory (8-10)	Good (11-13)	Excellent (14-15)
Presentation professionalism (15 points)	Clear lack of professionalism. Poor substance, style, organization, and correctness. Bad visual aids. Poor communication skills.	Information is presented with some lack of professionalism. Below average substance, style, organization, and correctness. Visual aids are insufficient. Mediocre verbal and written communication.	Adequate substance, style, organization, and correctness. Visual aids are adequate. Satisfactory verbal and written communication.	Good substance, style, organization, and correctness. Visual aids are mostly professional. Good verbal and written communication.	Excellent substance, style, organization, and correctness. Visual aids are professional. Excellent verbal and written communication.
			FINAL GRADE	/125	

Instructor Feedback:

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DESIGN PROJECT: IPR #2

NE350 Project New Start:

(Project Name)

(Date)

PANE

Researchers: ***(Names)***

Mentor: ***(Name)***

Client: ***(Company/Organization)***

Can Add Client Logo(s) if applicable (DTRA, DOE etc.)

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Agenda

- Project Overview
- Project Strategy
- Project Breakdown Structure (PBS)
- Preliminary Experimentation
- Project Schedule and Milestones
- Project Resources Required
- Project Risk Management Strategy
- Quad Chart

Project Overview

- **Purpose:** *A single statement that briefly explains what the project will cover as well as its major objectives (up to 3)*
- **Customers**: *List the current customers (up to 5) for major project deliverables*
- **Transition:** *List to which projects/customers/repositories the major deliverables are going or could be going*

Project Strategy (1 of 2)

- **Technical Strategy**
  - *Include up to 2 lines explaining the technical approach to solve the project*
  - *Include up to 2 lines explaining the technical approach to solve the project*
  - **Data Mining**
    - *What is already known going into this project*
    - *'Data Mining' is not limited to test data but to all preceding relevant work*
  - **Small Scale Experiments**
    - *List 0 to 2 small scale experiments necessary to close any known gaps that might impact developing the product*
  - **Large Scale Demonstrations**
    - *List up to 2 large scale demonstrations of the product*
  - **Operational Systems Design**
    - *Include up to 2 lines on the operational design (i.e., the final product)*

Project Strategy (2 of 2)

- **Acquisition Strategy**

  The preferred acquisition strategies that are applicable to the project (where would you get funding?):

  - *Research Grants*
  - *Defense Contract*
  - *Private Investment*
  - *MIPR*

- **Programmatic Strategy**

  List the high level programmatic deliverables (things you would need to provide your client) including, but not limited to:

  - *Source code and executables*
  - *Relevant software documents*
  - *Plans*
  - *Manuals*
  - *Reports*

Project Breakdown Structure (PBS)

Major tasks

- *List up to 5 tasks (this is similar to the specified, implied and essential tasks you had in IPR1)*
- *List up to 5 tasks (this is similar to the specified, implied and essential tasks you had in IPR1)*

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Preliminary Experimentation

Use at least 2 slides to show experiments or simulations you have performed up to this point to include results. Show calculations and governing equations leading you to discuss three options which you must ultimately recommend among them which will be the best option.

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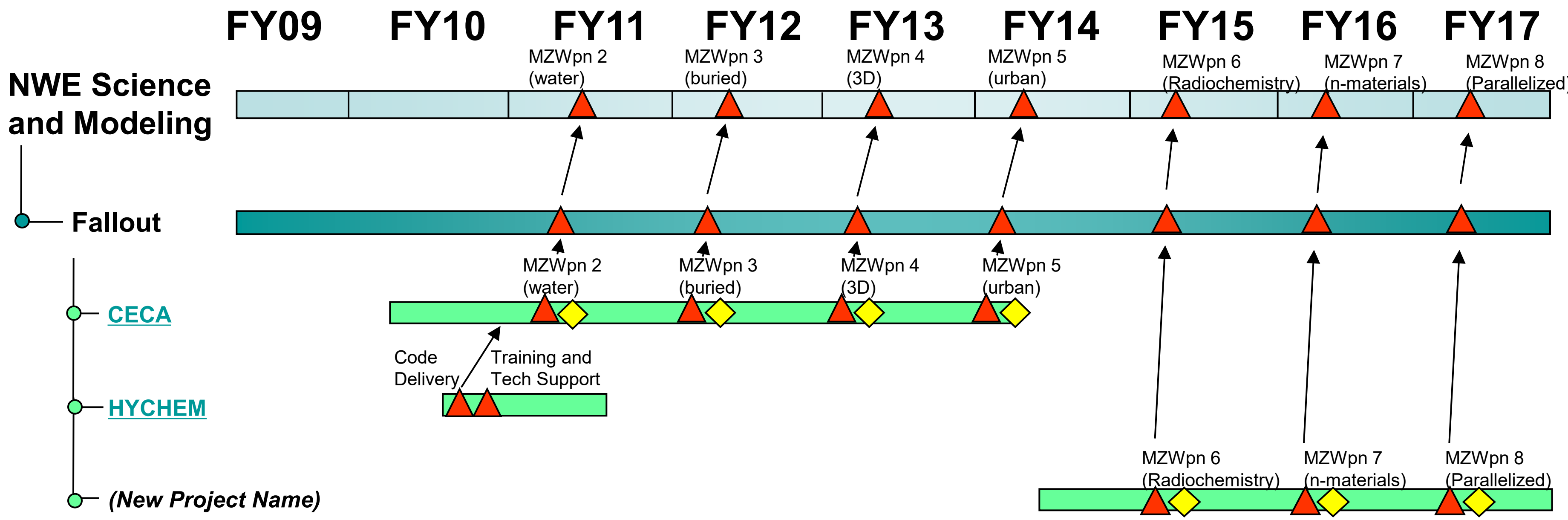
Project Schedule and Milestones



This is your updated Gantt Chart. It should be in real-time from now until December 2021. Change the "FYs" to weeks and graphics to suit the project schedule as appropriate

Color and Symbol Key

Program Bars-

Subprogram Bars-

Technology Area Bars-

Project Bars-

Key Deliverables/Milestones-

Decision Points-

Project Resources Required

$ in Millions	FY13	FY14	FY15	FY16	FY17	FY18	FY19	FY20
Primary Funds	***0.000***	***0.000***	***0.000***	***0.000***	***0.000***	***0.000***	***0.000***	***0.000***
Add'l Required	***0.000***	***0.000***	***0.000***	***0.000***	***0.000***	***0.000***	***0.000***	***0.000***

Modify this chart to suit your timeline, change the FYs as needed to fit your schedule. Make an estimate of the amount of $ to be spent on the development of your project

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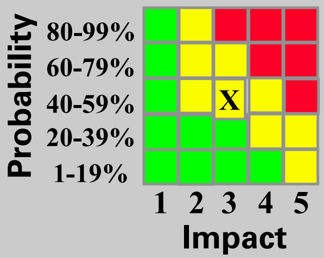
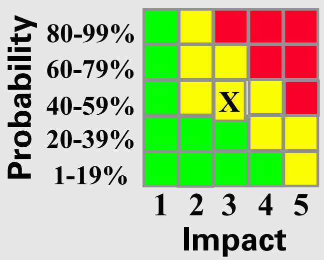
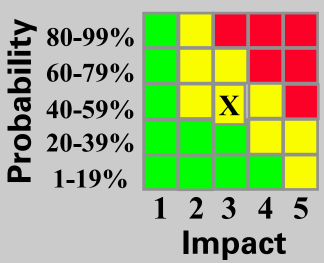
Project Risk Management Strategy

Risk Assessment	Risk and Impact	Mitigation Strategy
	**Risk # 1** Provide a high level description of the risk. This should be brief (no more than 20 words) and include metrics (sufficient to describe the risk). Identify the long term and short term impact(s) of each risk. Risk statements should be phrased in the positive sense "[Risk] will cause [effect] resulting in [project impact(s)]"	List 1 to 3 high level viable solutions available to mitigate each risk. Classify each mitigation strategy as one of these four: **Control:** • Deliberate use of the design process to lower risk • Examples include incremental development, increased testing before release, increased design margin and open systems design
	**Risk # 2** Classify all risks as "C/S/P" (chose one or two max for every risk) C – Cost: Risk that project will be unable to stay within cost estimate S – Schedule: Risk that project will be unable to stay on proposed schedule	**Avoid:** • Removal of requirement(s) that represent the majority of the risk probability • Example is lowering risk by accepting threshold vice objective performance
	**Risk # 3** P – Performance: Risk that project will be unable to meet proposed technical metrics Include no more than top 3 risks (and list in order)	**Accept:** • Living with the risk due to low probability or impact • Examples are deliberate use of budget and schedule reserves and continuous assessments of progress **Transfer:** • Moving the risk from one area of design to another • Examples are relying on hardware improvements to make up for software performance shortfalls, warranties and insurance

Contemporary Issues

Use 1-2 slides to explain how this design is relevant to a contemporary issue(s) in the news

Project Name

Overall Operational Concept Diagram (OV-1) Fill this gap with an appropriate Figure or Schematic of your Design

Key Milestones/Deliverables:

Anticipated Start Date – DD Mon 202#

- Key deliverable 1
- Key deliverable 2
- Milestone 1
- Key deliverable 3
- Key deliverable 4

Expected Completion Date – DD Mon 202#

Funding:

$ in Millions	FY13	FY14	FY15	FY16	FY17	FY18	FY19	FY20
Primary Funds	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*
Add'l Required	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*

Project Purpose:

A single statement that briefly explains what the project will cover as well as its major objectives (up to 3).

Requirements:

List of the high level requirements (up to 3).

Customers:

List the current customers (up to 5) for major project deliverables

Transition:

List to which projects/customers/repositories the major deliverables are going

COR:

PM: (Only include if different from COR)

Analyst: (Only government name, not contractor)

Key Performers:

- Original developer(s)
- Current developer(s)
- Support organization(s)
- Manufacturer(s)
- (Do not include a category if not applicable)

Contract Mechanism:

Current Status:

Detailed Drawings

Use 1-2 slides to show more detailed drawings in your selected design to highlight anything not shown in the Overall Concept Diagram on the previous slide. Use SolidWorks, Paint, or other appropriate modeling software to achieve this.

Feasibility of Design

Use at least 2 slides to discuss in detail the feasibility of your selected design. Include advantages and disadvantages and any other information to present to the client that may or may not require modification.

NE350 NEDP IPR 2 – Preliminary Design Review
Grading Rubric

Group Members: ______________________________
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Task	Failure (0-1)	Marginal (2)	Satisfactory (3)	Good (4)	Excellent (5)
Revised Problem Statement (5 points)	Fails to develop a revised problem statement. Does not incorporate new data or requirements.	Presents a marginal iteration of the problem statement that specifies client's needs. Incorporates new data or requirements.	Presents a satisfactory iteration of the problem statement that specifies client's needs. Incorporates new data or requirements.	Presents an iteration of the problem statement that succinctly specifies client's needs. Incorporates new data or requirements.	Presents an iteration of the problem statement that quantifies desired results to best meet the client's needs. Incorporates new data or requirements.
Task	**Failure (0-10)**	**Marginal (11-13)**	**Satisfactory (14-16)**	**Good (17-18)**	**Excellent (19-20)**
Functional Analysis (20 Points)	Fails to conduct functional analysis.	Poorly performs functional analysis of design project. Overlooks many client requirements.	Performs adequate functional analysis of design project. Overlooks some client specified requirements	Performs good functional analysis of design project with client's specified requirements	Performs excellent functional analysis of design project; incorporates all client specified requirements
Task	**Failure (0-13)**	**Marginal (14-17)**	**Satisfactory (18-20)**	**Good (21-23)**	**Excellent (24-25)**
Decision-Making (25 Points)	Fails to conduct any decision-making in the determination of a best design choice.	Relatively poor employment of a decision-making method. Design choice presentation has some issues. Missing advantages, disadvantages, and other appropriate factors.	Adequately employs a decision-making method to select the best design choice. Selected design choice is presentation is satisfactory. Includes advantages, disadvantages and other appropriate factors.	Employs a quality decision-making method to select the best design choice. Selected design choice is presented well. Includes most advantages, disadvantages and other appropriate factors.	Expertly employs a decision-making method to select the best design choice. Selected design choice is expertly presented. Includes all relevant advantages, disadvantages and other appropriate factors.
Task	**Failure (0-10)**	**Marginal (11-13)**	**Satisfactory (14-16)**	**Good (17-18)**	**Excellent (19-20)**
Initial Design Concept Drawing (20 Points)	Fails to show basic drawings of the selected preliminary design.	Shows marginal detailed drawings that may include relevant dimensions, features, and other essential information in the preliminary design.	Shows satisfactory detailed drawings that include most of the relevant dimensions, features, and other essential information in the preliminary design.	Shows detailed drawings of good quality that include most of the relevant dimensions, features, and other essential information in the preliminary design.	Expertly shows detailed drawings of the selected preliminary design that accounts for specifications beyond expectation of IPR 2.
Task	**Failure (0-13)**	**Marginal (14-17)**	**Satisfactory (18-20)**	**Good (21-23)**	**Excellent (24-25)**
Sample Calculations (25 Points)	Fails to show preliminary experimentation or any indication of design calculations performed.	Attempts to perform calculations, but uses inappropriate inputs, assumptions, or equations. Major math errors.	Correct calculations with several minor errors.	Correct and relevant calculations with few minor errors based off appropriate inputs, assumptions, and equations.	Professionally presented calculations. Correct, relevant, and complete based off correct inputs, assumptions and equations.
Task	**Failure (0-5)**	**Marginal (6-7)**	**Satisfactory (8)**	**Good (9)**	**Excellent (10)**
Gantt Chart (10 points)	Fails to submit a Gantt Chart.	Includes some tasks. Not properly formatted.	Includes most appropriate tasks and milestones with mostly appropriate time duration.	Includes all appropriate tasks and milestones with appropriate time duration. Properly formatted IAW NEDS.	Includes all appropriate tasks and milestones with appropriate time duration. Properly formatted IAW NEDS.

Task	Failure (0-5)	Marginal (6-7)	Satisfactory (8)	Good (9)	Excellent (10)
Formatting (10 Points)	Fails to follow the format specified IAW NEDS, Appendix 4 (IPR 2) to the Nuclear Engineering Design Supplement	Marginally uses correctly formatted Acknowledgement, Works Cited, Bibliography pages, tables, charts, drawings, and other documentation	Adequate formatting of all documents/slides.	Good, professional formatting of all documents/slides.	Excellent and professional formatting of all documents/slides.
Task	**Failure (0-5)**	**Marginal (6-7)**	**Satisfactory (8)**	**Good (9)**	**Excellent (10)**
Presentation and Professionalism (10 points)	Clear lack of professionalism. Poor substance, style, organization, and correctness. Bad visual aids. Poor communication skills.	Information is presented with some lack of professionalism. Below average substance, style, organization, and correctness. Visual aids are insufficient. Mediocre verbal and written communication.	Adequate substance, style, organization, and correctness. Visual aids are adequate. Satisfactory verbal and written communication.	Good substance, style, organization, and correctness. Visual aids are professional. Good verbal and written communication.	Excellent substance, style, organization, and correctness. Visual aids are professional. Excellent verbal and written communication.
				TOTAL	/125
				FINAL GRADE	

FEEDBACK/NOTES:

DESIGN PROJECT: IPR #3

NE350 Design Project IPR 3

(Project Name)

(Date)

PANE

Researchers: *(Names)*

Mentor: *(Name)*

Client: *(Company/Organization)*

Can Add Client Logo(s) if applicable (DTRA, DOE etc.)

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
UNITED STATES MILITARY ACADEMY

Agenda

- Design Description
- Design Specifications
- Design Calculations
- Graphs and Charts
- Works Cited

Design Description (1 of 3)

Project Name	
Amount of Source Used (mass)	Indicate how much mass (g) of what source was used and total activity (Bq or Ci)
Structural Component(s)	Indicate the major structural component materials used
Shielding Material(s)	Indicate what shielding materials were used
Major Features	Identify any major features or design requirements needed to be met (portability, security dose limits, uncertainty measurements, etc.)
Final Cost	Indicate how much the final design will cost in $

Provide the main basic 3D drawing of your detailed final design here. Fill out the table to the left.

Design Description (2 of 3)

This slide:
Provide detailed schematics with dimensions, labels of parts, views (side, front, top, etc. as needed).

Be sure to incorporate all aspects of the design to include location of source, location of detector (as required), shielding mechanisms

Design Description (3 of 3)

This slide:
Depict any additional drawings as needed to describe the use of the apparatus you designed.

This may include operation, loading/unloading, placement, etc. You may also include an animation if that helps your presentation.

Design Specifications (1 of 6)

Specified and Essential Requirements:

Use bullet points to provide a discussion here of how the final design incorporates the specified and essential requirements. Include a discussion of significant implied tasks as appropriate.

Design Specifications (2 of 6)

Results of Optimized Parameters:

Use bullet points to provide a discussion here of your results of the optimized parameters. You may use tables or fundamental equations here as needed.

Design Specifications (3 of 6)

Impact on assumptions, constraints, and limitations:

Use bullet points to provide a discussion here of the impact on assumptions to produce your design, and the constraints and limitations placed on your design. Ensure that you have a design that meets these expectations.

Design Specifications (4 of 6)

Impact of environmental factors on the design

Environmental Factor	Impact Description
Technical	Insert Descriptions here
Social	Insert Descriptions here
Political	Insert Descriptions here
Economic	Insert Descriptions here
Heath and Safety	Insert Descriptions here
Ethical	Insert Descriptions here
Security	Insert Descriptions here
Global Issues	Insert Descriptions here

Design Specifications (5 of 6)

Material Requirements:
Insert information here (use bullets or tables if necessary) to show what types of materials are used for what parts of the design, how much they cost, and any concerns for the acquisition of these materials

Test Requirements:
Discuss the test requirements from the preliminary experimentation, test requirements for the final design, and test requirements post-design prior to full commercial implementation

Design Specifications (6 of 6)

Methodology to Assess Success:

Use bullet points to provide a discussion here of how you plan to assess the success of the design and how the results will be fed back into the redesign through feedback mechanisms

Calculations (1 of 5)

Equations Used in the Design:

Number all of your equations used for your design here (i.e. Eqn (1)) and prose should be included that refer to numbered equations. For example, “Substitue Eqn (1) into Eqn (2). You may need more than one slide if you run out of space.

Calculations (2 of 5)

Computational Input:

Insert any mathematical model governing calculations using computational software here (Mathematica, MATLAB, Excel, MCNP, ORIGEN, etc.). Group must submit a separate computational software file in addition to these slides electronically to Blackboard. You may need more than one slide if you run out of space.

Calculations (3 of 5)

Computational Output:

Insert any mathematical model output calculations using computational software here (Mathematica, MATLAB, Excel, MCNP, VisED, ORIGEN, etc.). Group must submit a separate computational software file in addition to these slides electronically to Blackboard. You may need more than one slide if you run out of space, but you will not need to show all of the pages of your output. Only show what is critical to the design.

Calculations (4 of 5)

Relevant Graphs and Charts:

Insert any relevant properly labeled confidence intervals, charts, graphs/plots, etc. depicting your calculations and optimization, and application of your design; if you change a variable and iterate it, what effect does it have on another variable? Consider dose equivalent rate, thickness, cost, or other important parameters that can be varied.

NE Majors must provide a VISED plot showing particle distributions. Adhere to the guidelines as set by the "Standards for Technical Reports". You may need more than one slide for this.

Calculations (5 of 5)

Error Analysis:

Insert a discussion of your results and any error analysis performed. If using MCNP, present the error analysis of your output file to include analysis of R, statistical checks, FOM, etc.

Works Cited:

Insert any works cited here

NE350 NEDP IPR 3 – Design Optimization

Grading Rubric

Group Members: ______________________________

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Task	Failure (0-3)	Marginal (4-5)	Satisfactory (6-7)	Good (8-9)	Excellent (10)
Design Presentation (10 points)	Fails to present a final design.	Presents an insufficient final design that is presented unprofessionally and lacks sufficient details necessary to meet the client's needs.	Presents the final design. Includes acceptable sketches, CAD drawings, models, etc.	Presents the final design with good sketches, CAD drawings, models, etc. Provides good detail in a logical manner	Completely and clearly presents the final design with superb sketches, CAD drawings, models, etc. Provides excellent detail in a logical manner; showcases the iterations of the design over the design period.
Task	**Failure (0-6)**	**Marginal (7-10)**	**Satisfactory (11-14)**	**Good (15-18)**	**Excellent (19-20)**
Calculations (20 Points)	Fails to perform calculations to support the design.	Attempts calculations to support the design. Major math errors or clearly inappropriate assumptions. Format is difficult to follow.	Performs calculations to support the design with several minor errors.	Performs calculations to support the design with few errors. Work is professionally presented and easy to follow	Accurately performs appropriate calculations to support design specifications. Appropriate units and sig. figs. Presented professionally in an easy to follow format with appropriate assumptions specified.
Task	**Failure (0-3)**	**Marginal (4-5)**	**Satisfactory (6-7)**	**Good (8-9)**	**Excellent (10)**
Graphs/Charts (10 Points)	Fails to include a relevant graph	Poorly formatted graph that only marginally represents an important characteristic of the design and how it is affected by changing a variable.	Includes a good graph that doesn't accurately represent an important characteristic of the design and how it is affected by changing a variable.	Relevant graphs that accurately represent important characteristics of the design and how they are affected by changing a variable.	Professional, relevant graphs that accurately represent an important characteristic of the design and how it is affected by changing a variable. Follows all graphing standards in the "Standards for Technical Reports."
Task	**Failure (0-1)**	**Marginal (2)**	**Satisfactory (3)**	**Good (4)**	**Excellent (5)**
Gantt Chart (5 points)	Fails to submit a Gantt Chart.	Includes some tasks. Not properly formatted.	Includes most appropriate tasks and milestones with mostly appropriate time duration.	Includes all appropriate tasks and milestones with appropriate time duration. Properly formatted IAW NEDS.	Includes all appropriate tasks and milestones with appropriate time duration. Properly formatted IAW NEDS. Includes thoughtful, relevant tasks that would need to take place for real-world construction of the design.

Task	Failure (0-1)	Marginal (2)	Satisfactory (3)	Good (4)	Excellent (5)
Presentation, Formatting, and Professionalism (5 points)	Clear lack of professionalism. Poor substance, style, organization, and correctness. Bad visual aids. Poor communication skills.	Information is presented with some lack of professionalism. Below average substance, style, organization, and correctness. Visual aids are insufficient. Mediocre verbal and written communication.	Adequate substance, style, organization, and correctness. Visual aids are adequate. Satisfactory verbal and written communication.	Good substance, style, organization, and correctness. Visual aids are professional. Good verbal and written communication.	Excellent substance, style, organization, and correctness. Visual aids are professional. Excellent verbal and written communication. All citations correct.
				TOTAL	/50
				FINAL GRADE	

FEEDBACK/NOTES:

DESIGN PROJECT: FINAL SUBMISSION

NE350 FINAL DESIGN SUBMISSION

AY 24

(TOTAL OF 300 POINTS)

1. [100 pts] Project Presentation. (Group Submission) You will submit this as a digital pdf file to Canvas using the Final Submission Slide Template. Most of your slides in the presentation can be drawn from previous IPRs with little to no modification. Follow the notes in the slide template and modify accordingly. The slide presentation is essentially the Final Submission (see #3 below) in Powerpoint form. Review the Presentation Rubric to see how you can best brief this product.

2. [50 pts] Project Team Grade. (Individual Submission) Your instructor will require a Canvas quiz to gather this information. This is your opportunity to reveal how much of the **total** project you and your peers have done. Grades will be adjusted as deemed fit from the percentage of work perceived completed by the individual.

3. [150 pts] Project Final Submission. (Group Submission). You will submit this as a digital .pdf file to Canvas. This is the final design report, with guidelines drawn from the Nuclear Engineering Design Supplement (NEDS). **Review the Final Submission Rubric** for achievement of maximum points and formatting. Your final design report will incorporate all of the required information in aggregate including but not limited to: calculations, plots, tables, figures etc. into the design.

The following is some general guidance that we have used for writing articles to publish in peer-reviewed journal articles, with minor changes included to adhere to the NEDS. Each journal typically will have its own instructions to authors that you should use as the final guide when submitting your work. The Guide to Authors will normally include what type of reference style the journal uses such as APA, MLA or Chicago. As an example, you may use the following bulleted style in your final submission for NE350. You must still incorporate all of the required information annotated in the NEDS and rubric.

ABSTRACT

This is one paragraph that includes the statement of the problem, what you did to try to answer that problem, and a summary of your results. It should be sufficient for the casual reader to determine

HINTS FOR SUCCESS

1. Start completing the requirements immediately. Organize and divide up your time by completing the final submission report and presentation simultaneously.
2. Make all of your tables, figures, plots etc. first and then think about how you want to compile the final submission. Follow the rubrics!
3. About 80% is from IPRs 1, 2, and 3. If IPR 3 is done well, you will have most of your slides!
4. Put some thought into your recommendations for future work.

whether or not the work is relevant to what they are studying and if they want to read the rest of the paper.

1.0 INTRODUCTION

1.1 STATEMENT OF PURPOSE

What are you studying or designing?

1.2 BACKGROUND

This is your literature search. What have others done in the field? You may include some preliminary equations that support your calculations.

1.3 DESIGN STATEMENT

Simple problem statement. What do you intend to design/prove or disprove with your work? It should have some quantifiable value that can be measured so you can easily summarize if you did or did not succeed in your design.

1.4 SIGNIFICANCE OF THE WORK

Explain why the problem is important and how it will benefit science/man by designing it.

2.0 METHODS

- You can have several subtitles (2.1, 2.2 etc.) in line with the NEDS topics and Final Submission Rubric. This section should be complete enough that anyone who has access to the same materials and equipment could duplicate your results.
- Include any assumptions that you made and how they may have impacted the results. It is important to know if you are performing scoping calculations or if this is a preliminary or follow-on study.
- If you are doing calculations, you should provide the governing equations in the beginning of this section. The reader should be able to follow how you determined results from these equations. You don't have to include intermediate steps.
- If you are doing software simulations, you don't have to provide complete input files for computer simulations but you do need to provide enough information that someone could create the same input files you used. This includes which version of the code you are using and any updates that are installed. Provide tables that include all the important parameters for code input such as materials or critical dimensions.
- If you are doing an experiment, you have to be as precise as possible with the nomenclature of all equipment used, the masses and types of materials, and the procedures for all experimental setups and measurements.
- Be able to indicate any environmental factors that influenced the design.
- Discuss implementation of the final design.
- Be careful not to make it so blow-by-blow that will cause the reader to completely lose interest. It is a fine line between a narrative and a recipe that is still interesting.

3.0 RESULTS

- Again, if you have several experiments or calculations, you should have several subtitles (3.1, 3.2 etc.). Each set of results should be complete for that calculation or subsection. See Final Submission Rubric.
- Try to make the results as concise and clear as possible. Use tables and charts to illustrate the results as much as possible.
- Include error bars and estimates of uncertainty as required for measurements and computational processes.
- Do not spend a lot of time here discussing the validity of the results. That type of analysis will be done in the Conclusions.

4.0 CONCLUSIONS

- If you present results in several subsections, you may want to have the same subsections for the conclusions.

- Do not provide any new data or results in the conclusions. Your final design as a Figure should be shown.
- Discuss the validity of your results being sure to analyze the impact of any assumptions or simplifications you may have made.
- Finally, discuss how you have or have not answered the question you presented in the design question.

WORKS CITED

- Follow the rules that you have done in the past for any assistance given

APPENDICES

- As a minimum, you should include the files for all of your MCNP, MATLAB, Mathematica, or Excel work
- Can include spreadsheet calculations or other supporting calculations that would be too lengthy to include in the main body of the work.

5.0 RECOMMENDATIONS FOR FUTURE WORK

Finish the report by explaining to the reader what steps are next in the research and design project process. Do you need to perform the calculations again? Do you have to use different materials? Is this step in the research and design process adequately studied? If so, what would be the logical follow-on experiments to answer the broader question? Don't be afraid of being critical of your work in the conclusions or in your recommendations for the future.

GENERAL FORMATTING

Do not use "The figure below……" or "the following equation…" Always number your figures, tables, and equations and refer to them as such. "The criticality of the reactor over a period of 20 years is presented in Figure 1.0". The main reason for this is because when a paper is published, the journal will put the figures and charts where they best fit in the publication and they may not exactly fit following the sentence where you address them.

Don't use **anthropomorphism**- that is attributing human traits to anything that is not human. For example, "Figure 1.0 shows…." "Table II illustrates…." Etc. Figures and tables cannot show, present or illustrate *anything* because they are not human. Rather, "The results of the study are presented in Figure 1.0. From these results we determined…." Or, "From the data shown in Table II, it was determined….".

Try and be consistent with your person and voice. The Army writing style will state that you should never use 3rd person or passive voice. However, 3rd person and passive voice are often used in technical journals. There is an understanding that the researchers are the ones who performed the work. Most journals will not specify but you should keep the same person and voice throughout.

All references to work that you have completed should be in past-tense. Do not change tenses in the middle of a sentence or paragraph.

A specific journal may require you to put all of your figures and tables at the end of the manuscript. This is so they can reformat them for the final off-print. When submitting the report for NE350, place the figure, table or chart in the manuscript between paragraphs. It should be either right before or right after the paragraph where it is referenced.

NE350 NEDP Final Submission
Rubric (150 Marks)

Introductory Information -Acknowledgement Page (cover page) -Abstract -Table of Contents -List of Tables (as appropriate) -List of Figures (as appropriate) -List of Symbols (as appropriate)	**Failure (0-7)** 4 or more missing, inaccurate, or inappropriate	**Marginal (8-9)** 3 missing, inaccurate, or inappropriate	**Satisfactory (10-11)** 2 missing, inaccurate, or inappropriate. Ack. Page, abstract, and TOC required.	**Good (12-13)** 1 missing, inaccurate, or inappropriate. Ack. Page, abstract, and TOC required.	**Excellent (14-15)** Nothing missing. Presented superbly.
Introduction -Statement of Purpose -Background info with bottom line up front, refers to drawings, sketches, etc - Presents previous work in field showing current state-of-the art efforts and how design fits into this research - Discusses specs, requirements, constraints, limitations, restrictions.	**Failure (0-7)** Fails to introduce design in a meaningful way.	**Marginal (8-9)** 3 of 4 not addressed, incomplete, or poorly done.	**Satisfactory (10-11)** 2 of 4 not addressed, incomplete, or poorly done.	**Good (12-13)** 1 of 4 not addressed, incomplete, or poorly done.	**Excellent (14-15)** All aspects addressed. Presented superbly.
Assumptions -Discusses realistic assumptions essential to completeness and validity of report	**Failure (0-7)** Fails to discuss assumptions, or only discusses assumptions that are clearly wrong or unnecessary.	**Marginal (8-9)** Discusses assumptions that are related to the problem.	**Satisfactory (10-11)** Discusses some of the realistic assumptions that are considered essential to the completeness and validity of the report.	**Good (12-13)** Discusses most of the realistic assumptions that are considered essential to the completeness and validity of the report.	**Excellent (14-15)** Discusses complete list of realistic assumptions that are considered essential to the completeness and validity of the report.
General Approach -Develops general solution procedure -Explains approach to solving problems and optimized parameters -Appropriate equations and formulations -Explains methodology -Includes or references derivations -Calculations presented or referenced -Computer based drawings referenced -Discusses environmental factors (social, political, economic, health & safety, ethical, security, global, environmental) -Develops approach in prose (not chronology, blow-by-blow, or cookbook) -Detailed diagram of design using computer based drawing tool.	**Failure (0-7)** 7 or more items not addressed (if applicable), incomplete, or poorly done.	**Marginal (8-9)** 5 or 6 items not addressed (if applicable), incomplete, or poorly done.	**Satisfactory (10-11)** 3 or 4 items not addressed (if applicable), incomplete, or poorly done.	**Good (12-13)** 1 or 2 items not addressed (if applicable), incomplete, or poorly done.	**Excellent (14-15)** All aspects addressed (as applicable). Presented superbly.
Results and Discussion -Conspicuously presents end products and results -Discusses end products & results -Presents tables, graphs, charts as supporting evidence -Analyzes impacts of results -Results properly reported with uncertainties -Sensitivity analysis (if appropriate) Opinion on validity of results -Discusses effects of assumptions on results	**Failure (0-16)** 5 or more items not addressed (if applicable), incomplete, or poorly done.	**Marginal (17-19)** 4 items not addressed (if applicable), incomplete, or poorly done.	**Satisfactory (20-23)** 2 or 3 items not addressed (if applicable), incomplete, or poorly done.	**Good (24-27)** 1 item not addressed (if applicable), incomplete, or poorly done.	**Excellent (28-30)** All aspects addressed (as applicable). Presented superbly.

<u>Conclusions</u> -Summarizes findings -No new information presented Presents the answer to the problem -States the final choice for the design with compete specs referenced in this section	**Failure (0-7)**	**Marginal (8-9)**	**Satisfactory (10-11)**	**Good (12-13)**	**Excellent (14-15)**
	Fails to summarize findings or very poor summary.	Summarizes finding. Multiple deficiencies.	Acceptable summary of findings. 2 or 3 minor deficiency	Good summary of findings. 1 minor deficiency	Excellent summary of findings.
<u>Recommendations</u> -Recommends future work that have bearing on the problem and follow from conclusions -Recommends improvements -Discusses impacts of assumptions	**Failure (0-7)**	**Marginal (8-9)**	**Satisfactory (10-11)**	**Good (12-13)**	**Excellent (14-15)**
	Fails to provide appropriate recommendations or very poorly done.	Major deficiency or multiple minor deficiencies.	May have a few minor deficiencies.	May have 1 minor deficiency.	Excellent recommendations.
<u>Format</u> -Acknowledgement, works cited, bibliography pages correctly formatted -Documentation correctly formatted -Enclosures referenced from main body -Tables, charts, drawing correctly formatted -IAW *Standards for Technical Reports* format guidelines.	**Failure (0-7)**	**Marginal (8-9)**	**Satisfactory (10-11)**	**Good (12-13)**	**Excellent (14-15)**
	Very poor format showing very little effort.	Major deficiency or multiple minor deficiencies.	May have a few minor deficiencies.	May have 1 minor deficiency.	No deficiencies
<u>Grammar</u> -Substance -Style -Organization -Correctness -Proofread (spelling, typos)	**Failure (0-7)**	**Marginal (8-9)**	**Satisfactory (10-11)**	**Good (12-13)**	**Excellent (14-15)**
	Very poor grammar. No evidence of proofreading.	Major deficiency or multiple minor deficiencies.	May have a few minor deficiencies.	May have 1 minor deficiency.	No deficiencies

Outstanding Student Work

Shielded Shipping Container

By CDTs Hugh Baldwin, Jack Capriola, Samrath Singh, Michael Nguyen

UNITED STATES MILITARY ACADEMY
WEST POINT

Abstract

Faced with the task to safely transport an isotropic 650-Ci Co-60 source, our group presented three initial designs. We tested our designs through various metrics including shielding, weight and cost. Our calculations led us to select a spherical lead container with a 19.1675 cm radius. To meet safety standards and increase transportation ease, we integrated a copper outer case with foam padding that can easily be loaded on a pallet and moved with industry standard machinery.

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Introduction

Context:

Mashl Half-Life Labs is a Radioisotope Production Facility. They are facing a class-action lawsuit over the effects of radioactive materials on school children at a local playground during transport. They have tasked their top engineering team, tHingaJigS, to design a new shipping container that can properly transport radioactive material.

Problem Statement

tHingaJigS designs a shielded shipping container that meets NRC/DOE/DOT regulations NLT 05 MAY 2026 IOT safely transport a 650-Ci isotropic point source of Co-60 to a gamma radiography equipment manufacturer.

Managing Our Constraints

1. **Budget ($20,000)**

-Most of our budget went into the outer copper shell. This shell lowers the risk to temperature from high thermal exposure IAW NRC/DOE/DOT regulations.

2. Transportation- Can we move it?

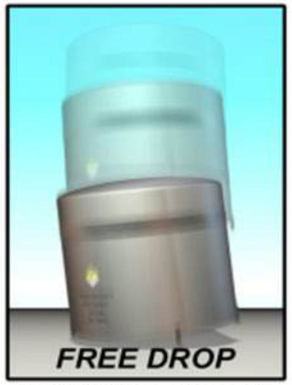
-Our package can be loaded onto a normal sized pallet with machinery, and transported upon a truck, plane, train, or boat. Its smaller size of ~.75 meters in diameter creates this ease.

3. Surface radiation (<200 mrem/hr)

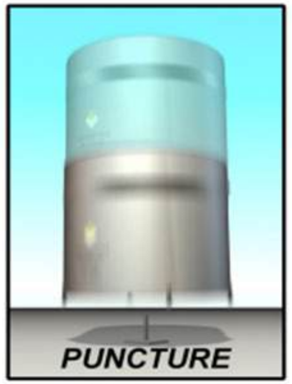
- Our innermost shielding, made of lead, prevents radiation from escaping higher than this limit.

4. Safety Tests

-Our package is specially designed to pass impact, puncture, temperature, and pressure stress testing.

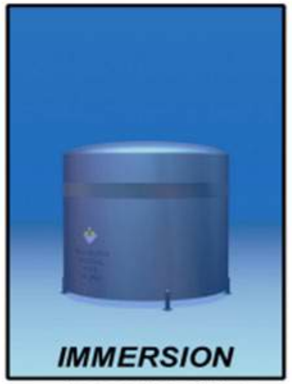


A 30-foot free drop onto a flat, essentially unyielding surface so that the package's weakest point is struck.



A 40-inch free drop onto a 6-inch diameter steel rod at least 8 inches long, striking the package at its most vulnerable spot.

Exposure of the entire package to 1475°F for 30 minutes.



Immersion of the package under 50 feet of water for at least 8 hours.

Source

In[1]:= $T_{1/2} = 5.2713\ \text{year} * \frac{365\ \text{day}}{1\ \text{year}} * \frac{24\ \text{hr}}{1\ \text{day}} * \frac{3600\ \text{s}}{1\ \text{hr}};$

$\lambda = \text{Log}[2] / T_{1/2};$

$A = 650\ \text{Ci} * \frac{3.7 * 10^{10}\ \text{Bq}}{1\ \text{Ci}} * \frac{1}{1\ \text{Bq}} * \frac{\text{atom}}{\text{s}};$

$Na = 6.02214199 * 10^{23} \frac{\text{atom}}{\text{mol}};$

$mm = 59.9338222 \frac{\text{g}}{\text{mol}};$

In[6]:= $m = \frac{A * mm}{\lambda * Na}$

Out[6]= 0.57403 g

In[7]:= $\frac{.57043\ \text{g}}{8.8\ \text{g} / \text{cm}^3}$ (*Volume of Source*)

Out[7]= $0.0648216\ \text{cm}^3$

Co-60 Source	
Activity (Ci)	650
Mass (g)	0.57403
Volume (cm^3)	0.0648

Isotropic Point source → Spherical Design

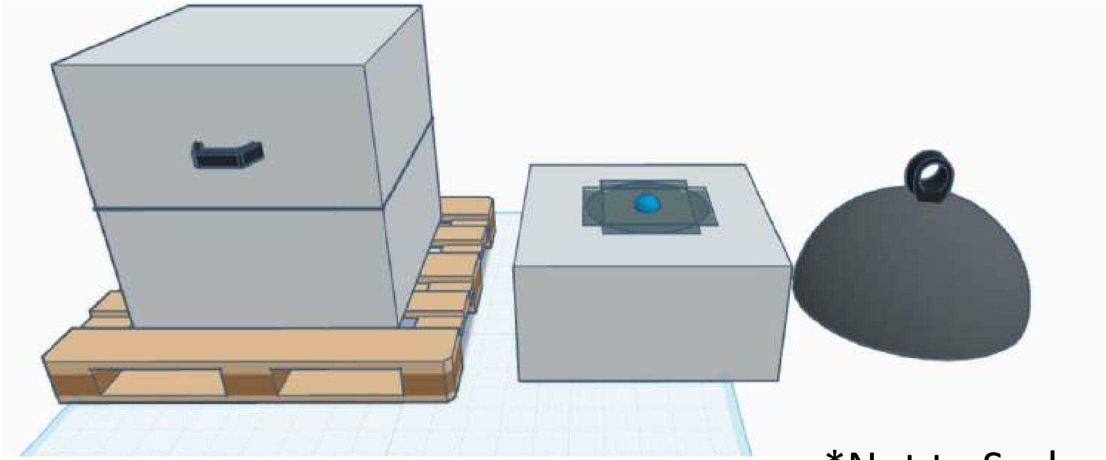
Material Selection

Decision Matrix

		Tin			Lead			Tungsten		
	Weight Factor	Value	Score	Rating	Value	Score	Rating	Value	Score	Rating
Weight (kg)	0.3	1118.68218	0.2	223.7364	338.8394	0.9	304.9554	172.647	0.7	120.8529
Volume (L)	0.1	153.28613	0.1	15.32861	8.99203	0.9	8.092827	29.88001	0.6	17.92801
Hardness (moh)	0.15	1.5	0.5	0.75	1.5	0.5	0.75	7.5	0.8	6
Therm. Conduct. (W/m K)	0.1	30	0.7	21	33	0.7	23.1	173	0.1	17.3
Cost ($)	0.35	48796.9166	0	0	704.7859	0.8	563.8287	54252.58	0	0
			Total	260.815		Total	900.727		Total	162.0809

Lead

- Good shielding
- Light-ish
- Cheap



*Not to Scale

Assumptions

1. Uniform material properties→ Constant μ and B

2. Constant source strength → Predictable radiation

3. Constant material properties → No Diff. Eqs.

Allow for simplification and flexibility

Optimization

Optimize radius for 200 mrem/hr surface radiation (r = 19.1675 cm)

```
:= (*Target Dose Rate*)
```

$$200\,\frac{\text{mrem}}{\text{hr}} * \frac{1\text{ rem}}{1000\text{ mrem}} * \frac{1\text{ rad}}{1\text{ rem}} * \frac{24\text{ hr}}{1\text{ day}} * \frac{365\text{ day}}{1\text{ yr}} * \frac{1 * \frac{\text{MeV}}{\text{g}*\text{s}}}{.5052\,\frac{\text{rad}}{\text{yr}}}$$

$$]= \frac{3467.93\text{ MeV}}{\text{g s}}$$

```
(*Material Constants*)

:= (*Lead*)
Clear[μ, x, r]
μ = 0.0569 * 11.34; (*cm^-1*)
x = 19.1675; (*cm*) (*change this value to determine final dose*)
r = x + 1; (*cm*)
μ0r = μ * r (*take this value into B calc*)

= BTin = Interpolation[TinData, InterpolationOrder → 1];
BLead = Interpolation[LeadData, InterpolationOrder → 1];
BTungsten = Interpolation[TungstenData, InterpolationOrder → 1];

(*Buildup Factor Formula*)

= (*B[energy,μ0r]*)
B = BLead[1.25, 13.012998705`];
```

$$E_\gamma = \frac{(1.173228 * 99.85) + (1.332492 * 99.9826)}{99.85 + 99.9826}; \;(*\text{MeV}*)$$

$$\mu_{en} = 0.0288; \;(*\frac{\text{cm}^2}{\text{g}}*)$$

$$S = f * A;$$

$$A = 650 * \frac{3.7 * 10^{10}}{1}; \;(*\frac{1}{\text{cm}^2\text{s}}*)$$

$$f = \frac{(.9985 + .999826)}{2};$$

```
(*Equations*)
```

$$n[\circ]:= \phi_u = \frac{S * \text{Exp}[-\mu * x]}{4 * \pi * r^2};$$

$$\phi = \phi_u * B;$$

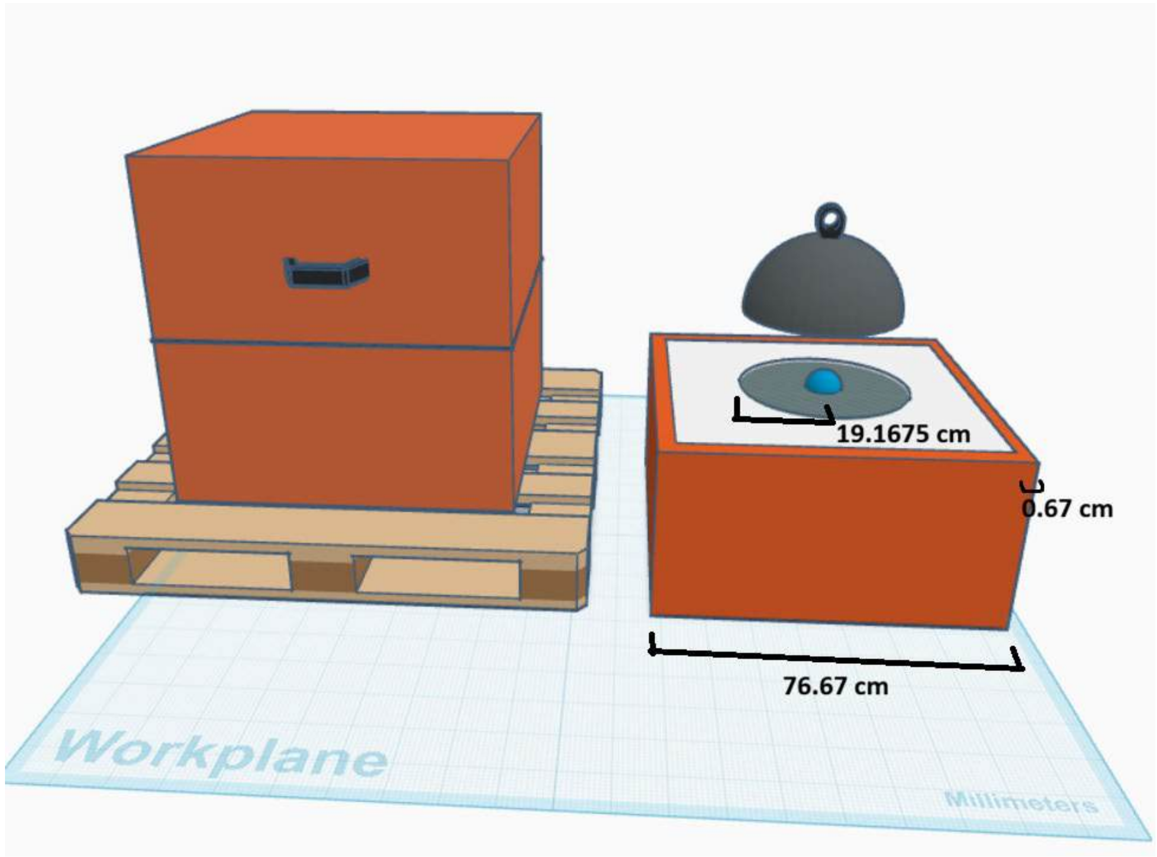
$$\text{soln} = \text{NSolve}[D\gamma == \phi * E_\gamma * \mu_{en},\, D\gamma] \;(*\frac{\text{MeV}}{\text{g}*\text{s}}*)$$

```
ut[∘]= {{Dγ → 3466.92}}
```

Final Design?

Spherical Lead Shield w Copper Case

Optimized Design Specs	
Radius (cm)	19.1675
Volume (L)	29.5
Weight (kg)	334.5
Cost ($)	696
Rad (mrem/hr)	199.941



*950 kg copper (~$12325) → outer case

Pre-Assembly price: ~$13,000

But can we make it cheaper and more functional?

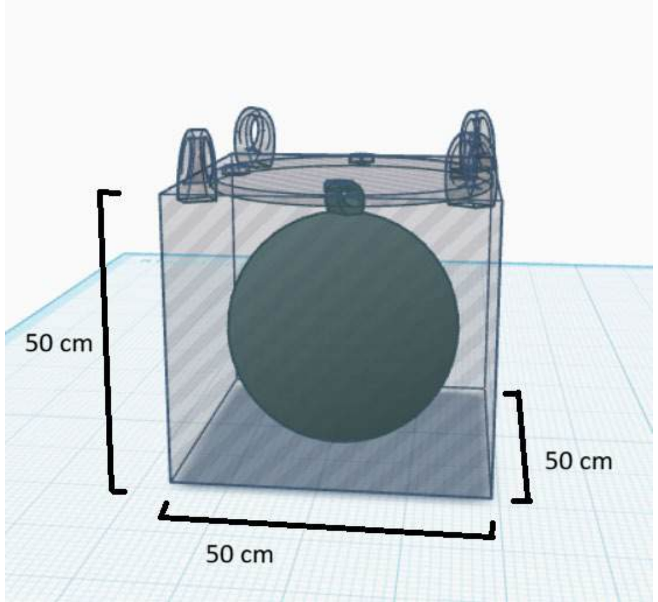
Final Design

Spherical Lead Shield

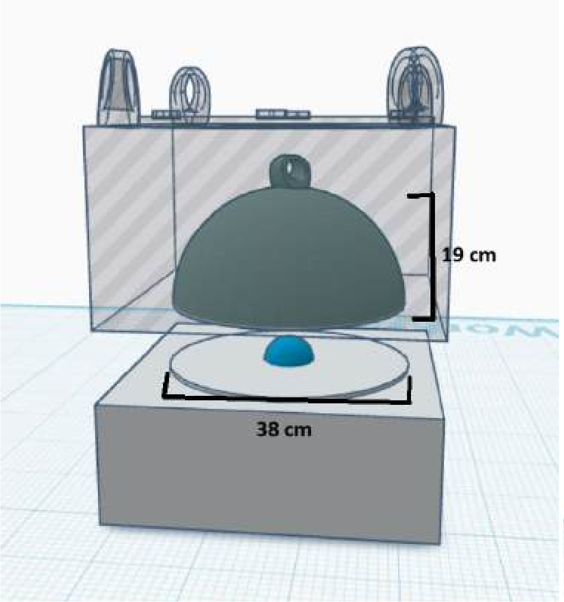
Radius (cm)	19.1675
Volume (L)	29.5
Weight (kg)	334.5
Cost of Lead ($)	696
Dose (mrem/hr)	199.941

Steel Case

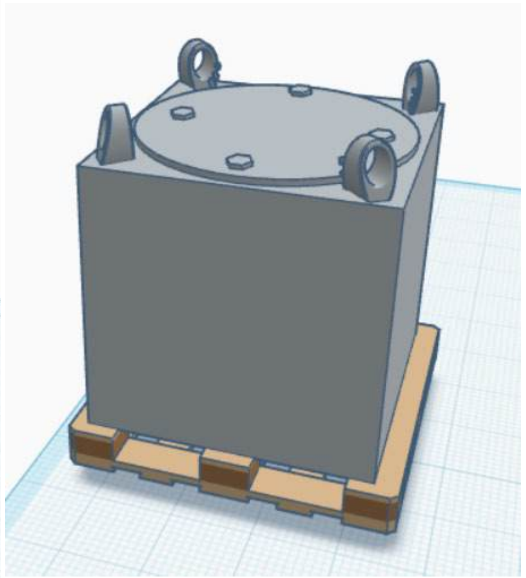
Melting Point (F)	~2500
Volume (cm^3)	100000
Weight (kg)	785
Cost of Steel ($)	785



Outer Dimensions Cross-section



Inner Dimensions Cross-section



Outer Appearance

In[21]:= $5\ cm * 50\ cm * 50\ cm$ (*Thicker Top Layer*)

Out[21]= $12\,500\ cm^3$

In[20]:= $(25\ cm * 50\ cm * 50\ cm) - \left(0.5 * \frac{4}{3} \pi * 19.1675\ cm^3\right)$ (*Bottom Half*)

Out[20]= $62\,459.9\ cm^3$

In[19]:= $4 * (25\ cm * 50\ cm * 5\ cm)$ (*Top Half Walls*)

Out[19]= $25\,000\ cm^3$

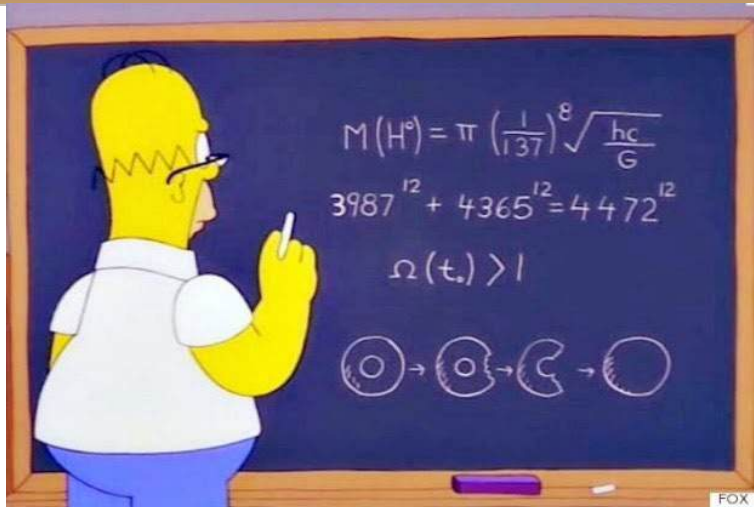
In[22]:= $(12\,500\ cm^3 + 25\,000\ cm^3 + 62\,459.8556\ cm^3) * 7.85 \frac{g}{cm^3} * \frac{1\ kg}{1000\ g}$

Out[22]= $784.685\ kg$

Price of Lead Shield ~ $696
Price of Steel Case ~ $785
Price of Steel Fittings ~ $200
Price of Bolts ~ $100

Pre-Assembly Price: ~ $1771

Design Summary



What does our design do?

- Passes all four major tests
- IAW NRC/DOE/DOT regulations
- Meets dose requirement of <200 mrem/hr
- Design cost is under budget
- Design is easy to transport

Future Work

- Model for different source geometries
- Iterate external container geometry to save more on materials
- Generate alloys to shield better for less volume and weight
- Scale design to carry more source/ activity

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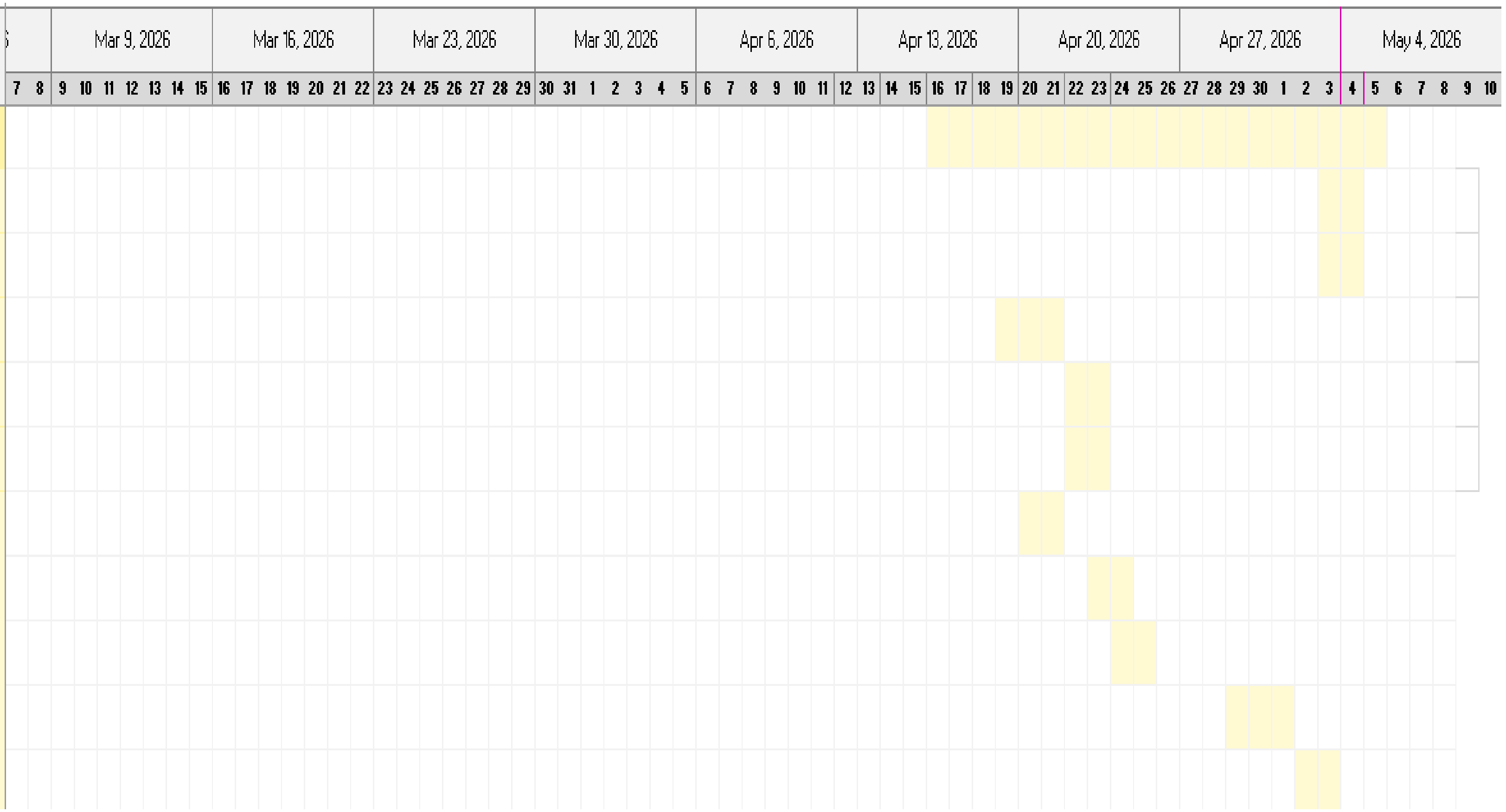
Gantt Chart

TASK	ASSIGNED TO	PROGRESS	START	END
Final Report Due (Milestone)		100%	4/16/26	5/5/26
Abstract	Hugh	100%	5/3/26	5/4/26
Introduction	Jack	100%	5/3/26	5/4/26
Managing Our Constraints	Jack	100%	4/19/26	4/21/26
Calculations- Source	Sam	100%	4/22/26	4/23/26
Calcultions- Material Selection	Sam	100%	4/22/26	4/23/26
Assumptions	Jack	100%	4/19/26	4/20/26
Optimization	Sam	100%	4/23/26	4/24/26
Design Drawings	Hugh	100%	4/24/26	4/25/26
Design Summary	Hugh	100%	4/29/26	5/1/26
Gantt Chart	Jack	100%	5/3/26	5/4/26



Sources/Acknowledgement

Price Calculation: https://www.metal.com/tungsten, https://www.metal.com/tin, https://www.metal.com/lead

Mohs Hardness Scale Reference: Metal Hardness Scale - A Chart of the Mohs Scale of Hardness - Alan's Factory Outlet

Thermal Resistance Reference: Thermal Conductivity of Metals and Alloys: Data Table & Reference Guide

Price Guide: Tin: 43.62 USD/kg, Lead: 2.08 USD/kg, Tungsten: 314.24 USD/kg

https://www.tinkercad.com/

https://thumbs.wbm.im/pw/small/0729a233ef6ef91d07522bf35ced6786.png

https://spydercrane.com/wp-content/uploads/2023/03/Lineup-PC-272x300.png

https://i.makeagif.com/media/8-13-2017/LpFV4H.gif

https://media1.giphy.com/media/v1.Y2lkPTZjMDliOTUycHViNWlmaGZvMDM4aGpwbWduYXpiOXgxand1NzJkYjd1ODRxdHZjZCZlcD12MV9naWZzX3NlYXJjaCZjdD1n/EOlw4aQjGBYkw/200w.gif

https://preview.redd.it/the-golden-posture-v0-3lz5kqsd885b1.jpg?auto=webp&s=3e438b0195697f4491f2d00179e467d8af41622b

https://usmetaldesigns.com/product/cb24-steel-crane-bolt-24in/

https://mepsinternational.com/gb/en/products/north-america-steel-prices

Poor Student Work

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Transporters Final Presentation

By Cadets Dolan, Grauberger, Macune, and Rodriguez

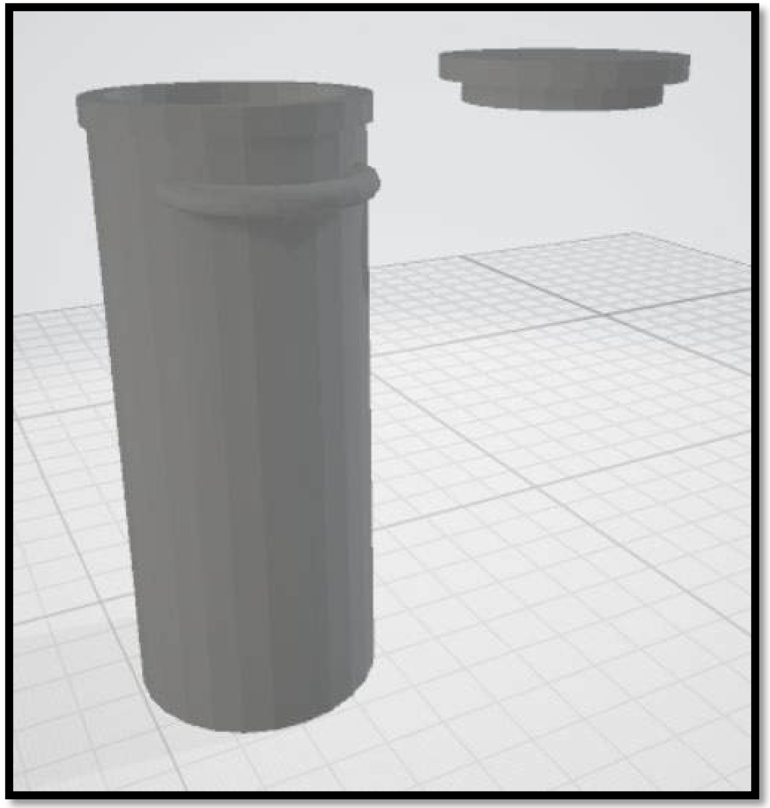
Problem Statement

Design a shipping container for a rod shaped 650-Ci Co-60 source that provides sufficient gamma radiation shielding to meet NRC/DOT transportation dose-rate limits: no more than 200 mrem/hour at the package surface and no more than 10 mrem/hour at 2 meters from the transport vehicle. The design will specify the container geometry, shielding material, needed wall thickness, dimensions, and handling features needed for safe loading, unloading, and transport of the source.

BLUF

We recommend a 21cm (diameter) 52cm (height) 95kg cylinder-container composed of lead, jacketed with 1cm of stainless-steel, and a detachable bolt-on lid.

Due to its tailored size for the given source, it is a more economical solution to transporting the given source at $3,951.



Limitations and Constraints

- Legal Shipping Dimensions (U.S. Highway Transport)
  - Maximum width: 8 ft 6 in
  - Maximum height: 13 ft 6 in
  - Typical trailer length: 48-53 ft
- Financial
  - Our financial constraints are not specified, but we still want to minimize the price of our design.
- Testing Limitations
  - We lack digital testing software.
  - We lack additional resources to test design.

Requirements and Assumptions

Requirements

- Withstand a 9 m drop onto a firm surface
- Withstand a 1 m drop onto a 6 in diameter steel spike
- Exposure to a temperature of 800°C for a duration of 30 minutes
- Submerged under 15 m of water for 8 hours
- Be light enough to not hinder the functions of the transport vehicle

Assumptions

- Co-60 comes contained in a rod-shape
- No shielding is provided by the trailer or pallet
- Buildup factor is assumed for the compound material

Design Description

Design Incorporation of Specified and Essential Requirements:
Our goal was to ensure that the design properly incorporated all of the specified and essential requirements that were asked from our client. Our design was able to do that by offering the proper shielding requirements.

Impact of Assumptions:
Our assumption of following the IAEA standards for safety helped us create a good guide as to what safety standards need to be followed when transporting radioactive material.

Constraints and Limitations:
Our engineering design knowledge falls short of what professional design companies would have optimized which limited our potential towards creating the best product. Making the storage device transportable was also a limitation as we could not make our dimensions unrealistic.

Other Key Factors:
Some other key factors we discussed was the safety and security of the design. We wanted to ensure that the safety of those transporting the Cobalt-60 was at the forefront of our design. Additionally, the security component was something we considered as well to ensure that the design could lock properly.

Material Requirements:
The material requirements for this project included a material that was good at shielding radiation. Additionally, we needed something that was durable to be able to withstand transportation errors.

Test Requirements:
Through our calculations on later slides, you will be able to see how we tested our overall design. There were certain modifications made throughout the process based on what our calculations gave us throughout the project.

Success of Design and Redesign Potential:
The success of our design is founded on the ability to shield a 650-Ci source of Cobalt-60 while creating a transportable container. Our results will be fed back into the redesign process by hopefully talking with industry experts on what we could do to improve while also refining our calculations to ensure we are maximizing every factor possible.

Future Work in the Field:
There is a lot of work left in this field to maximize the ability to transport highly radioactive substances. This includes experimenting with different materials and trying other transportation methods such as flight

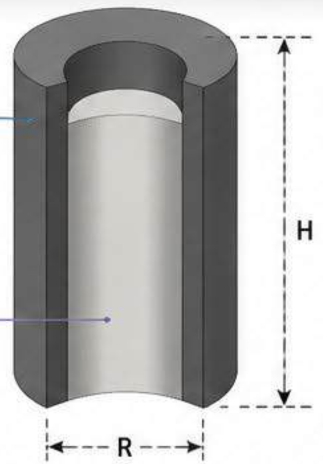
Decision Matrix

Criteria (1: worst - 5: best)	Effective shielding	Cost	Weight	Manufacturability	Durability	Toxicity	Total
Weight Factor	30	20	20	10	10	10	100
Option 1: Lead	4	4	3	5	5	2	3.8
Option 2: Tungsten	5	1	2	3	1	5	3
Option 3: Concrete	2	5	4	4	3	4	3.5

Based off the weighted decision matrix along with established industry standard, our team decided to use a stainless-steel enclosed lead shipping container to transport the Co-60 source.

Design Choice

EXAMPLE: COATED CYLINDRICAL SHIELD



Gamma rays enter from all directions.
Lead core attenuates. Steel coat protects and adds strength.

WHY LEAD ALONE IS NOT PREFERABLE

MECHANICALLY WEAK
Lead is soft (low tensile strength) and easily deforms, dents, or cracks under handling, impact, or load.

TOXIC EXPOSURE RISK
Uncoated lead surfaces can generate dust or be accidentally abraded, creating a toxic exposure hazard.

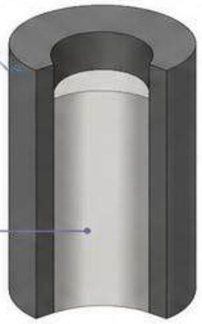
SURFACE DAMAGE RISK
Lead surfaces can corrode or be damaged, compromising long-term performance.

HIGH MAINTENANCE / REPLACEMENT COST
Damage or exposure issues can lead to higher maintenance, handling restrictions, and replacement cost.

DESIGN OPTION COMPARISON MATRIX

Design Option	Radiation Shielding (Effectiveness)	Cost (Material + Fabrication)	Structural Strength	Toxic Exposure Control	Overall Benefit
All Lead (Uncoated)	**High** Excellent attenuation	**Moderate** Lead material cost	**Low** Soft, prone to dents/cracks	**Poor** Lead surface exposed (toxic risk)	★☆☆☆☆ Not Preferred
All Steel/Iron	**Low–Moderate** Much more thickness needed	**High** More material and heavier	**High** Strong and durable	**Good** No toxic exposure	★★☆☆☆ Not Preferred
Lead Core with Thin Steel Coating	**High** Lead provides primary attenuation	**Good** Less steel used lower cost	**High** Steel adds strength	**High** Lead fully encapsulated	★★★★★ Best Option
Lead Core with Thick Steel Coating	**High** Similar attenuation to above	**Poor** More steel increases cost & weight	**High** Very strong	**High** Lead fully encapsulated	★★★☆☆ Works, but Not Optimal

RECOMMENDED DESIGN: LEAD CORE WITH THIN STEEL (IRON) COATING



- ✓ Achieves required attenuation with minimal total thickness
- ✓ Minimizes use of more expensive steel
- ✓ Provides mechanical strength and durability
- ✓ Prevents toxic lead exposure
- ✓ Best balance of performance, cost, and safety

KEY TAKEAWAY

Use lead for attenuation. Use steel (iron) wisely for strength and safety.

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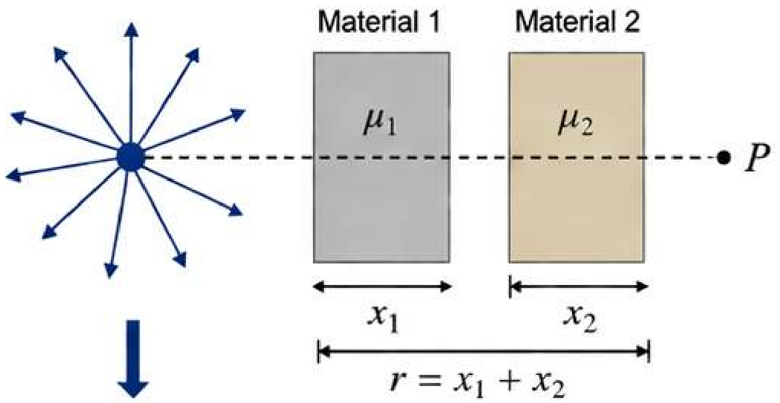
Computations

1. Geometry and Source

$$r = x_1 + x_2$$

$$I_0 = S = FA$$

$$G(r) = \frac{1}{4\pi r^2}$$



Unshielded intensity at distance r

$$I_{\text{unshielded}}(r) = I_0 G(r) = \frac{I_0}{4\pi r^2}$$

2. Attenuation Through Two Materials

After material 1 (thickness x_1):

$$I_1 = I_{\text{unshielded}}\, e^{-\mu_1 x_1}$$

After material 2 (thickness x_2):

$$I_2 = I_1\, e^{-\mu_2 x_2}$$

Substituting:

$$I_2 = \frac{I_0}{4\pi r^2} e^{-\mu_1 x_1} e^{-\mu_2 x_2}$$

$$I_2 = \frac{I_0}{4\pi r^2} e^{-(\mu_1 x_1 + \mu_2 x_2)}$$

$$I_2 = \frac{I_0}{4\pi (x_1 + x_2)^2} e^{-(\mu_1 x_1 + \mu_2 x_2)}$$

3. Including Buildup Factor (Constant)

$$I = B\,\frac{I_0}{4\pi (x_1 + x_2)^2} e^{-(\mu_1 x_1 + \mu_2 x_2)}$$

B = buildup factor (dimensionless), accounts for scattered radiation in the shielding materials.

4. Using $I_0 = S = FA$

$$I = B\,\frac{FA}{4\pi (x_1 + x_2)^2} e^{-(\mu_1 x_1 + \mu_2 x_2)}$$

This is the *transmitted intensity* behind two *shielding materials in series for an* isotropic source.

Computations

Constant / Parameter	Meaning / Description	Value	Units
Gamma Constant for ^{60}Co	Gamma constant for Cobalt-60. Relates the source strength and exposure rate at 1 meter (unshielded) in air.	**13.2**	mSv/(Ci·hr)
Source Activity (A)	Activity of the ^{60}Co gamma source.	**650**	Ci
Dose Rate Limit (D_{limit})	Maximum allowable dose rate at the point of interest.	**2**	mSv/hr
Mass Attenuation Coefficient Lead ($(\mu/\rho)_{Pb}$)	Mass attenuation coefficient for lead for ^{60}Co gamma rays.	**0.0569**	cm^2/g
Density of Lead (ρ_{Pb})	Density of lead.	**11.34**	g/cm^3
Attenuation Constant Lead (μ_{Pb})	Linear attenuation coefficient for lead. Calculated as $(\mu/\rho)_{Pb} \times \rho_{Pb}$.	**0.6455**	cm^{-1}
Mass Attenuation Coefficient Steel (Iron) ($(\mu/\rho)_{Fe}$)	Mass attenuation coefficient for steel (iron) for ^{60}Co gamma rays.	**0.0531**	cm^2/g
Density of Steel (Iron) (ρ_{Fe})	Density of steel (iron).	**7.87**	g/cm^3
Attenuation Constant Steel (Iron) (μ_{Fe})	Linear attenuation coefficient for steel (iron). Calculated as $(\mu/\rho)_{Fe} \times \rho_{Fe}$.	**0.4178**	cm^{-1}
Buildup Factor (B)	Buildup factor to account for scattered radiation in the shielding.	**6**	Dimensionless

Governing Equation Used

$$(\mu_{Pb}\, x) + (\mu_{Fe}\, y) + 2\ln(x+y) - \ln(B) = \ln\left(\frac{(\Gamma_{Co60} \times A)}{D_{limit}}\right)$$

Where:
- x = Lead shield thickness (cm)
- y = Steel (Iron) shield thickness (cm)
- Γ_{Co60} = 13.2 mSv/(Ci·hr)
- A = 650 Ci
- D_{limit} = 2 mSv/hr
- B = 6

Shielding Options to Reach Target Attenuation

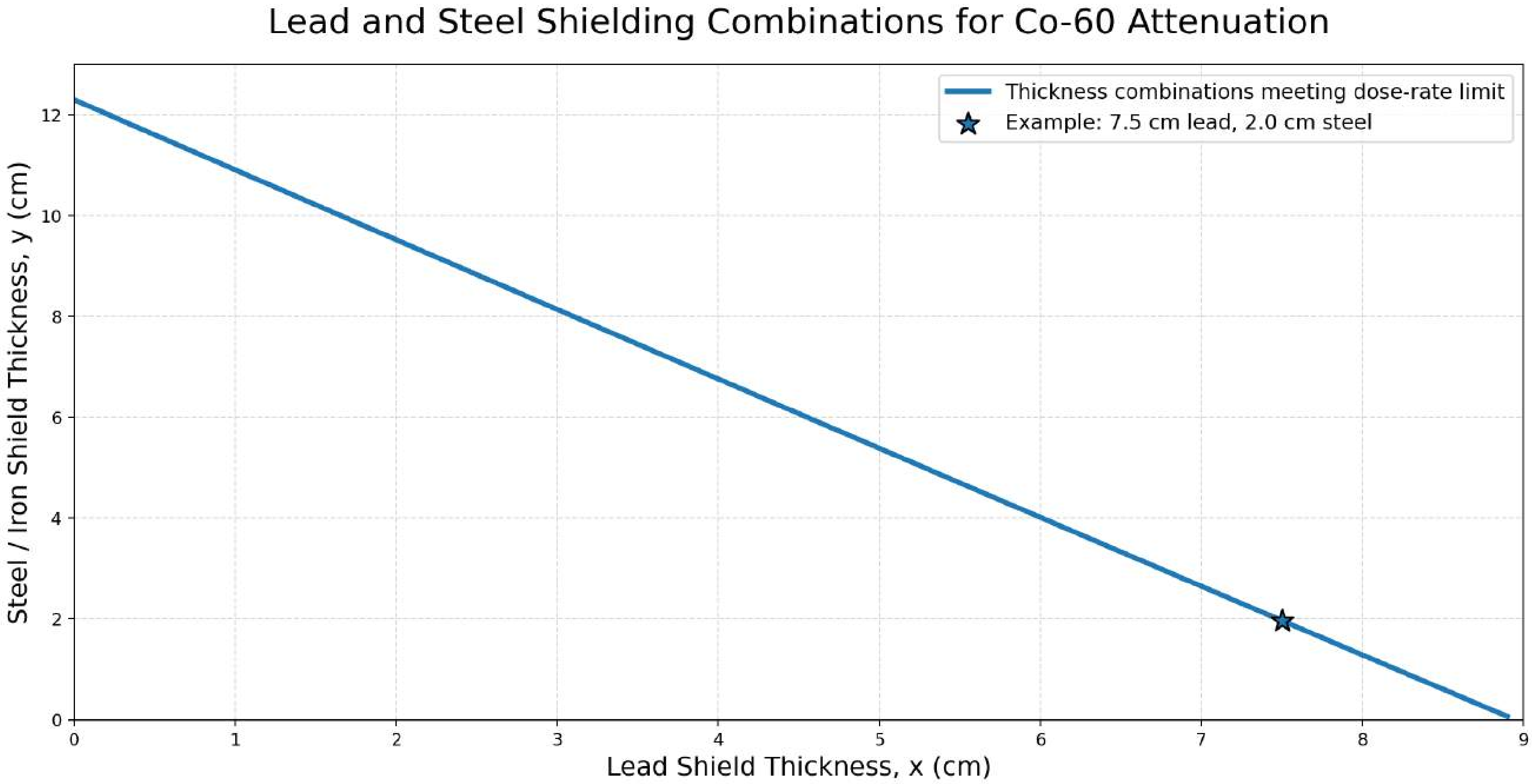
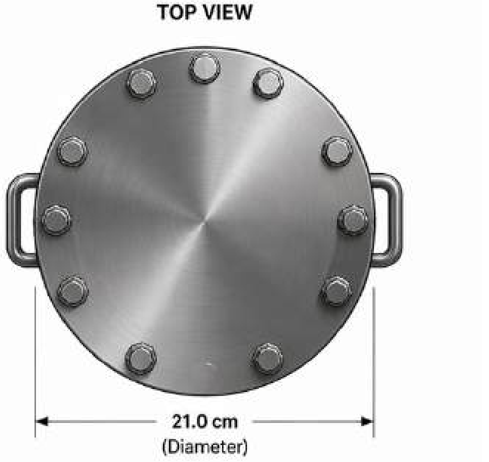


Figure 1. Combinations of lead and steel thickness that reduce the Co-60 dose rate to the desired limit using inverse-square attenuation, material attenuation, and buildup factor B = 6.

Sketches and Drawings

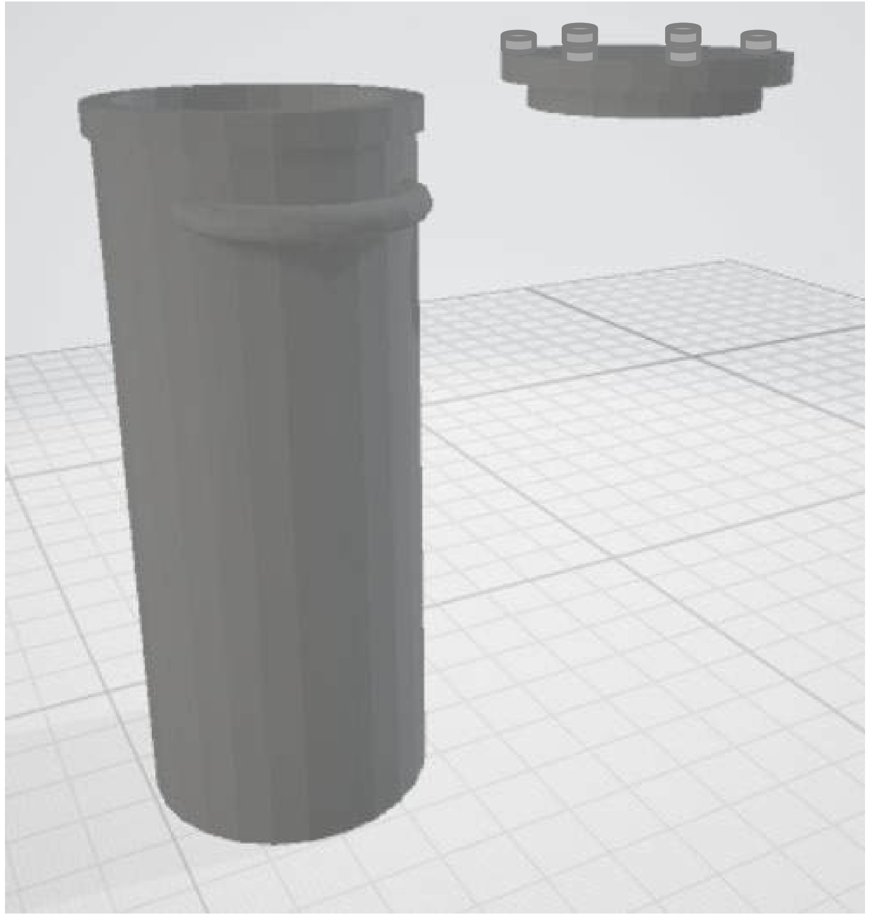


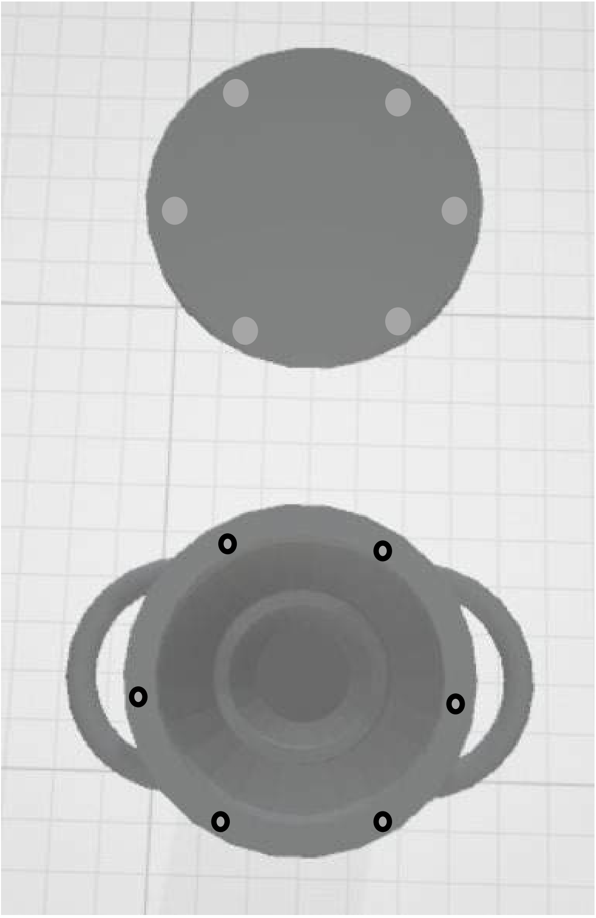
KEY DIMENSIONS	
Lead Shielding Radius (inner)	7.5 cm
Lead Shielding Height	20 cm
Outer Cylinder Radius	10.5 cm
Total Height (including lid)	52 cm
Outer Diameter	21.0 cm
Inner Liner Radius	(fits source capsule)
Material (Shell & Liner)	Stainless Steel
Shielding Material	Lead

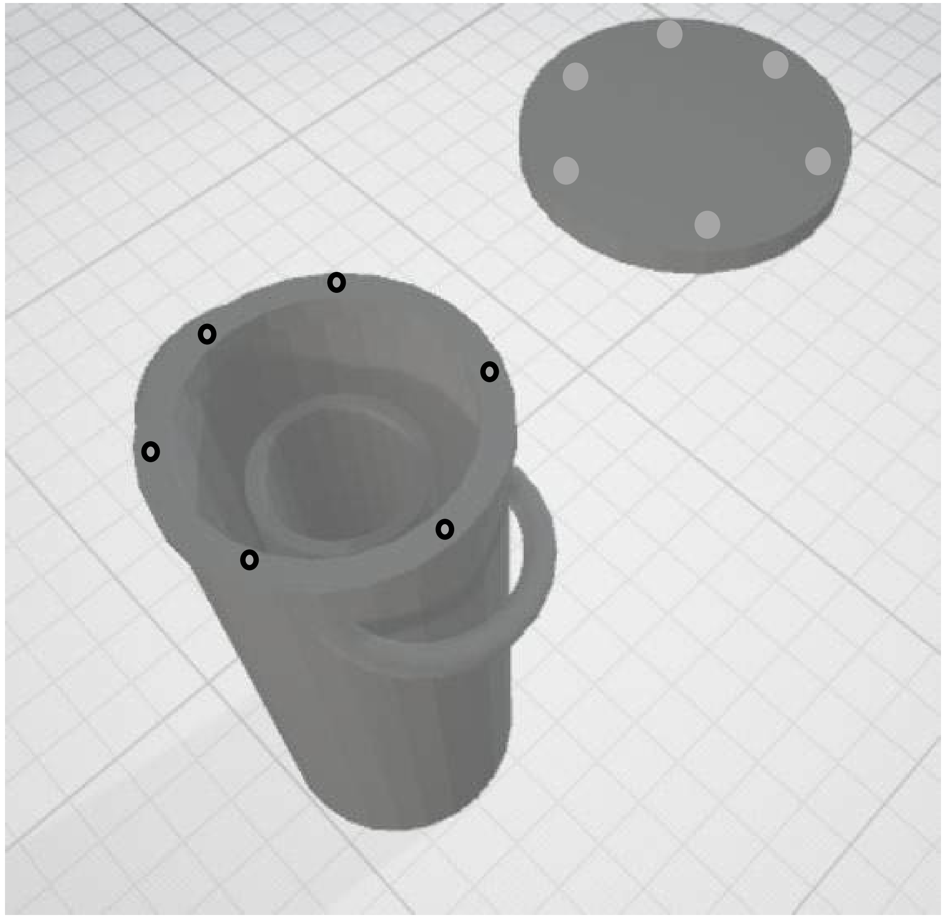
NOTES

- Lead shield is fully enclosed by stainless steel liner.
- All internal voids are lined with stainless steel.
- Lid secured with bolts.
- Co-60 source shown for representation.

Sketches and Drawings







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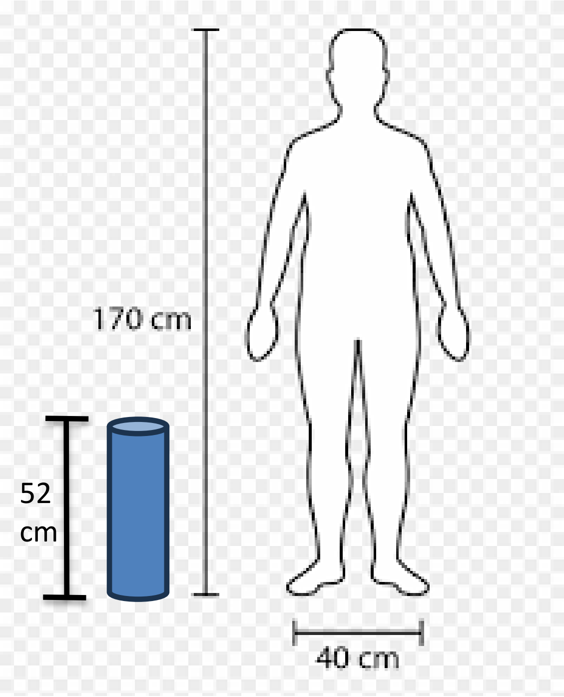
WEST POINT

Cost/Weight Breakdown

Component	Material	Outer radius	Inner radius	Height	Weight	Cost to Machine
Outer Lining	Stainless-steel	10.5 cm	9.5 cm	50.0 cm	24.9 kg	$600
Inner Lining	Stainless-steel	2.0 cm	1.0 cm	38.75 cm	2.9 kg	$1400
Lead Center	Lead	9.5 cm	2.0 cm	20 cm	61.5 kg	$1200
Lid	Stainless-Steel	11.0 cm	8.9 cm stepped radius	4.0 cm	5.5 kg	$750 (includes cost to thread bolt holes)
Bolts	Zinc plated hex bolt	0.3175 cm	N/A	1.905 cm	N/A	$1.08 (6x $0.18/bolt)

Total Cost and Weight

Given the costing estimates, the total price to manufacture the vessel is approximately $3,951.00, and the total weight is approximately 94.8 kg or 208.6 lbs.

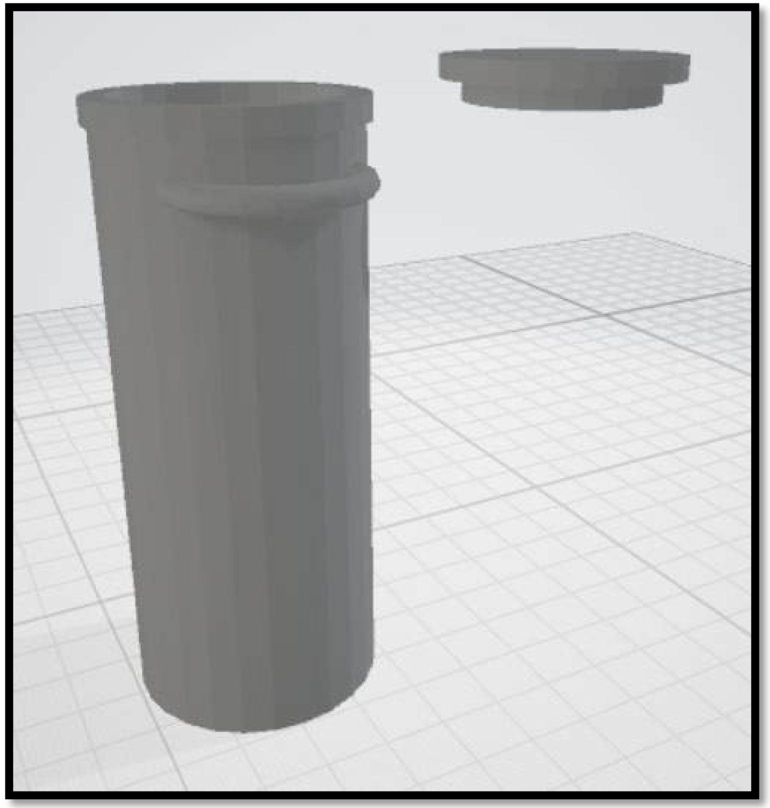


Implications

- Size
  - Allows for transportation of more than one source at a time on a single freight trailer
  - Light weight, able to be moved by 2-4 workers
- Convenience
  - Allows the company to purchase smaller quantities as needed without significant administrative burden.
  - Avoid use of crane
- Safety
  - Lower risk of mechanical accident
  - Smaller handling teams reduces exposure

We recommend a 21cm (diameter) 52cm (height) 95kg cylinder-container composed of lead, jacketed with 1cm of stainless-steel, and a detachable bolt-on lid.

Due to its tailored size for the given source, it is a more economical solution to transporting the given source at $3,951.



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Bibliography

ChatGPT: I used the following prompt into chatGPT. “I need a 3d model of the vessel. It is a stainless steel lined lead interior to house the cobalt. the total height is 52 cm including the lid. The lead cylinder is 7.5 cm radius with a height of 20 cm. The outer cylinder is 10.5 cm in radius” This allowed me to obtain a useful 3d product of the vessel used in transporting the product. I utilized this AI by giving it the dimensions we calculated for IPR2 and the hand sketches and then made a more refined model.

ChatGPT: I used the following prompt into chatGPT. "Given these dimensions, can you give me an accurate cost estimate to cast the two cylinders in stainless steel?" I then uploaded my CAD model into the AI and asked it to use industry standards to generate cost and weight breakdowns. I tried to get real quotes from manufacturing companies, but it proved to be a fruitless endeavor. ChatGPT gave our group reasonable estimations. https://chatgpt.com/c/69f7ce0f-18b8-83ea-a586-4d0e54bb56de.

We used Anderson Trucking Service to gather the dimension limits of freight transportation. Legal Freight Limits: Trucking Industry Weight, Width, Height, and Length Limits

The NRC was used to provide the requirements listed in this presentation.

Uptegrove, Yamato CDT C-2 '26. Assistance given to author, verbal discussion. I had no prior CAD experience, but CDT Uptegrove has lots of prior CAD experience. I asked him to help me design the model in CAD given all the dimensions. I inputted all the design criteria and CDT Uptegrove made it into a CAD prototype. West Point, NY. 3 May. 2026.

Computational Code Files

```
GammaConstantCo60 = 13.2; (*mSv/(Ci·hr)*)
A = 650; (*Ci*)
Dlimit = 2; (*mSv/hr*)
muOverRho1 = 0.0569;
rho1 = 11.34;
mu1 = muOverRho1 * rho1;
muOverRho2 = 0.0531;
rho2 = 7.87;
mu2 = muOverRho2 * rho2;
B = 6;
eq = (mu1 x) + (mu2 y) + 2 Log[x + y] - Log[B] == Log[GammaConstantCo60 * A / Dlimit];
Solve[eq, x]
Plot[y /. Solve[eq, y][[1]], {x, 0, 8.9}, AxesLabel → {"Shield Thickness lead(cm)", "Shield Thickness Steel (Iron)(cm)"}]
```

Thank you for your time, Questions?